



US012404340B2

(12) **United States Patent**
Pons et al.

(10) **Patent No.: US 12,404,340 B2**

(45) **Date of Patent: Sep. 2, 2025**

(54) **ANTIBODIES AGAINST
SIGNAL-REGULATORY PROTEIN ALPHA
AND METHODS OF USE**

(71) Applicant: **ALX Oncology Inc.**, South San Francisco, CA (US)

(72) Inventors: **Jaume Pons**, San Francisco, CA (US); **Bang Janet Sim**, South San Francisco, CA (US); **Hong Wan**, Foster City, CA (US); **Tracy Chia-Chien Kuo**, San Carlos, CA (US); **Steven Elliot Kauder**, San Mateo, CA (US); **William Don Harriman**, Alameda, CA (US); **Shelley Izquierdo**, Berkeley, CA (US)

(73) Assignee: **ALX Oncology Inc.**, South San Francisco, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 956 days.

(21) Appl. No.: **17/337,180**

(22) Filed: **Jun. 2, 2021**

(65) **Prior Publication Data**

US 2023/0018821 A1 Jan. 19, 2023

Related U.S. Application Data

(62) Division of application No. 15/710,798, filed on Sep. 20, 2017, now Pat. No. 11,401,338.

(60) Provisional application No. 62/515,480, filed on Jun. 5, 2017, provisional application No. 62/397,752, filed on Sep. 21, 2016.

(51) **Int. Cl.**

C07K 16/28 (2006.01)

A61K 39/00 (2006.01)

G01N 33/68 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 16/2896** (2013.01); **C07K 16/2803** (2013.01); **C07K 16/2818** (2013.01); **G01N 33/6854** (2013.01); **A61K 2039/505** (2013.01); **A61K 2039/507** (2013.01); **C07K 16/2827** (2013.01); **C07K 2317/23** (2013.01); **C07K 2317/24** (2013.01); **C07K 2317/30** (2013.01); **C07K 2317/31** (2013.01); **C07K 2317/34** (2013.01); **C07K 2317/52** (2013.01); **C07K 2317/56** (2013.01); **C07K 2317/565** (2013.01); **C07K 2317/622** (2013.01); **C07K 2317/76** (2013.01); **C07K 2317/92** (2013.01); **G01N 2333/70503** (2013.01); **G01N 2333/70596** (2013.01); **G01N 2500/02** (2013.01)

(58) **Field of Classification Search**

CPC **A61K 2039/505**; **C07K 16/2896**; **C07K 16/2803**; **G01N 33/6854**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

RE30,985 E	6/1982	Cartaya
4,560,655 A	12/1985	Baker
4,657,866 A	4/1987	Kumar
4,676,980 A	6/1987	Segal et al.
4,767,704 A	8/1988	Cleveland et al.
4,816,567 A	3/1989	Cabilly et al.
4,927,762 A	5/1990	Darfler
5,122,469 A	6/1992	Mather et al.
5,545,806 A	8/1996	Lonberg et al.
5,545,807 A	8/1996	Surani et al.
5,569,825 A	10/1996	Lonberg et al.
5,571,894 A	11/1996	Wels et al.
5,587,458 A	12/1996	King et al.
5,625,126 A	4/1997	Lonberg et al.
5,633,425 A	5/1997	Lonberg et al.
5,641,870 A	6/1997	Rinderknecht et al.
5,661,016 A	8/1997	Lonberg et al.
5,731,168 A	3/1998	Carter et al.
5,869,046 A	2/1999	Presta et al.

(Continued)

FOREIGN PATENT DOCUMENTS

EP	2242512 B1	4/2016
EP	3482772 A1	6/2017

(Continued)

OTHER PUBLICATIONS

Sim et al. (MAbs. Aug./Sep. 2019; 11 (6): 1036-1052).*

(Continued)

Primary Examiner — Janet L Epps-Smith

Assistant Examiner — Nada Ahmed Mahmoud Elmansy

(74) *Attorney, Agent, or Firm* — MORRISON & FOERSTER LLP

(57)

ABSTRACT

Provided herein, inter alia, are isolated antibodies that bind an extracellular domain of a human SIRP- α v1 polypeptide (e.g., the D1 domain), an extracellular domain of a human SIRP- α v2 polypeptide, or both. In some embodiments, the antibodies also bind an extracellular domain of a monkey SIRP- α polypeptide, an extracellular domain of a mouse SIRP- α polypeptide, an extracellular domain of a human SIRP- β polypeptide, and/or an extracellular domain of a human SIRP- γ polypeptide. In some embodiments, the antibodies block or do not binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, while in some embodiments, the antibodies reduce the affinity of a human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. Further provided herein are methods, polynucleotides, vectors, and host cells related thereto.

37 Claims, 85 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited**U.S. PATENT DOCUMENTS**

6,075,181 A 6/2000 Kucherlapati et al.
 6,143,559 A 11/2000 Michael et al.
 6,150,584 A 11/2000 Kucherlapati et al.
 6,180,370 B1 1/2001 Queen et al.
 6,248,516 B1 6/2001 Winter et al.
 7,575,893 B2 8/2009 Simmons
 8,088,896 B2 1/2012 Tesar et al.
 8,475,792 B2 7/2013 Dall'Acqua et al.
 8,592,644 B2 11/2013 Harriman et al.
 9,151,760 B2 10/2015 Weissman et al.
 9,352,037 B2 5/2016 van den Berg
 9,380,769 B2 7/2016 Leighton et al.
 9,399,682 B2 7/2016 Jaiswal et al.
 9,493,575 B2 11/2016 Jaiswal et al.
 9,605,076 B2 3/2017 Jaiswal et al.
 9,611,329 B2 4/2017 Jaiswal et al.
 9,623,079 B2 4/2017 Willingham et al.
 9,624,305 B2 4/2017 Jaiswal et al.
 9,765,143 B2 9/2017 Jaiswal et al.
 9,771,428 B2 9/2017 Weiskopf et al.
 9,790,275 B2 10/2017 van den Berg
 9,920,122 B2 3/2018 van den Berg
 10,064,925 B2 9/2018 Tseng et al.
 10,081,680 B2 9/2018 Weiskopf et al.
 10,184,004 B2 1/2019 Weiskopf et al.
 10,301,387 B2 5/2019 Willingham et al.
 10,329,354 B2 6/2019 Leeper et al.
 10,344,094 B2 7/2019 Weissman et al.
 10,611,842 B2 4/2020 Liu et al.
 10,618,976 B2 4/2020 Weissman et al.
 10,723,803 B2 7/2020 Weiskopf et al.
 10,780,117 B2 9/2020 Weissman et al.
 10,781,256 B2 9/2020 Weiskopf et al.
 10,851,164 B2 12/2020 Van Eenennaam et al.
 11,242,404 B2 2/2022 Pons et al.
 11,292,850 B2 4/2022 Pons et al.
 11,401,338 B2 8/2022 Pons et al.
 11,939,393 B2 3/2024 Pons et al.
 2002/0114807 A1 8/2002 Berg et al.
 2003/0026803 A1 2/2003 Barclay
 2003/0054415 A1 3/2003 Buhning et al.
 2004/0147731 A1 7/2004 Parkos
 2004/0213792 A1 10/2004 Clemmons et al.
 2006/0263356 A1 11/2006 Endl et al.
 2008/0160013 A1 7/2008 Clemmons et al.
 2010/0215640 A1 8/2010 Clemmons et al.
 2010/0239578 A1 9/2010 Danska et al.
 2013/0189253 A1 7/2013 Danska et al.
 2014/0065169 A1 3/2014 Jaiswal et al.
 2014/0161825 A1 6/2014 Jaiswal et al.
 2014/0242095 A1 8/2014 Wang et al.
 2016/0186150 A1 6/2016 Deming et al.
 2016/0333093 A1 11/2016 Weiskopf et al.
 2017/0114134 A1 4/2017 Clemmons et al.
 2017/0151282 A1 6/2017 Discher et al.
 2017/0247464 A1 8/2017 Poirier et al.
 2018/0105600 A1 4/2018 Pons et al.
 2018/0155424 A1 6/2018 van den Berg
 2018/0214524 A1 8/2018 Weissman et al.
 2018/0244748 A1 8/2018 Gill et al.
 2018/0312587 A1 11/2018 Van Eenennaam et al.
 2018/0312600 A1 11/2018 Poirier et al.
 2019/0119396 A1 4/2019 Liu et al.
 2019/0127477 A1 5/2019 Poirier et al.
 2019/0134089 A1 5/2019 Liu et al.
 2019/0153095 A1 5/2019 Matozaki et al.
 2019/0233515 A1 8/2019 Jaiswal et al.
 2019/0275150 A1 9/2019 Pincetic et al.
 2019/0322986 A1 10/2019 Keller et al.
 2019/0352419 A1 11/2019 Pons et al.
 2019/0359707 A1 11/2019 Pincetic et al.
 2020/0102387 A1 4/2020 Abbasian et al.
 2020/0129557 A1 4/2020 Shizuru et al.
 2020/0223923 A1 7/2020 Schnorr et al.
 2020/0262918 A1 8/2020 Liu et al.

2020/0297842 A1 9/2020 Puro et al.
 2020/0354469 A1 11/2020 Weiskopf et al.
 2022/0002434 A1 1/2022 Pons et al.
 2022/0324996 A1 10/2022 Pons et al.

FOREIGN PATENT DOCUMENTS

EP 3308641 B1 7/2019
 EP 2931752 B1 8/2019
 EP 3180363 B1 9/2019
 EP 3186395 B1 9/2019
 EP 3421601 B1 12/2019
 EP 3043181 B1 4/2020
 EP 3209691 B1 7/2020
 EP 3209769 B1 8/2020
 WO WO-1987/000195 A1 1/1987
 WO WO-1990/003430 A1 4/1990
 WO WO-1991/010741 A1 7/1991
 WO WO-1993/016185 A2 8/1993
 WO WO-1994/004690 A1 3/1994
 WO WO-1996/033735 A1 10/1996
 WO WO-1996/034096 A1 10/1996
 WO WO-1997/048723 A2 12/1997
 WO WO-1998/024893 A2 6/1998
 WO WO-1999/040940 A1 8/1999
 WO WO-2005/014653 A2 2/2005
 WO WO-2009/111014 A2 9/2009
 WO WO-2011/019844 A1 2/2011
 WO WO-2012/162422 A2 11/2012
 WO WO-2013/056352 A1 4/2013
 WO WO-2013/059159 A1 4/2013
 WO WO-2015/138600 A2 9/2015
 WO WO-2016/063233 A1 4/2016
 WO WO-2017/178653 A2 10/2017
 WO WO-2018/057669 A1 3/2018
 WO WO-2018190719 A2 10/2018
 WO WO-2018/210793 A2 11/2018
 WO WO-2018210795 A1 11/2018
 WO WO-2020013170 A1 1/2020
 WO WO-2020099653 A1 5/2020

OTHER PUBLICATIONS

Araghi et al. (J. Feline Med. Surg. Apr. 2014; 16 (4): 265-74).
 Lee et al. (J. Immunol. Dec. 1, 2007; 179 (11): 7741-50).
 Liu et al. (J. Mol. Biol. Jan. 19, 2007; 365 (3): 680-93).
 Barclay et al. (Annu. Rev. Immunol. 2014; 32: 25-50).
 Winkler et al. (J. Immunol. 2000; 165: 4505-14).
 "Functional Elements on SIRPalpha IgV Domain Mediate Cell Surface Binding to CD47," Liu et al., J. Mol. Biology, 365, 680-693 (Year: 2007).
 Almagro and Fransson (2008). "Humanization of antibodies," Front. Biosci. 13:1619-1633.
 Araghi et al., (2014). "Flow cytometric immunophenotyping of feline bone marrow cells and haematopoietic progenitor cells using anti-human antibodies," J. Feline Med. Surg. 16(4):265-74.
 Baca et al. (1997). "Antibody humanization using monovalent phage display," J. Biol. Chem., 272(16):10678-10684.
 Barclay et al., (2014). "The interaction between signal regulatory protein alpha (SIRPa) and CD47: structure, function, and therapeutic target," Annu. Rev. Immunol., 32:25-50.
 Barclay, A.N. and Brown, M.H. (2006). "The SIRP family of receptors and immune regulation," Nat. Rev. Immunol., 6(6):457-464.
 Barnes et al. (1980). "Methods for growth of cultured cells in serum-free medium," Anal. Biochem. 102:255-70.
 BioLegend. (2016). "Purified anti-human CD172a/b (SIRPa/b) Antibody," <https://www.biolegend.com/en-gb/global-elements/pdf-popup/purified-anti-human-cd172a-b-sirpalpha-beta-antibody-4028?filename=Purified%20anti-human%20CD172ab%20SIRPalpha%20Antibody.pdf&pdfgen=true>, Retrieved: Dec. 15, 2017.
 Boerner et al. (1991). "Production of antigen-specific human monoclonal antibodies from in vitro-primed human splenocytes," J. Immunol., 147(1):86-95.

(56)

References Cited

OTHER PUBLICATIONS

- Bravman, T. et al. (2006). "Exploring "one-shot" kinetics and small molecule analysis using the ProteOn XPR36 array biosensor," *Anal. Biochem.*, 358(2):281-288.
- Brennan et al. (1985). "Preparation of bispecific antibodies by chemical recombination of monoclonal immunoglobulin G1 fragments," *Science*, 229(4708):81-3.
- Bruggemann et al. (1993). "Designer mice: the production of human antibody repertoires in transgenic animals," *Year in Immunol.*, 7:33-40.
- Carter et al. (1992). "High level *Escherichia coli* expression and production of a bivalent humanized antibody fragment," *Biotechnology*, 10(2):163-167.
- Carter et al. (1992). "Humanization of an anti-p185HER2 antibody for human cancer therapy," *Proc. Natl. Acad. Sci. USA*, 89(10):4285-9.
- Champe et al. (1995). "Monoclonal antibodies that block the activity of leukocyte function-associated antigen 1 recognize three discrete epitopes in the inserted domain of CD11a," *J. Biol. Chem.*, 270(3):1388-1394.
- Chin et al., (2008). "Immune intervention with monoclonal antibodies targeting CD152 (CTLA-4) for autoimmune and malignant diseases," *Chang Gung Med J.*, 31(1):1-15.
- Chothia and Lesk, (1987). "Canonical structures for the hypervariable regions of immunoglobulins," *J. Mol. Biol.*, 196(4):901-917.
- Clackson et al. (1991). "Making antibody fragments using phage display libraries," *Nature*, 352: 624-628.
- Fellouse (2004). "Synthetic antibodies from a four-amino-acid code: a dominant role for tyrosine in antigen recognition," *Proc. Natl. Acad. Sci. USA*, 101(34):12467-12472.
- Fishwild et al. (1996). "High-avidity human IgG kappa monoclonal antibodies from a novel strain of minilocus transgenic mice," *Nature Biotechnol.*, 14(7):845-851.
- George et al. (1998). "Differential effects of anti-beta2-glycoprotein I antibodies on endothelial cells and on the manifestations of experimental antiphospholipid syndrome," *Circulation*, 97:900-906.
- Glanville, J. et al. (2009). "Precise determination of the diversity of a combinatorial antibody library gives insight into the human immunoglobulin repertoire," *Proc. Natl. Acad. Sci.*, 106(48):20216-20221.
- Graham et al. (1977). "Characteristics of a human cell line transformed by DNA from human adenovirus type 5," *J. Gen. Virol.*, 36(1):59-74.
- Griffiths et al. (1993). "Human anti-self antibodies with high specificity from phage display libraries," *EMBO J.*, 12(2):725-734.
- Gruber et al. (1994). "Efficient tumor cell lysis mediated by a bispecific single chain antibody expressed in *Escherichia coli*," *J. Immunol.*, 152(11):5368-74.
- Ham et al. (1979). "Media and growth requirements," *Meth. Enz.*, 58:44-93.
- Hamers-Casterman et al. (1993). "Naturally occurring antibodies devoid of light chains," *Nature*, 363(6428):446-448.
- Hatherley, D. et al. (2007). "The structure of the macrophage signal regulatory protein alpha (SIRPalpha) inhibitory receptor reveals a binding face reminiscent of that used by T cell receptors," *J. Biol. Chem.*, 282(19):14567-75.
- Hatherley, D. et al. (2008). "Paired receptor specificity explained by structures of signal regulatory proteins alone and complexed with CD47," *Mol. Cell*, 31(2):266-77.
- Hatherley, D. et al. (2009). "Structure of signal-regulatory protein alpha: a link to antigen receptor evolution," *J. Biol. Chem.*, 284(39):26613-9.
- Hollinger et al. (1993). "'Diabodies': small bivalent and bispecific antibody fragments," *Proc. Natl. Acad. Sci. USA*, 90(14):6444-6448.
- Hongo et al. (1995). "Development and characterization of murine monoclonal antibodies to the latency-associated peptide of transforming growth factor beta 1," *Hybridoma*, 14(3):253-260.
- Hoogenboom and Winter (1992). "By-passing signaling. Human antibodies from synthetic repertoires of germline VH gene segments rearranged in vitro," *J. Mol. Biol.*, 227(2):381-388.
- Hoogenboom, H.R. (2002). "Overview of antibody phage-display technology and its applications," *Methods in Molecular Biology*, 178:1-37.
- Hudson et al. (2003). "Engineered antibodies," *Nat. Med.*, 9(1):129-134.
- International Search Report and Written Opinion directed to PCT Application No. PCT/US2017/052592, dated Dec. 20, 2017, 22 pages.
- International Search Report and Written Opinion directed to PCT Application No. PCT/US2019/023238, dated May 20, 2019, 6 pages.
- Jakovovits et al. (1993). "Analysis of homozygous mutant chimeric mice: deletion of the immunoglobulin heavy-chain joining region blocks B-cell development and antibody production," *Proc. Natl. Acad. Sci. USA*, 90(36):2551-5.
- Jakovovits et al. (1993). "Germ-line transmission and expression of a human-derived yeast artificial chromosome," *Nature*, 362(6417):255-258.
- Jayaram, N. et al. (2012). "Germline VH/VL pairing in antibodies," *Protein Eng. Des. Sel.*, 25(10):523-529.
- Jiang et al., (2005). "A Novel Peptide Isolated from a Phage Display Peptide Library with Trastuzumab Can Mimic Antigen Epitope of HER-2," *J. Biol. Chem.*, 280(6):4656-4662.
- Jones et al. (1986). "Replacing the complementarity-determining regions in a human antibody with those from a mouse," *Nature*, 321(6069):522-525.
- Kharitonov, A. et al. (1997). "A family of proteins that inhibit signaling through tyrosine kinase receptors," *Nature*, 386(6621):181-6.
- Kim, E.J. et al. (2013). "SHPS-1 and a synthetic peptide representing its ITIM inhibit the MyD88, but not TRIF, pathway of TLR signaling through activation of SHP and PI3K in THP-1 cells," *Inflammation Research*, 62(4):377-86.
- Kohler and Milstein, (1975). "Continuous cultures of fused cells secreting antibody of predefined specificity," *Nature*, 256(5517):495-97.
- Kostelny et al. (1992). "Formation of a bispecific antibody by the use of leucine zippers," *J. Immunol.*, 148(5):1547-1553.
- Krieg et al. (2005). "Functional analysis of B and T lymphocyte attenuator engagement on CD4+ and CD8+ T cells," *Journal of Immunology* (Baltimore, MD. : 1950), vol. 175, No. 10, pp. 6420-6427, ISSN: 0022-1767.
- Lee et al. (2004). "Bivalent antibody phage display mimics natural immunoglobulin," *J. Immunol. Methods*, 284(1-2):119-132.
- Lee et al. (2004). "High-affinity human antibodies from phage-displayed synthetic Fab libraries with a single framework scaffold," *J. Mol. Biol.*, 340(5):1073-1093.
- Lee, W. Y. et al. (2010). "The role of cis dimerization of signal regulatory protein alpha (SIRPalpha) in binding to CD47," *J. Biol. Chem.*, 285(49):37953-63.
- Lee, W.Y. et al. (2007). "Novel structural determinants on SIRP alpha that mediate binding to CD47," *J. Immunol.*, 179(11):7741-50.
- Li et al. (2006). "Optimization of humanized IgGs in glycoengineered *Pichia pastoris*," *Nat. Biotech.*, 24(2):210-215.
- Liu et al., (2007). "Functional elements on SIRPalpha IgV domain mediate cell surface binding to CD47," *J. Mol. Biol.*, 365(3):680-93.
- Lloyd, C. et al. (2009). "Modelling the human immune response: performance of a 1011 human antibody repertoire against a broad panel of therapeutically relevant antigens," *Protein Eng. Des. Sel.*, 22(3):159-168.
- Lonberg (2008). "Fully human antibodies from transgenic mouse and phage display platforms," *Curr. Opin. Immunol.*, 20(4):450-459.
- Lonberg and Huszar (1995). "Human antibodies from transgenic mice," *Intern. Rev. Immunol.*, 13(1):65-93.
- Lonberg et al. (1994). "Antigen-specific human antibodies from mice comprising four distinct genetic modifications," *Nature*, 368(6474):856-859.

(56)

References Cited

OTHER PUBLICATIONS

- Ludwig et al. (2017). "Mechanisms of Autoantibody-Induced Pathology," *Front Immunol*, 8: 603.
- Majeti, R. et al. (2009). "CD47 is an adverse prognostic factor and therapeutic antibody target on human acute myeloid leukemia stem cells," *Cell*, 138(2):286-99.
- Marks et al. (1992). "By-passing immunization. Human antibodies from V-gene libraries displayed on phage," *J. Mol. Biol.*, 222(3): 581-597.
- Marks et al. (1992). "By-passing immunization: building high affinity human antibodies by chain shuffling," *Biotechnology*, 10(7):779-783.
- Mather (1980). "Establishment and characterization of two distinct mouse testicular epithelial cell lines," *Biol. Reprod.*, 23(1):243-252.
- Mather et al. (1982). "Culture of testicular cells in hormone-supplemented serum-free medium," *Annals N.Y. Acad. Sci.*, 383:44-68.
- Mettler Izquierdo, S. et al. (2016). "High-efficiency antibody discovery achieved with multiplexed microscopy," *Microscopy (Oxf)*, 341-52.
- Morimoto et al. (1992). "Single-step purification of F(ab')₂ fragments of mouse monoclonal antibodies (immunoglobulins G1) by hydrophobic interaction high performance liquid chromatography using TSKgel Phenyl-5PW," *Journal of Biochemical and Biophysical Methods*, 24(1-2):107-117.
- Morrison (1994). "Immunology. Success in specification," *Nature*, 368(6474):812-813.
- Morrison et al. (1984). "Chimeric human antibody molecules: mouse antigen-binding domains with human constant region domains," *Proc. Natl. Acad. Sci. USA*, 81(21):6851-6855.
- Murata, Y. et al. (2014). The CD47-SIRPα signaling system: its physiological roles and therapeutic application, *The Journal of Biochemistry*, 155(6):335-344.
- Nakaishi, A. et al. (2008). "Structural insight into the specific interaction between murine SHPS-1/SIRP alpha and its ligand CD47," *J. Mol. Biol.* 375:650-60.
- Nettleship, J. et al. (2013). "Crystal structure of signal regulatory protein gamma (SIRPγ) in complex with an antibody Fab fragment," *BMC Structural Biology*, 13(13):1-8.
- Neuberger (1996). "Generating high-avidity human Mabs in mice," *Nature Biotechnol.*, 14(7):826.
- Ochi, F. et al. (1997). "Epidermal growth factor stimulates the tyrosine phosphorylation of SHPS-1 and association of SHPS-1 with SHP-2, a SH2 domain-containing protein tyrosine phosphatase," *Biochem. Biophys. Res. Commun.*, 239(2):483-7.
- Presta (1992). "Antibody engineering," *Curr. Op. Struct. Biol.* 2:593-596.
- Presta et al. (1993). "Humanization of an antibody directed against IgE," *J. Immunol.*, 151(5):2623-32.
- Riechmann et al. (1988). "Reshaping human antibodies for therapy," *Nature*, 332(6162):323-327.
- Riemer et al., (2005). "Matching of trastuzumab (Herceptin) epitope mimics onto the surface of Her-2/neu—a new method of epitope definition," *Mol. Immunol.*, 42:1121-1124.
- Rudikoff et al. (1982). "Single amino acid substitution altering antigen-binding specificity," *Proc. Nat. Acad. Sci.*, vol. 79, pp. 1979-1983, *Immunology*.
- Seiffert, M. et al. (1999). "Human Signal-Regulatory Protein Is Expressed on Normal, But Not on Subsets of Leukemic Myeloid Cells and Mediates Cellular Adhesion Involving Its Counterreceptor CD47," *Blood*, 94(11):3633-3643.
- Seiffert, M. et al. (2001). "Signal-regulatory protein a (SIRPα) but not SIRPβ is involved in T-cell activation, binds to CD47 with high affinity, and is expressed on immature CD34+CD38-hematopoietic cells," *Blood*, 97(9):2741-2749.
- Sheriff et al. (1996). "Redefining the minimal antigen-binding fragment," *Nature Struct. Biol.*, 3(9):733-736.
- Sidhu et al. (2004). "Phage-displayed antibody libraries of synthetic heavy chain complementarity determining regions," *J. Mol. Biol.*, 338(2): 299-310.
- Sim et al. (2019). "Discovery of high affinity, pan-allelic, and pan-mammalian reactive antibodies against the myeloid checkpoint receptor SIRPα," *MAbs*. 11(6):1036-1052.
- Sims et al. (1993). "A humanized CD18 antibody can block function without cell destruction," *J. Immunol.*, 151(4):2296-308.
- Spieß, C. et al. (2015). "Alternative molecular formats and therapeutic applications for bispecific antibodies," *Mal. Immunol.*, 67(2 Pt A):95-106.
- Takenaka, K. et al. (2007). "Polymorphism in Sirpa modulates engraftment of human hematopoietic stem cells," *Nat. Immunol.*, 8(12):1313-23.
- Treffers et al., (2018). "Genetic variation of human neutrophil Fcγ receptors and SIRPα in antibody-dependent cellular cytotoxicity towards cancer cells," *European Journal of Immunology* 48(2):344-354.
- Urlaub et al. (1980). "Isolation of Chinese hamster cell mutants deficient in dihydrofolate reductase activity," *Proc. Natl. Acad. Sci. USA*, 77(7):4216-20.
- Van Der Heijden, J. (Jul. 1, 2014). "Genetic Variation in Human Fc Gamma Receptors: Functional Consequences of Polymorphisms and Copy Number Variation," located at <<https://dare.uva.nl/search?identifier=54e3332e-a8c8-4fec-a49d-833b35617f2f>> last visited on Sep. 7, 2017, pp. 115-135, 22 pages.
- Vollmers and Brandlein, (2005). "Death by stress: natural IgM-induced apoptosis," *Methods and Findings in Experimental and Clinical Pharmacology*, 27(3):185-91.
- Vollmers and Brandlein, (2005). "The 'early birds': natural IgM antibodies and immune surveillance," *Histology and Histopathology*, 20(3):927-937.
- Weiskopf, K. et al. (2013). "Engineered SIRPα variants as immunotherapeutic adjuvants to anticancer antibodies," *Science*, 341(6141):88-91.
- Willingham, S.B. et al. (2012). "The CD47-signal regulatory protein alpha (SIRPα) interaction is a therapeutic target for human solid tumors," 109(17):6662-7.
- Winkler et al. (2000). "Changing the antigen binding specificity by single point mutations of an anti-p24 (HIV-1) antibody," *J Immunol*. 165(8):4505-14.
- Winter et al. (1994). "Making antibodies by phage display technology," *Ann. Rev. Immunol.*, 12:433-455.
- Yanagita, T. et al. (2017). "Anti-SIRPα antibodies as a potential new tool for cancer immunotherapy," *JCI Insight*, 2(1):e89140.
- Yi et al. (2015). "Splenic dendritic cells survey red blood cells for missing self-CD47 to trigger adaptive immune responses," *Immunity*, 43(4):764-775.
- Zhang et al. (2020). "CD47 decline in pancreatic islet cells promotes macrophage-mediated phagocytosis in type I diabetes," *World J Diabetes*, 11(6): 239-251.
- Zhao, X.W. et al. (2011). "CD47-signal regulatory protein-α (SIRPα) interactions form a barrier for antibody-mediated tumor cell destruction," *Proc. Natl. Acad. Sci.*, 108(45):18342-7.
- U.S. Appl. No. 17/337,348, filed Jun. 2, 2021, titled "Antibodies Against Signal-Regulatory Protein Alpha and Methods of Use," inventor Pons et al., Applicant ALX Oncology Inc. (Copy not submitted herewith pursuant to the waiver of 37 C.F.R. § 1.98(a)(2)(iii)).
- U.S. Appl. No. 17/337,176, filed Jun. 2, 2021, titled "Antibodies Against Signal-Regulatory Protein Alpha and Methods of Use," inventor Pons et al., Applicant ALX Oncology Inc. (Copy not submitted herewith pursuant to the waiver of 37 C.F.R. § 1.98(a)(2)(iii)).
- Hlavacek et al. (1999). "Steric Effects on Multivalent Ligand-Receptor Binding: Exclusion of Ligand Sites by Bound Cell Surface Receptors," *Biophysical Journal*, vol. 76, Issue 6, pp. 3031-3043.
- Plückthun et al. (1997). "New protein engineering approaches to multivalent and bispecific antibody fragments," *Immunotechnology*, 3(2):83-105.
- Bostrom et al. (2009). "Improving antibody binding affinity and specificity for therapeutic development," *Methods Mol Biol*, 525:353-376.
- Gonzales et al. (2005). "Minimizing the immunogenicity of antibodies for clinical application," *Tumour Biol*, 26(1): 31-43.
- Janeway et al. (2001). "Immunobiology: the immune system in health and disease." 5th Editions, New York, Garland Science, 5 pages.

(56)

References Cited

OTHER PUBLICATIONS

Kipriyanov et al. (2004). "Generation and production of engineered antibodies." *Molecular biotechnology*, 26(1):39-60.

Kunik et al. (2012). "Structural consensus among antibodies defines the antigen binding site," 8(2):e1002388, 12 pages.

NCBI (2015). Genbank Accession No. EAX10606.1, protein tyrosine phosphatase, non-receptor type substrate 1, isoform CRA_a, 1 page.

Padlan (1994). "Anatomy of the antibody molecule," *Mol. Immunol.*, 31: 169-217.

Ratnikova et al. (2017). "CD47 receptor as a primary target for cancer therapy. *Molecular Biology*," 51(2):251-261 (translation of abstract only, in p. 261).

Silva et al. (2015.) "The S228P Mutation Prevents in Vivo and in Vitro IgG4 Fab-arm Exchange as Demonstrated using a Combination of Novel Quantitative Immunoassays and Physiological Matrix Preparation." *The Journal of Biological Chemistry*, 290(9): 5462-5469.

Wark et al. (2006). "Latest technologies for the enhancement of antibody affinity," *Adv Drug Deliv Rev*, 58(5-6): 657-670.

* cited by examiner

1 50
v1 (1) EEELQVIQPKSVLVAAGETATLRCTATSLIPVGPIQWFRGACPGRELIY
v2 (1) EEELQVIQPKSVLVAAGESAILHCTVTSLIPVGPIQWFRGACPARELIY
v3 (1) EEELQVIQPKSVLVAAGESAILLCTVTSLIPVGPIQWFRGACPARELIY
v4 (1) EEGLQVIQPKSVLVAAGESAILHCTATSLIPVGPIQWFRGACPGRELIY
v5 (1) EEELQVIQPKSVLVAAGETATLRCTATSLIPVGPIQWFRGACPGRELIY
v6 (1) EEELQVIQPKSVLVAAGETATLRCTATSLIPVGPIQWFRGACPGRELIY
v7 (1) EEELQVIQPKSVLVAAGESAILHCTVTSLIPVGPIQWFRGACPARELIY
v8 (1) EEELQVIQPKSVLVAAGETATLRCTATSLIPVGPIQWFRGACPARELIY
v9 (1) EEELQVIQPKSVLVAAGETATLRCTATSLIPVGPIQWFRGACPGRELIY
v10 (1) EEELQVIQPKSVLVAAGESAILHCTVTSLIPVGPIQWFRGACPARELIY *
51 100
v1 (51) NQKEGHFFRVTTVSDLTKRNNMDFSIRIGNITPADAGTYVCVKFKGSPD
v2 (51) NQKEGHFFRVTTVSESTKRNNMDFSISISNITPADAGTYVCVKFKGS-P
v3 (51) NQKEGHFFRVTTVSESTKRNNMDFSISISNITPADAGTYVCVKFKGS-P
v4 (51) NQKEGHFFRVTTVSDLTKRNNMDFSIRIGNITPADAGTYVCVKFKGSPD
v5 (51) NQKEGHFFRVTTVSDLTKRNNMDFSIRIGNITPADAGTYVCVKFKGSPD
v6 (51) NQKEGHFFRVTTVSDLTKRNNMDFIRIGNITPADAGTYVCVKFKGSPD
v7 (51) NQKEGHFFRVTTVSESTKRNNMDFSISISNITPADAGTYVCVKFKGS-P
v8 (51) NQKEGHFFRVTTVSESTKRNNMDFSISISNITPADAGTYVCVKFKGS-P
v9 (51) NQKEGHFFRVTTVSDLTKRNNMDFSIRISNITPADAGTYVCVKFKGSPD
v10 (51) NQKEGHFFRVTTVSESTKRNNMDFSISISNITPADAGTYVCVKFKGS-P **
101 119
v1 (101) DVEFKSGAGTELSVRKPS
v2 (100) DTEFKSGAGTELSVRKPS
v3 (100) DTEFKSGAGTELSVRKPS
v4 (101) DVEFKSGAGTELSVRKPS
v5 (101) DVEFKSGAGTELSVRKPS
v6 (101) DVEFKSGAGTELSVRKPS
v7 (100) DTEFKSGAGTELSVRKPS
v8 (100) DTEFKSGAGTELSVRKPS
v9 (101) DVEFKSGAGTELSVRKPS
v10 (100) DTEFKSGAGTELSVRKPS *

FIG. 1A

```

1      50
Human_v1      (1) EEELQVIQPKSVLVAAGETATLRCTATSLIPVGPIQWFERGAGPGRELIY
Human_v2      (1) EEELQVIQPKSVSVAAGESAILHCTVTSLLIPVGPIQWFERGAGPARELIY
Cyno          (1) EEELQVIQPEKSVSVAAGESATLNCATSLIPVGPIQWFERGVGPGRELIY
129mouse      (1) -KELKVTQPEKSVSVAAGDSTVLNCTLTSLIPVGPIKWIYRGVGGQSRLLIY
              ** * * * * **** * * * * * * * * * *
51      100
Human_v1      (51) NQKEGHFPRVTVSDLTKRNNMDFSIRIGNITPADAGTYTCVKFRKGS-
Human_v2      (51) NQKEGHFPRVTVSESTKRNNMDFSISISNITPADAGTYTCVKFRKGS--
Cyno          (51) HQKEGHFPRVTPVSDPTKRNNMDFSIRISNITPADAGTYTCVKFRKGS--
129mouse      (50) SFTGEHFPRVTNVSDATKRNNMDFSIRISNVTPEADAGTYTCVKFQKGPSE
              ***** * ** * * * * * * * * * *
101      120
Human_v1      (100) DDVEFKSCAGTELSVRAKPS
Human_v2      (99) PDTEFKSCAGTELSVRAKPS
Cyno          (99) PDVELKSCAGTELSVRAKPS
129mouse      (100) PDTEIQSCGGTEVYVLAKPS
              * * ** * ****
```

FIG. 1B


```

1      50
Human_v1      (1) EEELQVIQPKSVLVAAGETATLRCTATSLIPVCPQWFERGAGPGRELIY
Human_v2      (1) EEELQVIQPKSVSVAAGESAILHCTVTSLLIPVCPQWFERGAGPARELIY
      Cyno      (1) EEELQVIQPEKSVSVAAGESATLNCTATSLIEVGPQWFERGVGPARELIY
129mouse      (1) -KELKVTQPEKSVSVAAGDSTVLNCTLTSLLPVCPQWFERGVGQSRLLIY
chicken      (1) --DFKLQQPQSSVVVIKGDITLNLCTASCGPICA VKWVKWKGSDNQTVY
      ***** ** * ** ***** * ***** * ***** ** * *****
51      100
Human_v1      (51) NQKEGHFPRVTVSDLTKRNNMDFESIRIGNITPADAGTYVCVKFKGSP-
Human_v2      (51) NQKEGHFPRVTVSESTKRNNMDFESISNITPADAGTYVCVKFKGSGS--
      Cyno      (51) HQKEGHFPRVTPVSDPTKRNNMDFESIRISNITPADAGTYVCVKFKGSGS--
129mouse      (50) SFTGEHFPRVTNVSDATKRNNMDFESIRISNVTPEADAGTYVCVKFKGKPSE
chicken      (49) EHKGSFPRVMRAVPDPTN---DFTIRISNVSLDAGTYVCVKLRKGIV-
      ***** ***** *** ***** * * * ***** ** * *****
101      120
Human_v1      (100) DDVEFKSCAGTELSVRAKPS
Human_v2      (99) PDTEFKSCAGTELSVRAKPS
      Cyno      (99) PDVELKSCAGTELSVRAKPS
129mouse      (100) PDTEIQSGGGTEVVVLAKPS
chicken      (94) DDVVFTRGGGTEVSVHA---
      ***** * ** * *****
```

FIG. 2

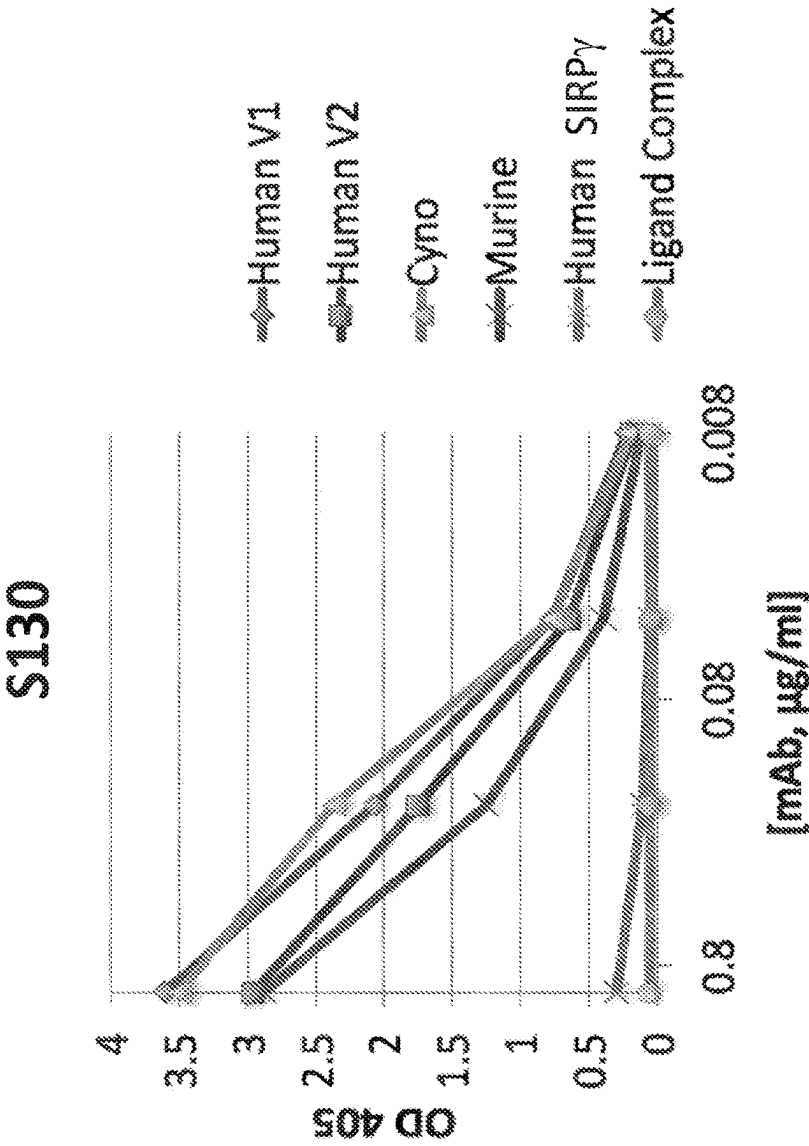


FIG. 3A

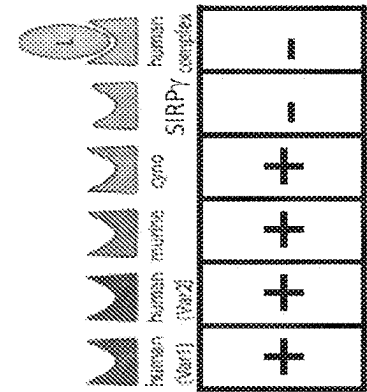


FIG. 3B

S121

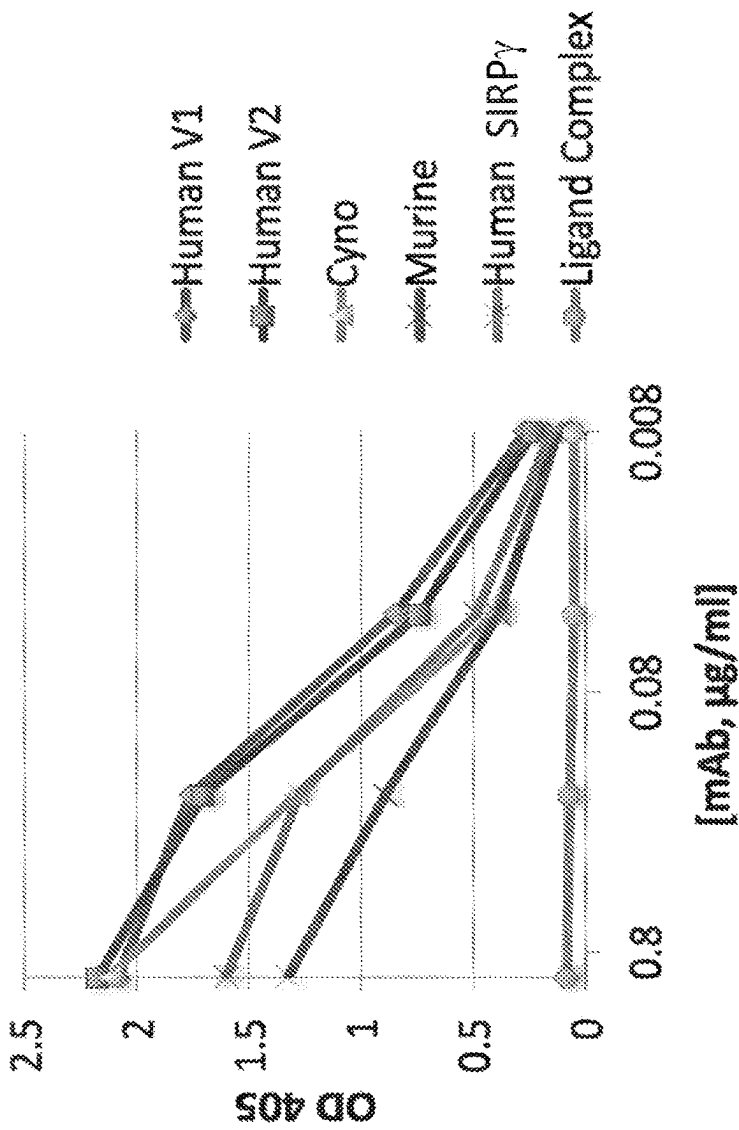


FIG. 4A

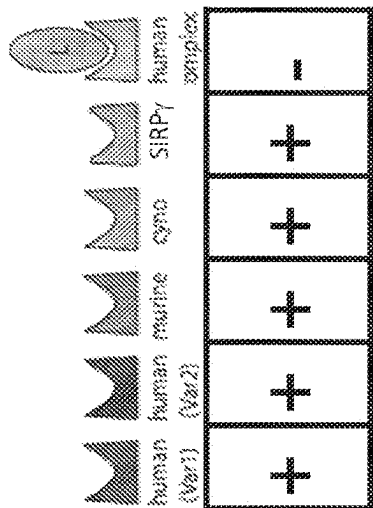


FIG. 4B

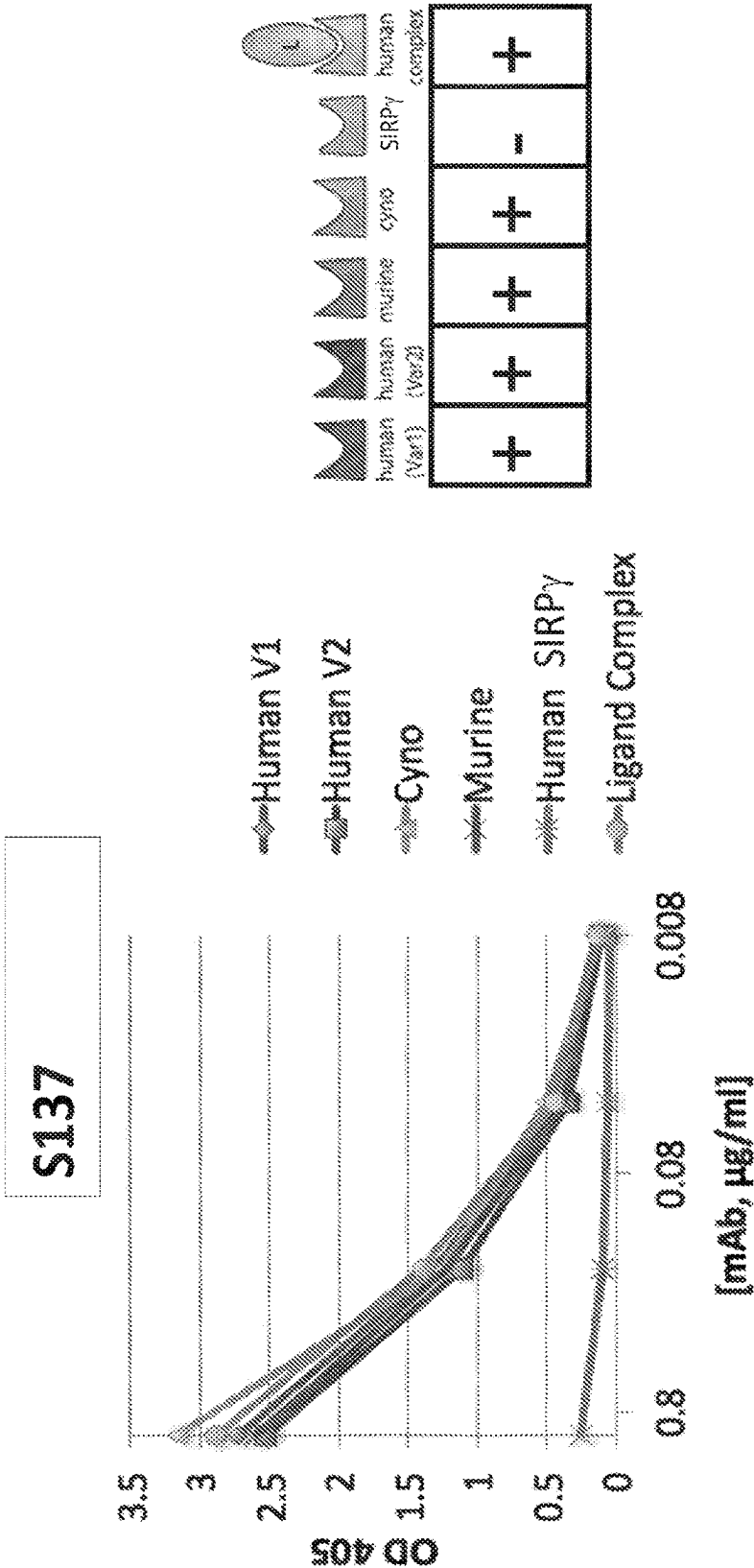


FIG. 5B

FIG. 5A

					
human (Var1)	human (Var2)	murine	cyno	SIRPy	human complex
+	+	+	+	-	+

S128

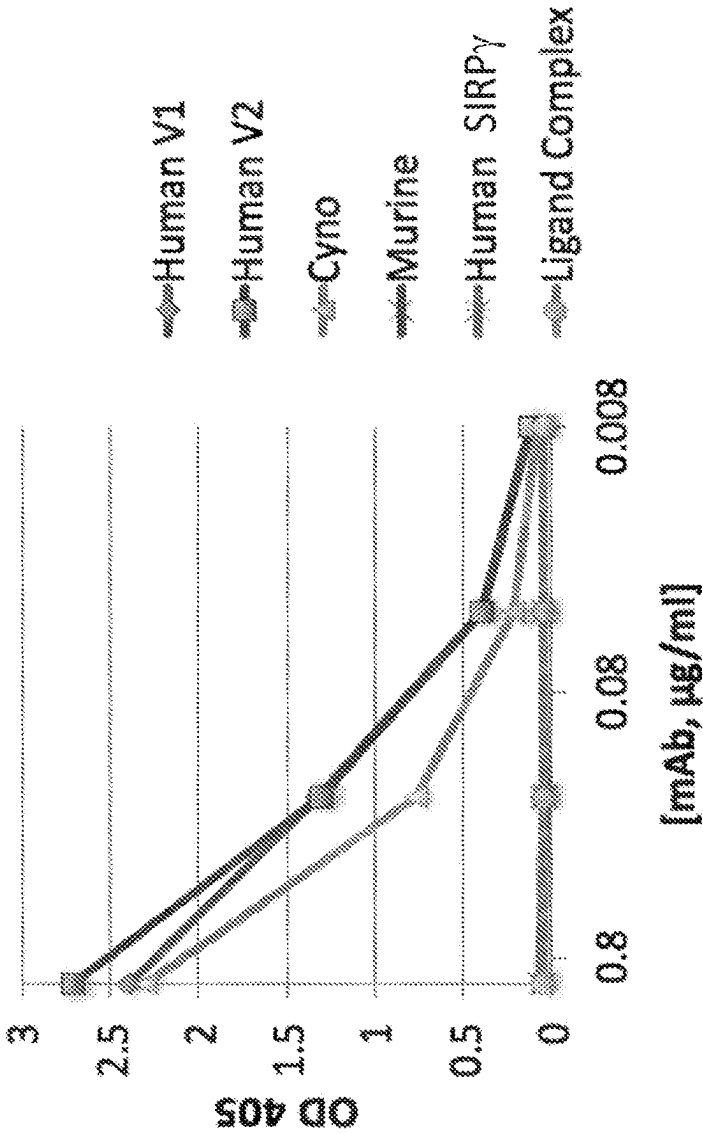


FIG. 6A

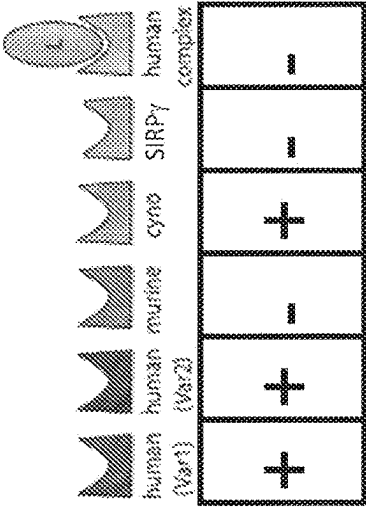


FIG. 6B

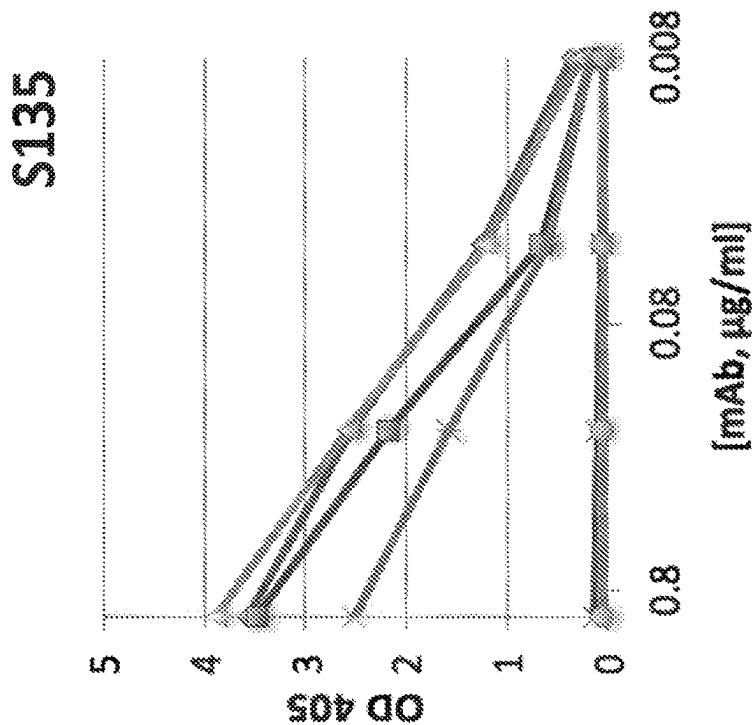


FIG. 7A

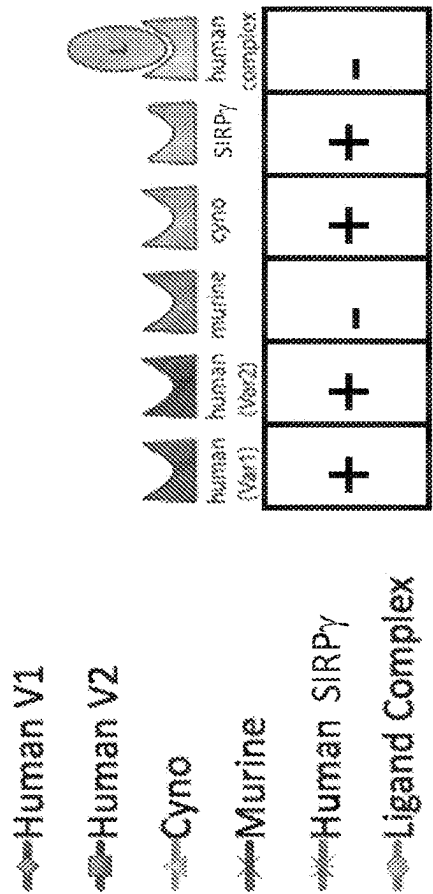


FIG. 7B

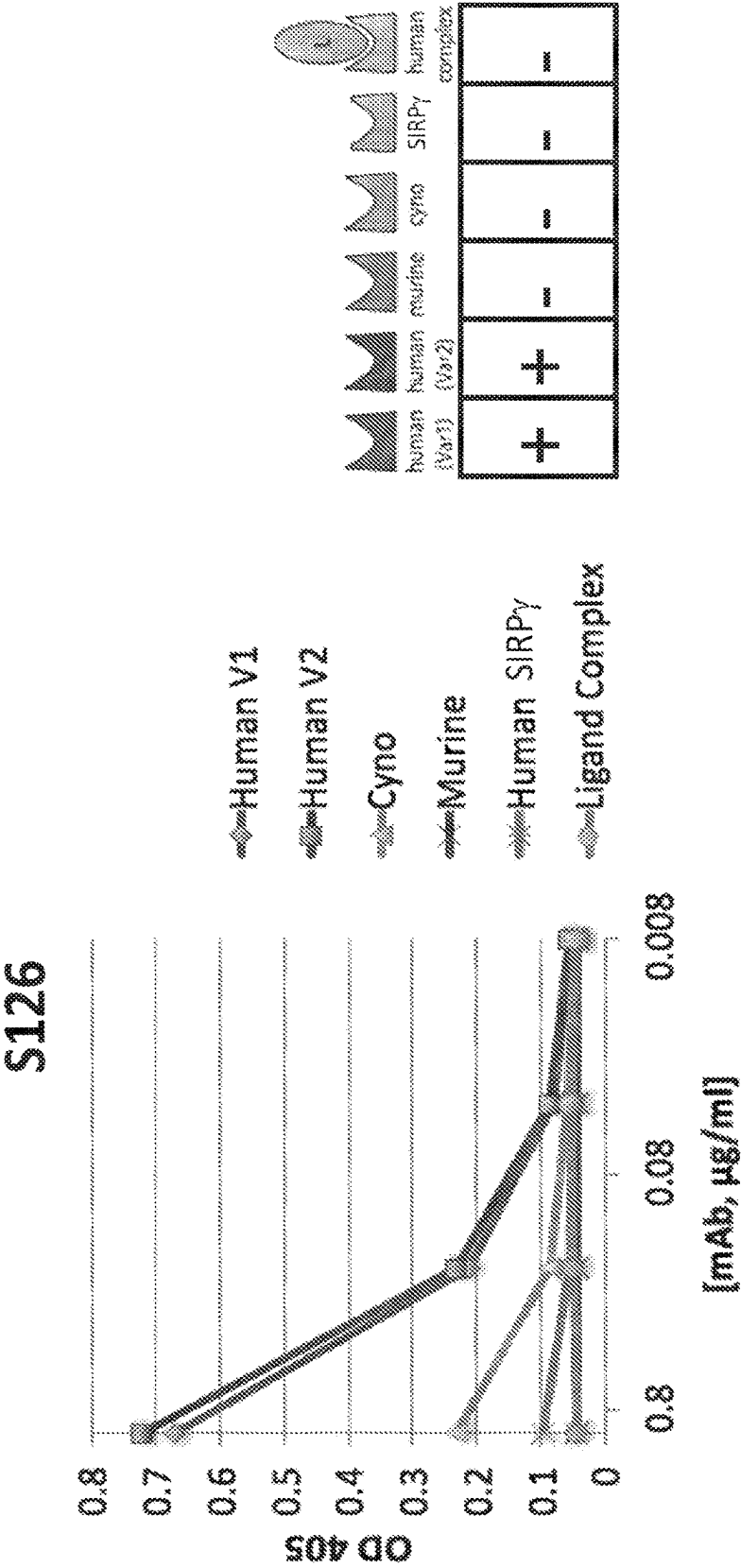


FIG. 8A

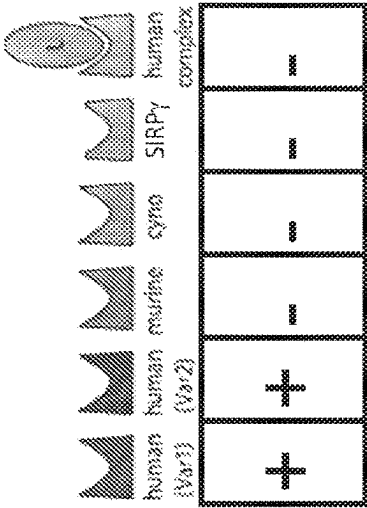


FIG. 8B

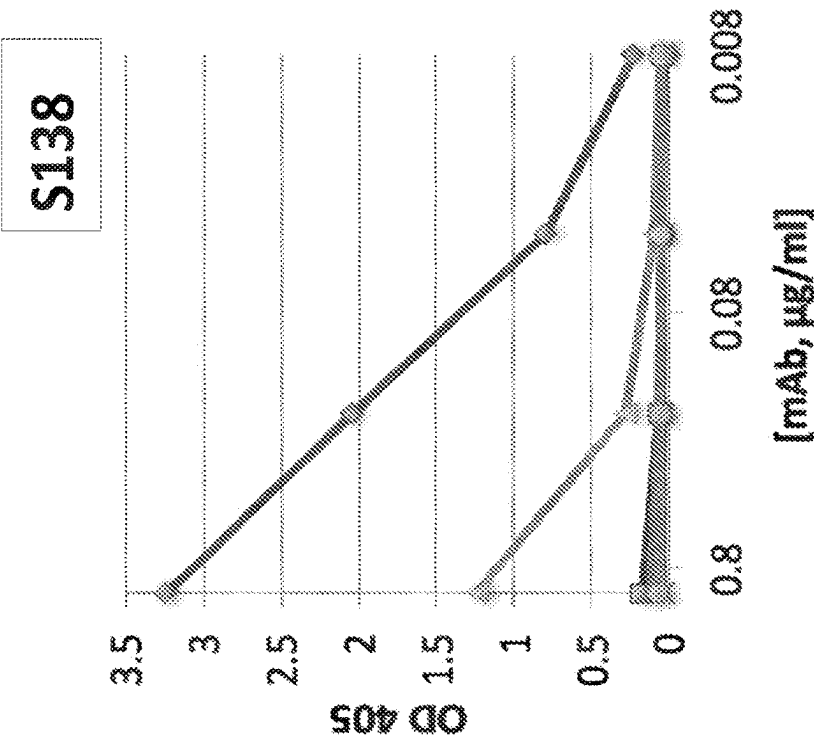


FIG. 9A

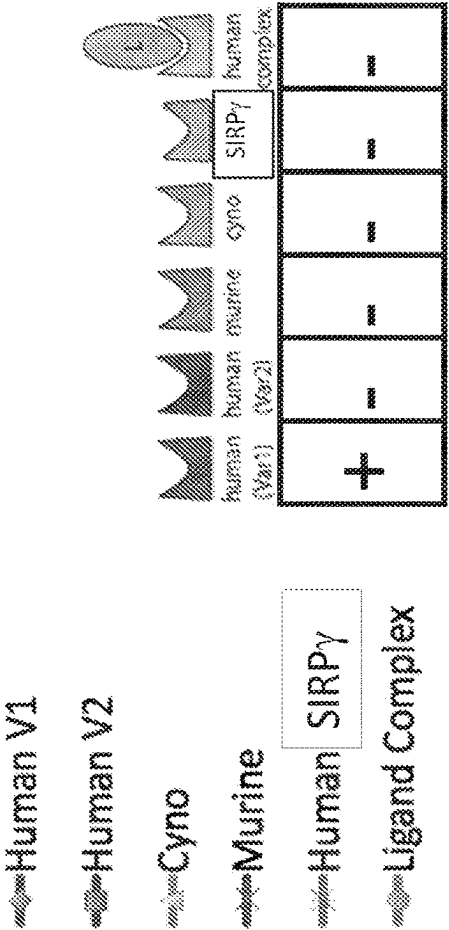


FIG. 9B

		linker	150
S1	(99)	101 GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
S2	(99)	GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
S8	(97)	GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
S9	(100)	GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
S11	(99)	GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
S12	(97)	GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
S13	(97)	GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
S14	(97)	GTTLTVLGQSSRSSGGSSGGGSMAAVTLD	150
		*** *	*
		H1	H2
S1	(149)	151 GSGFTFSSHAMNWVRQAPGKGLEWVAGISS	200
S2	(149)	ASGFDFSNFNMAWVRQGPCKGLEVVAEISD	200
S8	(147)	ASGFTFSSYNMGWVRQAPGKGLEFVAGIYASGS	200
S9	(150)	GSGFSSSYNMGWVRQAPGKGLEFVAGIYASGS	200
S11	(149)	ASGFTFSSFNMGWVRQAPGKGLEFVAAIYS	200
S12	(147)	ASGFTFSSYNMGWVRQAPGKGLEFVAGIYI	200
S13	(147)	ASGFTFSSYNMGWVRQAPDKGLEFVAGIYTG	200
S14	(147)	ASGFTFSSYNMGWVRQAPGKGLEFVAGIYS	200
		***** *	***** *

FIG. 10B

			<u>L1</u>	50
S115	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVG-SKLAWHQKPGQAPRLLIY		
S116	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVG-SKLAWHQKPGQAPRLLIY		
S117	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVG-SKLAWHQKPGQAPRLLIY		
S118	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVG-SKLAWHQKPGQAPRLLIY		
S119	(1)	EIVLTQSPATLSVSPGERATFSCRASQNVK-NDLAWYQQRPGQAPRLLIY		
S120	(1)	EIVLTQSPATLSVSPGERATFSCRASQNVK-NDLAWYQQRPGQAPRLLIY		
S122	(1)	EIVLTQSPATLSVSPGERATFSCRASQNVK-NDLAWYQQRPGQAPRLLIY		
S123	(1)	EIVLTQSPGTLSPGERVTLTTCRASQGIAG-KIAWYQKPGQAPRLLIY		
S126	(1)	EIVLTQSPGTLSPGERATLSCRASQSIGSSYLAWYQKPGQAPRLLIY		
S128	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVG-SKLAWHQKPGQAPRLLIY		
S130	(1)	EIVLTQSPGTLSPGERATLSCRASQNVK-SDLAWYQKKGQAPRLLIY		
S135	(1)	EIVLTQSPATLSVSPGERVTFSCRASQNVK-SDIAWYQKPGQAPRLLIY		
S137	(1)	ETVLTQSPGTLSPGERATLTCRASQSVY-TYLAWYQEKPGQAPRLLIY		
S138	(1)	EIVLTQSPGTLSPGERVILTTCRASQSVDTYNLAWYQKPGQAPRLLIY		
		* * * * *	*****	* * * *
		<u>L2</u>	<u>L3</u>	100
S115	(50)	DATNRATGISDRFSGSGGTDEFTLTISSLQTEDSAVYCCQYYWPPYRF		
S116	(50)	DATNRATGISDRFSGSRSGTDEFTLTISSLQTEDSAVYCCQYYWPPYRF		
S117	(50)	DATNRATGIPDRFSGSGGTDEFTLTISSLQTEDSAVYCCQYYWPPYRF		
S118	(50)	DASRRATGIPDRFSGSGGTDEFTLTISSLQTEDSAVYCCQYYWPPYRF		
S119	(50)	AARIRETGIPERFSGSGGTDEFTLTISLQSEDAVYCCQYYDWPPFTF		
S120	(50)	AARIRETGIPERFSGSGGTDEFTLTISLQSEDAVYCCQYYDWPPFTF		
S122	(50)	AARIRETGIPERFSGSGGTDEFTLTISLQSEDAVYCCQYYDWPPFTF		
S123	(50)	DASSRATGIPGRFSGSGGTDEFTLTISLQSEDAVYCCQHYDWSPLTF		
S126	(51)	DATNRATGIPDRFSGSGGTDEFTLTISSLQTEDSAVYCCQYYWPPYRF		
S128	(50)	DASNRRATGIPDRFSGSGGTDEFTLTISSLQTEDSAVYCCQYYWPPYRF		
S130	(50)	DANTRATDIPDRFSGSGGTDEFTLTISLQSEDAVYCCQHYDWPVTF		
S135	(50)	AASSRDTGIPDRFSGSGGTDEFTLTISSLQSEDFGVYCCQYYDWPPFTF		
S137	(50)	GASSRATGIPDRFSGSGGTDEFTLTISSLQSEDAVYCCQYYDRPPLTF		
S138	(51)	DLSTRATGIPDRFSGSGGTDEFTLTINSLEPEDEFAVYCHQYYDWPPYTF		
		*****	* * *	***

FIG. 10D

101 linker 150

S115 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S116 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S117 (100) GGGTKVEIKQSSRSSGGSSRGGSDVQLVESGGGVVRPGESLRLSCA
S118 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S119 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLLESGGGVQPGSLRLSCA
S120 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVQPGSLRLSCA
S122 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S123 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGLVQPGSLRLSCT
S126 (101) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S128 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S130 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S135 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S137 (100) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA
S138 (101) GGGTKVEIKQSSRSSGGSSGGGSDVQLVESGGGVVRPGESLRLSCA

151 H1 200

S115 (150) ASGFSSSYAMNVRQAPGEGLEWVSRINSGGGTDYAESVKGRFTISR
S116 (150) ASGFSSSYAMNVRQAPGEGLEWVSRINSGGGTDYAESVKGRFTISR
S117 (150) ASGFSSSYAMNVRQAPGEGLEWVSRINSGGGTDYAESVKGRFTISR
S118 (150) ASGFSSSYAMNVRQAPGEGLEWVSRINSGGGTDYAESVKGRFTISR
S119 (150) ASGFSSNFAMTVRQAPGEGLEWVSTIGSG--DTYYADSVKGRFTISR
S120 (150) ASGFSSNFAMTVRQAPGEGLEWVSTIGSG--DTYYADSVKGRFTISR
S122 (150) ASGFSSNFAMTVRQAPGEGLEWVSTIGSG--DTYYADSVKGRFTISR
S123 (150) ASGFTFERNYGMVVRQAPGEGLEWVSASSG-SGSTYYTDSVKGRFTISR
S126 (151) ASGFTFSNYDMTVRQAPGEGLEWVSGISGNGGSTYYADSVKGRFTISR
S128 (150) ASGFSSRSYAMNVRQAPGEGLEWVSRIDSGGGTDYADSVKGRFTISR
S130 (150) ASGFTFSNYAMSVVRQAPGEGLEWVSLITNGDGAYYADSVKGRFTISR
S135 (150) ASGFSSSIYAMSVVRQAPGEGLEWVSTIGAD--DTYYADSVKGRFTISR
S137 (150) ASGFTFSNYDMNVRQAPGEGLEWVSLISGGEIYYADSVKGRFTISR
S138 (151) ASGFTFSNYAMNVRQAPGEGLEWVSGISGRGGDTYYADSVKGRFTISR

* * * * *

FIG. 10E

	201	H3	250
S115	(200)	NSENTLYQMNSLRAEDTAVYYCAKQY---DWNSEFFDYWGLGALVTVSSS	
S116	(200)	NSENTLYQMNSLRAEDTAVYYCAKQY---DWNSEFFDYWGLGALVTVSSS	
S117	(200)	NSENTLYQMNSLRAEDTAVYYCAKQY---DWNSEFFDYWGLGALVTVSSS	
S118	(200)	NSENTLYQMNSLRAEDTAVYYCAKQY---DWNSEFFDYWGLGALVTVSSS	
S119	(198)	NSKNTLYQMNSLRAEDTAVYYCAKDSVSWSGDEFFDYWGLGTLVTVSSS	
S120	(198)	NSKNTLYQMNSLRAEDTAVYYCAKDSVSWSGDEFFDYWGLGTLVTVSSS	
S122	(198)	NSKNTLYQMNSLRAEDTAVYYCAKDSVSWSGDEFFDYWGLGTLVTVSSS	
S123	(199)	NSKNTLYQMNSLRAEDTAIYYCAKVT---WNN-FFDYWGLGTLVTVSSS	
S126	(201)	NSKNTLYQMNSLRAEDTAVYYCAMN---RWW-FFDYWGLGTLVTVSSS	
S128	(200)	NSKNTLYQMNSLRAEDTAVYYCAKQY---DWNSEFFDYWGLGAPVTVSSS	
S130	(200)	NSKNTLYQMNSLRAEDTAIYYCAKDGAAHYDIFFDYWGLGTPVTVSSS	
S135	(198)	NSKNTLYQMNSLRAEDTAVYYCAKDSVWWSGDEFFDYWGLGTLVTVSSS	
S137	(200)	NSKNTLYQMNSLRAEDTAVYYCAKEN---NRYRFFDDWGLGTLVTVSSS	
S138	(201)	NSKNTLYQMNSLRAEDTAIYYCAKGT---WNYGSEFFDYWGLGTLVTVSSS	
	*	*****	* **

FIG. 11C

		<u>L1</u>	<u>L2</u>	<u>54</u>
S16	1	ALTQASVSANLGETVKITCSGGDSSSHYYGWYQQKSPGSA	PVTVIYSDDERPS	
S59	(1)	ALTQASVSANLGETVKITCSGGDSSSHYYGWYQQKSPGSA	PVTVIYSDDERPS	
S27	(1)	ALTQASVSANLGETVKITCSGGDSSSHYYGWYQQKSPGSA	PVTVIYSDDERPS	
S55	(1)	ALTQASVSANLGETVEITCSGGDSSSHYYGWYQQKSPGSA	PVTVIYSDDERPS	
S29	(1)	ALTQASVSANLGETVKITCSGGDSSSHYYGWYQQKSPGSA	PVTVIYSDDERPS	
S26	(1)	ALTQASVSANLGGTVEITCSGGSS--YYGWYQQKSPGSA	PVTVIYSDNQRPS	
S21	(1)	ALTQASVSANPGETVKIACSGGDYYSYYGWYQQKAPGSA	LVTVIYSDDKRPS	
S24	(1)	ALTQASVSANPGETVKIACSGGSYYSYYGWYQQKSPGSA	LVTVIYSDDKRPS	
S74	(1)	ALTQASVSANPGETVKIACSGGSYYSYYGWYQQKSPGSA	LVTVIYSDDKRPS	
S25	(1)	ALTQASVSANPGETVEITCSGGSYSSYYAWYQQKSPGSA	PVTVIYSDDKRPS	
S73	(1)	ALTQASVSANPGETVKITCSGGIYSSYYAWYQQKSPGSA	PVTVIYSDDKRPS	
S30	(1)	ALTQASVSANPGETVEITCSGGSDSSYYGWYQQKSPGSA	PVTVIYSDNKRPS	
S28	(1)	ALTQASVSANPGETVKITCSGG--SDSYYGWHQKSPGSA	PVTVIYSDDQRP	
S201	(1)	ALTQASVSANPGETVKITCSGG--GTSSYYGWYQQKSPGSA	PVTLIHSDDKRPS	
S206	(1)	ALTQASVSANPGETVKITCSGG--GTSSYYGWYQQKSPGSA	PVTLIHSDDKRPS	
S22	(1)	ALTQASVSANPGETVKITCSGGDYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S60	(1)	ALTQASVSANPGETVKITCSGGDYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S65	(1)	ALTQASVSANPGETVKITCSGGDYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S69	(1)	ALTQASVSANPGETVKIICSGGDYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S23	(1)	ALTQASVSANPGETVKITCSGGDYSTYYAWYQQKSPGSA	PVTVIHSDDERPS	
S66	(1)	ALTQASVSANPGETVKITCSGGDYSTYYAWYQQKSPGSA	PVTVIHSDDKRPS	
S17	(1)	ALTQASVSANPGETVKITCSGGDWYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S70	(1)	ALTQASVSANPGETVKITCSGGDWYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S76	(1)	ALTQASVSANPGETVKITCSGGDWYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S202	(1)	ALTQASVSANPGETVKITCSGGGYSTYYGWYQQRSPGSA	PVTVIHSDDKRPS	
S56	(1)	ALTQASVSANPGETVKITCSGGGYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
S71	(1)	ALTQASVSANPGETVKITCSGGGYSTYYGWYQQKSPGSA	PVTVIHSDDKRPS	
		** * * *	*****	* * *

FIG. 11F

		L1	L2	54
1				
S136	(1)	ETVLTQSPGTLTSLSPGERATLTCRASQSVTYT	LA	WYQEKPGQAPRLLIYGASSR
S137	(1)	ETVLTQSPGTLTSLSPGERATLTCRASQSVTYT	LA	WYQEKPGQAPRLLIYGASSR
S175	(1)	ETVLTQSPGILSLSPGERATLTCRASQSVTYT	LA	WYQEKPGQAPRLLIYGASSR
S177	(1)	KTVLTQSPGTLTSLSPGERATLTCRADQSVTYT	LA	WYQERPGQAPRLLIYDASSR
S184	(1)	ETVLTQSPGTLTSLSPGERATLNCRASQSVTYT	LA	WYQEKPGQAPRLLIYDASSR
S185	(1)	ETVLTQSPGTLTSLSPGERATLNCRASQSVSY	LA	WYQERPGQAPRLLIYGASTR
S189	(1)	ETVLTQSPGTLTSLSPGERATLSCRASQSVTYT	LA	WYQQKPGQPRLLIHAARNR
S193	(1)	ETVLTQSPGTLTSLSPGERATLSCRASQSVTYT	LA	WYQEKPGQAPRLLIYAASR
S178	(1)	ETVLTQSPGTLTSLSPGERATLSCRASQSVTYT	LA	WYQEKPGQAPRLLIYGASSR
S180	(1)	ETVLTQSPGTLTSLSPGERATLTCRASQSVTYT	LA	WYQEKPGQAPRLLIYGASSR
S190	(1)	ETVLTQSPGTLTSLSPGERTTLTCRASQSVTYT	LA	WYQEKPGQAPRLLIYGASSR
	*	* * *	*	** *
55				
			L3	108
S136	(55)	ATGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S137	(55)	ATGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S175	(55)	AAGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S177	(55)	ATGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S184	(55)	ATGIPDRFSGSGGTEFTLT	ISSLESEDFAVY	CQQYDRPPLTFGGGKVEIK
S185	(55)	ATGIPDRFSGSGGTEFTLT	ISSLESDDFAVY	CQQYDRPPLTFGGGKVEIK
S189	(55)	AAGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S193	(55)	ATGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S178	(55)	ATGVPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S180	(55)	ATGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
S190	(55)	ATGIPDRFSGSGGTEFTLT	ISSLQSEDFAVY	CQQYDRPPLTFGGGKVEIK
	*	* *	*	** *

FIG. 11H

		<u>L1</u>	<u>L2</u>	<u>L3</u>
	1		54	
S115	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAWHQQKPGQAPRLLIYDATNR		
S116	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAWHQQKPGQAPRLLIYDATNR		
S117	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAWHQQKPGQAPRLLIYDATNR		
S118	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAWHQQKPGQAPRLLIYDASRR		
S132	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAWHQQKPGQAPRLLIYDASNR		
S191	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAWHQQKPGQAPRLLIYDATNR		
S198	(1)	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAWHQQKPGQAPRLLIYDATNR		**
	55			108
S115	(55)	ATGISDRFSGSGGTDFTLTISSLQTEDSAVYCCQYYWPPYRFGGGTKVEIK		
S116	(55)	ATGISDRFSGSGGTDFTLTISSLQTEDSAVYCCQYYWPPYRFGGGTKVEIK		
S117	(55)	ATGIPDRFSGSGGTDFTLTISSLQTEDSAVYCCQYYWPPYRFGGGTKVEIK		
S118	(55)	ATGIPDRFSGSGGTDFTLTISSLQTEDSAVYCCQYYWPPYRFGGGTKVEIK		
S132	(55)	ATGIPDRFSGSGGTDFTLTISSPQTEDSAVYCCQYYWPPYRFGGGTKVEIK		
S191	(55)	ATGIPDRFSGSGGTDFTLTISSGLQTEDSAVYCCQYYWPPYRFGGGTKVEIK		
S198	(55)	ATGIPDRFSGSGGTDFTLTISSLQTEDSAVYCCQYYWPPYRFGGGTKVEIK		**

FIG. 11I

		<u>H1</u>	<u>H2</u>	<u>60</u>
S173	(1)	DVQLVESGGGVVRPGE	SLRLSCAASGEAFSDH	MSWVRQGPGEGL
S174	(1)	DVQLVESGGGVVRPGE	SLRLSCAASGFTFSDH	MSWVRQGPGEGL
				WVAGISLRGGVTWY
				WVAGISLRGGVTWY
				*
		<u>H3</u>		<u>118</u>
S173	(61)	ADSVKGRFTISRDN	SKNTLYLRFLSLRT	EDAIYYCARESWNT
S174	(61)	ADSVKGRFTISRDN	SKNTLYLRFLSLRT	EDAIYYCARESWNT
				FFDYWGGLGLVTVSS
				FFDYWGGLGLVTVSS
		<u>L1</u>	<u>L2</u>	<u>55</u>
S173	(1)	EIVLTQSPGTL	SLSPGETATL	SCRASQNVRSN
S174	(1)	EIVLTQSPGTL	SLSPGETATL	SCRASQNVRSN
				LAWYQQKPKGQAPRLLIYDASSRA
				LAWYQQKPKGQAPRLLIYDASSRA
		<u>L3</u>		<u>108</u>
S173	(56)	TGIPDRFSGSGGTD	FTLTIS	SLQSEDAFVYCCQQYGN
S174	(56)	TGIPDRFSGSGGTD	FTLTIS	SLQSEDAFVYCCQQYGN
				PPPLTFGGG
				TKVEIK

FIG. 11J

		<u>H1</u>	<u>H2</u>	60
S213	(1)	DVQLVESGGVVRPAGESLRLSCEASGFTERNYYMTWVRQAPGEGLEWVSTISDTGDTAYY		
S214	(1)	DVQLVESEGGVVRPAGESLRLSCEASGFTERNSYMTWVRQAPGEGLEWVSTISDTGDTAYY		
		*		
		<u>H3</u>		119
S213	(61)	ADSVKGRFTISRDN SKNSLYLQMNSLRADDTAIYYCAKSWIWWTFEDYWGLGLTVTVSS		
S214	(61)	ADSVKGRFTISRDN SKNSLYLQMNSLRAEDTAIYYCAKSWIWWTFEDYWGLGLTVTVSS		
		*		
		<u>L1</u>	<u>L2</u>	55
S213	(1)	ALTQPASVSANLGGTVEITCSGGNSNHYGWYQQKSPGSA PVTLIYADTNRPSNIP		
S214	(1)	ALTQPASVSANLGGTVEITCSGGNSNHYGWYQQKSPGSA PVTLIYADTNRPSNIP		
			<u>L3</u>	103
S213	(56)	SRFSGTSGSTTLTITGVQAEDEAVYYCGSGSTGDGIFGAGTLLTVL		
S214	(56)	SRFSGTSGSTTLTITGVQAEDEAVYYCGSGSTGDGIFGAGTLLTVL		
		*		

FIG. 11K

>209_VH
DVQLVESGGAVVRPGESLRSLCKASGFTFTNFAMSWVRQAPGEGLEWVSGISGSDDTTYA
DSVKGRFTISRDNSESTLYLQMNSLRAEDTAVYYCVKDDSTVSWNTFFDYWGLGTLVTVSS

>209_VL
ALTQPASVSANLGGTVKITCSGGYGSDDGSSSYYGWYQQKSPGSAFVILIYWDDKRPSDIP
SRFSGSTGTTTLTITGVQAEDEAVYFCGTYDTSSGAIFGAGTTLTVL

FIG. 11L

		<u>H1</u>	<u>H2</u>	<u>59</u>
S123	(1)	DVQLVESGGGLVQPGGSLRLSCTASGFTFRNYGMSWVRQAPGEGLEWVSASSGSGSTYY		
S194	(1)	DVQLVESGGGVVRPGESLRLSCAASGFTFRNYGMSWVRQAPGEGLEWVSASSGSGSTYY		
		* * *		
		<u>H3</u>		<u>117</u>
S123	(60)	TDSVKGRFTISRDN SKNTLYLQMN SLRAEDTAIYYCAKVTWNNFEEDYWGLGLTVTVSS		
S194	(60)	ADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAIYYCAKVTWNNFEEDYWGLGLTVTVSS		
		*		
		<u>L1</u>	<u>L2</u>	
S123	(1)	EIVLTQSPGTLSPGERVTLTCRASQGIAGKIAWYQQKPGQAPRLLIYDASSRATGIP		
S194	(1)	EIVLAQSPDTLSVSPGERATLTCRASQDVAGKLAWYQQKPGQAPRLLIHATSSRADGIP		
		* * *	***	*
		<u>L3</u>		<u>108</u>
S123	(60)	GRFSGSGTEFTLTITSLQSEDAVYYCQQHYDWSPLTFGGGTKVEIK		
S194	(60)	ARESGSGTEFTLTITGLQSEDAVYYCQQHYDWSPLTFGGGTKVEIK		
		*		

FIG. 11M

		<u>H1</u>	<u>H2</u>	<u>H3</u>
	1		59	
S161	(1)	DVQLVESGGGVVRPGESLRSLSCAASGFTFSNFAMTWVRQAPGKGPENWSLVSVTATYY		
S162	(1)	DVQLVESGGGLVRPGESLRSLSCAASGFTFTNYAVTWVRQAPGEGLEWVSLISVTGTTY		
S163	(1)	DVQLVESGGGLVRPGESLRSLSCAASGFTFTNYAVTWVRQAPGEGLEWVSLISVTGTTY		
S164	(1)	DVQLVESGGGLVRPGESLRSLSCAASGFTFTNYAVTWVRQAPGEGLEWVSLISVTGTTY		
	1	* * *	* * *	*
	60			117
S161	(60)	ADSVKGRFTISRDNSTLYLQMNSLRAEDTAVYYCAKITWNNLFDYWGLGTLVTVSS		
S162	(60)	ADSVKGRFTISRDNSTLYLQMNSLRAEDTAVYYCAKVTWKNVFDYWGLGTLVTVSS		
S163	(60)	ADSVKGRFTISRDNSTLYLQMNGLRAEDTAVYYCAKVTWKNVFDYWGLGTLVTVSS		
S164	(60)	ADSVKGRFTISRDNSTLYLQMNGLRAEDTAVYYCAKVTWKNVFDYWGLGTLVTVSS		
		*	*	*
	1			59
S161	(1)	EIVLTQSPGTLSLSPGERATLSCRASQTVGSKLAWYQQKPGQAPRLLIYDSSSRASGIP		
S162	(1)	EIVLTQSPGTLSVSPGERATFSCRASQTVGSKLAWYQQKPGQAPRLLIYDANTRATGIP		
S163	(1)	EIVLTQSPGTLSVSPGERATFSCRASQTVGSKLAWYQQKPGQAPRLLIYDANTRATGIP		
S164	(1)	EIVLTQSPGTLSVSPGERATFSCRASQTVGSKLAWYQQKPGQAPRLLIYDANTRATGIP		
		*	***	*
	60			108
S161	(60)	DRFSGSGGTDFTLTISSLSQSEDSAVYYCQQHNDWPPYTFEGGKVEIK		
S162	(60)	ARESGSGGTDFTLTISSLSQSEDAVYYCQQHTDWPYTFEGGKVEIK		
S163	(60)	ARESGSGGTDFTLTISSLSQSEDAVYYCQQHTDWPYTFEGGKVEIK		
S164	(60)	ARESGSRSGTDFTLTISSLSQSEDAVYYCQQHTDWPYTFEGGKVEIK		
	*	*	*	*

FIG. 11N

>S149_VH
DVQLVESGGGVVRPGESLRLSCAASGFTFSNFAMNWRQAPGEGLEWVSLISVTATTYYA
DSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKVTWNNLFDYWGLGLTVTVSS
>S149_VL
EIVLTQSPGTLSLSPGERATLSCRASQPIDSYLAWYQQKPGQAPRLLIYNTVTRATGIPD
RFSGSGGTDFLTINRLEPEDFAVYYCQHQQYDWPPYIFGGGTKVEIK

>S218_VH
DVQLVESGGGVVRPGESLRLSCTASGFTFTSSTMNWRQAPGEGLDWVSSI STSGVITYY
ADSVKGRATISRDN SKNTLYLRLFSLRADDTAIYYCATDTFDHWGPGTLTVTVSS
>S218_VL
ALTQPASVSANPGETVKITCFGSSGNYGWFQQKSPGSAPVTIHYNNKRPSDIPSRFSGS
KSGSTGTLTITGVRAEDEAVYFCGAWETGSATFGAGTTLTVL

FIG. 110

	H1	H2	61
119	(1) DVQLLESGGVVQPGGSLRLSCAASGFSEFN <u>FAMT</u> WVRQAPGEGLEWVSTIGSGDTYYADS		
120	(1) DVQLVESGGGVVQPGGSLRLSCAASGFSEFN <u>FAMT</u> WVRQAPGEGLEWVSTIGSGDTYYADS		
121	(1) DVQLVESGGGVVQPGGSLRLSCAASGFSEFN <u>FAMT</u> WVRQAPGEGLEWVSTIGSGDTYYADS		
122	(1) DVQLVESGGGVVRPGE <u>SLRLSCAASGFSEFN</u> FAMT <u>WVRQAPGEGLEWVSTIGSGDTYYADS</u>		
	* * *		
	H3		120
119	(61) VKGRETTISRDN <u>SKNTLYLQMN</u> SLRAEDTAVYYCAK <u>DS</u> TVSWSGDEFFDYWGGLGLTLVTVSS		
120	(61) VKGRETTISRDN <u>SKNTLYLQMN</u> SLRAEDTAVYYCAK <u>DS</u> TVSWSGDEFFDYWGGLGLTRVTVSS		
121	(61) VKGRETTISRDN <u>SKNTLYLQMN</u> SLRAEDTAVYYCAK <u>DS</u> TVSWSGDEFFDYWGGLGLTLVTVSS		
122	(61) VKGRETTISRDN <u>SKNTLYLQMN</u> SLRAEDTAVYYCAK <u>DS</u> TVSWSGDEFFDYWGGLGLTLVTVSS		
			*

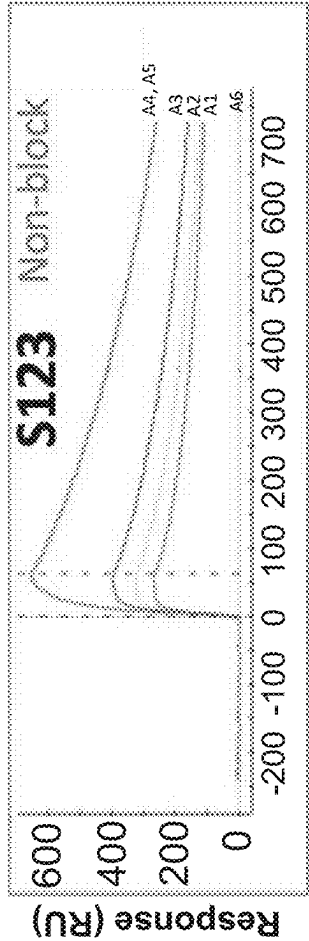
FIG. 11P

		L1	L2	54
119	(1)	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAWYQQRPQGAPRLLIYAARIR		
120	(1)	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAWYQQRPQGAPRLLIYAARIR		
121	(1)	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAWYQQRPQGAPRLLIYAARIR		
122	(1)	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAWYQQRPQGAPRLLIYAARIR		

		L3	108
119	(55)	ETGIPERFSGSGTEFTLTITSLQSEDFAVVYCQQYYDWPPEFTFGGGTKVEIK	
120	(55)	ETGIPERFSGSGTEFTLTITSLQSEDFAVVYCQQYYDWPPEFTFGGGTKVEIK	
121	(55)	ETGIPERFSGSGTEFTLTITSLQSEDFAVVYCQQYYDWPPEFTFGGGTKVEIK	
122	(55)	ETGIPERFSGSGTEFTLTITSLQSEDFAVVYCQQYYDWPPEFTFGGGTKVEIK	

- A1 = High affinity SIRPa (100nM)
- A2 = High affinity SIRPa (100nM) + 20nM CD47
- A3 = High affinity SIRPa (100nM) + 55nM CD47
- A4 = High affinity SIRPa (100nM) + 500 nM CD47
- A5 = High affinity SIRPa (100nM) + 1500 nM CD47
- A6 = PBST

FIG. 12A



Time (s)

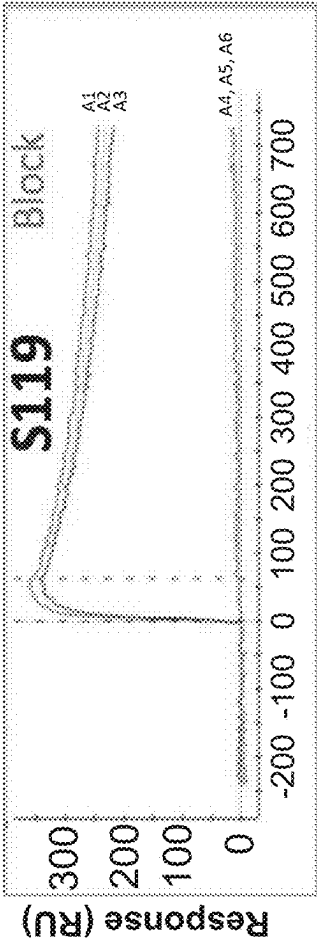


FIG. 12B

Time (s)

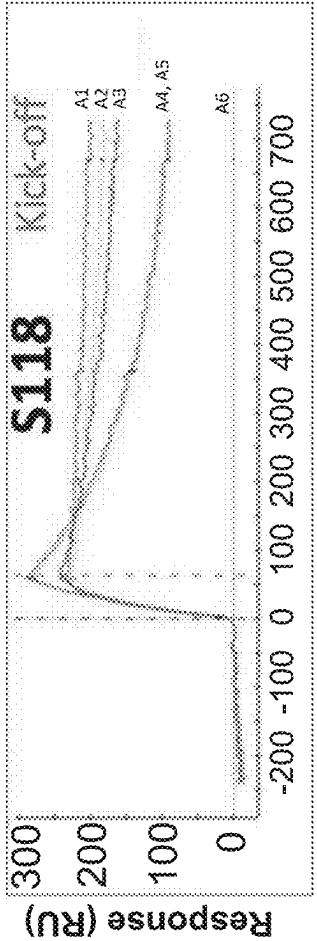


FIG. 12C

Time (s)

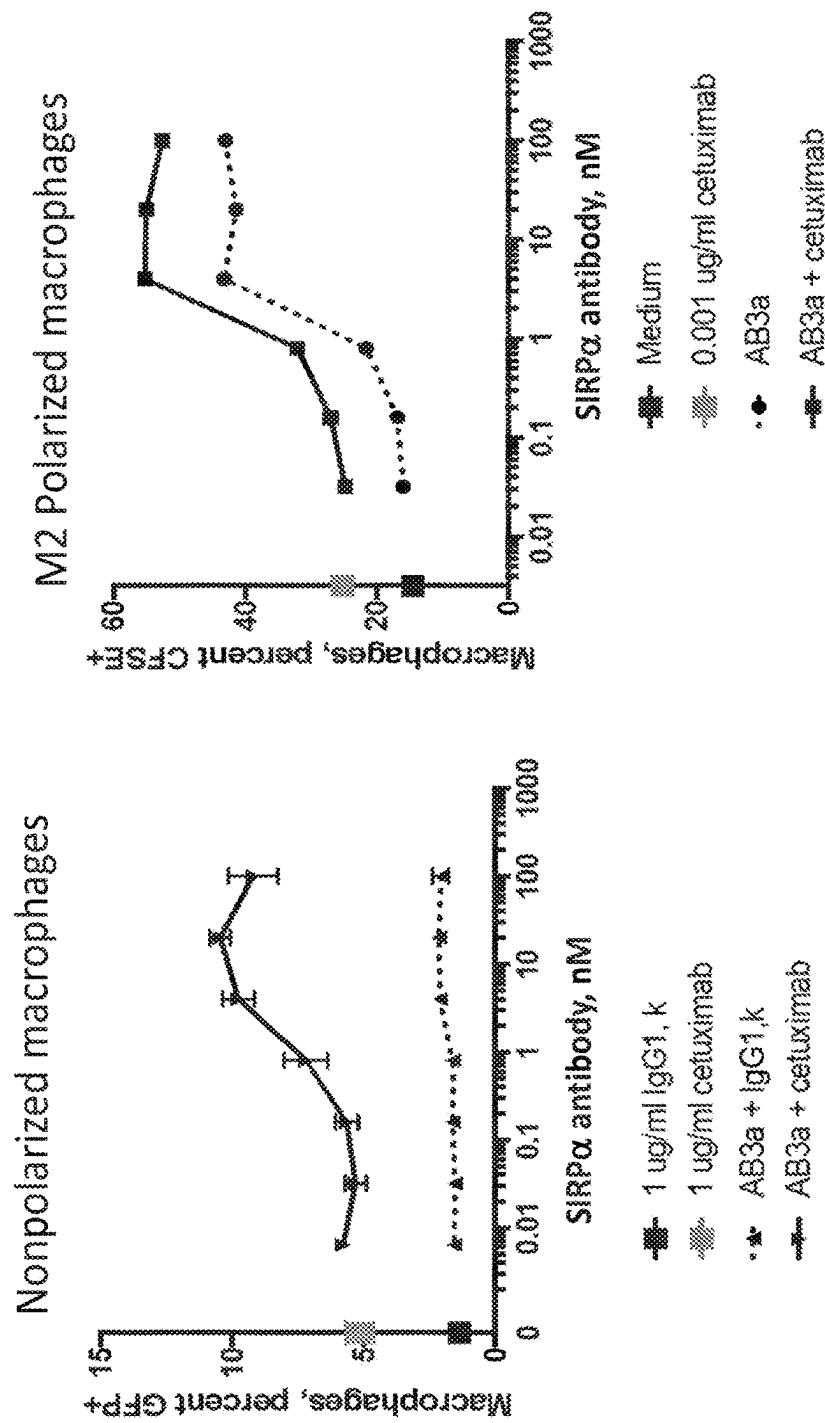


FIG. 13A

FIG. 13B

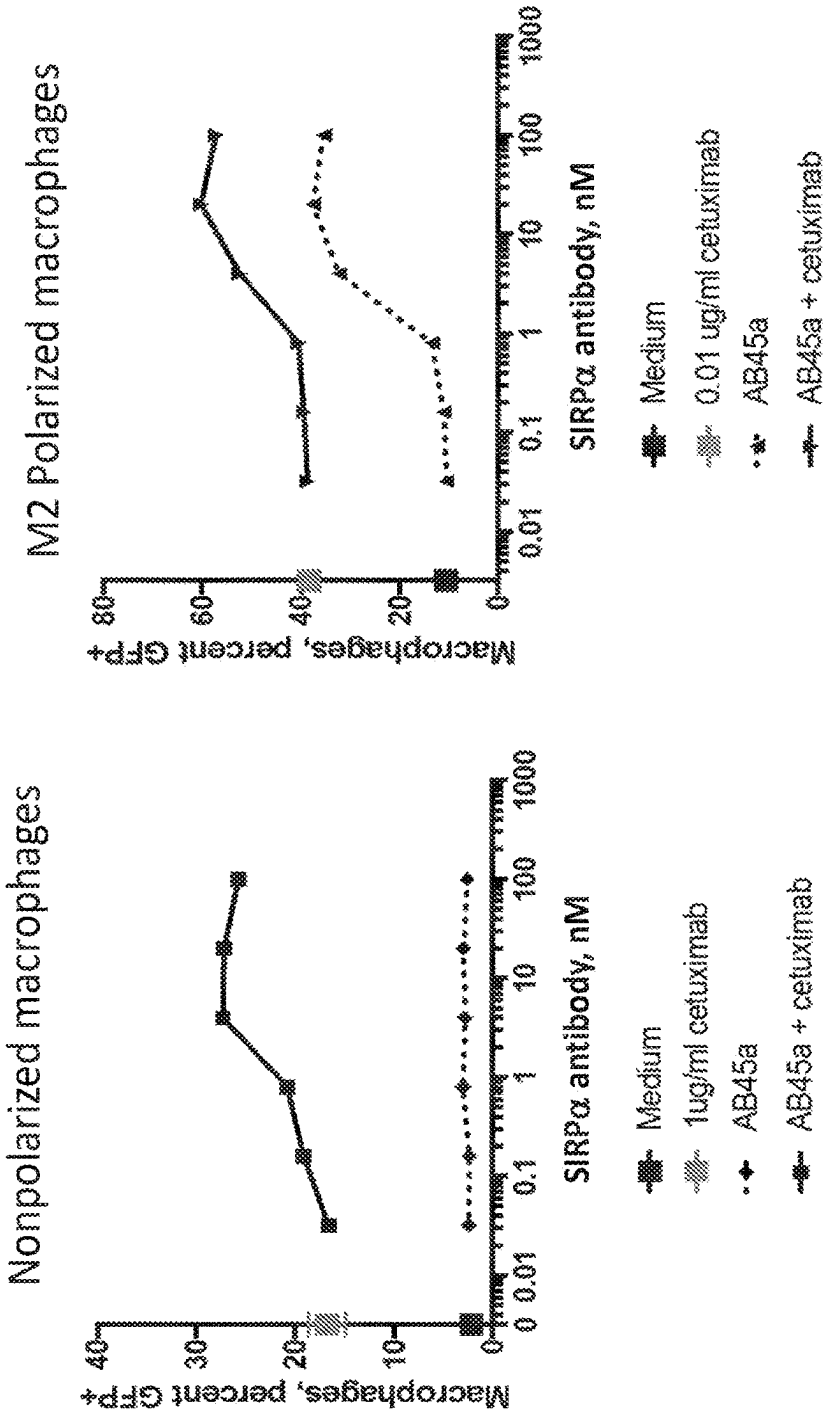


FIG. 13C

FIG. 13D

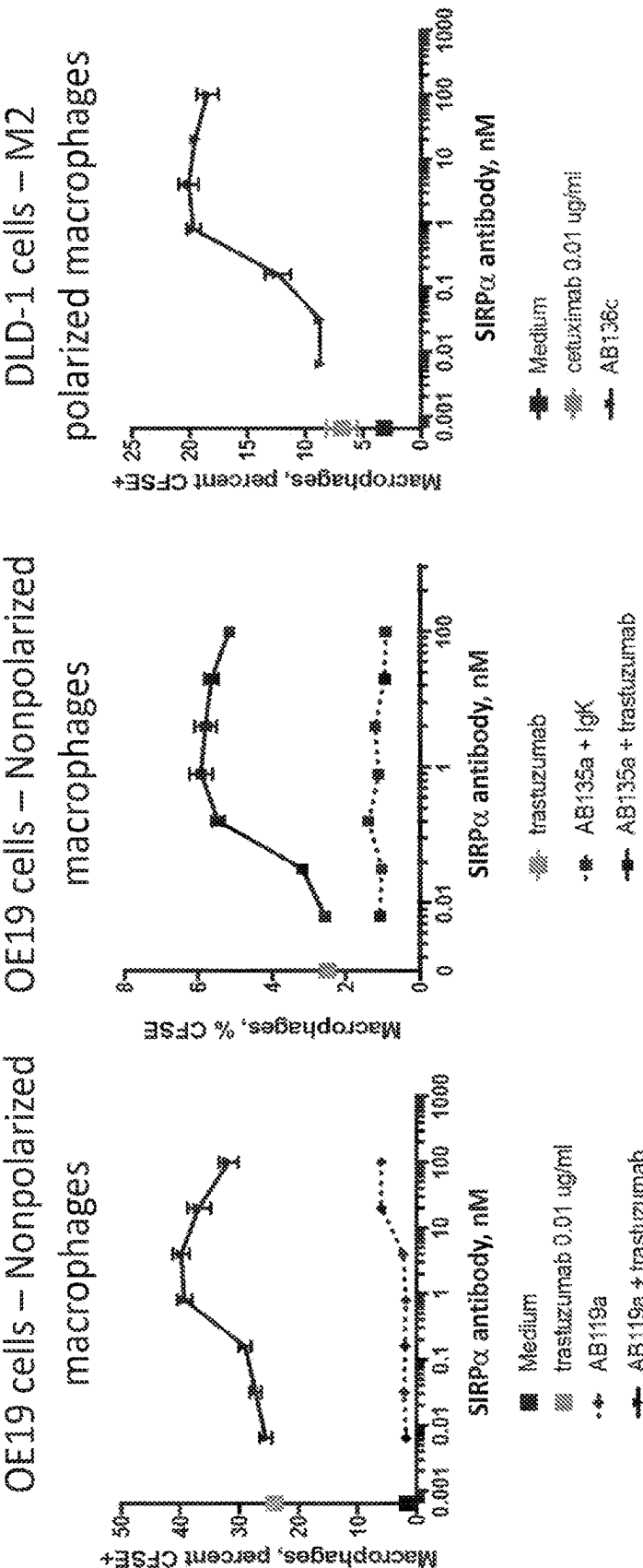
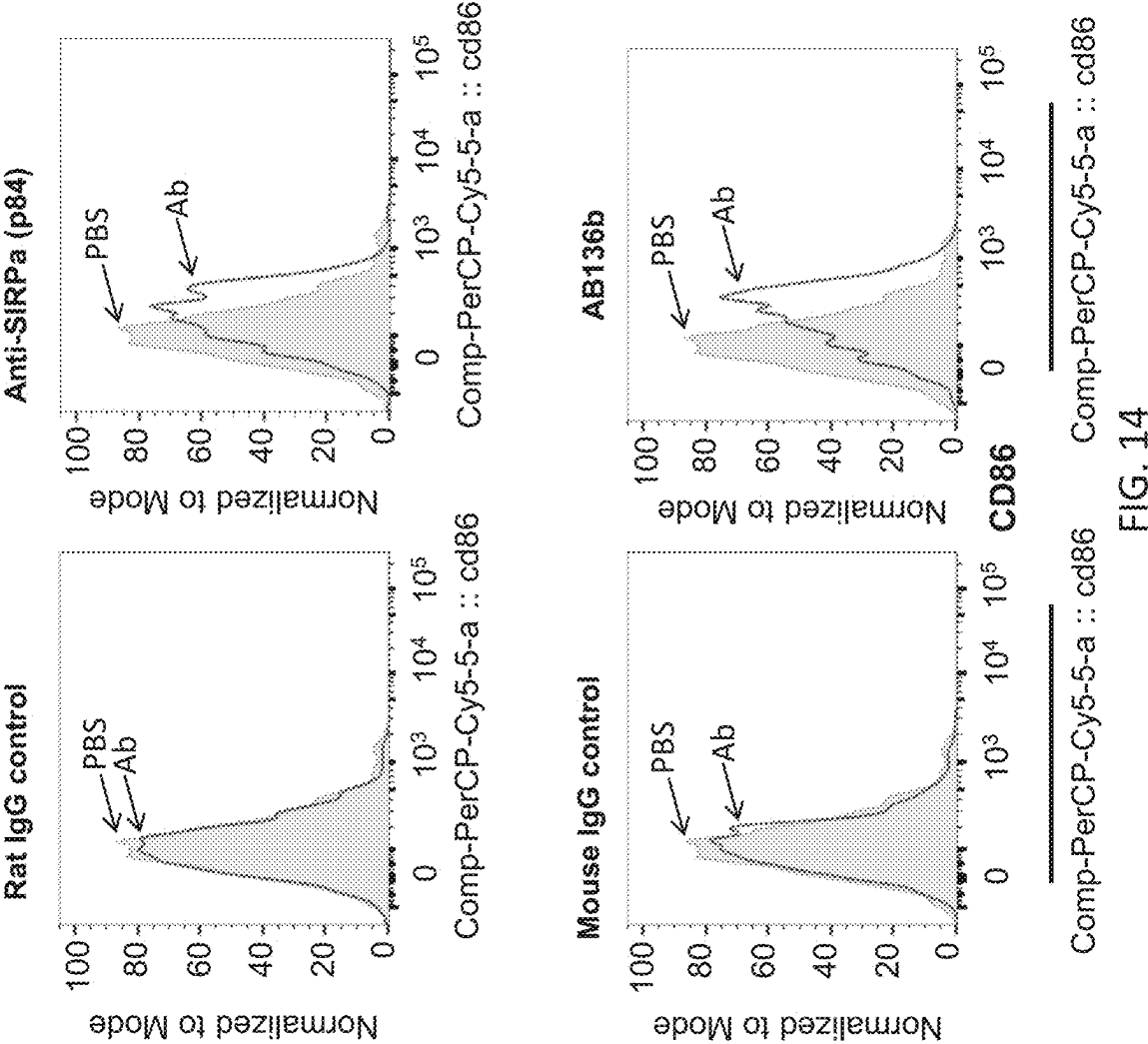
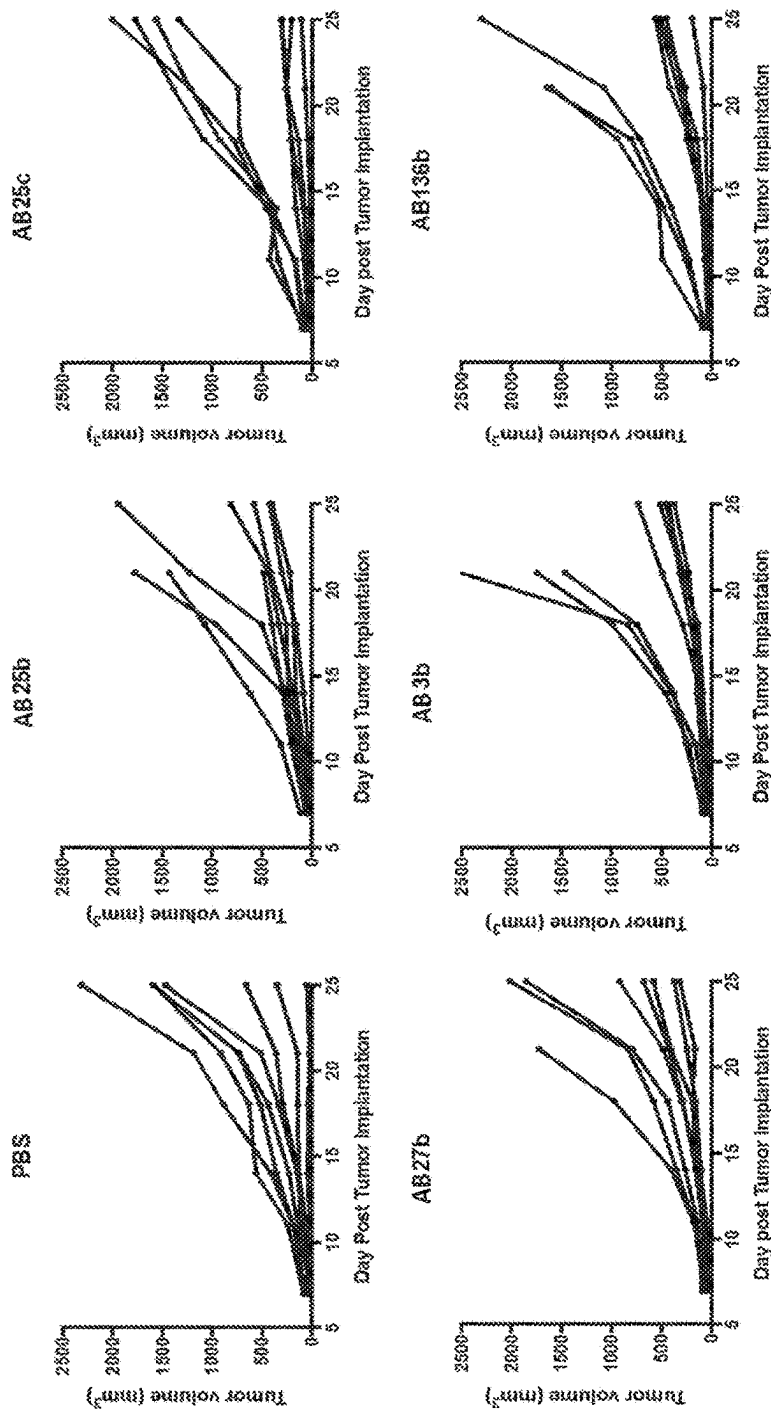


FIG. 13E

FIG. 13F

FIG. 13G





	PBS	AB 25b	AB 25c	AB 27b	AB 3b	AB 136b
Mice with tumor volume <600 mm ³	2	3	4	3	4	5

FIG. 15

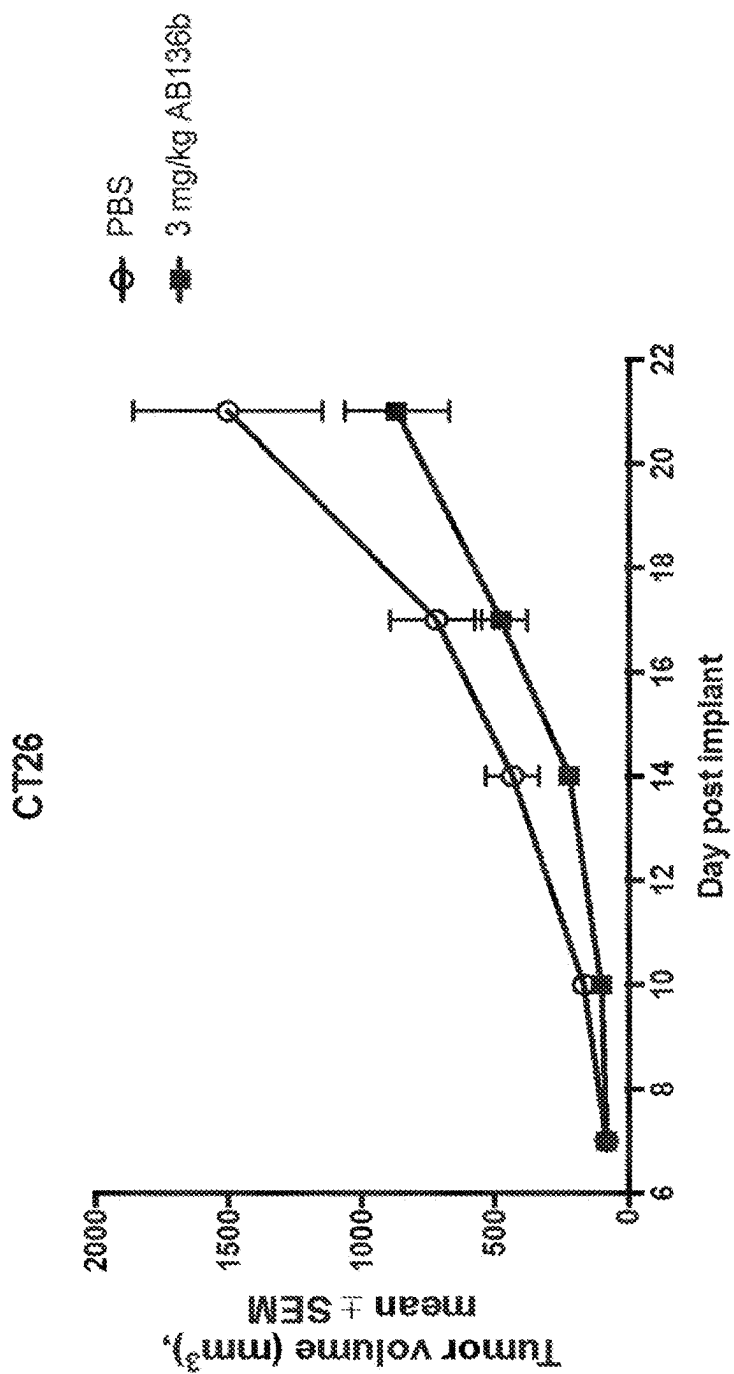


FIG. 16

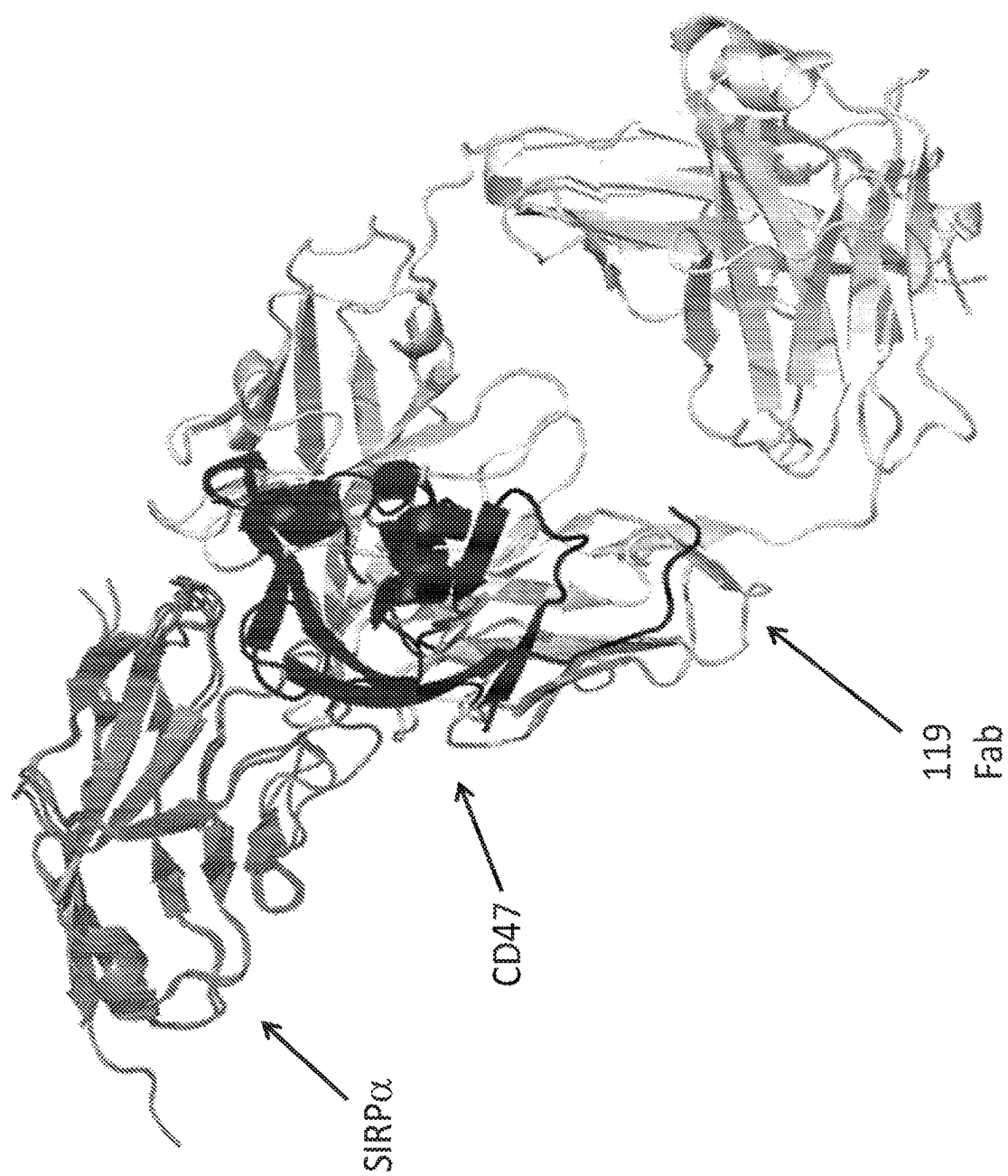


FIG. 17A

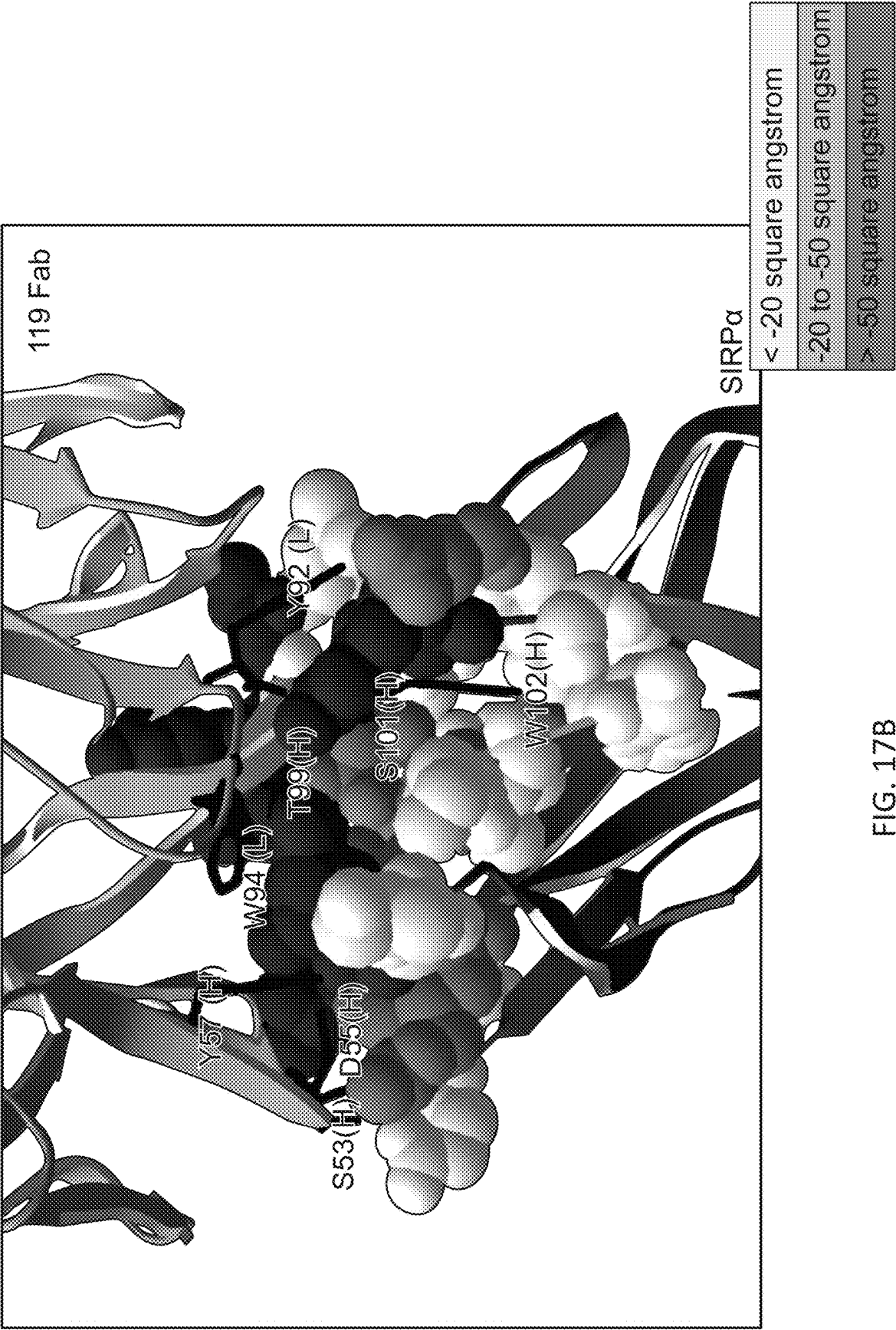


FIG. 17B

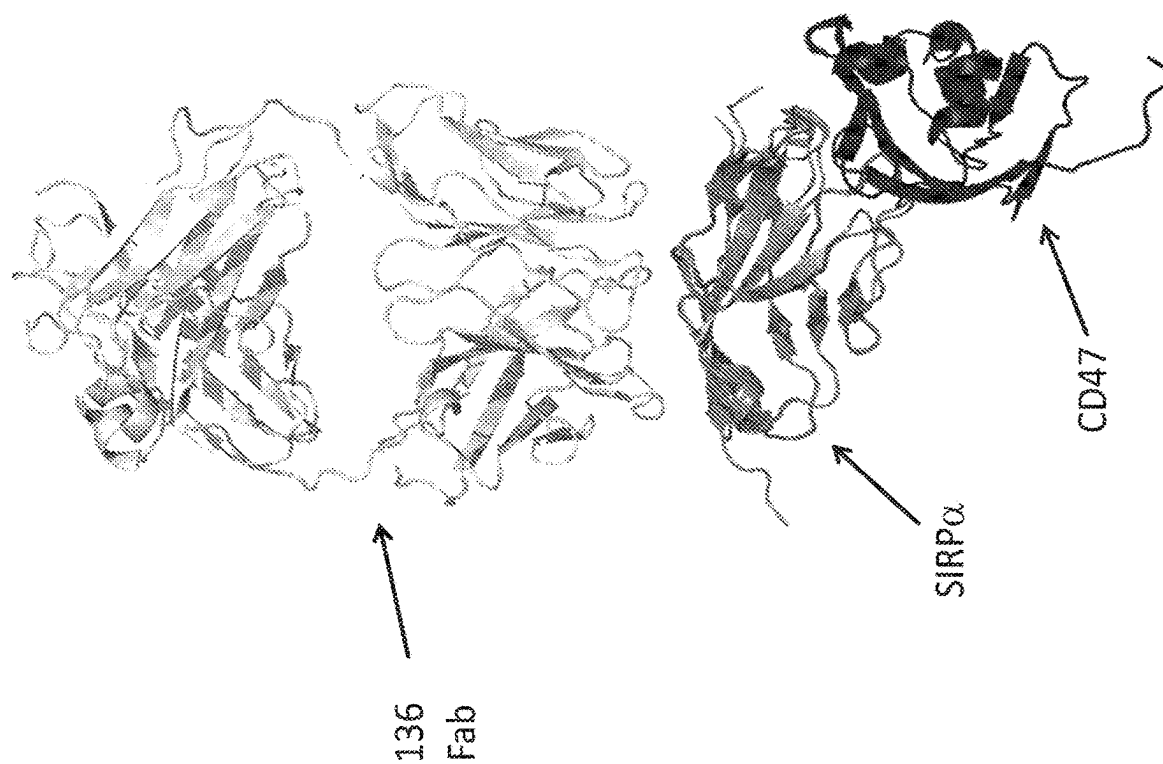


FIG. 18A

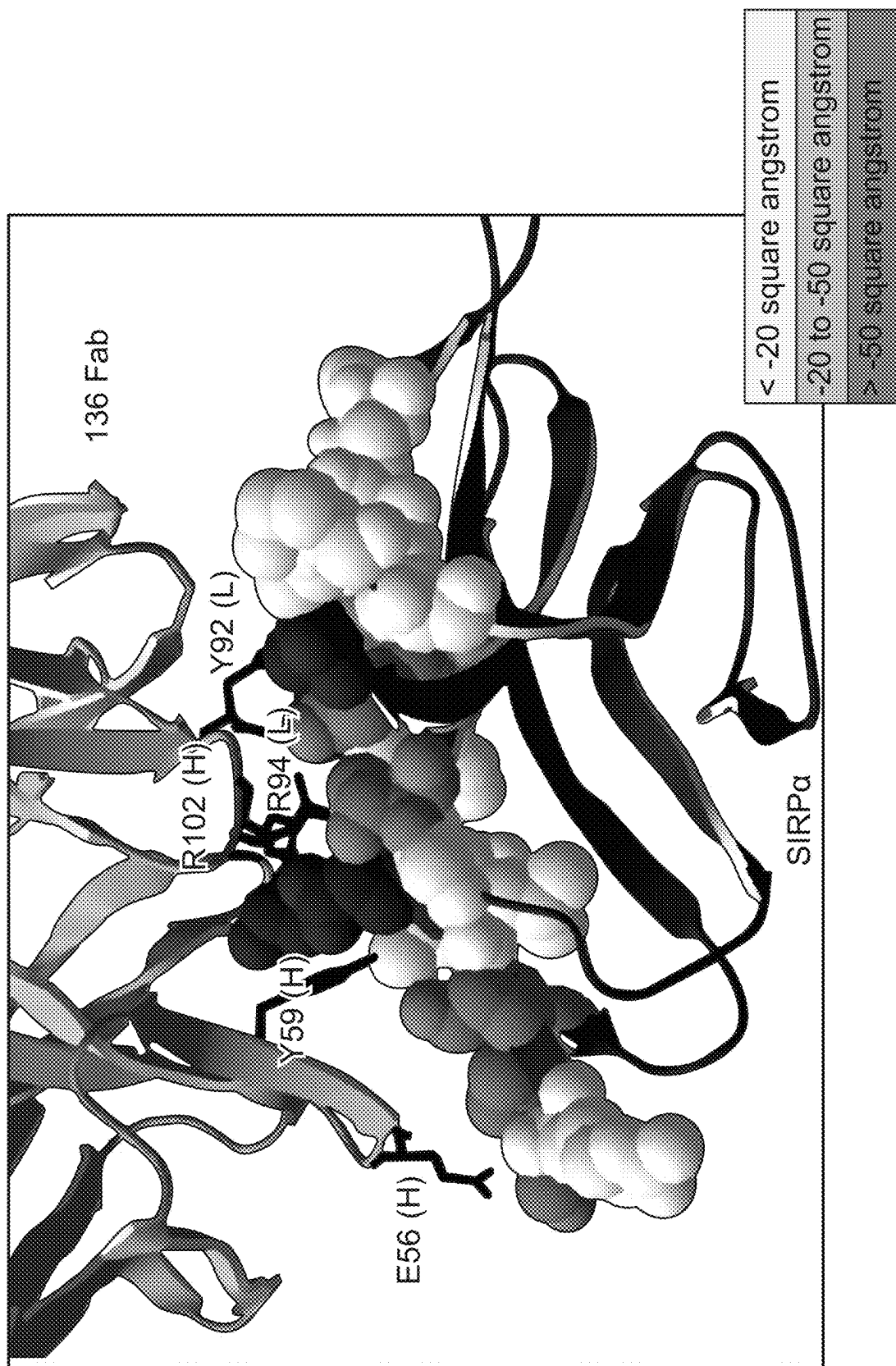


FIG. 18B

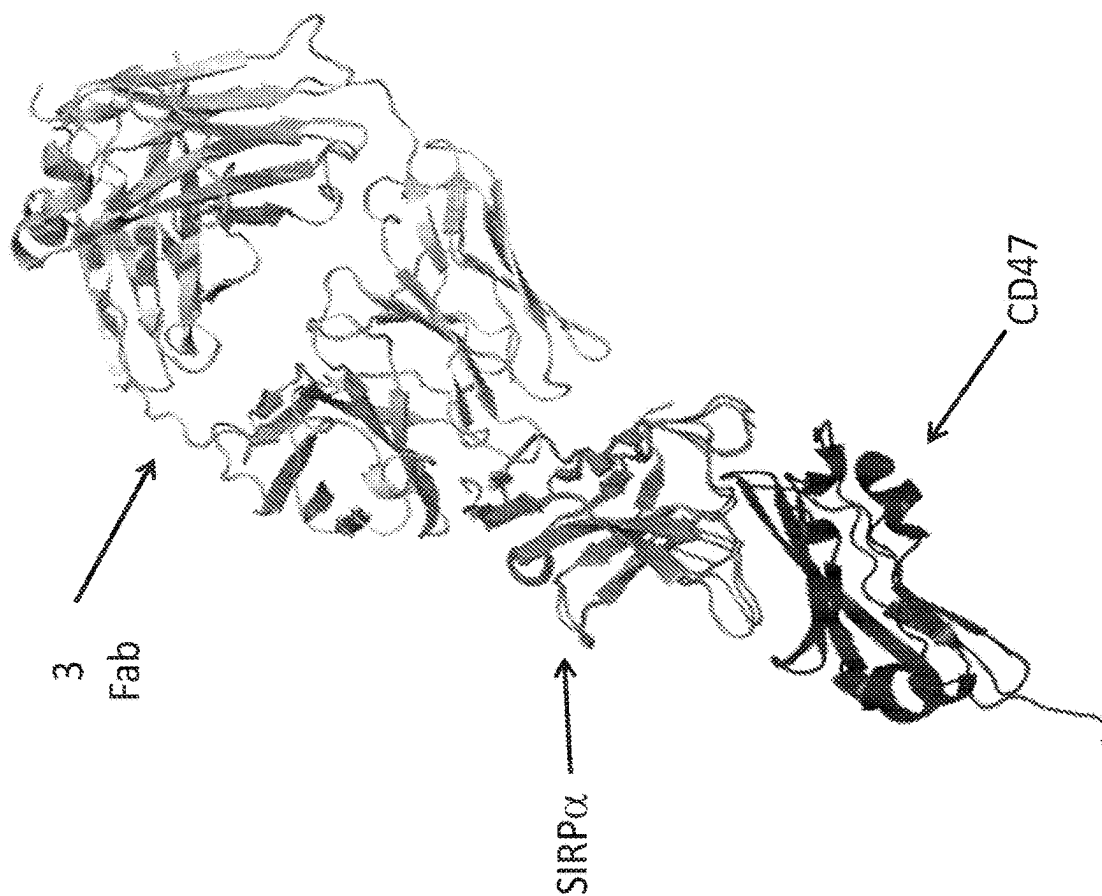


FIG. 19A

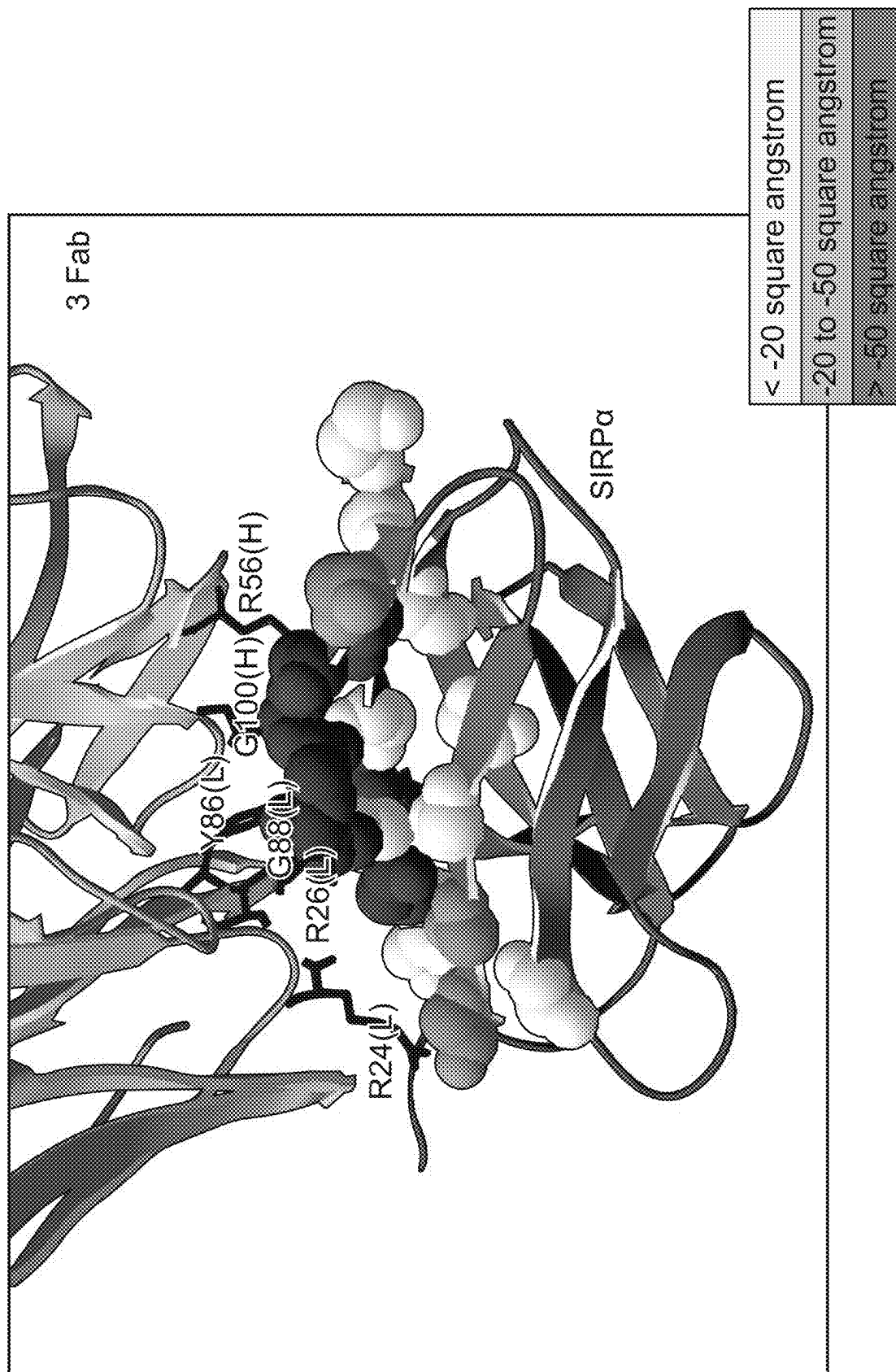


FIG. 19B

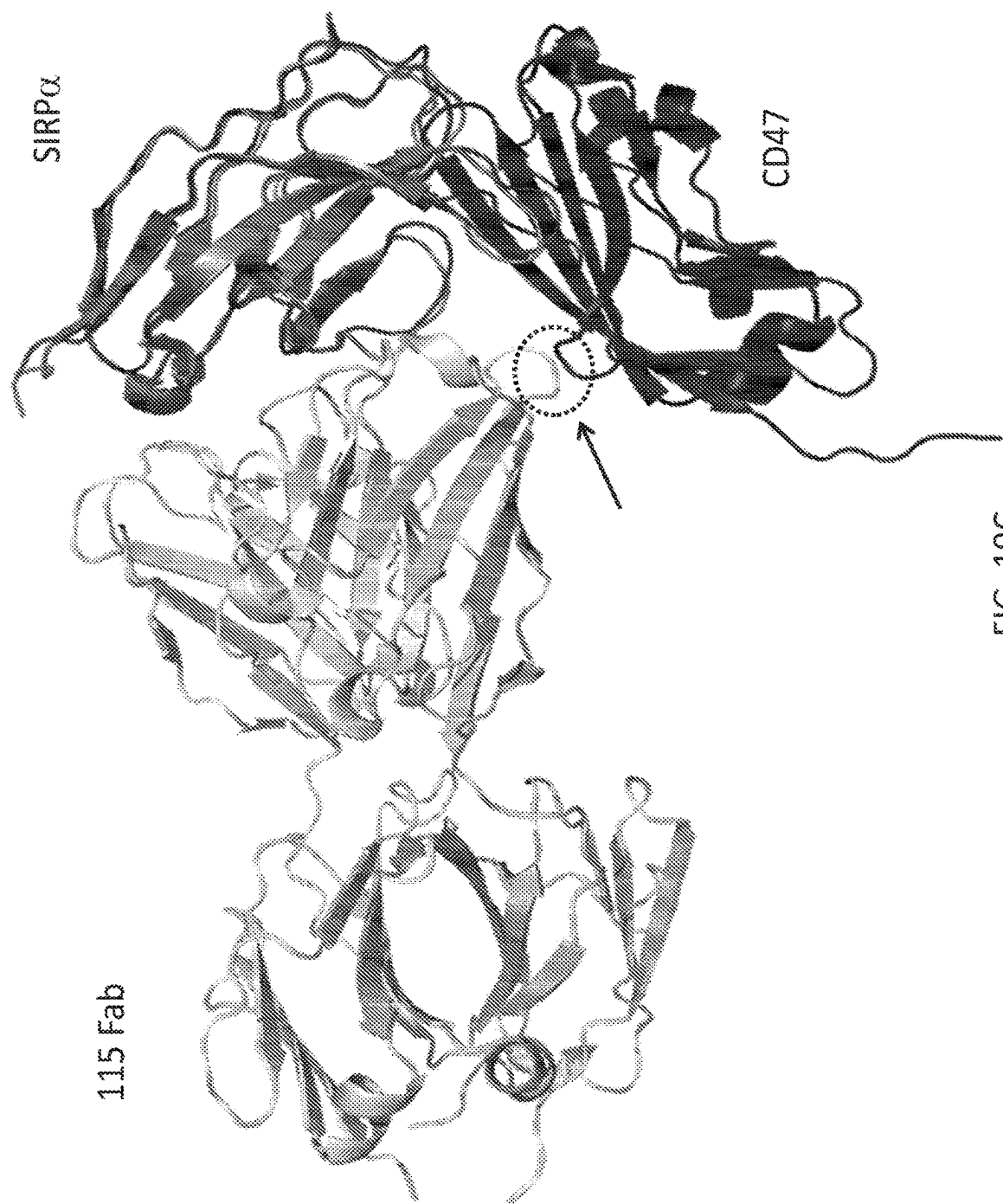


FIG. 19C

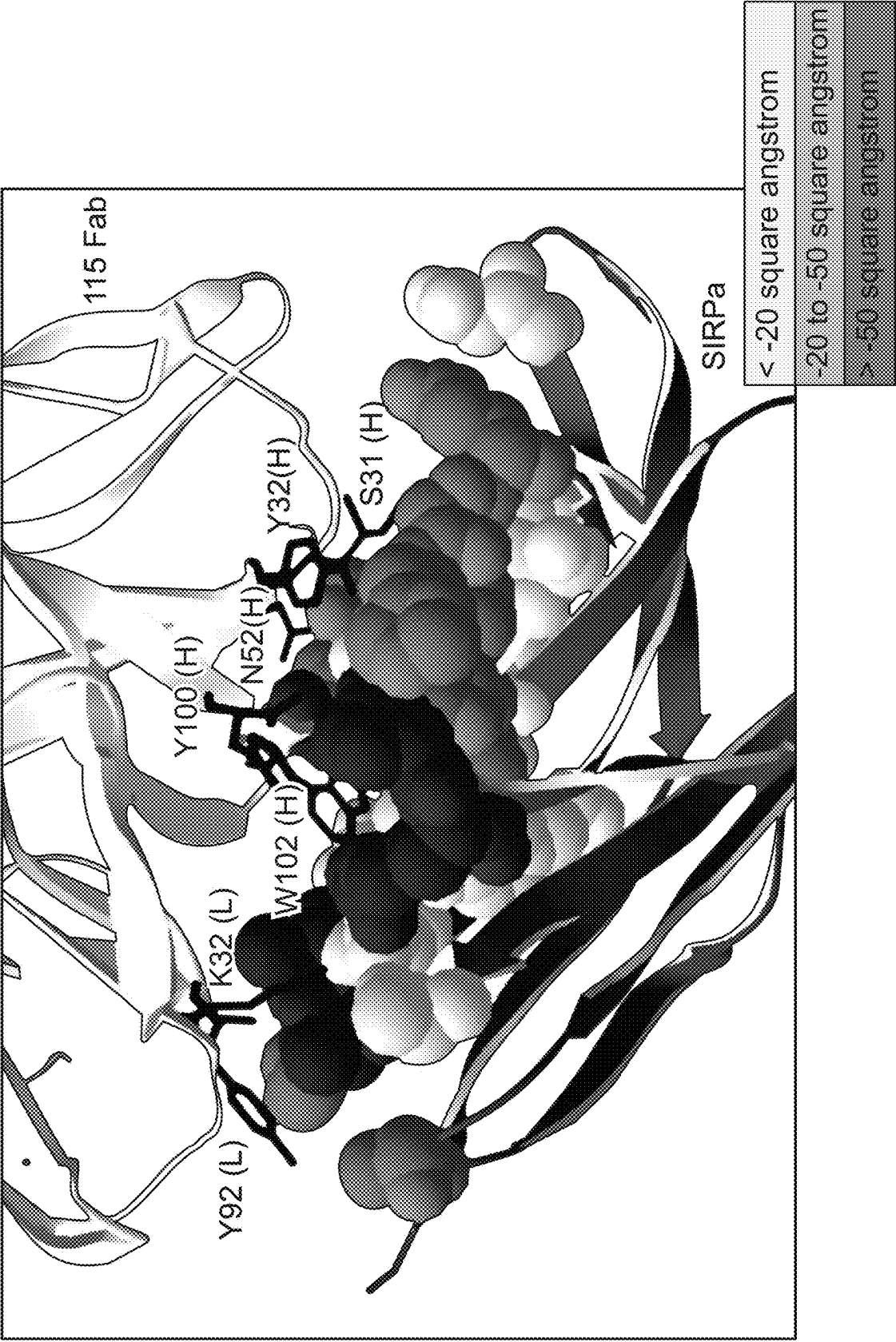


FIG. 19D

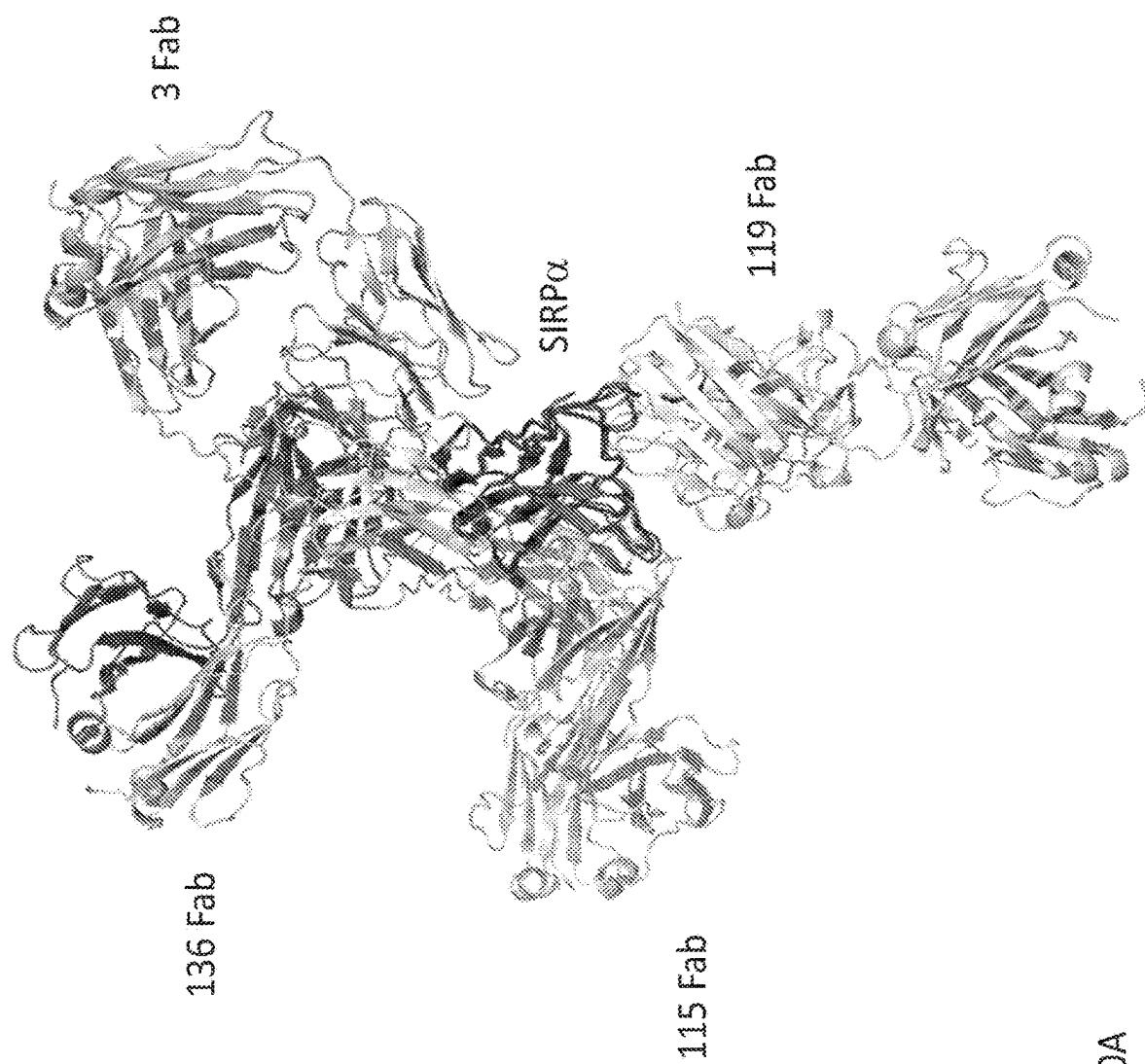


FIG. 20A

FIG. 20B

< -20 square angstrom
 -20 to -50 square angstrom
 > -50 square angstrom

Residue name	Residue number	Buried Surface Area (square angstrom)				
		CD47	Complex 1 (119)	Complex 2 (136)	Complex 3 (3)	Complex 4 (115)
GLU	3		-13.8			
GLN	5				-11.2	
VAL	6			-5.7	-7.2	
ILE	7				-61.9	
GLN	8			-13.2	-20.1	
PRO	9			-14.6	-7.2	
ASP	10			-1	50.3	
LYS	11			-157.6	-159.7	
SER	12			-17.6	-54.5	
LEU	14			-42.2	-29	
ALA	17					-19.7
ARG	24				-15.1	
THR	26				-28.1	
THR	28		-5.9		-33.5	
SER	29	-1.1	-17.3			
LEU	30	-44.3	-39.8			
ILE	31	-15.2	-106.5			
PRO	32		-32.2			
VAL	33	88.7	81.4			
GLY	34	-25				
PRO	35	-42	-6.4			
ILE	36	-10	-13.7			

FIG. 20C

Residue name	Residue number	Buried Surface Area (square angstrom)				
		CD47	Complex 1 (119)	Complex 2 (136)	Complex 3 (3)	Complex 4 (115)
GLN	37	-1.7				
TRP	38		-2.1			
PHE	39					-2
ARG	40					-18.2
ALA	42			-52.9		
GLY	43			-16.8		
PRO	44			-10.9		-44.1
GLY	45					-21.3
ARG	46			-6.8		-15.7
GLU	47					96.3
LEU	48					-70.3
ILE	49					-20.4
TYR	50	-5.1	-14.2			
ASN	51	-7.3				-3.3
GLN	52	-91.5	-81.9			
LYS	53	-115.8	-114.8			
GLU	54	-57.9	-35.3			-47.7
GLY	55					-19.8
HIS	56					-35.7
PHE	57					-20.6
PRO	58					-69.6
ARG	59					-69

< -20 square angstrom

-20 to -50 square angstrom

> -50 square angstrom

FIG. 20D

< -20 square angstrom
 -20 to -50 square angstrom
 > -50 square angstrom

Residue name	Residue number	Buried Surface Area (square angstrom)				
		CD47	Complex 1 (119)	Complex 2 (136)	Complex 3 (3)	Complex 4 (115)
THR	62		-21.5			
SER	64		-5.8			
ASP	65		-7.5			
LEU	66	-20.9	-11.6			
THR	67	-85.8	-62.4			
LYS	68	-24.6	-18.2			
ARG	69	-174	-134.3			
ASN	70		-73.7			
ASN	71		-28.4			
MET	72		-23.4		-12.5	
ASP	73		-1.3			
PHE	74	-11.8	-34.9			-11.2
SER	75		-2.2			
ILE	76		-4.5			
ALA	80					
THR	82					-52
PRO	83					-44.5
ALA	84					-84.6
ASP	85					-16.6
GLY	87			-4.1		
THR	88			-34.1	-12.3	
TYR	89					-1.5

FIG. 20E

< -20 square angstrom
 -20 to -50 square angstrom
 > -50 square angstrom

Residue name	Residue number	Buried Surface Area (square angstrom)				
		CD47	Complex 1 (119)	Complex 2 (136)	Complex 3 (3)	Complex 4 (115)
TYR	90			-48.8	-4.7	
LYS	93	-5.2				
ARG	95		-44.5			
LYS	96	-96.8	-83.2			
GLY	97	-22.7	-2.9			
SER	98	-56.1	-14.2			
PRO	99	-19.7				
ASP	100					-12.4
ASP	101	-9.9				
GLU	103			-20.1		
PHE	104			-6.9		
LYS	105			-13.1		
SER	106			-49.9		
GLY	107			-6.9		
ALA	108			-83.6	-54.9	
GLY	109			-12.5	-8.5	
GLU	111			-96.8	-77	
LEU	112			-3		
SER	113			-41.5	-3.7	
ARG	115			-4.9	-4.6	
ALA	116			-23.8		

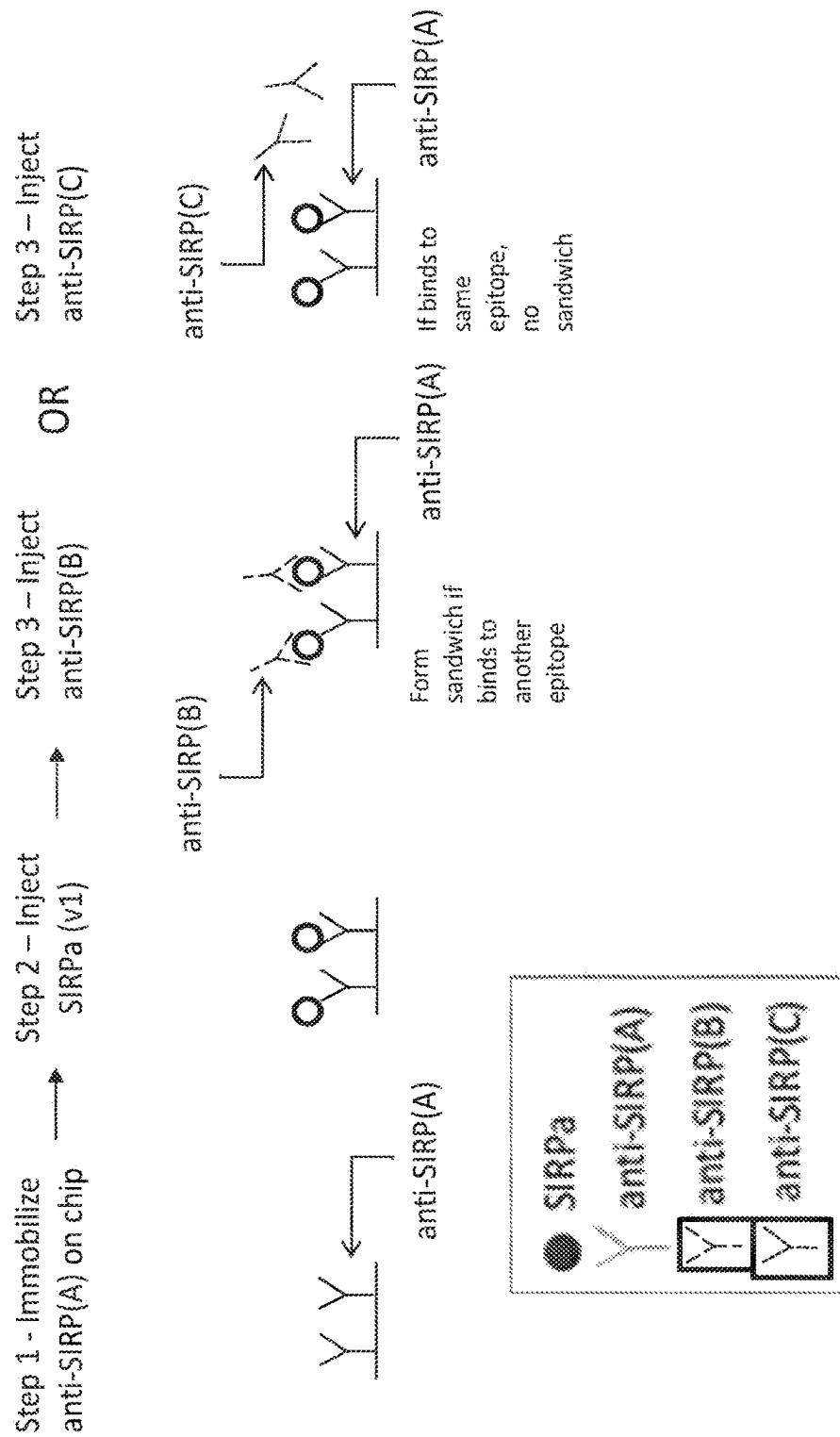


FIG. 21A

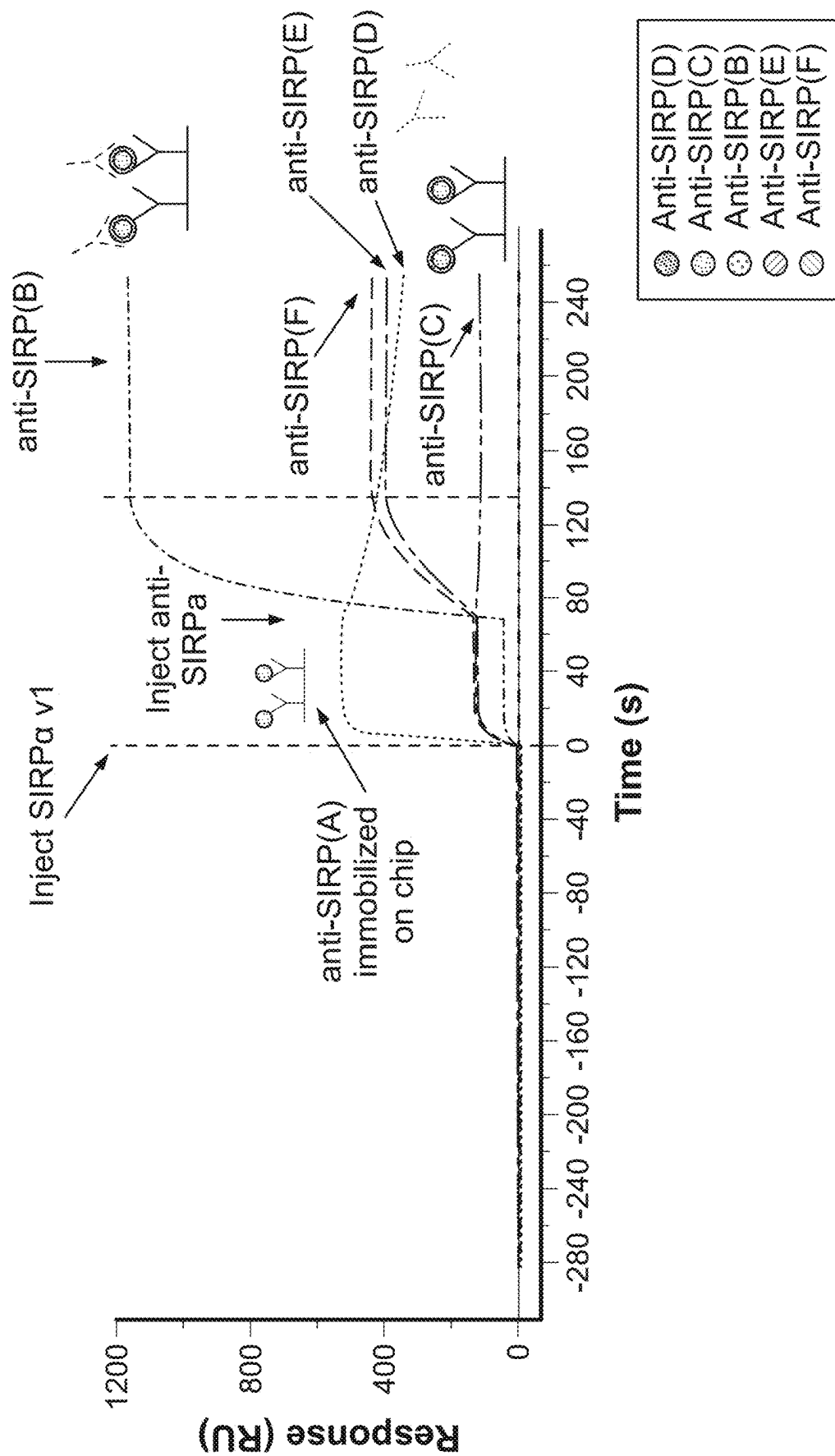


FIG. 21B

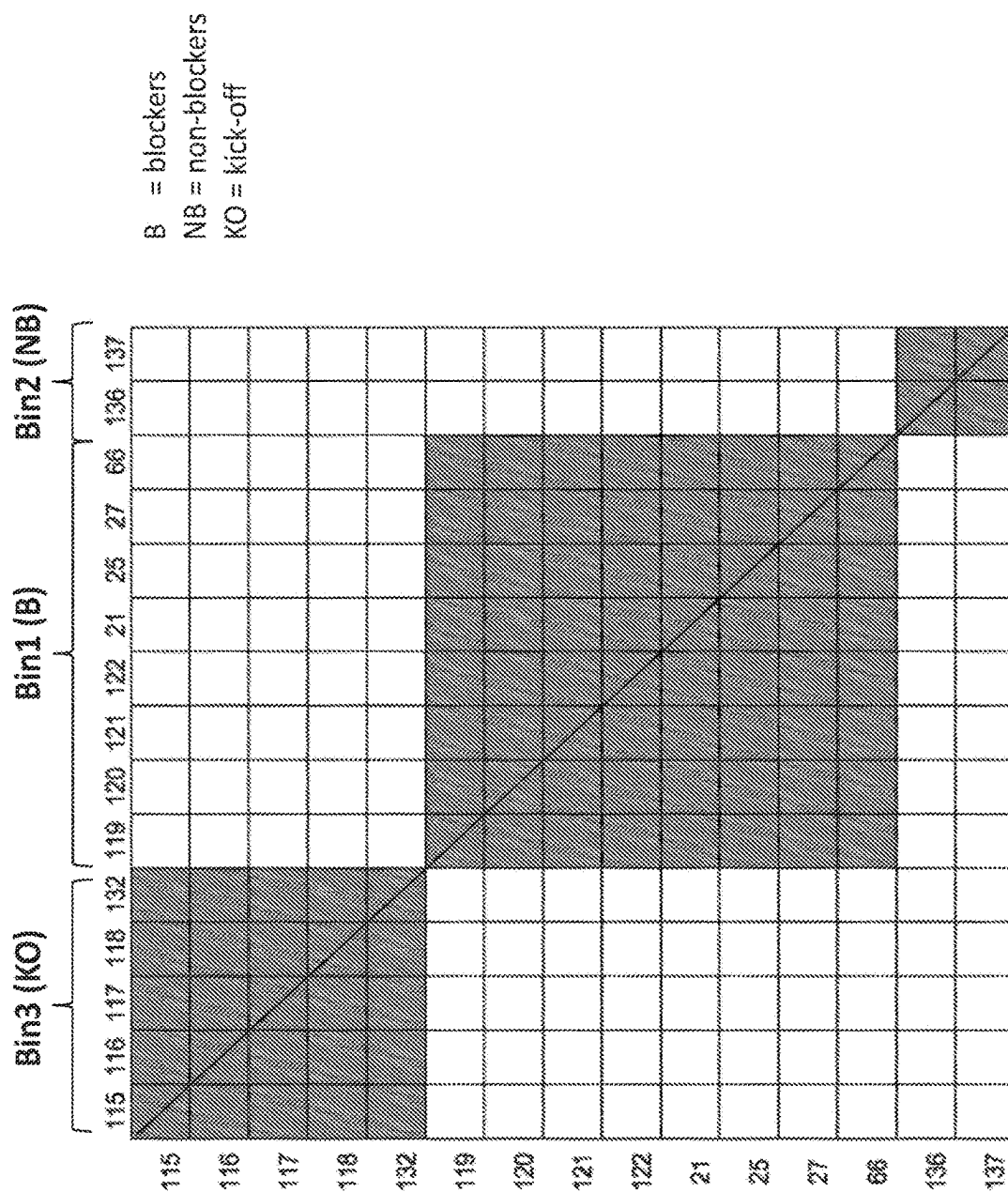
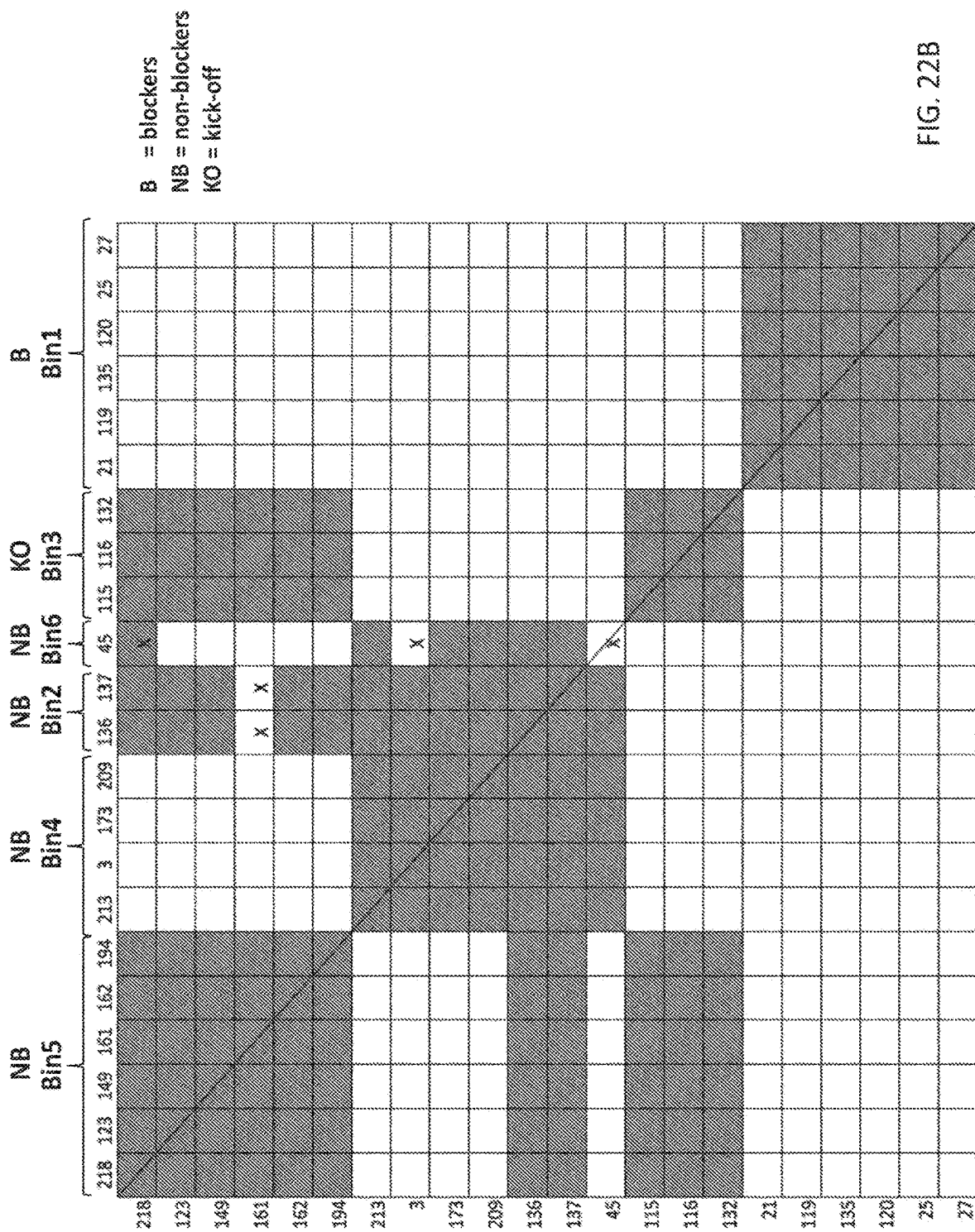


FIG. 22A



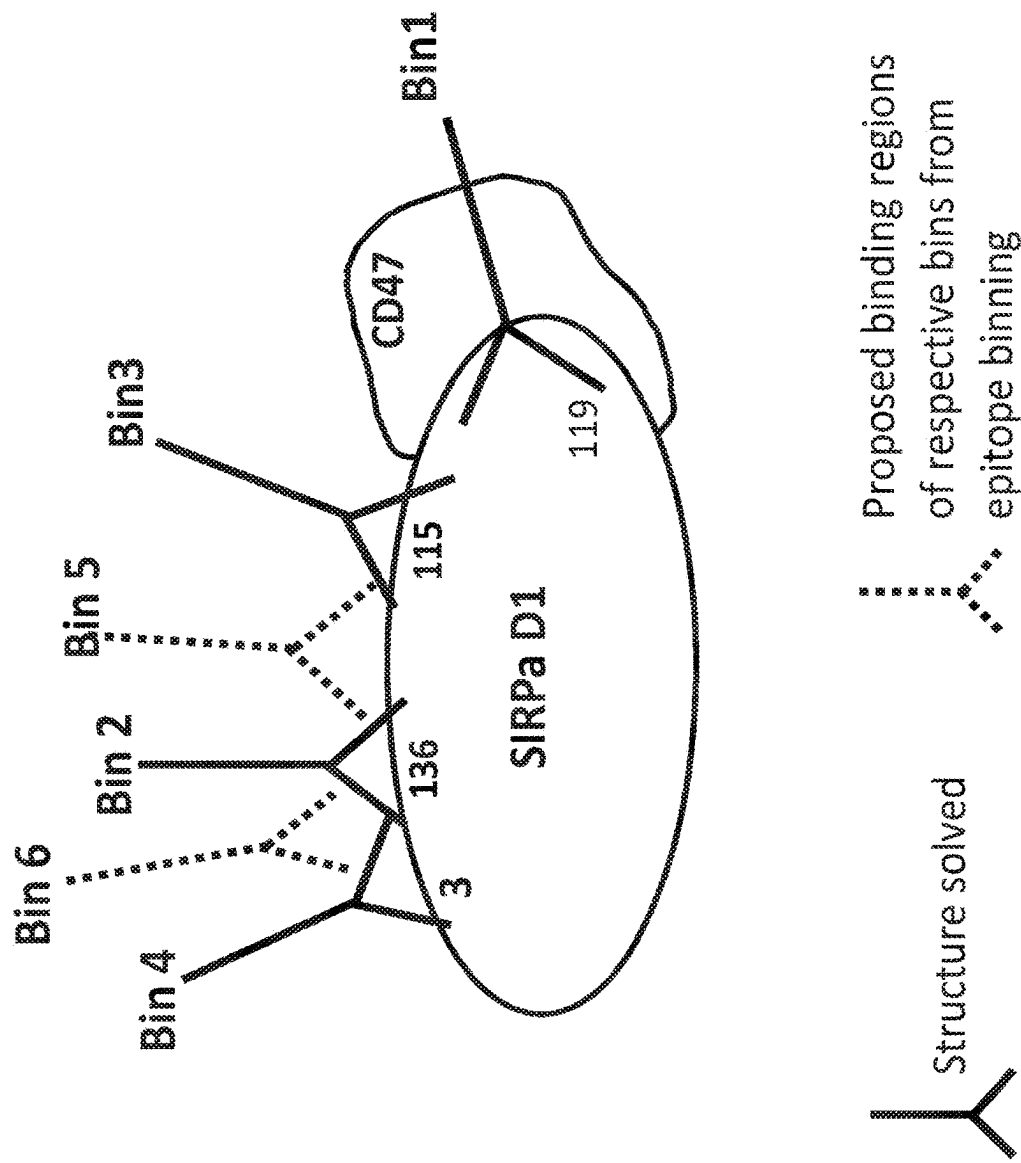


FIG. 23

1			H1	40
	119_VH_Wt	(1)	DVQLLESGGVVQPGGSLRLSCAASGFSNFAMTWVRQA	
	119_VH_Mutall	(1)	EVQLLESGGVVQPGGSLRLSCAASGFSNFAMTWVRQA	
	119_VH_Mutall_V34M	(1)	EVQLLESGGVVQPGGSLRLSCAASGFSNFAMTWVRQA	
	119_VH_Mutall_V34L	(1)	EVQLLESGGVVQPGGSLRLSCAASGFSNFAMTWVRQA	
		*		*
41			H2	80
	119_VH_Wt	(41)	PCEGLEWVSTIGSGDTYYADSVKGRFTISRDN SKNTLYLQ	
	119_VH_Mutall	(41)	PC KGLEWVSTIGSGDTYYADSVKGRFTISRDN SKNTLYLQ	
	119_VH_Mutall_V34M	(41)	PC KGLEWVSTIGSGDTYYADSVKGRFTISRDN SKNTLYLQ	
	119_VH_Mutall_V34L	(41)	PC KGLEWVSTIGSGDTYYADSVKGRFTISRDN SKNTLYLQ	
		*		*
81			H3	120
	119_VH_Wt	(81)	MNSLRAEDTAVYYCAKDS TVSN SGDEF DYWG LGTLVTVSS	
	119_VH_Mutall	(81)	MNSLRAEDTAVYYCAKDS TVSN SGDEF DYWG LGTLVTVSS	
	119_VH_Mutall_V34M	(81)	MNSLRAEDTAVYYCAKDS TVSN SGDEF DYWG LGTLVTVSS	
	119_VH_Mutall_V34L	(81)	MNSLRAEDTAVYYCAKDS TVSN SGDEF DYWG LGTLVTVSS	
		*		*

FIG. 24A

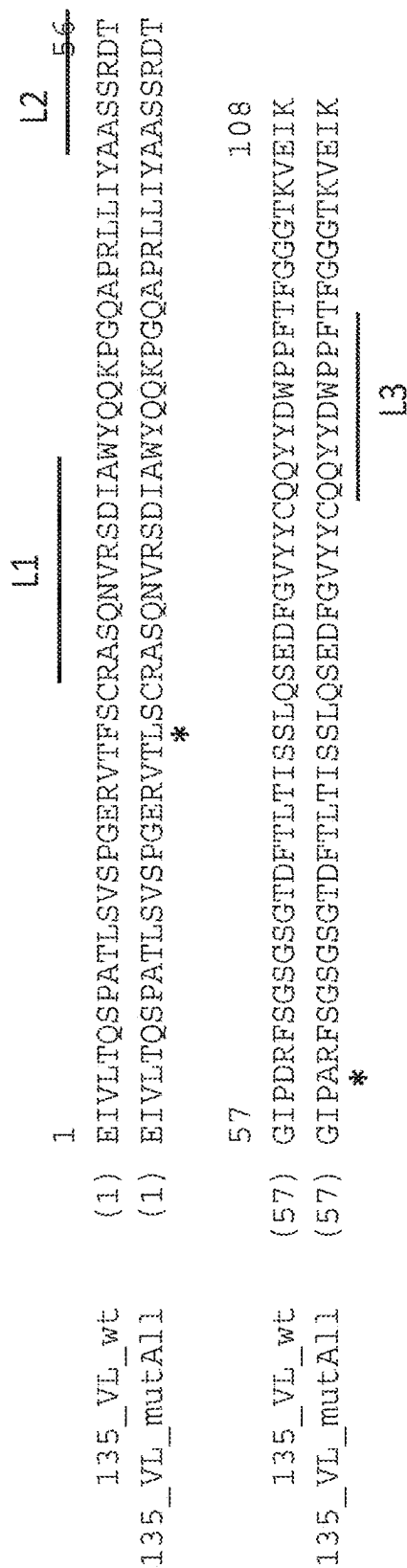


FIG. 25A

	<u>L1</u>	<u>L2</u> ⁵⁶
1		
136_VL_wt	(1) ETVLTQSPGTLTSLSPGERATLTCRASQSVVYTYLAWYQEKPGQAPRLLIYGASSRAT	
136_VL_mutall	(1) EIVLTQSPGTLTSLSPGERATLTCRASQSVVYTYLAWYQQKPGQAPRLLIYGASSRAT	
136_VL_Mutall_I2T	(1) ETVLTQSPGTLTSLSPGERATLTCRASQSVVYTYLAWYQQKPGQAPRLLIYGASSRAT*	
57		108
136_VL_wt	(57) GIPDRFSGSGSGTEFTLTISSLQSEDFAVVYCQQYVDRPPLTFGGGKVEIK	
136_VL_mutall	(57) GIPDRFSGSGSGTEFTLTISSLQSEDFAVVYCQQYVDRPPLTFGGGKVEIK	
136_VL_Mutall_I2T	(57) GIPDRFSGSGSGTEFTLTISSLQSEDFAVVYCQQYVDRPPLTFGGGKVEIK	
		<u>L3</u>

FIG. 26A

1	136_VH_wt	H1	40
(1)	136_VH_wt	DVQLVESGGGVVPPGESLRLSCAASGFTFSYDMNWVROA	
(1)	136_VH_mutall	EVQLVESGGGVVQPGRSLRLSCAASGFTFSYDVNWVROA	
(1)	136_VH_Mutall_V34M	EVQLVESGGGVVQPGRSLRLSCAASGFTFSYDMNWVROA	
(1)	136_VH_Mutall_V34L	EVQLVESGGGVVQPGRSLRLSCAASGFTFSYDLNWVROA	
*		*	*
41	136_VH_wt	H2	80
(41)	136_VH_wt	PGGLEWVSLISGGEIYYADSVKGRFTISRDN SKNTLY	
(41)	136_VH_mutall	PGKLEWVSLISGGEIYYADSVKGRFTISRDN SKNTLY	
(41)	136_VH_Mutall_V34M	PGKLEWVSLISGGEIYYADSVKGRFTISRDN SKNTLY	
(41)	136_VH_Mutall_V34L	PGKLEWVSLISGGEIYYADSVKGRFTISRDN SKNTLY	
*		*	*
81	136_VH_wt	H3	119
(81)	136_VH_wt	LQMNSLRAEDTAVYYCAKENNRYRFFDDWGGLTVTVSS	
(81)	136_VH_mutall	LQMNSLRAEDTAVYYCAKENNRYRFFDDWGGLTVTVSS	
(81)	136_VH_Mutall_V34M	LQMNSLRAEDTAVYYCAKENNRYRFFDDWGGLTVTVSS	
(81)	136_VH_Mutall_V34L	LQMNSLRAEDTAVYYCAKENNRYRFFDDWGGLTVTVSS	
*		*	*

FIG. 26B

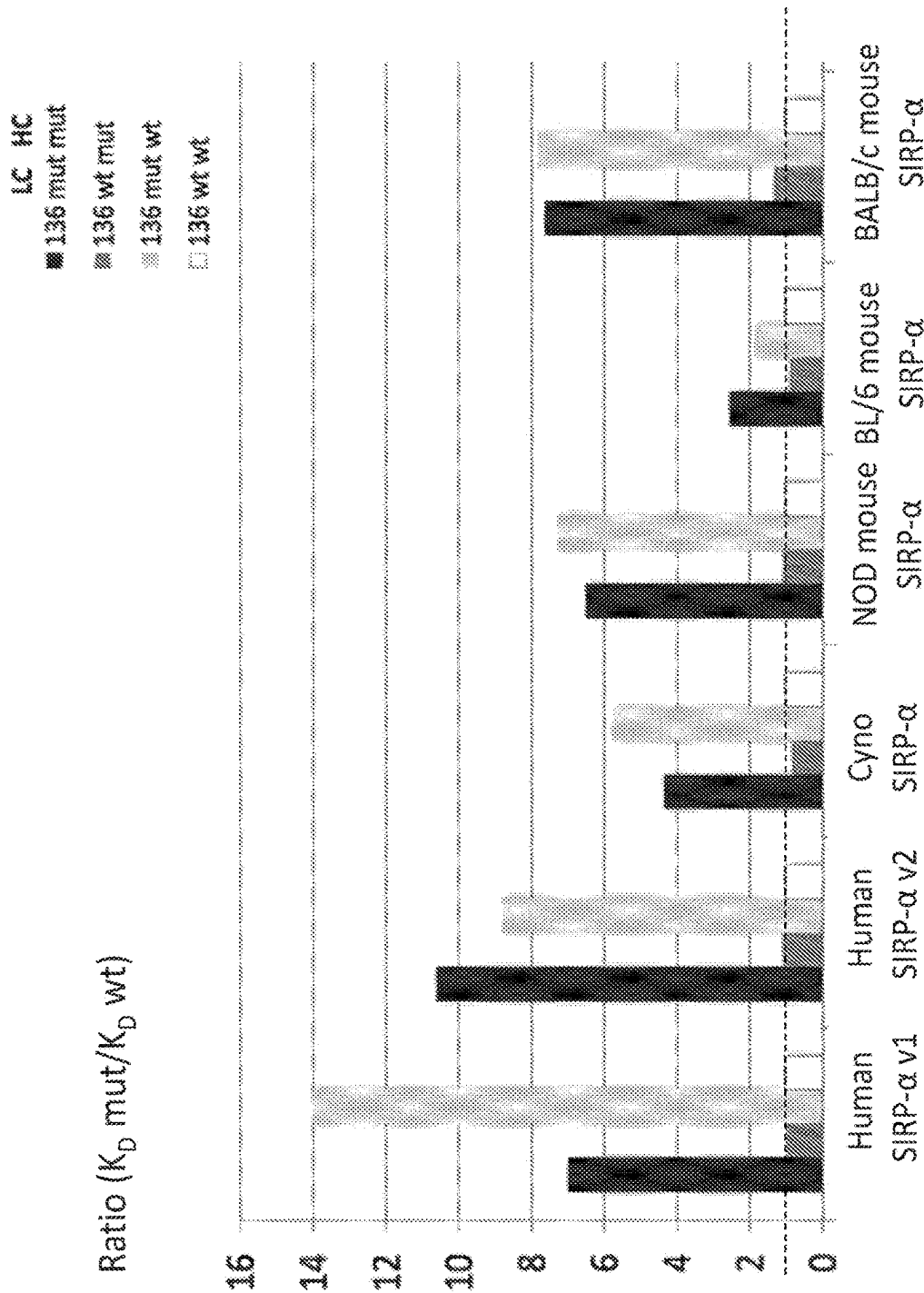
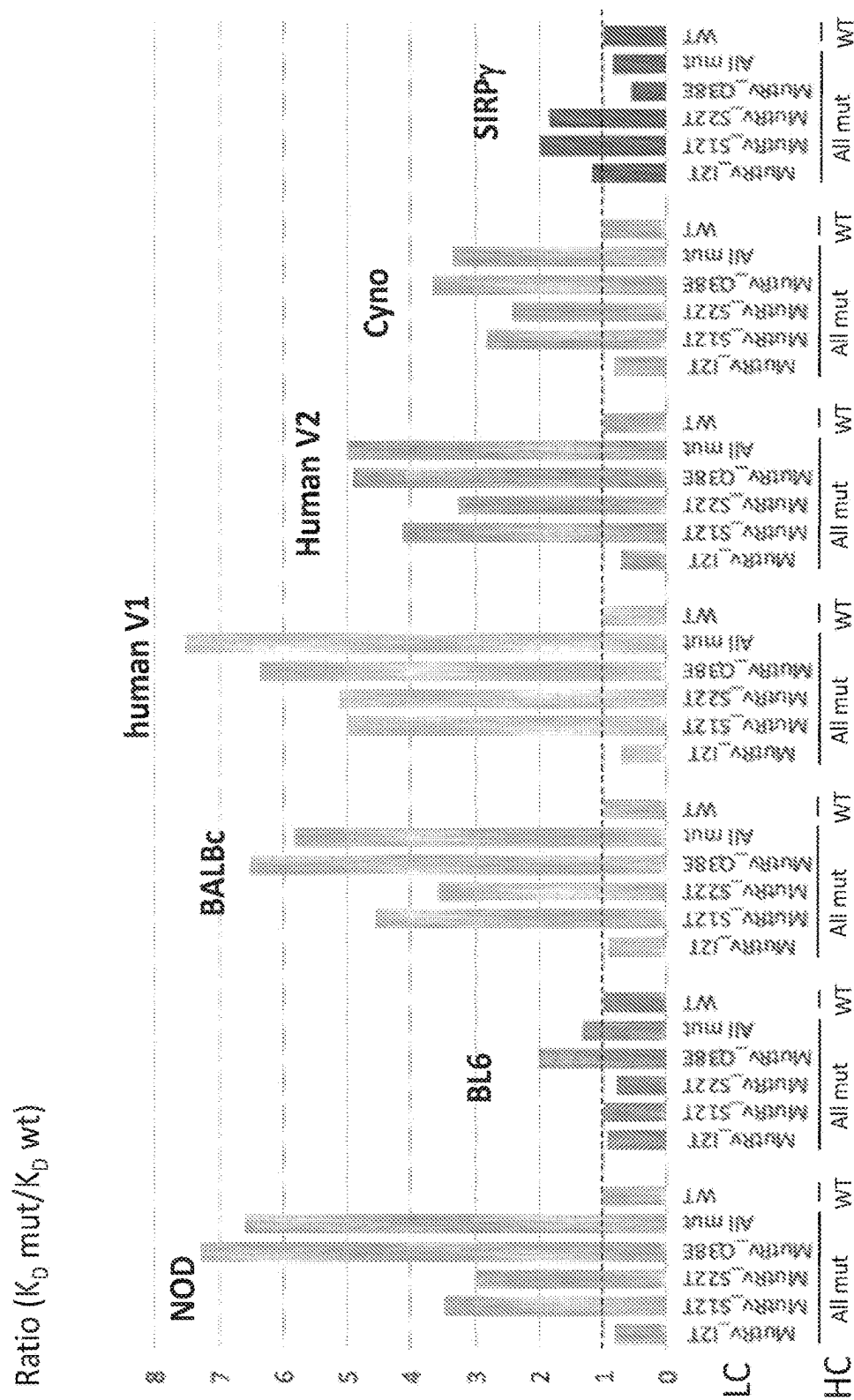
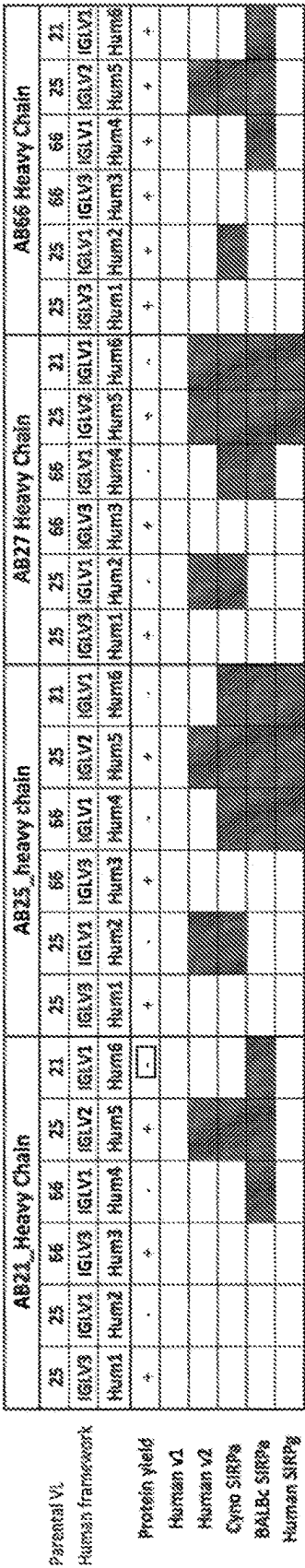


FIG. 27A





Protein Yield (mg/mL)

Binding affinity (Koff 1/s)

"+" – no change

"-" – reduced expression

• No change

• Reduced by 1-2log

FIG. 28

	<u>L1</u>	<u>L2</u>	<u>54</u>
Hum1	(1) SYELTQPPSVSPGQTARITCSGGSYSYYAWYQOKPGQAPVTLIYSDDKRP		
Hum8	(1) SYELTQPPSVSPGQTARITCSGCAYSSYYAWYQOKPGQAPVLIYSDSKRP		
hum9	(1) SYELTQPPSVSPGQTARITCSGCAYSSYYAWYQOKPGQAPVLIYSDDKRP		
	*	**	*
	55	<u>L3</u>	108
Hum1	(55) SNIPERFSGSSSGTTVTLTISGVQAEDEADYCGGYDQSSYTNPFGGGKLTVL		
Hum8	(55) SGIPERFSGSSSGTTVTLTISGVQAEDEADYCGGYDQSSYTNPFGGGKLTVL		
hum9	(55) SGIPERFSGSSSGTTVTLTISGVQAEDEADYCGGYDQSSYTNPFGGGKLTVL		
	*		

FIG. 29

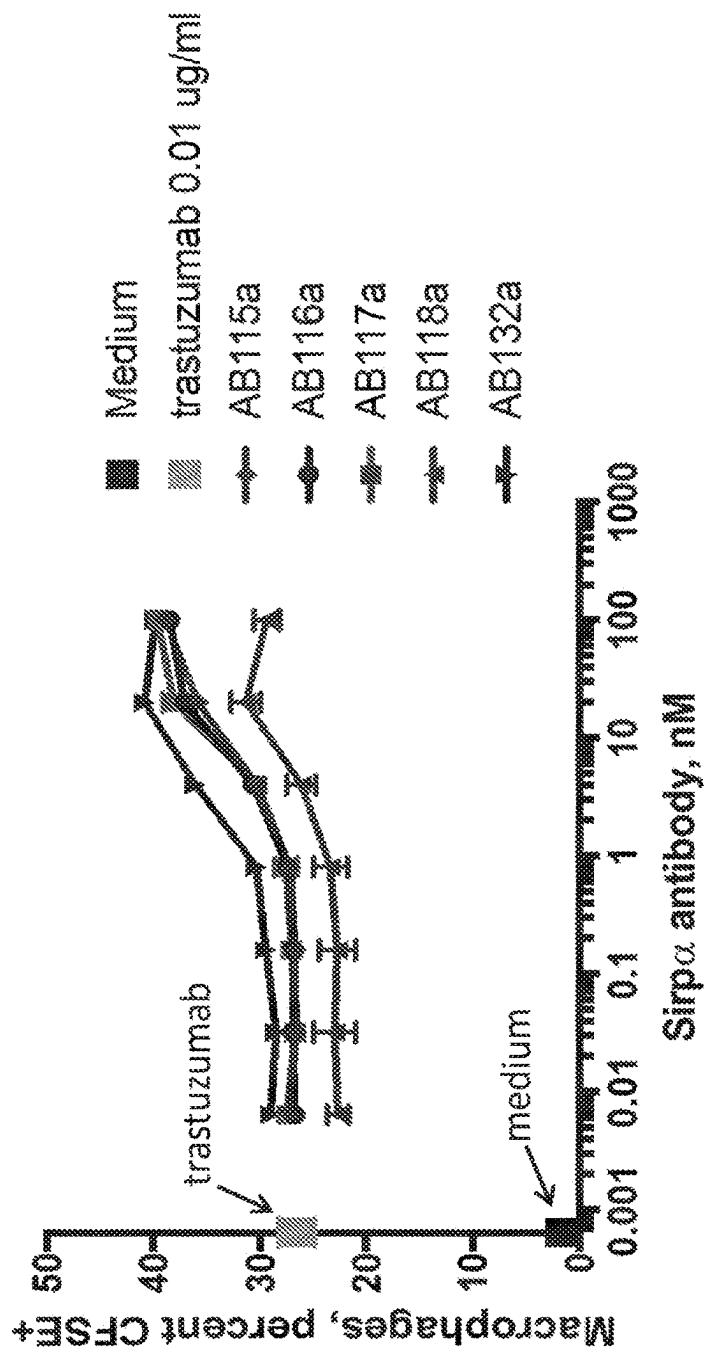


FIG. 31

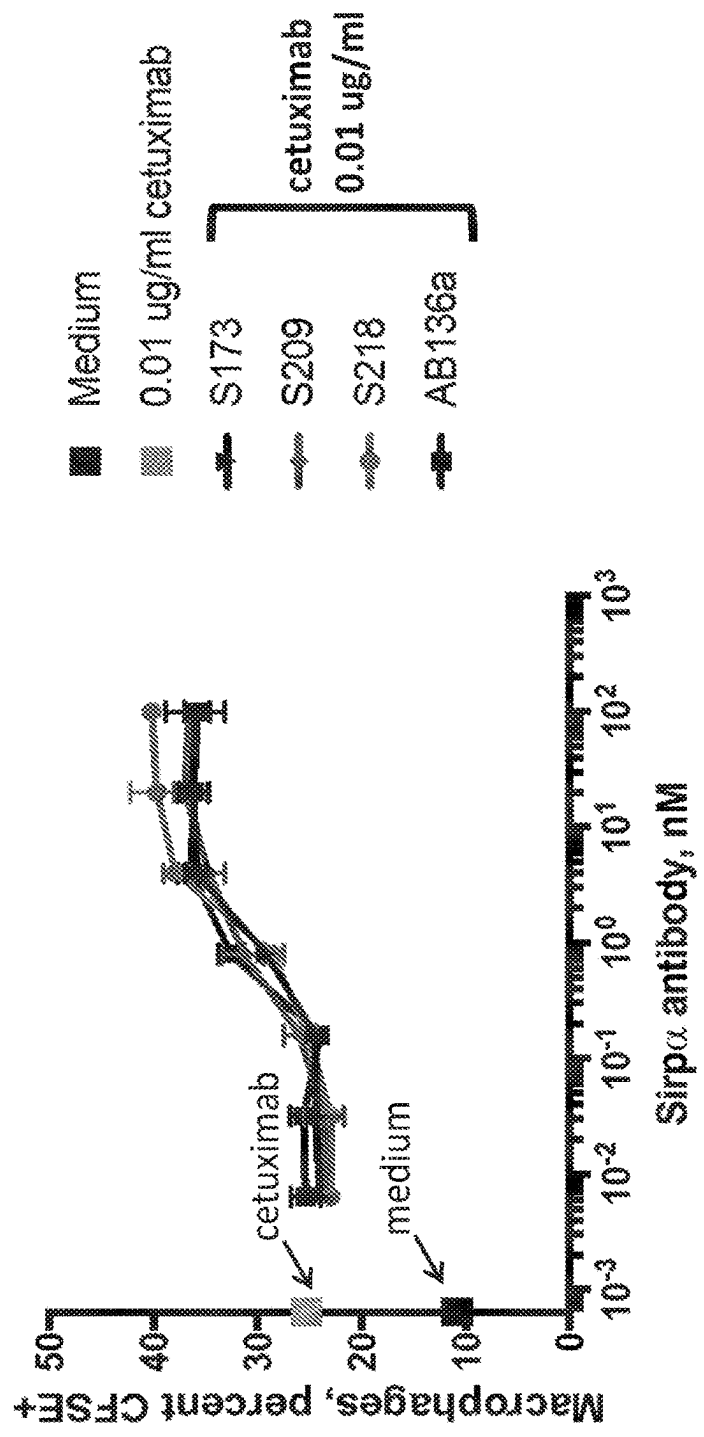


FIG. 32

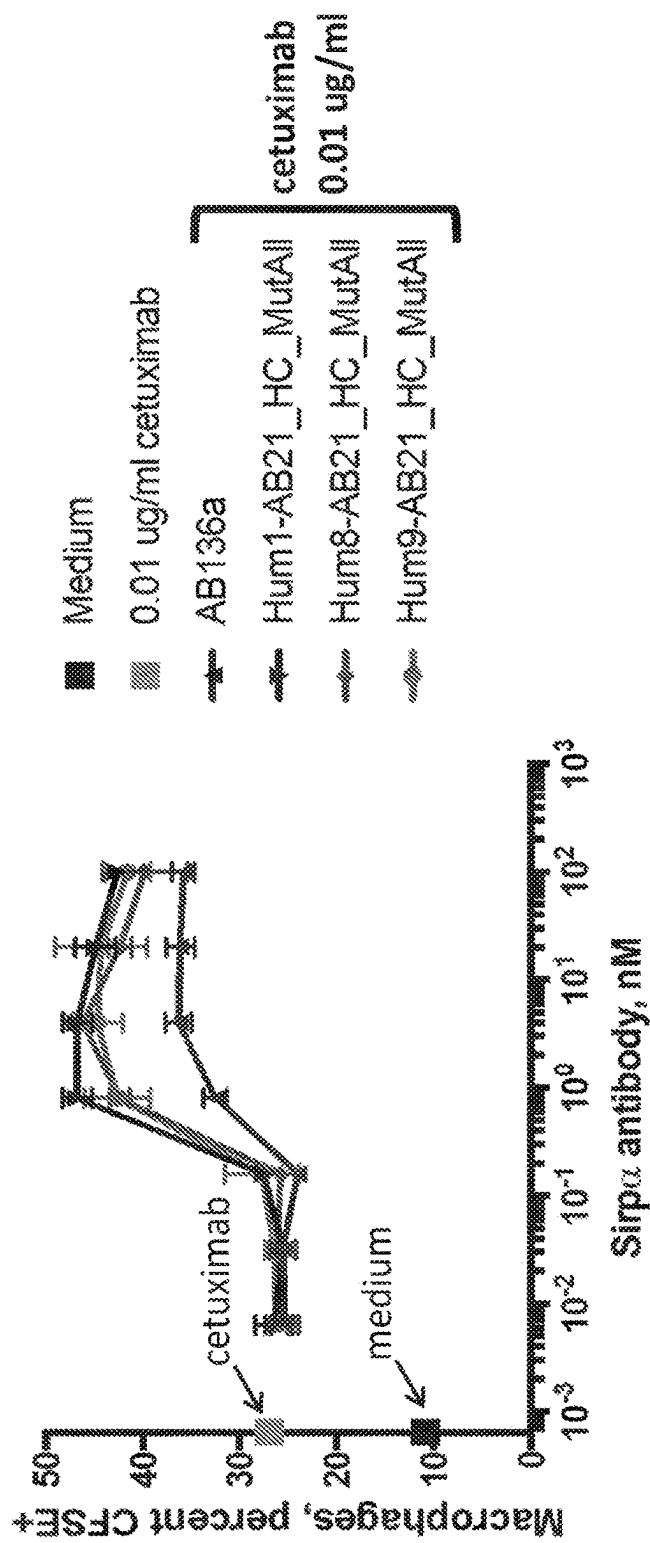


FIG. 33A

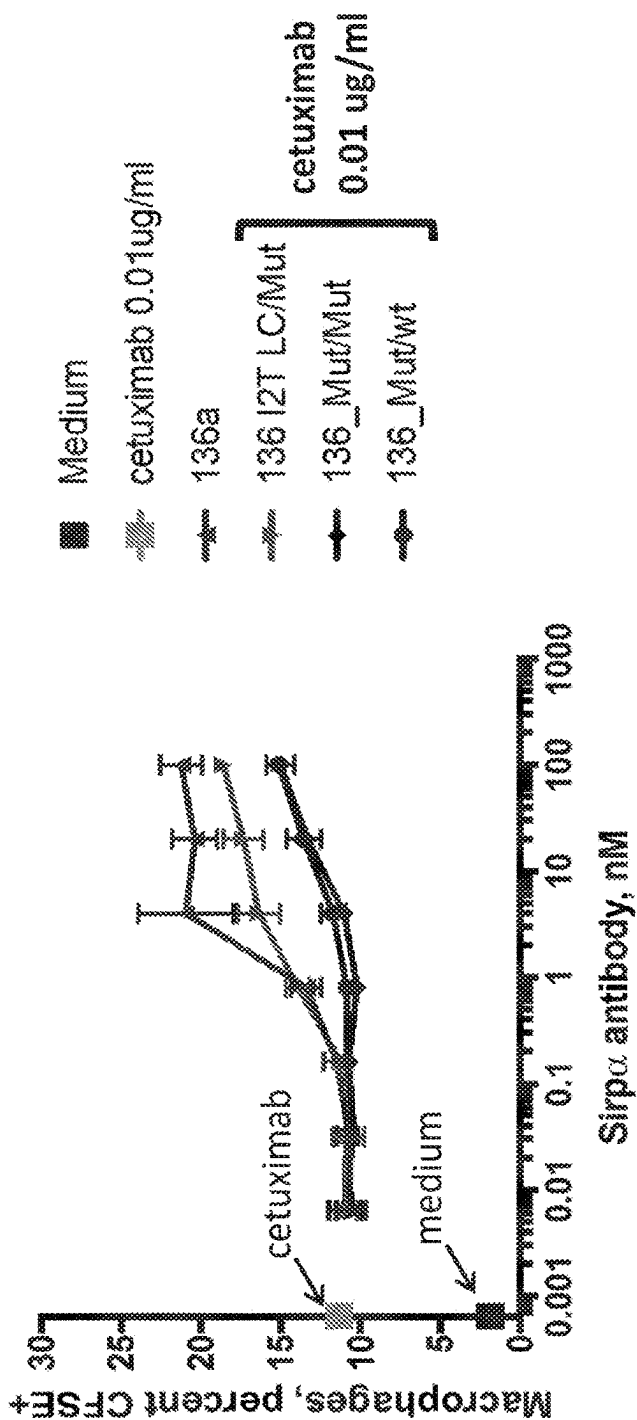


FIG. 33B

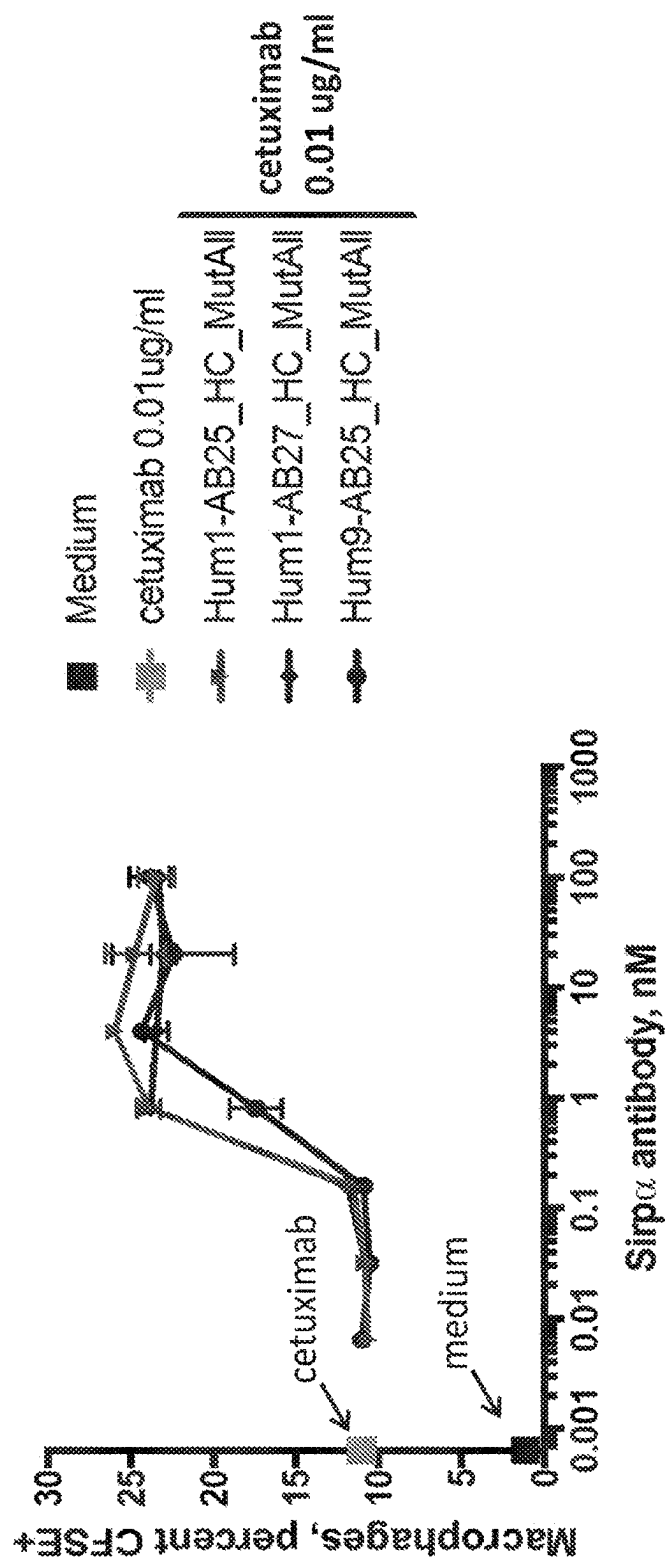


FIG. 33C

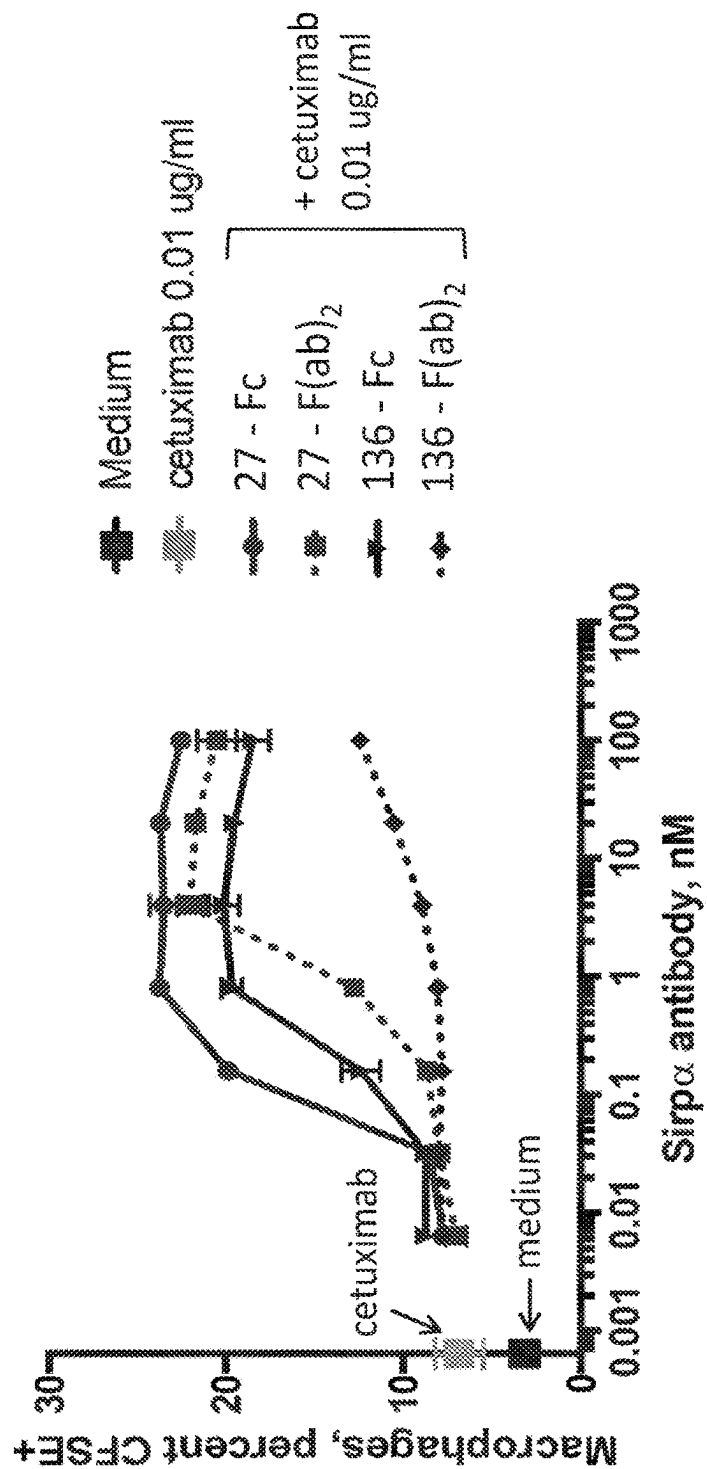


FIG. 34

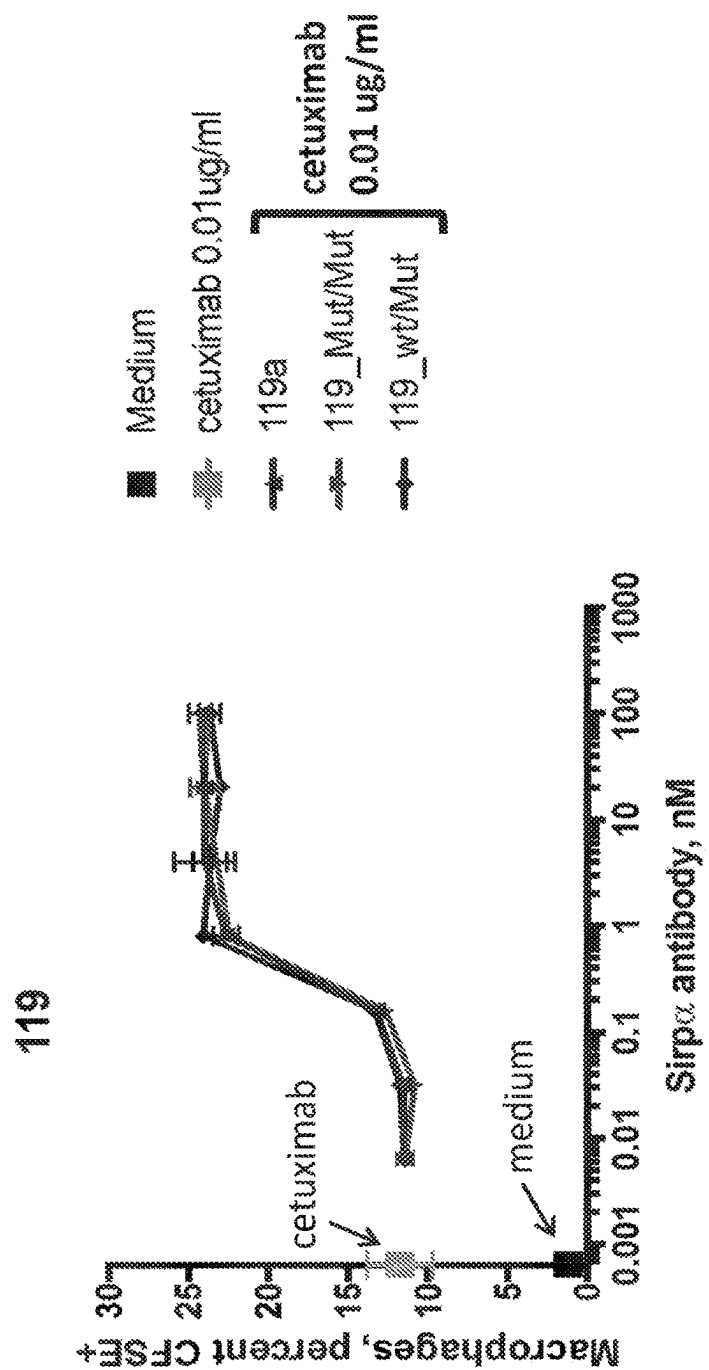


FIG. 35

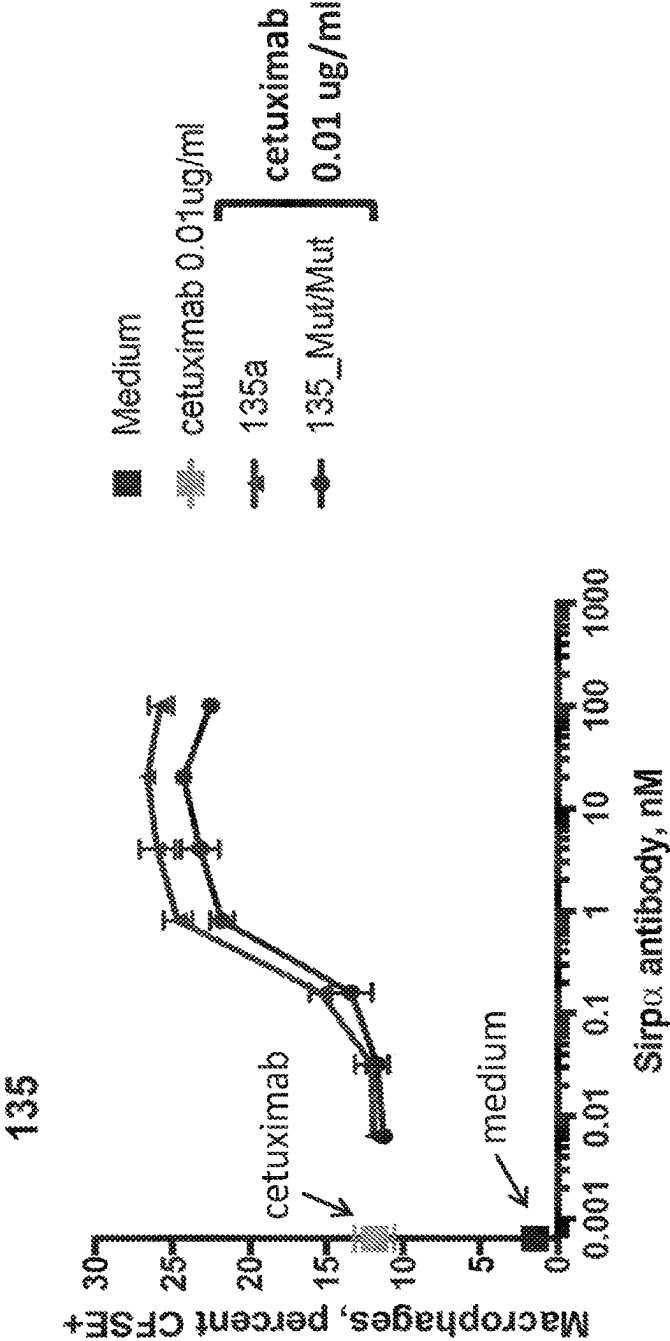


FIG. 36

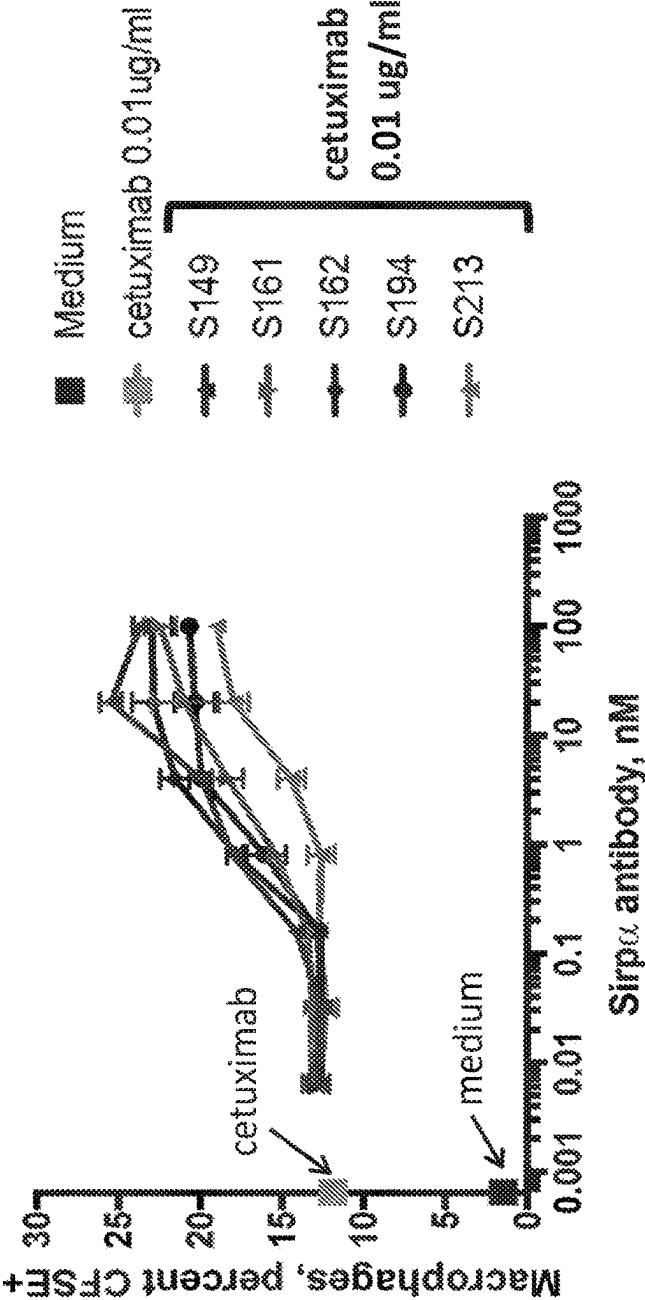


FIG. 37

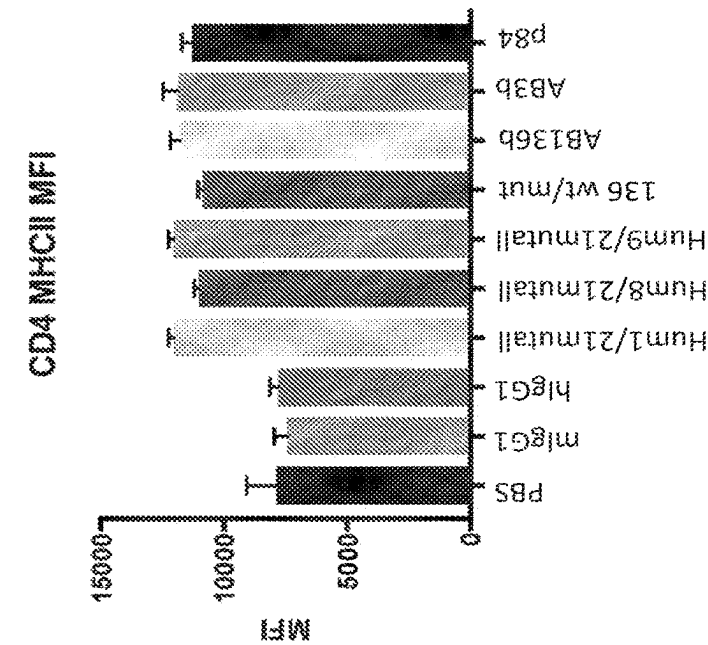


FIG. 38B

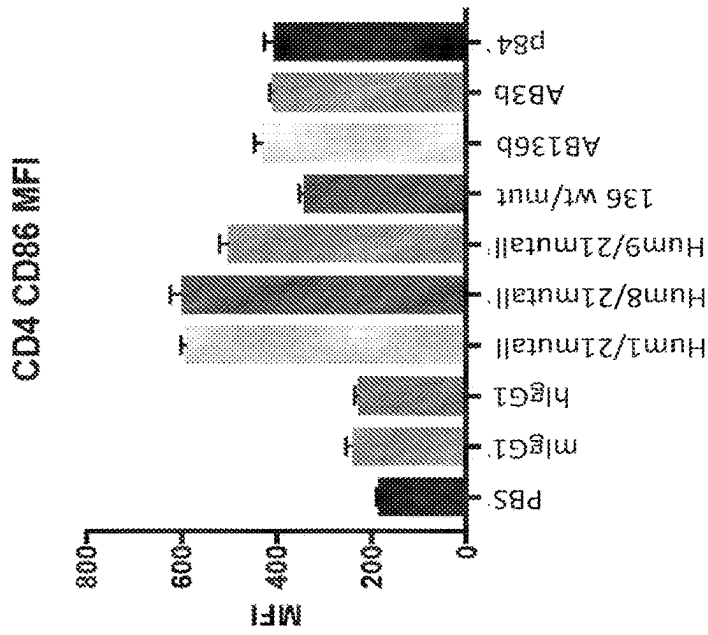


FIG. 38A

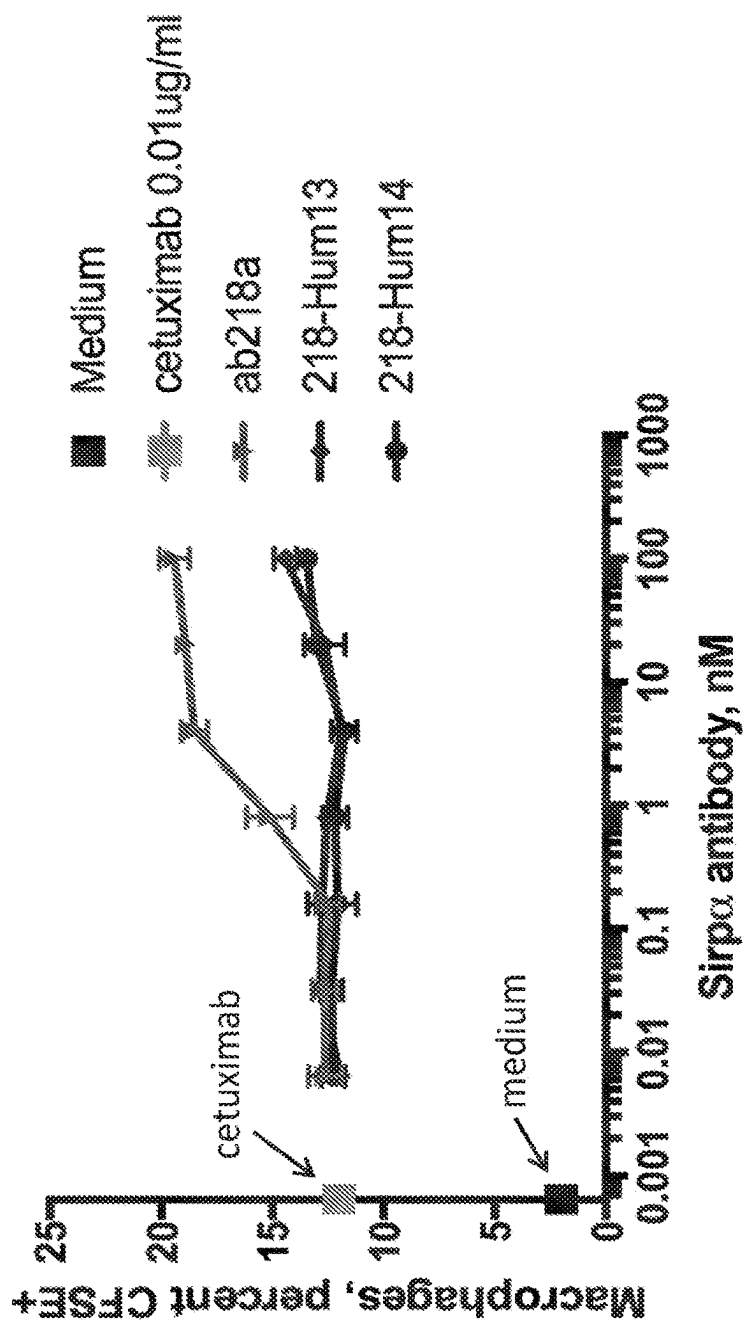


FIG. 39A

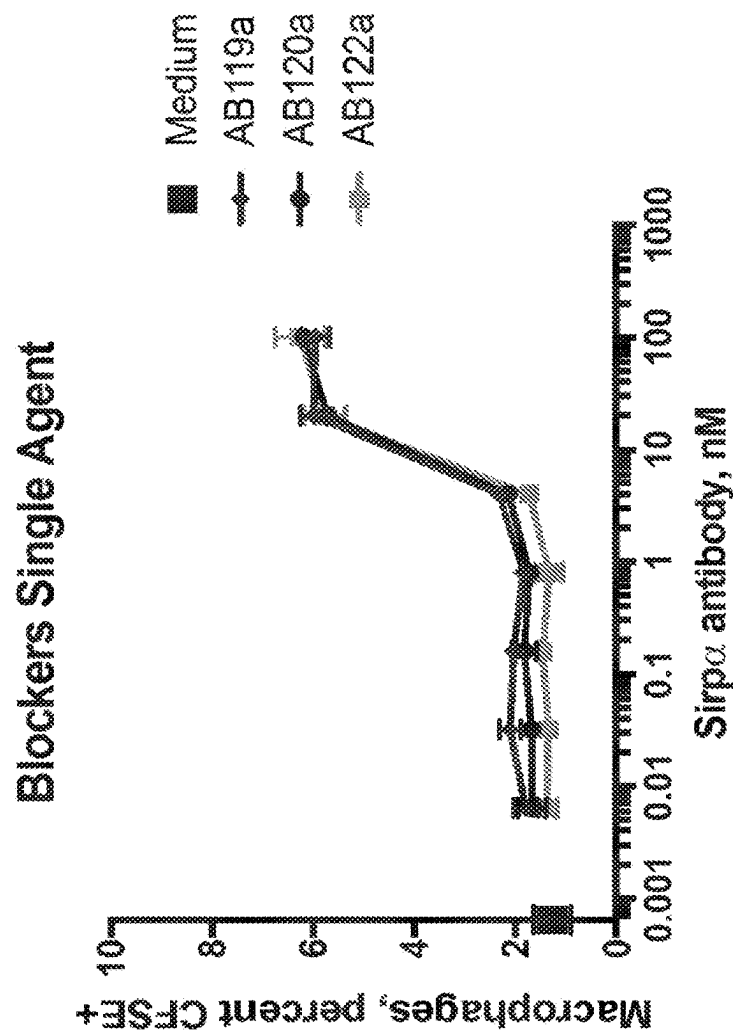


FIG. 39B

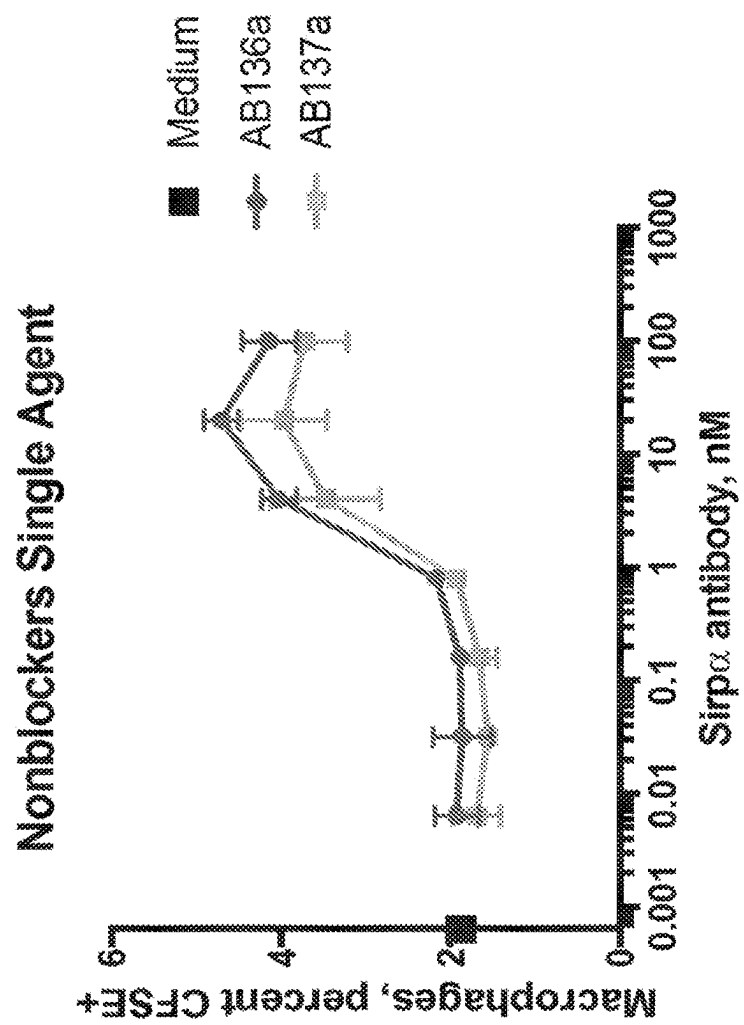


FIG. 39C

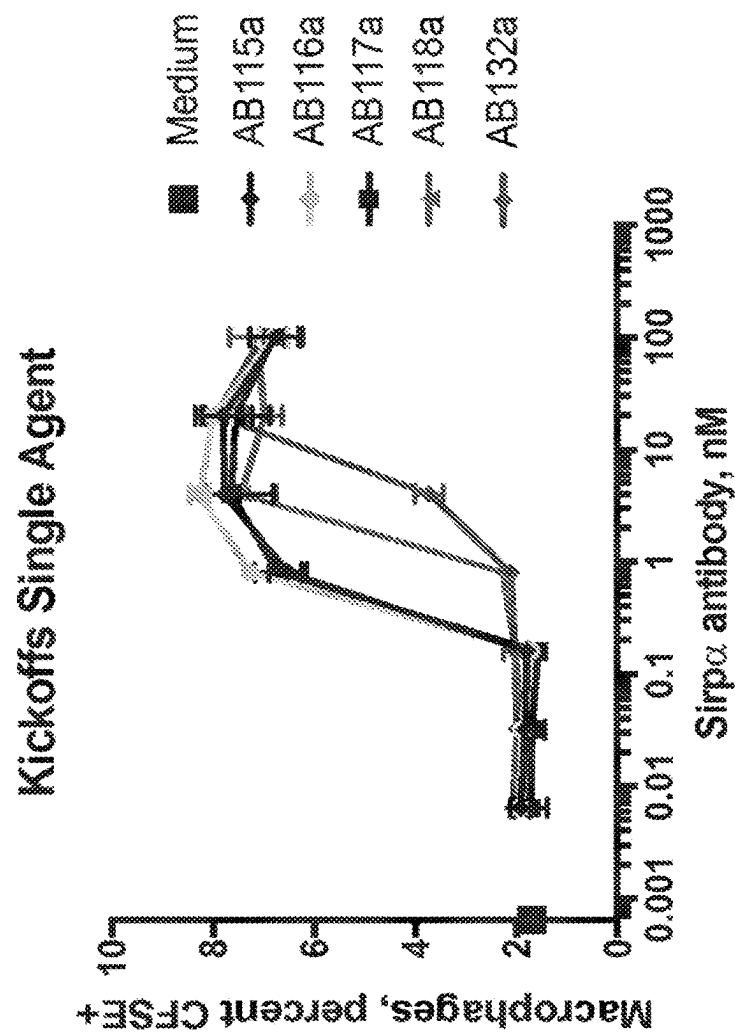
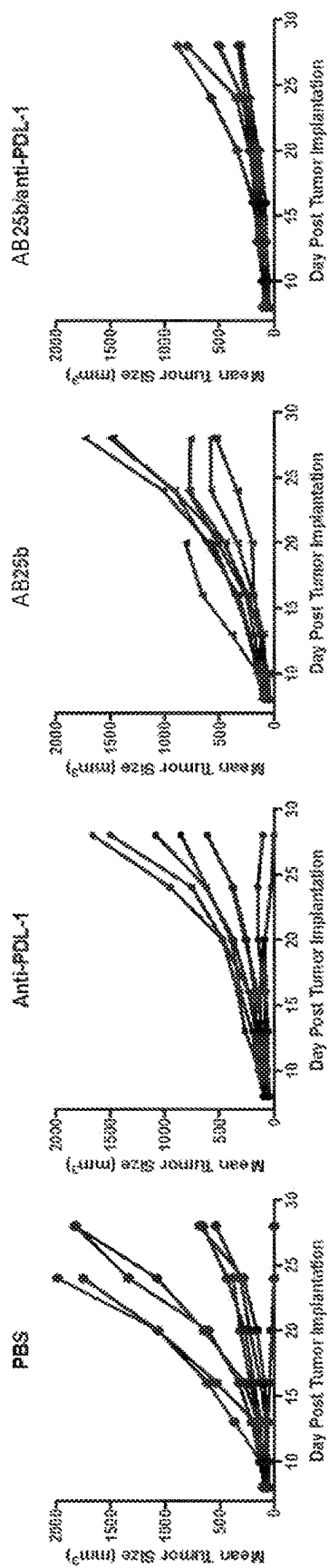
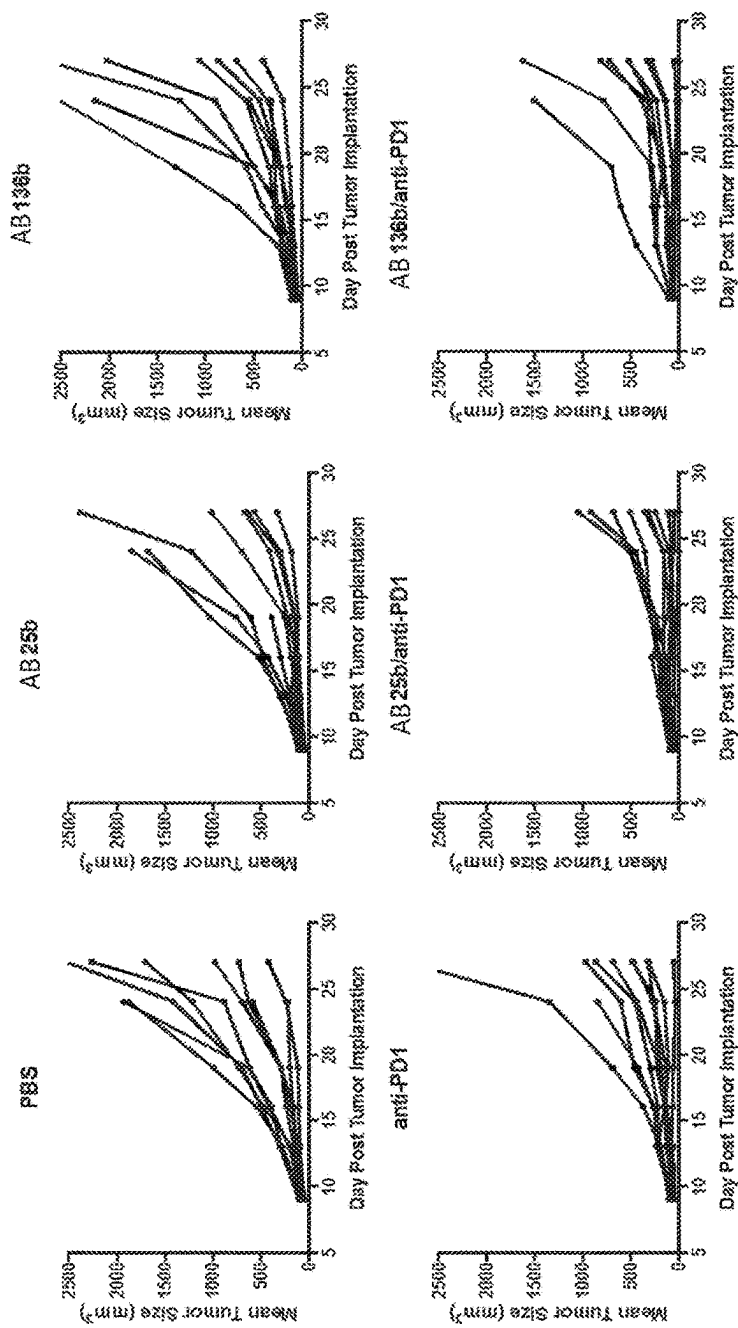


FIG. 39D



	PBS	Anti-PDL1	AB25b	AB25b/anti-PDL1
Mice with tumor volume <600 mm3	2	2	2	6

FIG. 40



	PBS	Anti-PD1	AB25b	AB25b/anti-PD1	AB136b	AB136b/anti-PD1
Mice with tumor volume <600 mm ³	1	5	2	7	1	6

FIG. 41

1

ANTIBODIES AGAINST SIGNAL-REGULATORY PROTEIN ALPHA AND METHODS OF USE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 15/710,798, filed Sep. 20, 2017, which claims the priority benefit of U.S. Provisional Application No. 62/397,752, filed Sep. 21, 2016, and U.S. Provisional Application No. 62/515,480, filed Jun. 5, 2017, each of which is hereby incorporated by reference in its entirety.

SUBMISSION OF SEQUENCE LISTING ON ASCII TEXT FILE

The content of the following submission on ASCII text file is incorporated herein by reference in its entirety: a computer readable form (CRF) of the Sequence Listing (file name: 757972000110SEQLIST.txt, date recorded: Jun. 1, 2021, size: 411 KB).

FIELD

The present disclosure relates to isolated antibodies that bind an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both, as well as polynucleotides, vectors, host cells, and methods related thereto.

BACKGROUND

Signal-regulatory protein alpha (SIRP- α) is part of a family of cell-surface receptors that plays critical roles in the regulation of the immune system (see, e.g., Barclay, A. N. and Brown, M. H. (2006) *Nat. Rev. Immunol.* 6:457-64). SIRP- α is expressed on the surface of various cells, including leukocytes such as dendritic cells, eosinophils, neutrophils, and macrophages. SIRP- α includes an extracellular domain that interacts with external stimuli such as ligands and an intracellular domain that mediates a variety of intracellular signals.

One of the major roles of SIRP- α is its regulation of the immune response through interactions with CD47. CD47 is expressed on the surface of a variety of cell types. When the IgSF domain of CD47 binds the extracellular domain (e.g., the D1 domain) of SIRP- α expressed on an immune cell (e.g., a macrophage), this transduces a SIRP- α -mediated signal in the immune cell that prevents phagocytosis of the CD47-expressing cell. Thus, CD47 serves to convey what has been termed a “don’t eat me” signal to the immune system that prevents phagocytosis of healthy cells (see, e.g., WO2015/138600 and Weiskopf, K. et al. (2013) *Science* 341:88-91). However, CD47 has also been shown to be highly expressed by a variety of cancers, and its interaction with SIRP- α in this context is thought to allow tumors to mimic the healthy “don’t eat me” signal in order to evade immune surveillance and phagocytosis by macrophages (see, e.g., Majeti, R. et al. (2009) *Cell* 138:286-99; Zhao, X. W. et al. (2011) *Proc. Natl. Acad. Sci.* 108:18342-7). As such, antibodies that block this interaction are highly desirable.

SIRP- α is known to be a highly polymorphic protein in humans, monkeys, and mice. For example, 20 amino acid differences have been identified between SIRP- α proteins in

2

the NOD and C57BL/6 mouse strains, and these polymorphisms lead to functional consequences related to CD47 binding and engraftment of human hematopoietic stem cells in these mouse strains. In humans, at least 10 distinct alleles of the SIRPA gene have been identified (Takenaka, K. et al. (2007) *Nat. Immunol.* 8:1313-23, Zhao, X. et al. (2011), *PNAS.* 108:18342-47; van der Heijden, J. (2014). Genetic variation in human Fc gamma receptors: Functional consequences of polymorphisms and copy number variation (Doctoral dissertation)).

Due to the importance of the SIRP- α -CD47 interaction in normal immune function and tumorigenesis, as well as the polymorphic nature of SIRP- α and the existence of other SIRP family receptors, the identification of antibodies having different binding specificities with intra- and/or inter-species cross-reactivity is of great interest for development of clinical candidates that are effective across human populations and the characterization of these candidates in various animal models. Thus, a need exists for both research tools and potential clinical candidates that modulate SIRP- α function, e.g., its binding interaction with CD47. A need also exists for methods of isolating antibodies with a variety of SIRP- α binding specificities and effects on CD47-SIRP- α binding in order to understand and effectively target this critical interaction.

All references cited herein, including patent applications, patent publications, non-patent literature, and UniProtKB/Swiss-Prot Accession numbers are herein incorporated by reference in their entirety, as if each individual reference were specifically and individually indicated to be incorporated by reference.

SUMMARY

To meet these and other needs, provided herein, inter alia, are isolated antibodies that bind an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both. In some embodiments, the antibody binds an extracellular domain of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPRV VTTVSDLTNRNNMDFSIRIGNIT-PADAGTYCYCVKFRKGSPDDVEFKSGAGTELSVRAKPS (SEQ ID NO:5). In some embodiments, the antibody binds an extracellular domain of a human SIRP- α v2 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRV TTVSESTKRENMDFSISISNIT-PADAGTYCYCVKFRKGSPDTEFKSGAGTELSVRAKPS (SEQ ID NO:6). In some embodiments, the antibody binds an extracellular domain of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPRV VTTVSDLTNRNNMDFSIRIGNIT-PADAGTYCYCVKFRKGSPDDVEFKSGAGTELSVRAKPS (SEQ ID NO:5) and binds an extracellular domain of a human SIRP- α v2 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRV TTVSESTKRENMDFSISISNIT-PADAGTYCYCVKFRKGSPDTEFKSGAGTELSVRAKPS

5

10

15

20

25

30

35

40

45

50

55

60

65

D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody binds a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide increases k_{off} of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell increases k_{off} of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and wherein the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and wherein the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and wherein binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the

5

10

15

20

25

30

35

40

45

50

55

60

65

of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from antibody S130. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and wherein the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from an antibody selected from the group consisting of S8, S13, S14, and S121. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, and a monkey SIRP- α polypeptide; wherein the antibody does not bind, or binds with reduced affinity to, at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and wherein the antibody does not block binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from antibody S137. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides and a monkey SIRP- α polypeptide; wherein the antibody does not bind a murine SIRP- α polypeptide; wherein the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and wherein the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from an antibody selected from the group consisting of S128. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; wherein the antibody does not bind a murine SIRP- α polypeptide; and wherein the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from an antibody selected from the group consisting of S9, S11, S119, S120, S122, and S135. In some embodiments, the antibody binds the extracellular domains (e.g., the D1

iff

cel

[illegible]

13

the antibody comprises a VH domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:135, 137, 139, or 141 and/or a VL domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:136, 138, 140, or 142.

In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:227 or 230, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:228 or 231, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:229; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:232, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:233, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:234. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:219 or 235, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:236 or 238, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:237; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:239, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:240, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:241. In some embodiments, the antibody comprises (a) an HVR-H1 sequence comprising the amino acid sequence of GFSFSX₁X₂AMX₃, wherein X₁ is N or I; X₂ is F or Y; and X₃ is T or S (SEQ ID NO:185); (b) an HVR-H2 sequence comprising the amino acid sequence of TIGX₄X₅DTYYADSVKG, wherein X₄ is S or A and X₅ is G or D (SEQ ID NO:186); (c) an HVR-H3 sequence comprising the amino acid sequence of DSTVX₆WSGDFFDY, wherein X₆ is S or G (SEQ ID NO:187); (d) an HVR-L1 sequence comprising the amino acid sequence of RASQNVX₇X₈DX₉A, wherein X₇ is K or R; X₈ is N or S; and X₉ is L or I (SEQ ID NO:188); (e) an HVR-L2 sequence comprising the amino acid sequence of AAX₁₀X₁₁RX₁₂T, wherein X₁₀ is R or S; X₁₁ is I or S; and X₁₂ is E or D (SEQ ID NO:189); and (f) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 119 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:335 and 97 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NO:335 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NO:97). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 135 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:127 and 104 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NO:127 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NO:104). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:97, 104, 120, 335, and 127 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain

14

sequence of SEQ ID NOs:335 and 127 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NOs:97 and 104). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:143-148 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:143-145 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:146-148). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:148-153 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:149-151 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:152, 153, and 148). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 136 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:155-160 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:155-157 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:158-160). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 21 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161-166 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:161-163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:164-166). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 25 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161, 163, 168, and 170-172 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:161, 168, and 163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:170-172). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 27 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:163, 173, 174, and 176-178 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:163, 173, and 174 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:176-178). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 66 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:162, 163, 179, and 182-184 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:162, 163, and 179 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:182-184). In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:143, 202, 204, or 205, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:144, 203, or 206, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:145 or 207; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:146 or 208, an HVR-L2 comprising the amino acid

16

sequence of SEQ ID NO:166. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:170, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:171, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:172. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:176, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:177, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:178. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:184. In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 3 (e.g., as listed in Table 2). In some embodiments, the antibody comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:242; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:243. In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 45 (e.g., as listed in Table 2). In some embodiments, the antibody comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:244; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:245. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, one, two, three, four, five, or six of the HVR sequences are defined by Kabat. In some embodiments, one, two, three, four, five, or six of the HVR sequences are defined by Chothia. In some embodiments, one, two, three, four, five, or six of the HVR sequences are defined by IMGT. In some embodiments, the antibody

comprises HVR sequences as defined by two or more of Kabat, Chothia, and IMGT (e.g., the antibody comprises one or more HVR sequences as defined by one delineation and one or more HVR sequences as defined by a different delineation).

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of NFAMT (SEQ ID NO: 175), NFAVT (SEQ ID NO:204), or NFALT (SEQ ID NO:305), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGSGDTYYADSVKG (SEQ ID NO:144), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVSWSGDFFDY (SEQ ID NO:145); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVKNDLA (SEQ ID NO:146), (ii) an HVR-L2 sequence comprising the amino acid sequence of AARIRET (SEQ ID NO:147), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:120, 335, 246, 258, or 327; and/or the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:97 or 312. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:335, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:335, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; or the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:312.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of IYAMS (SEQ ID NO:269), IYAVS (SEQ ID NO:213), or IYALS (SEQ ID NO:306), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGADDTYYADSVKG (SEQ ID NO:150), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVGWSGDFFDY (SEQ ID NO:151); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVRSIDIA (SEQ ID NO:152), (ii) an HVR-L2 sequence comprising the amino acid sequence of AASSRDT (SEQ ID NO:153), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises

an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:341, 247, 259, or 328; and/or the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:104 or 248. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; or the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:248.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of $X_1X_2DX_3N$, wherein X_1 is S or T; X_2 is Y or S; and X_3 is M, L, or V (SEQ ID NO:307); (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIX₁YYADSVKG, wherein X_1 is I or T (SEQ ID NO:308); and (iii) an HVR-H3 sequence comprising the amino acid sequence of $EX_1X_2X_3YRFFDX_4$, wherein X_1 is N or D; X_2 is N or D; X_3 is R or M; and X_4 is D or Y (SEQ ID NO:309); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of RAX_1QSVYX_2YLA , wherein X_1 is S or D; and X_2 is T or S (SEQ ID NO:310); (ii) an HVR-L2 sequence comprising the amino acid sequence of $X_1AX_2X_3RAX_4$, wherein X_1 is G, A, or D; X_2 is S or R; X_3 is S, N, or T; and X_4 is T or A (SEQ ID NO:311); and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, the antibody comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of SYDMN (SEQ ID NO:270), SYDVN (SEQ ID NO:221), or SYDLN (SEQ ID NO:313), (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIIYYADSVKG (SEQ ID NO:156), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ENNRYRFFDD (SEQ ID NO:157); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQSVYTYLA (SEQ ID NO:158), (ii) an HVR-L2 sequence comprising the amino acid sequence of GASSRAT (SEQ ID NO:159), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO: 249, 133, 260, or 329; and/or the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:134, 250, or 251. In some embodiments, the VH

domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; or the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:251.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: a heavy chain variable (VH) domain comprising: an HVR-H1 sequence comprising the amino acid sequence of $X_1X_2AX_3S$, wherein X_1 is S or T; X_2 is N, Y, H, or D; and X_3 is M, L, or V (SEQ ID NO:297); an HVR-H2 sequence comprising the amino acid sequence of $GISX_1X_2X_3X_4X_5X_6YYX_7X_8SX_9KG$, wherein X_1 is A or S; X_2 is G, S, or absent; X_3 is S, D or G; X_4 is G or S; X_5 is D, S, or G; X_6 is T or A; X_7 is P, G, V, I, A, or S; X_8 is A, D, or G; and X_9 is V or M (SEQ ID NO:298); and an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or a light chain variable (VL) domain comprising: an HVR-L1 sequence comprising the amino acid sequence of $SGGX_1X_2X_3SX_4YYX_5$, wherein X_1 is D, G, S, I, or absent; X_2 is S, W, G, Y, D, or absent; X_3 is S, Y, T, or D; X_4 is H, T, S, or Y; and X_5 is G or A (SEQ ID NO:299); an HVR-L2 sequence comprising the amino acid sequence of $SDX_1X_2RPX_3$, wherein X_1 is D or N; X_2 is E, K, or Q; and X_3 is S or P (SEQ ID NO:300); and an HVR-L3 sequence comprising the amino acid sequence of $X_1X_2YDX_3X_4X_5YX_6NX_7$, wherein X_1 is G or A; X_2 is G or A; X_3 is G, Y, Q, S, or A; X_4 is S, R, or T; X_5 is T or S; X_6 is A, I, V, L, or T; and X_7 is T, A, D, or P (SEQ ID NO:301). In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: a heavy chain variable (VH) domain comprising: a heavy chain variable (VH) domain comprising: an HVR-H1 sequence comprising the amino acid sequence of SX_1AX_2S , wherein X_1 is N or Y; and wherein X_2 is M, L, or V (SEQ ID NO:302); an HVR-H2 sequence comprising the amino acid sequence of $GISX_1GX_2X_3DTYXX_4X_5SVKG$, wherein X_1 is A or S; X_2 is G or absent; X_3 is S or G; X_4 is P, G, or V; and X_5 is A

or D (SEQ ID NO:303); and an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or a light chain variable (VL) domain comprising: an HVR-L1 sequence comprising the amino acid sequence of $SGGX_1YSSYYA$, wherein X_1 is S or A (SEQ ID NO:304); an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336); and an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISAGGSDTYYPASVKG (SEQ ID NO:195), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:135, 263, 264, or 330. In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISSGSDTYYGDSVKG (SEQ ID NO:197), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:137, 265, 266, or 331. In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SYAMS (SEQ ID NO:200), SYAVS (SEQ ID NO:272), or SYALS (SEQ ID NO:319), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISSGGDTYYVDSVKG (SEQ ID NO:201), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:139, 267, 268, or 332. In some embodiments, the VL domain comprises the sequence FW1-HVR-L1-FW2-HVR-L2-FW3-HVR-L3-FW4 (N-terminus to C-terminus), wherein FW1 comprises the amino acid sequence SYELTQPPSVSVSPGQTARITC (SEQ ID NO:314), FW2 comprises the amino acid sequence WYQQKPGQAPVTLIY (SEQ ID NO:315), FW3 comprises the amino acid sequence NIPERFSGSSSGTTVTLTISGVQAEDEADYYC (SEQ ID NO:316), and FW4 comprises the amino acid sequence FGGGTKLTVL (SEQ ID NO:317). In some embodiments, the VL domain comprises (i) an HVR-L1 sequence comprising the amino acid sequence of SGGSYSSYYA (SEQ ID NO:170), (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336), and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:252. In some embodiments, the VL domain comprises (i) an HVR-L1 sequence comprising the amino acid sequence of SGGAYSSYYA (SEQ ID NO:261), (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336), and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:262. In some embodiments, the VH domain

comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:330, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:137, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:268, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:330, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:268, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:332, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:137, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; or the VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:262.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of I31, V33, Q52, K53, T67, R69, N70, and K96, according to SEQ ID NO:296. In some embodiments, the antibody binds

to the human SIRP- α v1 polypeptide at I31, V33, Q52, K53, T67, R69, N70, and K96, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of L30, P32, E54, T62, N71, M72, F74, and R95, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L30, P32, E54, T62, N71, M72, F74, and R95, according to SEQ ID NO:296.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of 17, P9, D10, K11, S12, A42, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at K11, A42, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at 17, P9, D10, K11, S12, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of L14, T26, T28, T88, Y90, S106, S113, and A116, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T88, Y90, S106, S113, and A116 of human SIRP- α v1, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T26, and T28, according to SEQ ID NO:296.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of E47, L48, P58, R59, T82, and A84, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at E47, L48, P58, R59, T82, and A84, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of A17, P44, G45, 149, E54, G55, H56, F57, and P83, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at A17, P44, G45, 149, E54, G55, H56, F57, and P83 of human SIRP- α v1, according to SEQ ID NO:296.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds the extracellular domain of a human SIRP- α v1 polypeptide with a dissociation constant (K_D) of less than 100 nM, and wherein the antibody blocks binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain of a human SIRP- α v2 polypeptide with a dissociation constant (K_D) of less than 100 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain of a

human SIRP- γ polypeptide. In some embodiments, the antibody binds an extracellular domain of a murine SIRP- α polypeptide.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds the D1 domain of a human SIRP- α polypeptide, and wherein the antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the D1 domain of a human SIRP- α with a dissociation constant (K_D) of less than 100 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide with a dissociation constant (K_D) of less than 100 nM and/or binds the D1 domain of a human SIRP- α v2 polypeptide with a dissociation constant (K_D) of less than 100 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain of a murine SIRP- α polypeptide.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 119, 120, 121, 122, 21, 25, 27, 66, and 135. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:120 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:97; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:121 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:98; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:130 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:107; (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:122 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:99; (e) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:135 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:136; (f) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:137 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:138; (g) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:139 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:140; (h) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:141 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:142; or (i) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:127 and

a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:104.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 136 and 137. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:133 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:134; or (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:128 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:105.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 3, 213, 173, and 209. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:242 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:243; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:275 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:276; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:278 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:279; or (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:280 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:281.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 115, 116, 117, 118, and 132. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:116 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:93; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:117 and

a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:94; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:118 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:95; (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:119 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:96; or (e) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:282 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:283.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 218, 123, 149, 161, 162, and 194. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:284 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:285; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:123 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:100; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:286 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:287; (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:288 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:289; (e) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:290 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:291; or (f) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:292 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:293.

In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of antibody 45. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:244 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:245.

In some embodiments of any of the above embodiments, the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide. In some embodiments, the antibody enhances activation of a dendritic cell expressing a human SIRP- α polypeptide. In some embodiments, the antibody inhibits in vivo growth of a tumor that

expresses CD47. In some embodiments, the antibody does not prevent interactions between a CD47-expressing cell and a T cell.

In some embodiments of any of the above embodiments, the antibody is a monoclonal antibody. In some embodiments, the antibody is a scFv-Fc, single domain antibody, single heavy chain antibody, or single light chain antibody. In some embodiments, the antibody comprises a light chain constant region comprising the amino acid sequence of SEQ ID NO:325, 326, or 426. In some embodiments, the antibody comprises an amino acid sequence selected from the group consisting of SEQ ID NOs:320-324. In some embodiments, the antibody comprises an Fc region. In some embodiments, the Fc region is a human Fc region selected from the group consisting of an IgG1 Fc region, an IgG2 Fc region, and an IgG4 Fc region. In some embodiments, the Fc region comprises a human IgG1 Fc region comprising one or more mutations selected from the group consisting of L234A, L235A, L235E, G237A, and N297A, according to EU numbering. In some embodiments, the Fc region comprises a human IgG2 Fc region comprising one or more mutations selected from the group consisting of A330S, P331S and N297A, according to EU numbering. In some embodiments, the Fc region comprises a human IgG4 Fc region comprising one or more mutations selected from the group consisting of S228P, E233P, F234V, L235A, L235E, delG236, and N297A, according to EU numbering. In some embodiments, the antibody is an antibody fragment selected from the group consisting of a Fab, F(ab')₂, Fab'-SH, Fv, and scFv fragment. In some embodiments, the antibody is conjugated to a cytotoxic agent or label.

In some embodiments, the antibody is a bispecific antibody. In some embodiments, the antibody comprises a first antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and a second antigen binding domain that binds an antigen expressed by a cancer cell. In some embodiments, the antigen expressed by the cancer cell is selected from the group consisting of CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PD-L1, PTK7, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le^x, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. In some embodiments, the antibody is a chicken, humanized, chimeric, or human antibody. In some embodiments, the antibody is generated by or derived from a chicken.

Further provided herein are polynucleotides comprising the antibody according to any one of the above embodiments. Further provided herein are vectors comprising the polynucleotide according to any one of the above embodiments. Further provided herein are host cells comprising the polynucleotide or vector according to any one of the above embodiments. Further provided herein are methods of producing an antibody, comprising culturing the host cell

according to any one of the above embodiments such that the antibody is produced. In some embodiments, the methods further include recovering the antibody from the host cell.

Further provided herein are methods of treating or delaying progression of cancer in an individual, the methods comprising administering to the individual an effective amount of the antibody according to any one of the above embodiments. In some embodiments, the methods further comprise administering to the individual an effective amount of a second antibody. In some embodiments, the second antibody binds an antigen expressed by a cancer cell. In some embodiments, the antigen expressed by the cancer cell is selected from the group consisting of CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PTK7, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le^x, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. In some embodiments, the methods further comprise administering to the individual an effective amount of an immunotherapeutic agent. In some embodiments, the immunotherapeutic agent comprises a second antibody. In some embodiments, the second antibody binds to an antigen selected from the group consisting of PD-1, PD-L1, OX40, CTLA-4, CD137/4-1BB, TNFR2, B7-H3, FZD7, CD27, CCR4, CSF1R, CSF, TIM-3, LAG-3, VISTA, ICOS, CCR2, IDO, A2R, CD39, CD73, TIGIT, CD80, CD47, arginase, TDO, and PVRIG. In some embodiments, the first antibody binds the extracellular domain of a human SIRP- α v1 polypeptide, the extracellular domain of a human SIRP- α v2 polypeptide, or the extracellular domains of both a human SIRP- α v1 polypeptide and a human SIRP- α v2 polypeptide with a dissociation constant (K_D) of less than 100 nM, wherein the first antibody blocks binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-1. In some embodiments, the first antibody binds the D1 domain of a human SIRP- α polypeptide, wherein the first antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-1. In some embodiments, the first antibody binds the extracellular domain of a human SIRP- α v1 polypeptide with a dissociation constant (K_D) of less than 100 nM, wherein the first antibody blocks binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-L1. In some embodiments, the first antibody binds the D1 domain of a human SIRP- α polypeptide, wherein the first antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-L1. In some embodiments, the individual is a human.

Further provided herein are methods of treating or delaying progression of an autoimmune disease or an inflammatory disease in an individual, the methods comprising administering to the individual an effective amount of the antibody according to any one of the above embodiments. In some embodiments, the autoimmune disease or inflammatory disease is selected from the group consisting of multiple sclerosis, rheumatoid arthritis, a spondyloarthropathy, systemic lupus erythematosus, an antibody-mediated inflammatory or autoimmune disease, graft versus host disease, sepsis, diabetes, psoriasis, psoriatic arthritis, atherosclerosis, Sjogren's syndrome, progressive systemic sclerosis, scleroderma, acute coronary syndrome, ischemic reperfusion, Crohn's Disease, ulcerative colitis, endometriosis, glomerulonephritis, IgA nephropathy, polycystic kidney disease, myasthenia gravis, idiopathic pulmonary fibrosis, asthma, atopic dermatitis, acute respiratory distress syndrome (ARDS), vasculitis, and inflammatory autoimmune myositis. In some embodiments, the individual is a human.

Further provided herein are methods of identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide, the methods comprising (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. Further provided herein are methods of identifying an antibody or antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide, the methods comprising contacting an antibody or antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. Further provided herein are methods of identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide, the methods comprising (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1

variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. Further provided herein are methods of identifying an antibody or antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide, the methods comprising contacting an antibody or antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and (d) detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the SIRP- α D1 variant comprises an amino acid sequence selected from the group consisting of SEQ ID NOs:17-52. In some embodiments, the IgSF domain of CD47 comprises the amino acid sequence of SEQ ID NO:16. In some embodiments, the polypeptide comprising the IgSF domain of CD47 comprises a human CD47 extracellular domain. In some embodiments, the polypeptide comprising the IgSF domain of CD47 further comprises an antibody Fc region.

Further provided herein are methods of producing an anti-SIRP- α antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide, the methods comprising: (a) immunizing a chicken with a peptide comprising at least a portion of a human SIRP- α extracellular domain (e.g., the D1 domain); (b) obtaining an antibody from an antibody-producing cell from the immunized chicken; and (c) detecting binding between the antibody obtained from the cell and the extracellular domains (e.g., the D1 domains) of a human SIRP- α polypeptide, wherein binding between the antibody and the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide indicates that the antibody is an anti-SIRP- α antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α variant polypeptide. In some embodiments, the antibody is a chicken, humanized, chimeric, or human antibody. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVG-PIQWFRGAGPGRELIYNQKEGHFPR VTTVSDLTNRNNMDFSIKIGNIT-PADAGTYCYCVKFRKGSPDDVEFKSGAGTELSVRAP PS (SEQ ID NO:5). In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a

human SIRP- α v2 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVSVAAGESAILHCTVT-SLIPVGPIQWFRGAGPARELIYNQKEGHFPRV TTVSESTKRENMDFSISISNIT-PADAGTYCYCVKFRKGSPDTEFKSGAGTELSVRAPKS (SEQ ID NO:6). In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVG-PIQWFRGAGPGRELIYNQKEGHFPR VTTVSDLTNRNNMDFSIKIGNIT-PADAGTYCYCVKFRKGSPDDVEFKSGAGTELSVRAP PS (SEQ ID NO:5) and binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVG-PIQWFRGAGPARELIYNQKEGHFPRV TTVSESTKRENMDFSISISNIT-PADAGTYCYCVKFRKGSPDTEFKSGAGTELSVRAPKS (SEQ ID NO:6). In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of three, four, five, six, seven, eight, nine or ten different human SIRP- α variant polypeptides. In some embodiments, each of the three, four, five, six, seven, eight, nine or ten different human SIRP- α variant polypeptides comprises an extracellular domain (e.g., the D1 domain) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 5, 6, and 76-83. In some embodiments, the methods further comprise detecting binding between the antibody obtained from the cell and an extracellular domain (e.g., the D1 domain) of one or more SIRP- α polypeptides selected from the group consisting of a monkey SIRP- α polypeptide, a murine SIRP- α polypeptide, a human SIRP- β polypeptide, and a human SIRP- γ polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide. In some embodiments, the monkey SIRP- α polypeptide is a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of at least two different monkey SIRP- α variant polypeptides. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:11, an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:12, or both. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different murine SIRP- α variant polypeptides. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of one or more murine SIRP- α polypeptides, and wherein the one or more murine SIRP- α polypeptides each comprise an amino acid sequence selected from the group consisting of SEQ ID NOs: 7-10. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody does not bind an extracellular domain (e.g., the D1 domain) of a human SIRP-0 polypeptide. In some embodiments, the antibody does not bind an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP-0 polypeptide comprising the amino acid sequence of

SEQ ID NO:13, the extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide comprising the amino acid sequence of SEQ ID NO:14, or both. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide comprising the amino acid sequence of SEQ ID NO:15. In some embodiments, the methods further comprise detecting binding or a lack of binding between the antibody obtained from the cell and a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, the antibody binds a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody does not bind a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell increases k_{off} of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell increases k_{off} of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody modulates SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, the cell is a leukocyte

selected from the group consisting of a macrophage, a dendritic cell, a neutrophil, an eosinophil, and a myeloid-derived suppressor cell (MDSC). In some embodiments, the antibody inhibits SIRP- α signaling in a macrophage expressing a human SIRP- α polypeptide. In some embodiments, the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide.

It is to be understood that one, some, or all of the properties of the various embodiments described herein may be combined to form other embodiments of the present invention. These and other aspects of the invention will become apparent to one of skill in the art. These and other embodiments of the invention are further described by the detailed description that follows.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A shows an alignment among the D1 domains of 10 different human SIRP- α variant polypeptides. Sequences shown correspond to SEQ ID NOs: 5, 6, and 76-83 (from top to bottom). Amino acid differences are indicated by asterisks.

FIG. 1B shows an alignment between human v1, human v2, cynomolgus monkey, and 129 mouse SIRP- α D1 domains. Sequences shown correspond to SEQ ID NOs: 5, 6, 11, and 7 (from top to bottom). Amino acid differences are indicated by asterisks.

FIG. 1C shows alignments between various human and mouse SIRP- α D1 domains, with R1, R2 and R3 loops indicated. Shown is an alignment between human v1, human v2, 129 mouse, NOD mouse, C57BL/6 mouse, and BALB/c mouse SIRP- α D1 domains. Sequences shown correspond to SEQ ID NOs: 5-10 (from top to bottom). Amino acid differences are indicated by asterisks.

FIG. 2 shows an alignment between human v1, human v2, cynomolgus monkey, 129 mouse, and chicken SIRP- α D1 domains. Sequences shown correspond to SEQ ID NOs: 5, 6, 11, 7, and 84 (from top to bottom). Amino acid differences are indicated by asterisks.

FIGS. 3A & 3B show binding specificity of antibody clone S130 for a variety of SIRP peptides. FIG. 3A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP- α D1 domains, as well as human SIRP- γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants bound to the IgSF domain of CD47 (SEQ ID NO:16). The pre-formed complex was generated by mixing two high affinity human SIRP- α v1 and v2 polypeptides (SEQ ID NOs: 17 and 19) in a 1:1 ratio and combining the mixture with CD47 to generate the SIRP- α :CD47 complex. The pre-formed complex SIRP- α :CD47 complex for FIGS. 4A-9B are also prepared similarly. FIG. 3B summarizes the binding specificity of the clone against each of these targets ("+" indicates binding; "-" indicates non-binding).

FIGS. 4A & 4B show binding specificity of antibody clone S121 for a variety of SIRP peptides. FIG. 4A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP- α D1 domains, as well as human SIRP- γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 4B summarizes the binding specificity of the clone against each of these targets ("+" indicates binding; "-" indicates non-binding).

FIGS. 5A & 5B show binding specificity of antibody clone S137 for a variety of SIRP peptides. FIG. 5A shows the ELISA binding curves for the antibody against the

human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 5B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).

FIGS. 6A & 6B show binding specificity of antibody clone S128 for a variety of SIRP peptides. FIG. 6A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 6B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).

FIGS. 7A & 7B show binding specificity of antibody clone S135 for a variety of SIRP peptides. FIG. 7A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 7B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).

FIGS. 8A & 8B show binding specificity of antibody clone S126 for a variety of SIRP peptides. FIG. 8A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 8B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).

FIGS. 9A & 9B show binding specificity of antibody clone S138 for a variety of SIRP peptides. FIG. 9A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 9B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).

FIGS. 10A-10C show an alignment of VH and VL domains of the scFv-Fc clones obtained from a wild-type chicken. SEQ ID NOs:53-60 are shown (in order from top to bottom in the alignment). CDR and linker sequences are indicated by lines. Amino acid differences are indicated by asterisks.

FIGS. 10D-10F show an alignment of VH and VL domains of the scFv-Fc clones obtained from a chicken that produces human antibodies. SEQ ID NOs:61-74 are shown (in order from top to bottom in the alignment). CDR and linker sequences are indicated by lines. Amino acid differences are indicated by asterisks.

FIGS. 11A-11D show an alignment of VH and VL domains of Family 2 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 294, 139, 358, 362, 354, 380, 384, 350, 137, 374, 356, 352, 135, 348, 376, 346, 342, 344, 141, 360, 370, 382, 364, 366, 368, 372, and 378 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 295, 363, 140, 359, 355, 351, 136, 349, 377, 138, 375, 357, 353, 381,

385, 345, 365, 367, 369, 347, 142, 343, 371, 379, 383, 361, and 373 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIGS. 11E-11F show an alignment of VH and VL domains of Family 3 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 133, 128, 396, 386, 398, 402, 392, 388, 390, 394, and 400 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 134, 105, 387, 389, 395, 397, 399, 403, 391, 393, and 401 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIGS. 11G & 11H show an alignment of VH and VL domains of Family 4 clones. The amino acid sequences of the VH domains are SEQ ID NOs:116, 117, 118, 119, 282, 404, and 406 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 93, 94, 95, 96, 283, 405, and 407 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIG. 11I shows an alignment of VH and VL domains of Family 5, Bin 4 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 278 and 412 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 279 and 413 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIG. 11J shows an alignment of VH and VL domains of additional Family 5, Bin 4 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 275 and 414 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs: 276 and 415 (in order from top to bottom in the alignment). HVR and linker sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIG. 11K shows the sequences of VH and VL domains (SEQ ID NOs 280 and 281, respectively of Family 5, Bin 4 clone S209. The HVR sequences are underlined.

FIG. 11L shows an alignment of VH and VL domains of Family 5, Bin 5 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 123 and 292 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 100 and 293 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIG. 11M shows an alignment of VH and VL domains of additional Family 5, Bin 5 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 288, 290, 408, and 410 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs: 289, 291, 409, and 411 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIG. 11N shows the sequences of VH and VL domains of clone 149 (SEQ ID NOs 286 and 287, respectively) and clone 218 (SEQ ID NOs 284 and 285, respectively). HVR sequences are underlined.

FIGS. 11O & 11P show an alignment of VH and VL domains of Family 1 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 120, 121, 130, and 122

35

(in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs: 97, 98, 107, and 99 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

FIGS. 12A-12C show surface plasmon resonance (SPR) binding profiles of representative antibody clones binding a pre-formed complex of a high affinity SIRP- α variant (SEQ ID NO:18) mixed with increasing concentrations of the IgSF domain of CD47 (SEQ ID NO:16). FIG. 12A shows the binding curve of an antibody clone (S123) that does not block CD47 binding to SIRP- α (e.g., a non-blocking antibody). FIG. 12B shows the binding curve of an antibody clone (S119) that blocks CD47 binding to SIRP- α (e.g., a blocking antibody). FIG. 12C shows the binding curve of an antibody clone (S118) that binds to SIRP- α and reduces its affinity for binding CD47 (e.g., a "kick off" antibody).

FIGS. 13A-13G show the results of in vitro tumor cell phagocytosis assays using macrophages treated with anti-SIRP- α antibody (at indicated series of concentrations), cetuximab or trastuzumab, anti-SIRP- α antibody plus cetuximab or trastuzumab, or control antibody (IgG1, κ), as indicated. Macrophages that had phagocytosed tumor cells were identified as cells positive for CD33, CD206, and CFSE by flow cytometry. Tumor cells assayed were DLD-1 (FIGS. 13A-13D & 13G) or OE19 cells (FIGS. 13E & 13F). Anti-SIRP- α antibodies tested were AB3a (FIGS. 13A & 13B), AB45a (FIGS. 13C & 13D), AB119a (FIG. 13E), AB135a (FIG. 13F), and AB136c (FIG. 13G).

FIG. 14 shows the results of in vivo dendritic cell activation assays on dendritic cells isolated from the spleens of Balb/c mice treated with anti-SIRP- α antibody AB136b, control rat anti-mouse anti-SIRP- α antagonistic antibody (clone p84), rat IgG control, or mouse IgG control, as indicated. Mice were intravenously injected with the indicated antibody at 10 mg/kg, and spleens were harvested five hours after injection. Activation marker CD86 on dendritic cells was measured by flow cytometry.

FIG. 15 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess single agent activity. MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8 mice/group). Mice were treated with vehicle (PBS), CD47 blocking anti-SIRP- α antibody AB25b, CD47 blocking anti-SIRP- α antibody AB25c, CD47 blocking anti-SIRP- α antibody AB27b, CD47 non-blocking anti-SIRP- α antibody AB3b, or CD47 non-blocking anti-SIRP- α antibody 136b. Treatment was initiated when tumors were an average of 60 mm³, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks with anti-SIRP α antibodies. Animals were sacrificed when tumors reached a volume of ~2000 mm³.

FIG. 16 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess single agent activity. CT26 cells were implanted subcutaneously in BALB/c mice (8-9 mice were used per group) that were treated with AB136b or vehicle (PBS), as indicated. Treatment was initiated when tumors were an average of 80 mm³, day 7 post implant. Mice were dosed intraperitoneally (IP) at 3 mg/kg or 10 mg/kg twice a week for three weeks with anti-SIRP α antibodies. Animals were sacrificed when tumors reached a volume of ~2000 mm³.

FIG. 17A shows a comparison of CD47 and anti-SIRP- α antibody clone 119 Fab binding to SIRP- α , as determined by X-ray crystallography.

36

FIG. 17B shows the interaction site between anti-SIRP- α antibody clone 119 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 119 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

FIG. 18A shows a comparison of CD47 and anti-SIRP- α antibody clone 136 Fab binding to SIRP- α , as determined by X-ray crystallography.

FIG. 18B shows the interaction site between anti-SIRP- α antibody clone 136 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 136 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

FIG. 19A shows a comparison of CD47 and anti-SIRP- α antibody clone 3 Fab binding to SIRP- α , as determined by X-ray crystallography.

FIG. 19B shows the interaction site between anti-SIRP- α antibody clone 3 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 3 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

FIG. 19C shows a comparison of CD47 and anti-SIRP- α antibody clone 115 Fab binding to SIRP- α , as determined by X-ray crystallography.

FIG. 19D shows the interaction site between anti-SIRP- α antibody clone 115 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 115 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

FIG. 20A shows a comparison of CD47, anti-SIRP- α antibody clone 119 Fab, anti-SIRP- α antibody clone 136 Fab, anti-SIRP- α antibody clone 3 Fab, anti-SIRP- α antibody clone 115 Fab binding to SIRP- α , as determined by X-ray crystallography.

FIGS. 20B-20E show the epitopes for CD47, anti-SIRP- α antibody clone 119 Fab, anti-SIRP- α antibody clone 136 Fab, anti-SIRP- α antibody clone 3 Fab, and anti-SIRP- α antibody clone 115 Fab binding to SIRP- α , as determined by X-ray crystallography. Values indicate difference between surface accessible area of each residue atom in the Fab/CD47 when analyzed alone vs. when analyzed in complex with SIRP- α , expressed as buried surface area ($\text{\AA}^2$). Residue numbering according to SEQ ID NO: 296.

FIG. 21A shows a flowchart for epitope binning of anti-SIRP- α antibodies.

FIG. 21B shows results of an exemplary assay for epitope binning of anti-SIRP- α antibodies A, B, C, D, E, and F.

FIGS. 22A & 22B show the results of epitope binning of the indicated anti-SIRP- α antibodies. The clone number for the ligand (anti-SIRP α) bound to the chip is indicated as rows, and the clone number for the analytes (anti-SIRP α) injected over the chip is indicated as columns. White boxes indicate antibodies that form sandwiches (and are considered to bind different epitopes). Gray boxes indicate antibodies that did not form sandwiches (and are considered to bind the same epitope). "X" indicates scenarios where the data from one orientation disagrees with the other.

FIG. 23 provides a model for anti-SIRP- α antibody and CD47 binding to the SIRP- α D1 domain based on epitope

37

binning. Representative antibody clones for each bin are provided (and labeled by number).

FIG. 24A shows alignments between the parental 119 heavy chain ("119_VH_Wt"), the 119 variant heavy chain with 4 mutations (3 back-mutations to germline sequence in framework and one mutation in CDR-H1 removing a potential oxidation hot spot; "VH_MutALL"), the 119 variant heavy chain with 3 mutations (3 back-mutations to germline sequence in framework only; "VH_MutAll_V34M"), and the 119 variant heavy chain with 3 mutations and an M34L mutation (3 back-mutations to germline sequence in framework; "VH_MutAll_V34L"). Sequences depicted are: SEQ ID NO:335 for 119_VH_Wt, SEQ ID NO:246 for VH_MutALL, SEQ ID NO:258 for VH_MutAll_V34M, and SEQ ID NO:327 for VH_MutAll_V34L. CDR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 24B shows alignments between the parental 119 light chain ("119_VL_Wt"), and the 119 variant light chain with 4 mutations (4 back-mutations to germline sequence in framework; "VL_mutAll"). Sequences depicted are: SEQ ID NO:97 for 119_VL_Wt and SEQ ID NO:312 for VL_mutAll. CDR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 25A shows alignments between the parental 135 light chain ("VL_wt") and the 135 variant light chain with 2 mutations (2 back-mutations to germline sequence in framework; "VL_mutALL"). Sequences depicted are: SEQ ID NO:104 for 135_VL_wt and SEQ ID NO:248 for 135_VL_MutALL. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 25B shows alignments between the parental 135 heavy chain ("VH_wt") the 135 variant heavy chain with 6 mutations (5 back-mutations to germline sequence in framework and one mutation in CDR-H1 removing a potential oxidation hot spot; "VH_MutAll"), the 135 variant heavy chain with 5 back-mutations to germline sequence in framework ("VH_MutAll_V34M"), and the 135 variant heavy chain with 5 back-mutations to germline sequence in framework and M34L mutation ("VH_MutAll_V34L"). Sequences depicted are: SEQ ID NO:341 for VH_wt, SEQ ID NO:247 for VH_MutAll, and SEQ ID NO:259 for VH_MutAll_V34M, and SEQ ID NO:328 for VH_MutAll_V34L. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 26A shows alignments between the parental 136 light chain ("VL_wt"), the 136 variant light chain with 4 mutations (4 back-mutations to germline sequence in framework; "VL_mutaLL"), the 136 variant light chain with a single I2T back-mutation reverted to wild-type sequence in an otherwise "all mut" background ("VL_Mutall_I2T"). Sequences depicted are: SEQ ID NO: 134 for VL_wt, SEQ ID NO:250 for VL_mutaLL, and SEQ ID NO:251 for VL_Mutall_I2T. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 26B shows alignments between the parental 136 heavy chain ("VH_wt"), the 136 variant heavy chain with 6 mutations (5 back-mutations to germline sequence in framework and one mutation in CDR-H1 removing a potential oxidation hot spot; "VH_mutall"), the 136 variant heavy chain with 5 back-mutations to germline sequence in framework ("VH_Mutall_V34M"), and the 136 variant heavy chain with 5 back-mutations to germline sequence in framework and M34L mutation ("VH_Mutall_V34L"). Sequences depicted are: SEQ ID NO:133 for VH_wt, SEQ ID NO:249 for VH_mutall, SEQ ID NO:260 for VH_Mutall_V34M,

38

and SEQ ID NO:329 for VH_Mutall_V34L. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 27A shows binding affinities of antibody 136 variants to six SIRP- α proteins: human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), NOD mouse SIRP- α (SEQ ID NO:8), BL/6 mouse SIRP- α (SEQ ID NO:9), and BALB/c mouse SIRP- α (SEQ ID NO:10). Antibody variants had mutant ("mut") or parental ("wt") light chains and mutant or parental heavy chains, as indicated in the order light chain/heavy chain. On the graph, y-axis indicates the ratio of K_D mut/ K_D wt. A ratio of 1 means antibody had equivalent K_D to wt/wt antibody (indicated by dotted line); ratio of >1 indicates lower affinity than wt/wt; and ratio of <1 indicates higher affinity than wt/wt.

FIG. 27B shows binding affinities of antibody 136 variants to seven SIRP- α proteins: BL/6 mouse SIRP- α (SEQ ID NO:9), NOD mouse SIRP- α (SEQ ID NO:8), BALB/c mouse SIRP- α (SEQ ID NO:10), human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), and human SIRP- γ v1 (SEQ ID NO:15). In addition to testing the wt 136 and all mutant 136, variants were constructed eliminating each individual mutation in an all mutant light chain background. On the graph, y-axis indicates the ratio of K_D mut/ K_D wt. A ratio of 1 means antibody had equivalent K_D to wt/wt antibody (indicated by dotted line); ratio of >1 indicates lower affinity than wt/wt; and ratio of <1 indicates higher affinity than wt/wt.

FIG. 28 compares expression yield and binding affinity of antibodies having indicated human heavy chains and humanized light chains in FreeStyle™ 293-FS cells (Thermo Fisher).

FIG. 29 shows an alignment between Hum1, Hum 8, and Hum9 VL domains. Hum8 was generated based on Hum1 but with 5 amino acid substitutions near or in HVR-L1 and -L2 that increase humanness. Hum9 was generated based on Hum1 but with 4 amino acid substitutions near or in HVR-L1 and -L2 that increase humanness. SEQ ID NOs: 252 (Hum1), 416 (Hum 8), and 262 (Hum9) are depicted. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 30 shows alignments between the antibody 21 variant with germline back-mutations ("HC_MutAll"), the antibody 21 variant with germline back-mutations and mutation in CDR-H1 removing a potential oxidation hot spot M34V ("HC_MutAll_M34V"), the antibody 21 variant with germline back-mutations and M34L mutation ("HC_MutAll_M34L"), the antibody 25 variant with germline back-mutations ("HC_MutAll"), the antibody 25 variant with germline back-mutations and mutation in CDR-H1 removing a potential oxidation hot spot M34V ("HC_MutAll_M34V"), the antibody 25 variant with germline back-mutations and M34L mutation ("HC_MutAll_M34L"), the antibody 27 variant with germline back-mutations ("HC_MutAll"), the antibody 27 variant with germline back-mutations and mutation in CDR-H1 removing a potential oxidation hot spot M34V ("HC_MutAll_M34V"), and the antibody 27 variant with germline back-mutations and M34L mutation ("HC_MutAll_M34L"). Sequences depicted are SEQ ID NOs:263, 264, 330, 267, 268, 332, 265, 266, and 331 (top to bottom). HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

FIG. 31 shows the results of in vitro phagocytosis assays using HER2(+) OE19 cells as the target and M2 macrophages as the phagocytosing cell. "Kick off" anti-SIRP- α

antibodies were tested at the indicated concentrations in combination with the anti-HER2 antibody trastuzumab. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 32 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Non-blocking anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

FIGS. 33A-33C show the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 34 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Non-blocking anti-SIRP- α antibodies 27 and 136 were each tested as a full-length antibody (with Fc region) or F(ab)₂ fragment at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 35 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Blocking anti-SIRP- α antibody 119 variants were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 36 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Blocking anti-SIRP- α antibody 135 variants were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 37 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Non-blocking anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

FIGS. 38A-38B show the results of in vivo dendritic cell activation assays with the indicated anti-SIRP- α antibodies. Mice were intravenously injected with the indicated antibody at 10 mg/kg, and spleens were harvested five hours after injection. Activation markers CD86, MHCII and CCR7 on CD4+ dendritic cells were measured by flow cytometry.

FIG. 39A shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. 218a and 218 variant anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 39B shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Exemplary blocking anti-SIRP- α antibodies 119a, 120a, and 122a were tested at the indicated concentrations. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 39C shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Exemplary non-blocking

anti-SIRP- α antibodies 136a and 137a were tested at the indicated concentrations. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 39D shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Exemplary kick off anti-SIRP- α antibodies 115a, 116a, 117a, 118a, and 132a were tested at the indicated concentrations. Phagocytosis was measured by percentage of CFSE+ cells.

FIG. 40 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess activity of combining anti-SIRP- α treatment with PD-L1/PD-1 pathway inhibition. CT26 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8 mice/group). Mice were treated with vehicle (PBS), anti-PD-L1 antibody, CD47 blocking anti-SIRP- α antibody AB25b, or AB25b and PD-L1. Treatment was initiated when tumors were an average of 60 mm³, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks and sacrificed when tumors reach a volume of ~2000 mm³.

FIG. 41 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess activity of combining anti-SIRP- α treatment with PD-L1/PD-1 pathway inhibition. MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8 mice/group). Mice were treated with vehicle (PBS), anti-PD-1 antibody, CD47 blocking anti-SIRP- α antibody AB25b, AB25b and anti-PD-1, CD47 non-blocking anti-SIRP- α antibody AB136b, or AB136b and anti-PD-1. Treatment was initiated when tumors were an average of 60 mm³, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks and sacrificed when tumors reach a volume of ~2000 mm³.

DETAILED DESCRIPTION

The present disclosure describes antibodies that bind the extracellular domains (e.g., the D1 domains) of one or more human SIRP- α polypeptides and have a variety of SIRP- α binding profiles of potential interest. These unique SIRP- α binding profiles include one or more of the following binding capabilities, which are combined in a multifactorial manner to yield a multitude of unique specificities. For instance, the antibody can bind the extracellular domains (e.g., the D1 domains) of human SIRP- α v1, human SIRP- α v2, or both; the antibody can bind the extracellular domains (e.g., the D1 domains) of one or more monkey SIRP- α polypeptides, or it can lack binding thereto; the antibody can bind the extracellular domains (e.g., the D1 domains) of one or more murine SIRP- α polypeptides, or it can lack binding thereto; the antibody can bind the extracellular domain (e.g., the D1 domain) of a human SIRP β polypeptide, or it can lacking binding thereto; and/or the antibody can bind the extracellular domain (e.g., the D1 domain) of a human SIRP γ polypeptide, or it can lacking binding thereto. In addition, the present disclosure describes antibodies that block binding of CD47 to SIRP- α , antibodies that do not block binding of CD47 to SIRP- α , and antibodies that do not block binding of CD47 to SIRP- α but decrease SIRP- α 's affinity for CD47, leading to more rapid dissociation of the CD47/SIRP- α complex. In addition, the present disclosure describes anti-SIRP- α antibodies with one or more in vitro and/or in vivo biological properties of interest, such as the ability to enhance macrophage phagocytosis, enhance dendritic cell activation, inhibit in vivo growth of a tumor that expresses CD47, and/or the ability to accomplish one or

more of these activities without preventing interactions between a CD47-expressing cell and a T cell.

The methods described herein may be used to identify antibodies with unique combinations of the above binding specificities. Without wishing to be bound to theory, it is thought that the ability to identify unique anti-SIRP- α antibodies with different binding profiles as described above allows for the identification of antibodies (e.g., those described herein) with desirable clinical properties and advantages for pre-clinical research.

In one aspect, provided herein are isolated antibodies that bind the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both. Further provided herein are polynucleotides and vectors encoding the antibodies of the present disclosure, as well as methods of antibody production related thereto.

In another aspect, provided herein are methods for treating or delaying progression of cancer in an individual, comprising administering to the individual an effective amount of an antibody of the present disclosure.

In another aspect, provided herein are methods for treating or delaying progression of an autoimmune or inflammatory disease in an individual, comprising administering to the individual an effective amount of an antibody of the present disclosure.

In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the methods include (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the methods include contacting an antigen binding domain or antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide.

In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the method includes (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the method includes contacting an antigen binding domain or antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide.

In another aspect, provided herein are methods for producing an anti-SIRP- α antibody that binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides. In some embodiments, the method includes (a) immunizing a chicken with a peptide comprising at least a portion of a human SIRP- α extracellular domain (e.g., the D1 domain); (b) obtaining an antibody from an antibody-producing cell from the immunized chicken; and (c) detecting binding between the antibody obtained from the cell and the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, wherein binding between the antibody and the extracellular domains (e.g., the D1 domains) of the two or more different human SIRP- α variant polypeptides indicates that the antibody is an anti-SIRP- α antibody that binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides.

Definitions

Before describing the disclosed embodiments in detail, it is to be understood that the present disclosure is not limited to particular compositions or biological systems, which can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

As used in this specification and the appended claims, the singular forms "a", "an" and "the" include plural referents

unless the content clearly dictates otherwise. Thus, for example, reference to “a molecule” optionally includes a combination of two or more such molecules, and the like.

The term “about” as used herein refers to the usual error range for the respective value readily known to the skilled person in this technical field. Reference to “about” a value or parameter herein includes (and describes) embodiments that are directed to that value or parameter per se.

It is understood that aspects and embodiments of the present disclosure include “comprising,” “consisting,” and “consisting essentially of” aspects and embodiments.

A “SIRP- α polypeptide” as used herein may refer to any endogenous or naturally occurring SIRP- α polypeptide encoded by a genome from any vertebrate, including mammals such as humans, monkeys, rodents (e.g., mouse or rat), and birds, such as chickens. The term also includes naturally occurring variants, e.g., alternatively spliced variants, allelic variants, or polymorphisms (e.g., those described herein). The term may further refer to full-length, unprocessed SIRP- α polypeptides as well as SIRP- α polypeptides that result from cellular processing, e.g., removal of a signal sequence, etc. Exemplary SIRP- α polypeptide sequences are described herein. In some embodiments, a human SIRP- α polypeptide is one encoded by a human SIRPA gene, e.g., as described by NCBI Gene ID No. 140885. As described herein, SIRP- α polypeptides are highly polymorphic within and among species. For example, at least 10 human variants with amino acid polymorphisms in the extracellular domain have been identified.

SIRP- α polypeptides include an extracellular domain that binds ligands/partners, e.g., CD47. SIRP- α polypeptides comprise 3 highly homologous immunoglobulin (Ig)-like extracellular domains—D1, D2, and D3. The SIRP- α D1 domain (“D1 domain”) refers to the membrane distal, extracellular domain of SIRP- α and mediates binding of SIRP- α to CD47 (see, e.g., Hatherley, D. et al. (2008) *Mol. Cell* 31:266-77; Hatherley, D. et al. (2007) *J. Biol. Chem.* 282:14567-75; Hatherley, D. et al. (2009) *J. Biol. Chem.* 284:26613-9; and Lee, W. Y. et al. (2010) *J. Biol. Chem.* 285:37953-63). The extracellular domain generally refers to the entire extracellular portion of SIRP- α , e.g., as expressed on a cell surface, and may include distinct SIRP- α domains, such as the D1 domain. The D1 domain contains residues shown to be critical for mediating CD47 binding (see, e.g., Lee, W. Y. et al. (2007) *J. Immunol.* 179:7741-50). In some embodiments, an antibody that binds an extracellular domain of a SIRP- α polypeptide binds one or more residues of the D1 domain. Exemplary human SIRP- α D1 domain sequences are described throughout the present disclosure and include without limitation SEQ ID NOs:5, 6, and 76-83. Human SIRP- α D2 and D3 domain sequences are also known and include, without limitation, APVVS GPAARATPQHTVSFT-
CESHGFSPRDITLKWFKNGNELSDFQTNVDPVGESV-
SYSI HSTAKVVLTRDVDHSQVICE-
VAHVTLQGDPLRGTTANLSETIR (SEQ ID NO:131) for the D2 domain and VPPTLEVTQQPVRAE-
NQVNVTCQVRKFPYQRLQLTWLENGNVSRTE-
TASTVTENKDGTYNWMWSLLVNVSAH-
RDDVKLTQVEHGDQPAVSKSHDLKVS (SEQ ID NO:132) for the D3 domain.

As used herein, “CD47” (also known as integrin associated protein (IAP), MER6, and OA3) refers to a polypeptide that, among other roles, serves as a binding partner for SIRP- α polypeptides. In some embodiments, CD47 refers to a human CD47 polypeptide, e.g., a polypeptide encoded by a human CD47 gene, such as that described by NCBI Ref

Seq ID No. 961. Exemplary human CD47 amino acid sequences are known (see, e.g., NCBI Reference Sequence Accession No. NP_001768). In particular, the IgSF domain of CD47 refers to the N-terminal extracellular domain of CD47 that is known to be critical for SIRP- α binding (see, e.g., Barclay, A. N. and Brown, M. H. (2006) *Nat. Rev. Immunol.* 6:457-64 and Hatherley, D. et al. (2009) *J. Biol. Chem.* 284:26613-9). In some embodiments, an IgSF domain of CD47 comprises the amino acid sequence of QLLFNKTKSVETFSNDTVVIPCFVTNMEAQNTTE-
VYVKWKFGRDIYTFDGLNKSTV PTFSSAKI-
EVSQLLKGDASLKMDKSDAVSHTGNYTCEVTEL-
TREGETIIEELKYRVVS (SEQ ID NO:16). The term “CD47” may also include modified CD47 polypeptides that are able to bind SIRP- α , e.g., a polypeptide comprising an IgSF domain of CD47 conjugated to another polypeptide or other moiety, e.g., an Ig Fc region.

As used herein, a “SIRP- α epitope” may refer to the amino acids of a SIRP- α polypeptide that form the binding site for an anti-SIRP- α antibody of the present disclosure and/or a SIRP- α binding partner, including without limitation CD47. Binding of an antibody or other polypeptide to an epitope can be characterized and/or mapped using a variety of assays known in the art, including without limitation a cross-blocking assay (see *Antibodies, A Laboratory Manual*, Cold Spring Harbor Laboratory, Ed Harlow and David Lane (1988)), epitope mapping (see Champe et al., *J. Biol. Chem.* 270:1388-1394 (1995)), X-ray co-crystallography, epitope binning, site-directed mutagenesis, oligo-peptide scanning, high-throughput mutagenesis mapping, hydrogen/deuterium exchange, limited proteolysis, and so forth.

As used herein “modulating SIRP- α signaling” may refer to antagonizing, agonizing, or otherwise interfering with one or more aspects of SIRP- α signaling in a cell expressing a SIRP- α polypeptide. SIRP- α signaling may refer to one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, eosinophils, neutrophils, dendritic cells, and myeloid-derived suppressor cells (MDSCs).

As used herein, the term “antibody” may refer to intact antibodies; antibody fragments (including without limitation Fab, F(ab')₂, Fab'-SH, Fv, diabodies, scFv, scFv-Fc, single domain antibodies, single heavy chain antibodies, and single light chain antibodies), provided that they exhibit the desired biological activity (e.g. epitope binding); monoclonal antibodies; polyclonal antibodies; monospecific antibodies; multi-specific antibodies (e.g., bispecific antibodies); and antibody-like proteins.

As used herein, the term “bispecific” when used in reference to an antibody or antibody fragment includes an antibody or antibody fragment that possesses two different binding specificities. For example, each binding specificity may recognize a different antigen, or each binding specificity may recognize the same antigen with different affinity and/or precise epitope. In some embodiments, each different binding specificity comprises one or more different antibody antigen binding domains (e.g., variable domains), such that the bispecific antibody or antibody fragment comprises at least a first antigen binding domain with a first binding

specificity and a second antigen binding domain with a second binding specificity. A variety of exemplary bispecific antibody formats are described herein and known in the art.

An “isolated” antibody may refer to an antibody that has been separated and/or recovered from a component of its natural environment, e.g., a host cell or organism. In some embodiments, an antibody is purified to a desired purity by weight (e.g., at least 95%); and/or homogeneity by SDS-PAGE using, for example, staining by silver, Coomassie, etc. In some embodiments, an isolated antibody is obtained following one or more purification steps.

As is known in the art, “native” antibodies refer to typically heterotetrameric complexes including two identical light (L) chains and two identical heavy (H) chains. Variable numbers of disulfide bonds connect the two heavy chains, and one connects each light chain to a heavy chain, in addition to intrachain disulfide bridges. The heavy chains include a variable domain (VH) followed (N-terminus to C-terminus) by three or four constant domains. The light chains include a variable domain (VL) followed by a constant domain (CL). Typically, mammalian light chains fall into one of two categories based on amino acid sequence: kappa and lambda.

A “constant domain” may refer to the more conserved portion of the antibody or fragment, e.g., outside the variable domains. The term may include the CL domain as well as heavy chain constant domains CH1, CH2, CH3 and optionally CH4.

Constant domains of the heavy chain can be assigned to one of 5 major types: IgA, IgD, IgE, IgG, and IgM. Several subtypes exist for many of these major types. The subunit structures and three-dimensional configurations of different classes of immunoglobulins are well known and described generally in, for example, Abbas et al. *Cellular and Molecular Immunology*, 4th ed. (W.B. Saunders, Co., 2000).

As used herein, the term “antibody variable domain” refers to the portions of the light and heavy chains of an antibody that include the complementary determining regions (CDRs, e.g., CDR L1, CDR L2, CDR L3, CDR H1, CDR H2, and CDR H3) and framework regions (FRs).

The term “variable” refers to the fact that subsequences of the variable domains differ substantially in sequence between antibodies and are critical to the binding specificity of a particular antibody for its antigen. Variability is concentrated in three hypervariable regions (HVRs) in both VH and VL domains. The more conserved portions of variable domains are called the framework regions (FR) in which the HVRs are interspersed. The variable domains of native heavy and light chains each comprise four FR regions connected by three HVRs that form loops (see Kabat et al., *Sequences of Proteins of Immunological Interest*, Fifth Edition, National Institute of Health, Bethesda, Md. (1991)).

The term “hypervariable region (HVR)” may refer to the subregions of the VH and VL domains characterized by enhanced sequence variability and/or formation of defined loops. These include three HVRs in the VH domain (H1, H2, and H3) and three HVRs in the VL domain (L1, L2, and L3). H3 is believed to be critical in imparting fine binding specificity, with L3 and H3 showing the highest level of diversity. See Johnson and Wu, in *Methods in Molecular Biology* 248:1-25 (Lo, ed., Human Press, Totowa, N.J., 2003).

A number of HVR delineations are known. The Kabat Complementarity Determining Regions (CDRs) are based on sequence variability and are the most commonly used

Health, Bethesda, Md. (1991)). Chothia refers instead to the location of the structural loops (Chothia and Lesk *J. Mol. Biol.* 196:901-917 (1987)). The AbM HVRs represent a compromise between the Kabat HVRs and Chothia structural loops, and are used by Oxford Molecular's AbM antibody modeling software. The “contact” HVRs are based on an analysis of the available complex crystal structures. The residues from each of these HVRs are noted below. “Framework” or “FR” residues are those variable domain residues other than the HVR residues.

Loop	Kabat	AbM	Chothia	Contact
L1	L24-L34	L24-L34	L26-L32	L30-L36
L2	L50-L56	L50-L56	L50-L52	L46-L55
L3	L89-L97	L89-L97	L91-L96	L89-L96
H1	H31-H35B	H26-H35B	H26-H32	H30-H35B (Kabat Numbering)
H1	H31-H35	H26-H35	H26-H32	H30-H35 (Chothia Numbering)
H2	H50-H65	H50-H58	H53-H55	H47-H58
H3	H95-H102	H95-H102	H96-H101	H93-H101

“Extended” HVRs are also known: 24-36 or 24-34 (L1), 46-56 or 50-56 (L2) and 89-97 or 89-96 (L3) in the VL and 26-35 (H1), 50-65 or 49-65 (H2) and 93-102, 94-102, or 95-102 (H3) in the VH (Kabat numbering).

“Numbering according to Kabat” may refer to the numbering system used for heavy chain variable domains or light chain variable domains of the compilation of antibodies in Kabat et al., supra. The actual linear amino acid sequence may contain fewer or additional amino acids corresponding to a shortening of, or insertion into, a FR or HVR of the variable domain. The Kabat numbering of residues may be determined for a given antibody by alignment at regions of homology of the sequence of the antibody with a “standard” Kabat numbered sequence. Typically, the Kabat numbering is used when referring to a residue in the variable domains (approximately residues 1-107 of the light chain and residues 1-113 of the heavy chain), whereas the EU numbering system or index (e.g., the EU index as in Kabat, numbering according to EU IgG1) is generally used when referring to a residue in the heavy chain constant region.

“Full length” or “intact” antibodies typically include heavy chains with an Fc region, e.g., as opposed to an antibody fragment. Antigen-binding “Fab” fragments with a single antigen binding site may be released from the residual Fc fragment by papain digestion. F(ab')₂ fragments include two antigen-binding sites produced by pepsin treatment of an antibody. Antibody fragments will, however, include one or more antibody variable regions.

An “Fv” fragment contains a complete antigen-binding site. A single chain Fv (scFv) can include a VH and a VL domain linked by a peptide linker such that the VH and VL domains associate, e.g., as in an antibody or Fab fragment, such that the HVRs form an antigen binding site. See Pluckthün, in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds., (Springer-Verlag, New York, 1994), pp. 269-315. In some embodiments, the scFv is fused to an antibody Fc domain (e.g., scFv-Fc). While six HVRs typically comprise an antigen binding site, a single variable domain with three HVRs is still capable of binding an antigen, albeit at a lower affinity. See Hamers-Casterman et al., *Nature* 363:446-448 (1993); Sheriff et al., *Nature Struct. Biol.* 3:733-736 (1996). Single domain antibodies (e.g., camelid antibodies) typically include a single, monomeric variable domain for antigen binding. Single heavy

chain (VHH) and single light chain antibodies are also known. A Fab' fragment typically includes a few more residues at the C-terminal end than a Fab fragment. A Fab'-SH includes cysteine residues with a free thiol. Various chemical couplings of antibody fragments are known in the art.

A "diabody" includes antibody fragments with two antigen-binding sites. These include a VH and VL domain connected by a linker, which is typically too short to facilitate pairing of domains in the same chain. Diabodies may be bivalent or bispecific. Tribodies and tetrabodies, or other numbers of VH/VL domains are known. See Hudson et al., *Nat. Med.* 9:129-134 (2003).

As used herein, a "monoclonal" antibody refers to an antibody obtained from a population of substantially homogeneous antibodies, e.g., substantially identical but allowing for minor levels of background mutations and/or modifications. "Monoclonal" denotes the substantially homogeneous character of antibodies, and does not require production of the antibody by any particular method. In some embodiments, a monoclonal antibody is selected by its HVR, VH, and/or VL sequences and/or binding properties, e.g., selected from a pool of clones (e.g., recombinant, hybridoma, or phage-derived). A monoclonal antibody may be engineered to include one or more mutations, e.g., to affect binding affinity or other properties of the antibody, create a humanized or chimeric antibody, improve antibody production and/or homogeneity, engineer a multispecific antibody, resultant antibodies of which are still considered to be monoclonal in nature. A population of monoclonal antibodies may be distinguished from polyclonal antibodies as the individual monoclonal antibodies of the population recognize the same antigenic site. A variety of techniques for production of monoclonal antibodies are known; see, e.g., the hybridoma method (e.g., Kohler and Milstein, *Nature*, 256:495-97 (1975); Hongo et al., *Hybridoma*, 14 (3): 253-260 (1995), Harlow et al., *Antibodies: A Laboratory Manual*, (Cold Spring Harbor Laboratory Press, 2nd ed. 1988); Hammerling et al., in: *Monoclonal Antibodies and T-Cell Hybridomas* 563-681 (Elsevier, N.Y., 1981)), recombinant DNA methods (see, e.g., U.S. Pat. No. 4,816,567), phage-display technologies (see, e.g., Clackson et al., *Nature*, 352: 624-628 (1991); Marks et al., *J. Mol. Biol.* 222: 581-597 (1992); Sidhu et al., *J. Mol. Biol.* 338(2): 299-310 (2004); Lee et al., *J. Mol. Biol.* 340(5): 1073-1093 (2004); Fellouse, *Proc. Natl. Acad. Sci. USA* 101(34): 12467-12472 (2004); and Lee et al., *J. Immunol. Methods* 284(1-2): 119-132 (2004), and technologies for producing human or human-like antibodies in animals that have parts or all of the human immunoglobulin loci or genes encoding human immunoglobulin sequences (see, e.g., WO 1998/24893; WO 1996/34096; WO 1996/33735; WO 1991/10741; Jakobovits et al., *Proc. Natl. Acad. Sci. USA* 90: 2551 (1993); Jakobovits et al., *Nature* 362: 255-258 (1993); Bruggemann et al., *Year in Immunol.* 7:33 (1993); U.S. Pat. Nos. 5,545,807; 5,545,806; 5,569,825; 5,625,126; 5,633,425; and 5,661,016; Marks et al., *Bio/Technology* 10: 779-783 (1992); Lonberg et al., *Nature* 368: 856-859 (1994); Morrison, *Nature* 368: 812-813 (1994); Fishwild et al., *Nature Biotechnol.* 14: 845-851 (1996); Neuberger, *Nature Biotechnol.* 14: 826 (1996); and Lonberg and Huszar, *Intern. Rev. Immunol.* 13: 65-93 (1995).

"Chimeric" antibodies may refer to an antibody with one portion of the heavy and/or light chain from a particular isotype, class, or organism and another portion from another isotype, class, or organism. In some embodiments, the

variable region will be from one source or organism, and the constant region will be from another.

"Humanized antibodies" may refer to antibodies with predominantly human sequence and a minimal amount of non-human (e.g., mouse or chicken) sequence. In some embodiments, a humanized antibody has one or more HVR sequences (bearing a binding specificity of interest) from an antibody derived from a non-human (e.g., mouse or chicken) organism grafted onto a human recipient antibody framework (FR). In some embodiments, non-human residues are further grafted onto the human framework (not present in either source or recipient antibodies), e.g., to improve antibody properties. In general, a humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the hypervariable loops correspond to those of a non-human immunoglobulin, and all or substantially all of the FRs are those of a human immunoglobulin sequence. The humanized antibody optionally will also comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. See Jones et al., *Nature* 321:522-525 (1986); Riechmann et al., *Nature* 332:323-329 (1988); and Presta, *Curr. Op. Struct. Biol.* 2:593-596 (1992).

A "human" antibody may refer to an antibody having an amino acid sequence which corresponds to that of an antibody produced by a human and/or has been made using any of the techniques for making human antibodies as disclosed herein. Human antibodies can be produced using various techniques known in the art, including phage-display libraries. Hoogenboom and Winter, *J. Mol. Biol.*, 227:381 (1991); Marks et al., *J. Mol. Biol.*, 222:581 (1991); preparation of human monoclonal antibodies as described in Cole et al., *Monoclonal Antibodies and Cancer Therapy*, Alan R. Liss, p. 77 (1985); Boerner et al., *J. Immunol.*, 147(1):86-95 (1991); and by administering the antigen to a transgenic animal that has been modified to produce such antibodies in response to antigenic challenge, but whose endogenous loci have been disabled, e.g., immunized xenomice (see, e.g., U.S. Pat. Nos. 6,075,181 and 6,150,584 regarding XENOMOUSE™ technology) or chickens with human immunoglobulin sequence(s) (see, e.g., WO2012162422, WO2011019844, and WO2013059159).

As used herein, the term "linker" refers to a linkage between two elements, e.g., protein domains. In some embodiments, a linker can be a covalent bond or a spacer. The term "spacer" refers to a moiety (e.g., a polyethylene glycol (PEG) polymer) or an amino acid sequence (e.g., a 1-200 amino acid sequence) occurring between two polypeptides or polypeptide domains to provide space or flexibility (or both space and flexibility) between the two polypeptides or polypeptide domains. In some embodiments, an amino acid spacer is part of the primary sequence of a polypeptide (e.g., joined to the spaced polypeptides or polypeptide domains via the polypeptide backbone).

The term "cytotoxic agent" as used herein may refer to any agent that inhibits cellular proliferation or induces cell death. Cytotoxic agents include, but are not limited to, chemotherapeutic agents; radioactive isotopes; growth inhibitory agents; and toxins such as small molecule toxins or enzymatically active toxins, including fragments and/or variants thereof. Exemplary cytotoxic agents include without limitation metabolic inhibitors, anti-microtubule agents, platinum containing compounds, alkylating agents, proteasome inhibitors, topoisomerase II inhibitors, antimetabolites, topoisomerase I inhibitors, signal transduction pathway inhibitors, non-receptor tyrosine kinase angiogenesis inhibitors, hormones and hormonal analogues, proapoptotic

agents, inhibitors of LDH-A, cell cycle inhibitors, HDAC inhibitors, and antibiotic agents.

As used herein, a "label" may include any moiety that serves as a detection agent, e.g., of binding between a labeled antibody of the present disclosure and a macromolecule or cell. Exemplary labels include without limitation fluorescent (e.g., compounds or proteins), radioactive, or enzymatic moieties, as well as affinity purification tags.

As used herein, an antibody may be said to "bind" an antigen with an affinity sufficient to render the antibody useful for in vitro and/or in vivo manipulation of the antigen. In some embodiments, an antibody that "binds" an antigen has a dissociation constant (K_D) for the antigen that is less than or equal to 1 μ M at 25° C.

As used herein, the term "affinity" or "binding affinity" refers to the strength of the binding interaction between two molecules. Generally, binding affinity refers to the strength of the sum total of non-covalent interactions between a molecule and its binding partner, such as a high affinity SIRP- α D1 variant and CD47. Unless indicated otherwise, binding affinity refers to intrinsic binding affinity, which reflects a 1:1 interaction between members of a binding pair. The binding affinity between two molecules is commonly described by the dissociation constant (K_D) or the association constant (K_A). Two molecules that have low binding affinity for each other generally bind slowly, tend to dissociate easily, and exhibit a large K_D . Two molecules that have high affinity for each other generally bind readily, tend to remain bound longer, and exhibit a small K_D . In some embodiments, the K_D of two interacting molecules is determined using known methods and techniques, e.g., surface plasmon resonance (SPR). K_D can be calculated as the ratio of k_{off}/k_{on} .

As used herein, the term " K_D less than" refers to a numerically smaller K_D value and an increasing binding affinity relative to the recited K_D value. As used herein, the term " K_D greater than" refers to a numerically larger K_D value and a decreasing binding affinity relative to the recited K_D value.

As used herein, "treatment" may refer to therapeutic administration of a molecule, compound, formulation, composition, etc. so as to alter one or more pathological symptoms in an individual or cell being treated. Desirable effects of treatment can include without limitation decelerating disease progression, ameliorating or palliating a pathological symptom or disease state, improving prognosis, and/or achieving disease remission. For example, an individual's cancer is successfully "treated" if one or more symptoms associated with cancer are mitigated or abolished, such as, without limitation, reducing the proliferation of cancer cells, eliminating cancer cells or tumor burden, decreasing symptoms resulting from the cancer, increasing the quality of life of the individual, lessening the dose of other medication(s), and/or prolonging survival of the individual. As another example, an autoimmune or inflammatory disease may be successfully "treated" if one or more symptoms associated with the autoimmune or inflammatory disease are mitigated or abolished, such as, without limitation, reducing autoreactive immune cells and/or inflammatory immune cells or cytokines, decreasing immune activation and/or inflammation, slowing or mitigating organ damage resulting from the disease, decreasing symptoms resulting from the disease, increasing the quality of life of the individual, lessening the dose of other medication(s), and/or prolonging survival of the individual.

As used herein, "delaying progression" of a disease may refer to slowing, retarding, deferring, postponing develop-

ment of, stabilizing, or otherwise hindering the pathological course of the disease. In some embodiments, the term may refer to a delay sufficient to effectively encompass prevention, e.g., in preventing the individual from developing the disease. In some embodiments, e.g., an advanced cancer, delaying progression may include delaying metastasis. One of skill in the art will appreciate that the precise length of delay may depend, e.g., upon the specific disease, condition of the individual, and the like.

The terms "cancer" and "cancerous" may describe dysregulated or unregulated cell growth/proliferation by a cell or cells in a mammal. Any cancer type known in the art may be included, such as but not limited to carcinoma, sarcoma, lymphoma, leukemia, lymphoma, and blastoma. More particular examples of such cancers include, but are not limited to, lung cancer, squamous cell cancer, brain tumors, glioblastoma, head and neck cancer, hepatocellular cancer, colorectal cancer (e.g., colon or rectal cancers), liver cancer, bladder cancer, gastric or stomach cancer, pancreatic cancer, cervical cancer, ovarian cancer, cancer of the urinary tract, breast cancer, peritoneal cancer, uterine cancer, salivary gland cancer, kidney or renal cancer, prostate cancer, vulval cancer, thyroid cancer, anal carcinoma, penile carcinoma, melanoma, multiple myeloma and B-cell lymphoma (including non-Hodgkin's lymphomas (NHL)); acute lymphoblastic leukemia (ALL); chronic lymphocytic leukemia (CLL); acute myeloid leukemia (AML); Merkel cell carcinoma; hairy cell leukemia; chronic myeloblastic leukemia (CML); and associated metastases.

As used herein, the term "effective amount" may refer to an amount of an antibody of the present disclosure or a pharmaceutical composition containing an antibody of the present disclosure that is sufficient and effective in achieving a desired therapeutic effect in treating or delaying progression of a patient having a disease, such as a cancer, e.g., solid tumor or hematological cancer. In some embodiments, a therapeutically effective amount will avoid adverse side effects, and/or such side effects will be outweighed by beneficial effects. An effective amount may depend upon the individual being treated, e.g., age, weight, sex, disease state, as well as the ability of the agent to produce a desired response. An effective amount can be administered in one or more administrations. As in the clinical context, an effective amount of a drug, compound, or pharmaceutical composition may or may not be achieved in conjunction with another drug, compound, or pharmaceutical composition, such as another therapeutic agent. Thus, an "effective amount" may also be considered in the context of administering one or more additional therapeutic agents, and a single agent may be considered to be given in an effective amount if, in conjunction with one or more other agents, a desirable result may be or is achieved.

As used herein, the term "pharmaceutical composition" may refer to a medicinal or pharmaceutical formulation that includes an active ingredient as well as excipients or diluents (or both excipients and diluents) and enables the active ingredient to be administered by suitable methods of administration. In some embodiments, the pharmaceutical compositions disclosed herein include pharmaceutically acceptable components that are compatible with one or more antibodies of the present disclosure. In some embodiments, the pharmaceutical composition is in tablet or capsule form for oral administration or in aqueous form for intravenous or subcutaneous administration, for example by injection.

As used herein, the terms "subject," "individual," and "patient" are used interchangeably to refer to a vertebrate,

51

for example, a mammal. Mammals include, but are not limited to, murines, simians, humans, farm animals, sport animals, and pets.

As used herein, “in conjunction with” or “in combination with” may refer to administration of one therapeutic in addition to (e.g., before, during, and/or after) another therapeutic.

Antibodies

Certain aspects of the present disclosure relate to antibodies that bind the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain of a human SIRP- α v2 polypeptide, or both. As demonstrated herein, antibodies have been characterized that specifically bind to v1 or v2, as well as antibodies that bind to both proteins. In humans, at least 10 distinct alleles of SIRPA have been identified (see FIG. 1A; see also Takenaka, K. et al. (2007) *Nat. Immunol.* 8:1313-23). Antibodies that bind one or more human SIRP- α polypeptides and possess one or more of the other binding specificities described herein are an advantageous discovery of the present disclosure. Further, the present disclosure demonstrates methods for producing and identifying antibodies representing a surprising diversity of novel SIRP- α binding specificity profiles.

In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELINQKEGHFPRV TTVSDLTNRNMDFSIRIGNIT-PADAGTYTCVKFRKGSPPDVEFKSGAG-TELSVRAPKS (SEQ ID NO:5). In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELINQKEGHFPRV TVSESTKRENMDFSISINIT-

52

PADAGTYTCVKFRKGSPPDTEFKSGAGTELSVRAPKS (SEQ ID NO:6). In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAG-ETAWTLRCTATSLIPVGPIQWFRGAGPGREL- IYNQKEGHFPRV TTVSDLTNRNMDFSIRIGNIT-PADAGTYTCVKFRKGSPPDVEFKSGAGTELSVRAPKS (SEQ ID NO:5) and an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide comprising the amino acid sequence of

(SEQ ID NO: 6)

EEELQVIQPDKSVLVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRVTVSESTKRENMDFSISINITPADAGTYTCVKFRKGSPPDTEFKSGAGTELSVRAPKS.

In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 different human SIRP- α variant polypeptides. As used herein, a “human SIRP- α variant polypeptide” may refer to a naturally occurring human SIRP- α variant polypeptide or polymorphism found expressed in a human, e.g., as opposed to a variant bearing one or more engineered mutations. For example, in some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of one or more human SIRP- α variant polypeptides comprising a sequence shown in the Table 1. In some embodiments, an antibody of the present disclosure binds to an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide and/or an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, and binds to an extracellular domain (e.g., the D1 domain) of one or more human SIRP- α polypeptides selected from v3, v4, v5, v6, v7, v8, v9, and v10.

TABLE 1

Human SIRP- α variant polypeptide sequences corresponding to the D1 domain.		
Variant	SEQ ID NO	Sequence
v1	5	EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELINQKEGHFPRVTTVSDLTNRNMDFSIRIGNITPADAGTYTCVKFRKGSPPDVEFKSGAGTELSVRAPKS
v2	6	EEELQVIQPDKSVLVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRVTVSESTKRENMDFSISINITPADAGTYTCVKFRKGSPPDTEFKSGAGTELSVRAPKS
v3	76	EEELQVIQPDKSVLVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRVTVSESTKRENMDFSISINITPADAGTYTCVKFRKGSPPDTEFKSGAGTELSVRAPKS
v4	77	EEELQVIQPDKSVLVAAGESAILHCTATSLIPVGPIQWFRGAGPGRELINQKEGHFPRVTTVSDLTNRNMDFSIRIGNITPADAGTYTCVKFRKGSPPDVEFKSGAGTELSVRAPKS
v5	78	EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELINQKEGHFPRVTTVSDLTNRNMDFSIRIGNITPADAGTYTCVKFRKGSPPDVEFKSGAGTELSVRAPKS
v6	79	EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELINQKEGHFPRVTTVSDLTNRNMDFSIRIGNITPADAGTYTCVKFRKGSPPDVEFKSGAGTELSVRAPKS
v7	80	EEELQVIQPDKSVLVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRVTVSESTKRENMDFSISINITPADAGTYTCVKFRKGSPPDTEFKSGAGTELSVRAPKS

TABLE 1-continued

Human SIRP- α variant polypeptide sequences corresponding to the D1 domain.			
Variant	SEQ ID NO	Sequence	
v8	81	EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRG AGPARELIYNQKEGHFPRVTTVSESTKRENMDFSISISNITP ADAGTYTCVKFRKGSPDTEFKSGAGTELSVRAKPS	
v9	82	EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRG AGPGRELIYNQKEGHFPRVTTVSDLTNRNNMDFSIRISNIT PADAGTYTCVKFRKGSPDDVEFKSGAGTELSVRAKPS	
v10	83	EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRG AGPARELIYNQKEGHFPRVTTVSESTKRENMDFSISISNITP ADAGTYTCVKFRKGSPDTEFKSGAGTELSVRAKPS	

In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide (e.g., the D1 domain of a monkey SIRP- α polypeptide). In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide (e.g., found in the organism *Macaca fascicularis*). In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of at least two different monkey SIRP- α variant polypeptides. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of at least two different cynomolgus SIRP- α variant polypeptides. For example, in some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of EEELQVIQPEKSVSVAAGESATLNCTATSLIPVGPIQWFRGVGPGRELIYHQKEGHFPRVTPVSDPTKRNNMDFSIRISNITPADAGTYTCVKFRKGSPDVELKSGAGTELSVRAKPS (SEQ ID NO:11), an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of EEELQVIQPEKSVSVAAGDSATLNCTVSSLPVGPIQWFRGAGPGRELIYNLKEGHFPRVTVASDPTKRNNMDFSIRISNITPADAGTYTCVKFRKGSPDVELKSGAGTELSVRAKPS (SEQ ID NO:12), or both.

In some embodiments, an antibody of the present disclosure binds an extracellular domain of a murine or mouse SIRP- α polypeptide (e.g., found in the organism *Mus musculus*; e.g., the D1 domain of a murine or mouse SIRP- α polypeptide). In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different murine SIRP- α variant polypeptides. A variety of murine SIRP- α variant polypeptides from different mouse strains are known. In some embodiments, the murine SIRP- α variant polypeptide comprises an amino acid sequence selected from

(SEQ ID NO: 7; from 129 mouse strain)
KELKVTQPEKSVSVAAGDSTVLNCTLTSLLPVGPIKWYRGVQSRLLIYS
FTGEHFPRTVNSDATKRNNMDFSIRISNVTPEAGTYTCVKFKGSPD
DTEIQSGGGTEVYVLAKPS,

(SEQ ID NO: 8; from NOD mouse strain)
TEVKVIQPEKSVSVAAGDSTVLNCTLTSLLPVGPIRWYRGVQSRLLIYS

-continued

20 FTTEHFPRTVNSDATKRNNMDFSIRISNVTPEAGTYTCVKFKGSPD
EIQSGGGTEVYVLAK,

(SEQ ID NO: 9; from C57BL/6 mouse strain)
KELKVTQPEKSVSVAAGDSTVLNCTLTSLLPVGPIRWYRGVQSRLLIYS

25 PAGEYVPRIRNVSDTTKRNNMDFSIRISNVTPEAGTYTCVKFKGSPD
DTEIQSGGGTEVYVLAK,
and

(SEQ ID NO: 10; from BALB/c mouse strain)
30 TEVKVTQPEKSVSVAAGDSTVLNCTLTSLLPVGPIRWYRGVQSRLLIYS
FTGEHFPRTVNSDATKRNNMDFSIRISNVTPEAGTYTCVKFKGSPD
DTEIQSGGGTEVYVLAK.

35 In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP family protein. In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, a human SIRP- β polypeptide refers to a polypeptide encoded by a human SIRPB1 gene, e.g., as described by NCBI Ref Seq ID No. 10326. In some embodiments, the extracellular domain (e.g., the D1 domain) of the human SIRP- β polypeptide comprises the amino acid sequence of

(SEQ ID NO: 13)
EDELQVIQPEKSVSVAAGESATLRCAMTSLIPVGPIWFRGAGAGRELIY
50 NQKEGHFPRVTTVSELTKRNNMDFSIRISNITPADAGTYTCVKFRKGSPD
DVEFKSGAGTELSVRAKPS
or

(SEQ ID NO: 14)
55 EEELQVIQPDKSVSVAAGESATLHCTVTSLIPVGPIQWFRGAGPGRELIY
NQKEGHFPRVTTVSDLTNRNNMDFSIRISNITPADAGTYTCVKFRKGSPD
HVEFKSGAGTELSVRAKPS.

60 In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, a human SIRP- γ polypeptide refers to a polypeptide encoded by a human SIRPG gene, e.g., as described by NCBI Ref Seq ID No. 55423. In some embodiments, the extracellular domain (e.g., the D1 domain) of the human SIRP- γ polypeptide comprises the amino acid sequence of

(SEQ ID NO: 15)
 EEELQMIQPEKLLLVTVGKTATLHCTVTSLLPVGVLWFRGVGPGRRELIY
 NQKEGHFPRVTTVSDLTNRNNMDFSIIRISSITPADVGTYYCVKFRKGSPE
 NVEFKSGPGTEMALGAKPS.

In addition to antibodies that bind one or more of the polypeptides described above, the present disclosure contemplates antibodies that do not bind one or more of the polypeptides described above. Stated another way, the binding profile of an antibody of the present disclosure may be characterized by positively or negatively reciting any of the binding specificities and/or properties described herein.

In some embodiments, an antibody of the present disclosure modulates SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, an antibody of the present disclosure antagonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, an antibody of the present disclosure interferes with SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, an antibody of the present disclosure agonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, SIRP- α signaling includes one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), and nitric oxide production. Various assays for measuring SIRP- α signaling in a cell include without limitation SIRP- α phosphorylation, SHP1 and SHP2 co-immunoprecipitation, PI3-kinase signaling, cytokine production (both inflammatory IL-12, IL-23, TNF α , IFN and suppressive cytokines IL-10, IL-4, IL-13, cell surface markers levels for M1 and M2 macrophage markers) or dendritic cell activation and function; Kharitonov, A. et al. (1997) *Nature* 386: 181-6; Ochi, F. et al. (1997) *Biochem. Biophys. Res. Commun.* 239:483-7; Kim, E. J. et al. (2013) *Inflammation Research* 62:377-86; Yi, T. et al. (2015) *Immunity* 43:764-75.

In some embodiments, the cell expressing a human SIRP- α polypeptide is a leukocyte. In some embodiments, the cell is a macrophage, dendritic cell, neutrophil, eosinophil, or myeloid-derived suppressor cell (MDSC). In some embodiments, an antibody of the present disclosure decreases or antagonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the SIRP- α signaling assays described herein or otherwise known in the art. In some embodiments, an antibody of the present disclosure increases or agonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the SIRP- α signaling assays described herein or otherwise known in the art.

In some embodiments, an antibody of the present disclosure modulates an intercellular phenotype mediated by SIRP- α . In some embodiments, an antibody of the present disclosure enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide. For example, phagocytic activity of a macrophage treated or contacted with an antibody of the present disclosure can be compared with phagocytic activity of a macrophage not treated or contacted with the antibody, or phagocytic activity of a macrophage

that expresses a human SIRP- α polypeptide and is treated or contacted with an antibody of the present disclosure can be compared with phagocytic activity of a macrophage that does not express a human SIRP- α polypeptide and is treated or contacted with the antibody. Exemplary phagocytosis assays may be found, e.g., in Wieskopf, K. et al (2013) *Science* 341: 88 and Willingham, S. B. et al. (2012) *Proc. Natl. Acad. Sci.* 109:6662-7. In some embodiments, an antibody of the present disclosure increases phagocytosis by a macrophage expressing a human SIRP- α polypeptide by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the phagocytosis assays described herein or otherwise known in the art.

In some embodiments, an antibody of the present disclosure enhances activation of dendritic cell(s) expressing a human SIRP- α polypeptide (e.g., an increased level of activation of individual dendritic cells, or an increased proportion of dendritic cells that are activated within a sample population). For example, activation of dendritic cell(s) treated or contacted with an antibody of the present disclosure can be compared with activation of dendritic cell(s) not treated or contacted with the antibody, or activation of dendritic cell(s) that express a human SIRP- α polypeptide and are treated or contacted with an antibody of the present disclosure can be compared with activation of dendritic cell(s) that do not express a human SIRP- α polypeptide and are treated or contacted with the antibody. Exemplary dendritic cell activation assays are described herein. In some embodiments, an antibody of the present disclosure increases dendritic cell (e.g., expressing a human SIRP- α polypeptide) activation by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the dendritic cell activation assays described herein or otherwise known in the art.

In some embodiments, an antibody of the present disclosure inhibits in vivo growth of a tumor that expresses CD47. For example, in vivo growth of a tumor that expresses CD47 and is treated with an antibody of the present disclosure can be compared against in vivo growth of a tumor that expresses CD47 and is not treated with an antibody of the present disclosure. Exemplary in vivo tumor growth assays are described herein. In some embodiments, an antibody of the present disclosure inhibits in vivo growth of a tumor that expresses CD47 by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the in vivo tumor growth assays described herein or otherwise known in the art.

In some embodiments, an antibody of the present disclosure blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (e.g., a "blocking" antibody). For example, the antibody and the CD47 polypeptide may "compete" for the same SIRP- α epitope, and/or antibody binding to SIRP- α may be mutually exclusive with CD47 binding to SIRP- α . The binding interface between SIRP- α and CD47, as well as residues of both proteins that participate in binding, are known; see Hatherley, D. et al. (2007) *J. Biol. Chem.* 282:14567-75 and Nakaishi, A. et al. (2008) *J. Mol. Biol.* 375:650-60. Exemplary assays for determining whether an antibody blocks CD47 binding to SIRP- α are described herein. In some embodiments, an antibody of the present disclosure blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47

polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. For example, in vitro ELISA and SPR assays are described herein, although this is not meant to be limiting, as other in vitro binding assays may also be used. In some embodiments, antibody binding to a complex comprising a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein) bound to an IgSF domain of CD47 is used to screen for blocking, non-blocking, and/or kick off antibodies. In some embodiments, “blocking” and/or “non-blocking” antibodies can be tested via surface plasmon resonance (SPR; e.g., as described in Example 1). For example, a complex can be formed between an IgSF domain of CD47 and a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein), then binding of a test antibody to the complex can be measured. For an antibody that blocks binding of SIRP- α to CD47, at increasing concentrations of CD47, one would expect fewer molecules of SIRP- α to be available to bind to the antibody since the antibody competes for the same binding site as CD47 and most/all SIRP- α is complexed with CD47. Therefore, one would expect the resonance (RU) to decrease with increasing concentration of CD47 in the mixture. In some embodiments, a “blocking” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain of a SIRP- α polypeptide (e.g., the D1 domain) at one or more residues of the binding interface between CD47 and SIRP- α , i.e., the blocking antibody and CD47 share partially or completely overlapping epitopes. In some embodiments, a “blocking” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain of a SIRP- α polypeptide (e.g., the D1 domain) at one or more amino acid positions that are also bound by CD47 in the CD47:SIRP- α complex. The binding interfaces between SIRP- α and exemplary anti-SIRP- α antibodies or CD47 are described in Example 4. In some embodiments, an antibody of the present disclosure blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, the in vivo assay may assess binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell by assaying one or more aspects of SIRP- α signaling, e.g., one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, neutrophils, dendritic cells, eosinophils, and myeloid-derived suppressor cells (MDSCs).

In some embodiments, an antibody of the present disclosure does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (e.g., a “non-blocking” antibody). For example, the antibody and

the CD47 polypeptide may bind distinct and/or non-overlapping epitopes of SIRP- α . In some embodiments, an antibody of the present disclosure does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. For example, in vitro ELISA and SPR assays are described herein, although this is not meant to be limiting, as other in vitro binding assays may also be used. In some embodiments, antibody binding to a complex comprising a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein) bound to an IgSF domain of CD47 is used to screen for blocking, non-blocking, and/or kick off antibodies. In some embodiments, “blocking” and/or “non-blocking” antibodies can be tested via surface plasmon resonance (SPR; e.g., as described in Example 1). For example, a complex can be formed between an IgSF domain of CD47 and a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein), then binding of a test antibody to the complex can be measured. For an antibody that does not block binding of SIRP- α to CD47, the antibody would be expected to bind to SIRP- α /CD47 complex and form a sandwich. Therefore, at increasing concentrations of CD47, the resonance would increase accordingly due to increased sandwich formation. In some embodiments, a “non-blocking” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain of a SIRP- α polypeptide (e.g., the D1 domain) at one or more residues that are distinct from the binding interface between CD47 and SIRP- α , i.e., the non-blocking antibody and CD47 share completely non-overlapping epitopes. The binding interfaces between SIRP- α and exemplary anti-SIRP- α antibodies or CD47 are described in Example 4. In some embodiments, an antibody of the present disclosure does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, the in vivo assay may assess binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell by assaying one or more aspects of SIRP- α signaling, e.g., one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, neutrophils, dendritic cells, eosinophils, and myeloid-derived suppressor cells (MDSCs). It is a surprising finding of the present disclosure that antibodies that do not block SIRP- α interaction with CD47 are able to increase phagocytosis and block in vivo tumor growth. Without wishing to be bound to theory, it is thought that non-blocking anti-SIRP- α antibodies can modulate one or more functions of SIRP- α independent of CD47 binding.

In some embodiments, binding of an antibody of the present disclosure to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide (e.g., a “kick off” antibody). In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide increases k_{off} of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human high affinity SIRP- α polypeptide (e.g., as described herein) increases the k_{off} of the human high affinity SIRP- α polypeptide (e.g., as described herein) for binding an IgSF domain of a human CD47 polypeptide to greater than about 1×10^{-3} 1/s. For example, the antibody and the CD47 polypeptide may have adjacent or partially overlapping SIRP- α epitopes, such that the antibody is able to bind SIRP- α when it is bound to CD47, but the antibody: SIRP- α promotes dissociation of the SIRP- α . CD47 complex. In some embodiments, an antibody of the present disclosure reduces affinity of an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide increases k_{off} of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. For example, in vitro ELISA and SPR assays are described herein, although this is not meant to be limiting, as other in vitro binding assays may also be used. In some embodiments, antibody binding to a complex comprising a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein) bound to an IgSF domain of CD47 is used to screen for blocking, non-blocking, and/or kick off antibodies. In some embodiments, an antibody of the present disclosure reduces affinity of an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, a “kick off” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain (e.g., the D1 domain) of a SIRP- α polypeptide at one or more residues of the binding interface between CD47 and SIRP- α , i.e., the kick off antibody and CD47 share partially overlapping epitopes. In some embodiments, a “kick off” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain (e.g., the D1 domain) of a SIRP- α polypeptide at 1 or more residues that are also bound by CD47 in the CD47:SIRP- α complex. For example, a “kick off” anti-SIRP- α antibody can bind to the extracellular domain (e.g., the D1 domain) of a SIRP- α polypeptide at 2 or more residues that are also bound by CD47 in the CD47:SIRP- α complex and are at the periphery of the CD47 binding epitope of SIRP- α . The binding interfaces between SIRP- α and exemplary anti-SIRP- α antibodies or CD47 are described in Example 4. In some embodiments, binding of an antibody of the present disclosure to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell increases k_{off} of the human SIRP- α polypeptide for binding an IgSF domain

of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, the in vivo assay may assess binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell by assaying one or more aspects of SIRP- α signaling, e.g., one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, neutrophils, dendritic cells, eosinophils, and myeloid-derived suppressor cells (MDSCs).

In some embodiments, an antibody of the present disclosure modulates one or more immune cell functions by binding to two or more (or all three) of SIRP- α , SIRP β , and SIRP γ .

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody blocks

binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey (e.g., cynomolgus) SIRP- α polypeptide; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey (e.g., cynomolgus) SIRP- α polypeptide; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey (e.g., cynomolgus) SIRP- α polypeptide; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide.

In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, and a monkey SIRP- α polypeptide; the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 3A & 3B).

In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody blocks binding

between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 4A & 4B).

In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, and a monkey SIRP- α polypeptide; the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody does not block binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 5A & 5B).

In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides and a monkey SIRP- α polypeptide; the antibody does not bind a murine SIRP- α polypeptide; the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 6A & 6B).

In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; the antibody does not bind a murine SIRP- α polypeptide; and the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 7A & 7B).

In some embodiments, an antibody of the present disclosure comprises three CDRs from a VH domain comprising a sequence set forth in Table 2 and/or three CDRs from a VL domain comprising a sequence set forth in Table 2 (VH and VL sequences with CDRs highlighted are provided in FIGS. 10A-10F & 11A-11P). For example, in some embodiments, an antibody of the present disclosure comprises three CDRs from a VH domain comprising a sequence selected from SEQ ID NOs: 294, 139, 358, 362, 354, 380, 384, 350, 137, 374, 356, 352, 135, 348, 376, 346, 342, 344, 141, 360, 370, 382, 364, 366, 368, 372, 378, 133, 128, 396, 386, 398, 402, 392, 388, 390, 394, 400, 116, 117, 118, 119, 282, 404, 406, 278, 412, 275, 414, 280, 123, 292, 288, 290, 408, 410, 286, 284, 120, 121, 130, and 122 and/or three CDRs from a VL domain comprising a sequence selected from SEQ ID NOs: 295, 363, 140, 359, 355, 351, 136, 349, 377, 138, 375, 357, 353, 381, 385, 345, 365, 367, 369, 347, 142, 343, 371, 379, 383, 361, 373, 134, 105, 387, 389, 395, 397, 399, 403, 391, 393, 401, 93, 94, 95, 96, 283, 405, 407, 279, 413, 276, 415, 281, 100, 293, 289, 291, 409, 411, 287, 285, 97, 98, 107, and 99. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising a sequence set forth in Table 2 and/or a VL domain comprising a sequence set forth in Table 2 (see FIGS. 10A-10F & 11A-11P). For example, in some embodiments, an antibody of the present disclosure comprises a VH domain comprising an amino acid sequence selected from SEQ ID NOs: 294, 139, 358, 362, 354, 380, 384, 350, 137, 374, 356, 352, 135, 348,

63

376, 346, 342, 344, 141, 360, 370, 382, 364, 366, 368, 372, 378, 133, 128, 396, 386, 398, 402, 392, 388, 390, 394, 400, 116, 117, 118, 119, 282, 404, 406, 278, 412, 275, 414, 280, 123, 292, 288, 290, 408, 410, 286, 284, 120, 121, 130, and 122 and/or a VL domain comprising an amino acid sequence selected from SEQ ID NOs: 295, 363, 140, 359, 355, 351, 136, 349, 377, 138, 375, 357, 353, 381, 385, 345, 365, 367, 369, 347, 142, 343, 371, 379, 383, 361, 373, 134, 105, 387, 389, 395, 397, 399, 403, 391, 393, 401, 93, 94, 95, 96, 283, 405, 407, 279, 413, 276, 415, 281, 100, 293, 289, 291, 409, 411, 287, 285, 97, 98, 107, and 99. In some embodiments, an antibody of the present disclosure comprises six CDR sequences from an antibody described in Table 2 (see FIGS.

64

10A-10F & 11A-11P). In some embodiments, an antibody of the present disclosure comprises a VL domain comprising the VL domain sequence of VL domains Hum1-Hum9 as described herein. In some embodiments, the antibody is a humanized antibody. In some embodiments, the antibody is a human antibody. In some embodiments, an antibody of the present disclosure comprises a VH domain and/or a VL domain from an antibody described in Table 2. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising a D or E residue followed by (e.g., in the direction of N-terminus to C-terminus) a VH domain sequence selected from SEQ ID NOs:116-130.

TABLE 2

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
S1	Chicken	VL	85	ALTQPASVSANPGETVEITCSGGGSNNA YGFQKSPGSAPLTVIYDNGKRPSDIPS RFSGSKSDTGTLTITRVQAEDEAVYYC GSADNSGAGVFGAGTTTLTVL
S2	Chicken	VL	86	AVTQPASVSANPGETVRITCSGDSSSYYS WHQKSPGSAPVSVIYSNTDRPSDIPSRF SGSASGSTATLTI TGVQAEDEAVYFCGA YDSSSDSDIFGAGTTTLTVL
S8	Chicken	VL	87	AVTQPSSVSANPGETVEITCSGSSTYYGW YQKSPGSAPVTVIYDNDKRPSDIPSRFS GSKSGSTHTLTIITGVQVEDEAVYFCGNED NNYVAIFGAGTTTLTVL
S9	Chicken	VL	88	ALTQPSSVSANPGETVKITCSGDNSAHY YYGWYQKSPGSAPVTVIYDNDKRPSGI PSRFSGSASGSTATLTIITGVQVEDEAVYF CGSADSSNPAIFGAGTTTLTVL
S11	Chicken	VL	89	AVTQPASVSANPGETVKITCSGSSSGSYG WYQKSPGSAPVTLIYETNKRPSNIPSRF SGSKSGSTATLTI TGVQADDEAVYYCGS EDSSTYLSIFGAGTTTLTVL
S12	Chicken	VL	90	AVTQPASVSANPGETVKITCSGDSSYYG WYQKSPGSAPVTVIYDNDKRPSNIPSRF SGSKSGSTGTLTITGVQADDEAVYFCGN EDNSYVAIFGAGTTTLTVL
S13	Chicken	VL	91	AVTQPASVSANPGETVKITCSGSSSYYG WYRQKSPGSAPVTLIYDNDKRPSGIPSRF SGSKSGSTNTLTITGVQADDEAVYYCGN EDNSYVGIFGAGTTTLTVL
S14	Chicken	VL	92	AVTQPASVSANLGETVKITCSGDSSYYG WYQKAPGSAPVTLIYDNDKRPSNIPSRF SGSKSGSTATLTI TGVQADDEAVYYCGN EDMNYVGIFGAGTTTLTVL
S115	Human	VL	93	ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDATNRAT GISDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK
S116	Human	VL	94	ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDATNRAT GISDRFSGSRSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK
S117	Human	VL	95	ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDATNRAT GIPDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK
S118	Human	VL	96	ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDASRRAT

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
				GIPDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYWPPYRFGGGTKVEIK
S119	Human	VL	97	EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPQGAPRLLIYAARIRETG IPERFSGSGSGTEFTLTITSLQSEDAVYY CQQYYDWPPFTFGGGTKVEIK
S120	Human	VL	98	EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPQGAPRLLIYAARIRETG IPERFSGSGSGTEFTLTITSLQSEDAVYY CQQYYDWPPFTFGGGTKVEIK
S122	Human	VL	99	EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPQGAPRLLIYAARIRETG IPERFSGSGSGTEFTLTITSLQSEDAVYY CQQYYDWPPFTFGGGTKVEIK
S123	Human	VL	100	EIVLTQSPGTLSPGERVLTLCRASQGIA GKIAWYQQKPGQAPRLLIYDASSRATGIP GRFSGSGSGTEFTLTITSLQSEDAVYYC QQHYDWSPLTFGGGTKEIK
S126	Human	VL	101	EIVLTQSPGTLTLSPGERATLSCRASQSIG SSYLAWYQQKPGQAPRLLIYDATNRATG IPDRFSGSGSGTDFTLTISLQTEDSAVYY CQQYYWPPYRFGGGTKVEIK
S128	Human	VL	102	ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDASNRA GIPDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYWPPYRFGGGTKVEIK
S130	Human	VL	103	EIVLTQSPGTLSPGERATLSCRASQNV RSDLAWYQQKLGQAPRLLIYDANTRAT DIPDRFSGSGSGTEFTLTISLQSEDAVY YCQHYDWPVTFGGGTKEIK
S135	Human	VL	104	EIVLTQSPATLSVSPGERVTFSCRASQNV RSDIAWYQQKPGQAPRLLIYAASSRDTGI PDRFSGSGSGTDFTLTISLQSEDFGVYY CQQYYDWPPFTFGGGTKVEIK
S137	Human	VL	105	ETVLTQSPGTLTLSPGERATLTCRASQSV YTYLAWYQEKPGQAPRLLIYGASSRATG IPDRFSGSGSGTVFTLTISLQSEDAVYY CQQYYDRPPLTFGGGTKEIK
S138	Human	VL	106	EIVLTQSPGTLSPGERVILTLCRASQSVD TYNLAWYQQKPGQAPRLLIYDLSTRATG IPDRFSGSGSGTEFTLTINSLEPEDAVYY CHQYYDWPPYTFGGGTKEIK
S121	Human	VL	107	EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPQGAPRLLIYAARIRETG IPERFSGSGSGTEFTLTITSLQSEDAVYY CQQYYDWPPFTFGGGTKVEIK
S1	Chicken	VH	108	AVTLDESGGGLQTPGGALSLVCKGSGFT FSSHAMNWVRQAPGKGLEWVAGISSDG RFTYYGAAVQGRATISRDNQGSTVRLQL NNLRAEDTATYYCTKNGGCGSGDLDC IDAWGHGTEVIVSS
S2	Chicken	VH	109	AVTLDESGGGLQTPGGALSLVCKASGFD FSNFMMAWVRQGPGLVYVAEISDTGS TPYYGSAVQGRATISRDNQGSTVRLQLN NLRAEDTGTYYCTRNFGSSVSSIDAWGH GTEVIVSS
S8	Chicken	VH	110	AVTLDESGGGLQTPGGALSLVCKASGFT FSSYNMGWVRQAPGKLEFVAGIYASG SSTDTDTTYGPAVAGRATISRDNQGSTV RLQLNNLRAEDTGTYYCAKAAGCSTH TCTAYIADSIDAWGHGTEVIVSS

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
S9	Chicken	VH	111	AVTLDESGGGLQTPGRALSLVCRGSGFSI SSYNMGWVRQAPGKGLEFIASIGSDGSS THYAPAVKGRATITRDVGQSTVRLQLNN LRAEDTGTIFYCAKDAYQCSYATCNDYL DTIDAWGHGTEIVSS
S11	Chicken	VH	112	AVTLDESGGGLQTPGGALSLVCKASGFT FSSYNMGWVRQAPGKGLEFVAIYSGNS AEYGAAVQGRATISRDNQGSTVRLQLN NLRAEDTGIYFCAKDAGSGCYSGVCAGT SSIDAWGHGTEIVSS
S12	Chicken	VH	113	AVTLDESGGGLQTPGGALSLVCKASGFT FSSYNMGWVRQAPGKGLEFVAGIYIASG DLGTTYGAAVQGRATISRDDGQSTVRLQ LNNLRAEDTGTIFYCAKSAGGCSAHSCDT YIADSIDAWGHGTEIVSS
S13	Chicken	VH	114	AVTLDESGGGLQTPGGALSLVCKASGFT FSSYNMGWVRQAPDKGLEFVAGIYTG DAGLSTTYGAAVQGRATISRDNQGSTVR LQLNNLGAEDTGIYFCTKSAGGCSYDNC DAYIADSIDAWGHGTEIVSS
S14	Chicken	VH	115	AVTLDESGGGLQTPGGALSLVCKASGFT FNSYNMGWVRQAPGKGLEFVAGIYSAG GDTSTTYGAAVNGRATISRDNQGSTVRL QLNNLRAEDTGIYFCAKAAGGCTAHNC DAYIADSIDAWGHGTEIVSS
S115	Human	VH	116	VQLVESGGGVVRPGESLRLSCAASGFSFS SYAMNWVRQAPGEGLEWVSRINSGGGG TDYAESVKGRFTISRDNSENTLYLQMNS LRAEDTAVYYCAKQYDWNSEFFDYWGL GALVTVSS
S116	Human	VH	117	VQLVESGGGVVRPGESLRLSCAASGFSFS SYAMNWVRQAPGEGLEWVSRINSGGGG TDYAESVKGRFTISRDNSENTLYLQMNS LRAEDTAVYYCAKQYDWNSEFFDYWGL GALVTVSS
S117	Human	VH	118	VQLVESGGGVVRPGESLRLSCAASGFSFS SYAMNWVRQAPGEGLEWVSRINSGGGG TDYAESVKGRFTISRDNSENTLYLQMNS LRAEDTAVYYCAKQYDWNSEFFDYWGL GALVTVSS
S118	Human	VH	119	VQLVESGGGVVRPGESLRLSCAASGFSFS SYAMNWVRQAPGEGLEWVSRINSGGGG TDYAESVKGRFTISRDNSENTLYLQMNS LRAEDTAVYYCAKQYDWNSEFFDYWGL GALVTVSS
S119	Human	VH	120	VQLVESGGGVVQPGGSLRLSCAASGFSF SNFAMTWVRQAPGEGLEWVSTIGSGDT YYADSVKGRFTISRDNSENTLYLQMNSL RAEDTAVYYCAKDSVSWSGDFFDYWG LGTLVTVSS
S120	Human	VH	121	VQLVESGGGVVQPGGSLRLSCAASGFSF SNFAMTWVRQAPGEGLEWVSTIGSGDT YYADSVKGRFTISRDNSENTLYLQMNSL RAEDTAVYYCAKDSVSWSGDFFDYWG LGTRVTVSS
S122	Human	VH	122	VQLVESGGGVVRPGESLRLSCAASGFRF SNFAMTWVRQAPGEGLEWVSTIGSGDT YYADSVKGRFTISRDNSENTLYLQMNSL RAEDTAVYYCAKDSVSWSGDFFDYWG LGTLVTVSS
S123	Human	VH	123	VQLVESGGGLVQPGGSLRLSCTASGFTF RNYGMSWVRQAPGEGLEWVSASSSGSGS

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
				TYYTDSVKGRFTISRDN SKNTLYLQMN LRAEDTAIYYCAKVTWNNFFDYWGLGT LVTVSS
S126	Human	VH	124	VQLVESGGGVVVRPGESLRLSCAASGFTF SNYDMTWVRQAPGEGLEWVSGISGNGG STYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAMNRWWFDYWGLGT LVTVSS
S128	Human	VH	125	VQLVESGGGVVVRPGESLRLSCAASGFSF RSYAMNWRQAPGEGLEWVSRIDSGGG GTDYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAKQYDWN SFFDYWGL GAPVTVSS
S130	Human	VH	126	VQLVESGGGVVVRPGESLRLSCAASGFTF SNYAMSWVRQAPGEGLEWVSLITNGD GAYYADSVKGRFTISRDN SKNTLYLQMN NSLRAEDTAIYYCAKDGAAHYDIFFDY WGLGTPVTVSS
S135	Human	VH	127	VQLVESGGGVVVRPGESLRLSCAASGFSF IYAMSWVRQAPGEGLEWVSTIGADDTY YADSVKGRFTISRDN SKNTLYLQMN AEDTAVYYCAKDSTVGW S GDFFDYWG LGTLVTVSS
S137	Human	VH	128	VQLVESGGGVVVRPGESLRLSCAASGFTF SSYDMNWRQAPGEGLEWVSLISGSGEI IYYADSVKGRFTISRDN SKNTLYLQMN LRAEDTAVYYCAKENNRYRFFDDWGLG TLVTVSS
S138	Human	VH	129	VQLVESGGGVVVRPGESLRLSCAASGFTF SNYAMNWRQAPGEGLEWVSGISGRGG DTYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAIYYCAKGTWNYG SFDYWGL GTLVTVSS
S121	Human	VH	130	VQLVESGGGVVQPGGSLRLSCAASGFSF SNFAMTWVRQAPGEGLEWVSTIGSGDT YYADSVKGRFTISRDN SKNTLYLQMN RAEDTAVYYCAKDSTVSW S GDFFDYWG LGTLVTVSS
AB136	Human	VH	133	DVQLVESGGGVVVRPGESLRLSCAASGFT FSSYDMNWRQAPGEGLEWVSLISGSGE IYYADSVKGRFTISRDN SKNTLYLQMN LRAEDTAVYYCAKENNRYRFFDDWGLG TLVTVSS
AB136	Human	VL	134	ETVLTQSPGTLTSLPGERATLTCSRQSV YTYLAWYQEKPGQAPRLLIYGASSRATG IPDRFSGSGSGTEFTLTISLQSEDFAVYY CQQYYDRPPLTFGGGTKVEIK
AB21	Human	VH	135	DVQLVESGGGVVVRPGESLRLSCAASGFT FSSNAMSWVRQAPGKGLEWLAGISAGG SDTYYPASVKGRFTISRDN SKNTLYLQMN NTLTAEDTAVYYCARETNH LFDYWGL GTLVTVSS
AB21	Chicken	VL	136	ALTQPASVSANPGETVKIACSGGDYYSY YYGWYQQKAPGSALVTVIYSDDKRPSDI PSRFSGSASGSTATLTITGVRAEDEAVYY CGGYDYSTYANAFGAGTTTLTVL
AB25	Human	VH	137	DVQLVESGGGVVVRPGESLRLSCEASGFT FSSNAMSWVRQAPGKGLEWVAGISSGS DTYYGDSVKGRFTISRDN SKNTLYLQMN SLTAEDTAVYYCARETNH LFDYWGLG TLVTVSS

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
AB25	Chicken	VL	138	ALTQPASVSANPGETVEITCSGGSYSSYY YAWYQQKSPGSAFVTLIYSDDKRPSNIPS RFGSGSASGSTATLTITGVRAEDEAVYFCG GYDQSSYTNFPFGAGTTLTVL
AB27	Human	VH	139	DVQLVESGGGVVRPGESLRLSCAVSGFR FSSYAMSWVRQAPGKGLEWVSGISSGG DTYYVDSVKGRFTISRDNKNTLYLQVN SLTAEDTAIYYCARETNWHLFDYWGLG TLVTVSS
AB27	Chicken	VL	140	ALTQPASVSADLGETVKITCSGGDSSSHY YGWYQQKSPGSAFVTIYSDDERPSDIPS RFGSGSASGSTATLTITGVRVEDEAIYYCG AYDGSYANTFGAGTTLTVL
AB66	Human	VH	141	DVQLVESGGGVVRPGESLRLSCAASGFT FSSYAMSWVRQAPGKGLEWLAGISAGG SDTYIIDSVKGRFTISRDNPKNSLYLQMS SLTAEDTAVYYCARETNWHLFDYWGLG TLVTVSS
AB66	Chicken	VL	142	ALTQPASVSANPGETVKITCSGGDYYST YYAWYQQKSPGSAFVTIHSDDKRPSDI PSRFGSGSASGSAATLIITGVRVEDEAVYY CGGYDGRTYINTFGAGTTLTVL
AB119	Human	HVR-H1	143	GFSESNFAMT
AB119	Human	HVR-H2	144	TIGSGDTYYADSVKG
AB119	Human	HVR-H3	145	DSTVSWSGDFFDY
AB119	Human	HVR-L1	146	RASQNVKNDLA
AB119	Human	HVR-L2	147	AARIRET
AB119	Human	HVR-L3	148	QQYYDWPPFT
AB135	Human	HVR-H1	149	GFSESIYAMS
AB135	Human	HVR-H2	150	TIGADDTYYADSVKG
AB135	Human	HVR-H3	151	DSTVGWSGDFFDY
AB135	Human	HVR-L1	152	RASQNVRSIDIA
AB135	Human	HVR-L2	153	AASSRDT
AB135	Human	HVR-L3	148	QQYYDWPPFT
AB136	Human	HVR-H1	155	GFTFSSYDMN
AB136	Human	HVR-H2	156	LISGSGEIIYYADSVKG
AB136	Human	HVR-H3	157	ENNRYRFFDD
AB136	Human	HVR-L1	158	RASQSVYTYLA
AB136	Human	HVR-L2	159	GASSRAT
AB136	Human	HVR-L3	160	QQYYDRPPLT
AB21	Human	HVR-H1	161	GFTFSSNA
AB21	Human	HVR-H2	162	ISAGGSDT
AB21	Human	HVR-H3	163	ARETNWHLFDY
AB21	Chicken	HVR-L1	164	SGGDYYSYYG
AB21	Chicken	HVR-L2	165	TVIYSDDKRPSD
AB21	Chicken	HVR-L3	166	GGDYDSTYANA

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
AB25	Human	HVR-H1	161	GFTFSSNA
AB25	Human	HVR-H2	168	ISSGSDT
AB25	Human	HVR-H3	163	ARETNWHLFDY
AB25	Chicken	HVR-L1	170	SGGSYSSYYA
AB25	Chicken	HVR-L2	171	TLIYSDDKRPSN
AB25	Chicken	HVR-L3	172	GGYDQSSYTNP
AB27	Human	HVR-H1	173	GFRFSSYA
AB27	Human	HVR-H2	174	ISSGGDT
AB27	Human	HVR-H3	163	ARETNWHLFDY
AB27	Chicken	HVR-L1	176	SGGDSSSHYYG
AB27	Chicken	HVR-L2	177	TVIYSDDERPSD
AB27	Chicken	HVR-L3	178	GAYDGSYANT
AB66	Human	HVR-H1	179	GFTFSSYA
AB66	Human	HVR-H2	162	ISAGGSDT
AB66	Human	HVR-H3	163	ARETNWHLFDY
AB66	Chicken	HVR-L1	182	SGGDYYSTYYA
AB66	Chicken	HVR-L2	183	TVIHSDDKRPSD
AB66	Chicken	HVR-L3	184	GGYDGRTYINT
AB3	Chicken	VH	242	AVTLDESGGGLQTPGGALSLVCKASGFIFS DYGMNWVRQAPGKGLEFVAQITSGSRTY YGA AVKGRATISRDNRQSTVKLQLNNLRA EDTGIYFCARDFGSGVGSIDAWGNGTEVIV SS
AB3	Chicken	VL	243	ALTQPASVSANLGGTVKITCSGSRGRYGW YQQRSPGSAPVTVIYRDNQRPSNIPSRFSS TSGSTSTLTITGVQADESVYFCGSYDGS DIFGAGTTLTVL
AB45	Chicken	VH	244	AVTLDESGGGLQTPGGALSLVCKASGFTF SSYAMGWVRQAPGKGLEWVAGIDDDGST ANYGPAVKGRATISRDNQSTVRLQLNNP RAEDSGTYFCAKASVTGWSAHISGRLDTW GHGTEVIVSS
AB45	Chicken	VL	245	ALTQPASVSANPGETVKITCSGGGIYYYG WYQQKSPGSAPVTLIYENDKRPSDIPSRFS GSTSGSTNTFTITGVQAEDEAVYICGGYD SNTTSGIFGAGTTLTVL
AB119 mut	Human with D1E, E43K, L112Q, and M34V mutations	VH	246	EVQLLESGGGVVQPGGSLRLSCAASGFSFS NFAVTWVRQAPGKGLEWVSTIGSGDTYY ADSVKGRFTISRDN SKNTLYLQMNSLRAE DTAVYYCAKDSVTSWSGDFFDYWGQGT LTVVSS
AB135 mut	Human with D1E, R13Q, E16G, E43K L112Q, and M34V mutations	VH	247	EVQLVESGGGVVQPGGSLRLSCAASGFSFS IYAVSWVRQAPGKGLEWVSTIGADDTYY ADSVKGRFTISRDN SKNTLYLQMNSLRAE DTAVYYCAKDSVTVWSGDFFDYWGQGT LTVVSS
AB136 mut all	Human with D1E, R13Q, E16R, E43K	VH	249	EVQLVESGGGVVQPGSLRLSCAASGFTFS SYDVNWVRQAPGKGLEWVSLISGSGEIIY YADSVKGRFTISRDN SKNTLYLQMNSLRA

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
	L111Q, and M34V mutations			EDTAVYYCAKENNRYRFFDDWGQGTLLVT VSS
AB136 mut all	Human with T2I, T12S, T22S, and E28Q mutations	VL	250	EIVLTQSPGTL ^S SLSPGERATL ^S SCRASQSVYT YLA ^W YQQKPGQAPRL ^I IYGASSRATGIPD RFSGSGSGTEFTLT ^I ISSLQSEDFAVYYCQQ YYDRPPLTFGGGK ^T KEIK
AB136 mut all_I2T	Human with T12S, T22S and E38Q mutations	VL	251	ETVLTQSPGTL ^S SLSPGERATL ^S SCRASQSVY TYLA ^W YQQKPGQAPRL ^I IYGASSRATGIP DRFSGSGSGTEFTLT ^I ISSLQSEDFAVYYCQ QYYDRPPLTFGGGK ^T KEIK
Hum1	Humanized	VL	252	SYELTQPPSVSVSPGQTARITCSGGSYSSY <u>YYAWYQQKPGQAPVTLIYSDDKRPSNIPE</u> RFSGSSSGTTVTLTISGVQAEDEADYYCGG <u>YDQSSYTNPF</u> GGGK ^T LTVL
Hum2	Humanized	VL	253	QSVLTQPPSVSAAPGQKVTISCSGGSYSSY <u>YYAWYQQLPGTAPKTLIYSDDKRPSNIPD</u> RFSGSKSGTSATLGITGLQTGDEADYYCG <u>GYDQSSYTNPF</u> GTGK ^T VTVL
Hum3	Humanized	VL	254	SYELTQPPSVSVSPGQTARITCSGGDYYST <u>YYAWYQQKPGQAPVTIYSDDKRPSDIPE</u> RFSGSSSGTTVTLTISGVQAEDEADYYCGG <u>YDGRTYINT</u> FGGK ^T LTVL
Hum4	Humanized	VL	255	QSVLTQPPSVSAAPGQKVTISCSGGDYYST <u>YYAWYQQLPGTAPKTVIYSDDKRPSDIPD</u> RFSGSKSGTSATLGITGLQTGDEADYYCG <u>GYDGRTYINT</u> FGTGT ^T VTVL
Hum5	Humanized	VL	256	QSALTQPASVSGSPGQSITISCTGTSSDVGS <u>YSSYYAWYQQHPGKAPKTLIYSDDKRPS</u> NVSNRFSGSKSGNTASLTISGLQAEDEADY <u>YCGGYDQSSYTNPF</u> GGGK ^T LTVL
Hum6	Humanized	VL	257	QSVLTQPPSVSAAPGQKVTISCSGGDYYSY <u>YYGWYQQLPGTAPKTVIYSDDKRPSDIPD</u> RFSGSKSGTSATLGITGLQTGDEADYYCG <u>GYDYSTYANAF</u> GTGK ^T VTVL
119 VH MutAll_V34M	Human with D1E, E43K, and L112Q mutations	VH	258	EVQLLES ^G GGGVVQPGGSLRLSCAASGFSFS NFAMTWVRQAPGKLEWVSTIGSGDTYY ADSVKGRFTISRDN ^S SKNTLYLQMNSLRAE DTAVYYCAKDSTVSWSGDFFDYWGQGTLL VTVSS
135 VH MutAll_V34M	Human with D1E, R13Q, E16G, E43K, and L112Q mutations	VH	259	EVQLVES ^G GGGVVQPGGSLRLSCAASGFSFS IYAMSWVRQAPGKLEWVSTIGADDTYY ADSVKGRFTISRDN ^S SKNTLYLQMNSLRAE DTAVYYCAKDSTVSWSGDFFDYWGQGTLL VTVSS
136 VH MutAll_V34M	Human with D1E, R13Q, E16R, E43K, and L111Q mutations	VH	260	EVQLVES ^G GGGVVQPG ^R SLRLSCAASGFTFS SYDMNWVRQAPGKLEWVSLISGSEIIY YADSVKGRFTISRDN ^S SKNTLYLQMNSLRA EDTAVYYCAKENNRYRFFDDWGQGTLLVT VSS
Hum 8	Humanized	VL	416	SYELTQPPSVSVSPGQTARITCSGGAYSSY YYAWYQQKPGQAPVLVIYSDSKRPSGIPE RFSGSSSGTTVTLTISGVQAEDEADYYCGG YDQSSYTNPFGGGK ^T LTVL
Hum9	Humanized	HVR-L1	261	SGGAYSSYYA
Hum9	Humanized	VL	262	SYELTQPPSVSVSPGQTARITCSGGAYSSY YYAWYQQKPGQAPVLVIYSDDKRPSGIPE RFSGSSSGTTVTLTISGVQAEDEADYYCGG YDQSSYTNPFGGGK ^T LTVL

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
AB21 Mut All	Human with germline back- mutations	VH	263	EVQLVESGGGVVQPGGSLRLSCAASGFTF SSNAMSWVRQAPGKGLEWVAGISAGGSD TYYPASVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCARETNHFLFDYWGQGT LVTSS
AB21 Mut All M34V	Human with germline back- mutations and liability mutation	VH	264	EVQLVESGGGVVQPGGSLRLSCAASGFTF SSNAVS WVRQAPGKGLEWVAGISAGGSD TYYPASVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCARETNHFLFDYWGQGT LVTSS
AB25 Mut All	Human with germline back- mutations	VH	265	EVQLVESGGGVVQPGGSLRLSCAASGFTF SSNAMS WVRQAPGKGLEWVAGISAGSDT YYGDSVKGRFTISRDN SKNTLYLQMNSLT AEDTAVYYCARETNHFLFDYWGQGT LVTSS
AB25 Mut All M34V	Human with germline back- mutations and liability mutation	VH	266	EVQLVESGGGVVQPGGSLRLSCAASGFTF SSNAVS WVRQAPGKGLEWVAGISAGSDTY YGD SVKGRFTISRDN SKNTLYLQMNSLTA EDTAVYYCARETNHFLFDYWGQGT LVTSS
AB27 Mut All	Human with germline back- mutations	VH	267	EVQLVESGGGVVQPGGSLRLSCAASGFRF SSYAMS WVRQAPGKGLEWVSGISSGDDT YYVDSVKGRFTISRDN SKNTLYLQMNSLR AEDTAVYYCARETNHFLFDYWGQGT LVTSS
AB27 Mut All M34V	Human with germline back- mutations and liability mutation	VH	268	EVQLVESGGGVVQPGGSLRLSCAASGFRF SSYAVS WVRQAPGKGLEWVSGISSGDDTY YVD SVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCARETNHFLFDYWGQGT LVTSS
135 V34M	Human with liability mutation	HVR-H1	269	IYAMS
136 V34M	Human with liability mutation	HVR-H1	270	SYDMN
21/25 M34V	Human	HVR-H1	271	SNAVS
27 M34V	Human	HVR-H1	272	SYAVS
115	Human	VH	273	DVQLVESGGGVVVRPGESLRLSCAASGFSFS SYAMN WVRQAPGEGLEWVSRINSGGGT DYAESVKGRFTISRDN SENTLYLQMNSLR AEDTAVYYCAKQYDWN SFDDYWG LGAL LVTSS
115	Human	VL	274	ETVLTQSPATLSVSPGERATLSCRASQT VSKLAWHQKPGQAPRLLIYDATNRATGIS DRFSGSGSGTDFTLTITSSLTQEDSAVYYCQ QYYWPPYRFGGGTKVEIK
213	Human	VH	275	DVQLVESGGGVVVRPGESLRLSCEASGFTF RNYMT WVRQAPGEGLEWVSTISDTGDT AYYADSVKGRFTISRDN SKNSLYLQMNSL RADDTAIYYCAKSWIWWTFDDYWG LGTL LVTSS
213	Chicken	VL	276	ALTQPASVSANLGGTVEITCSGGNSNHYG WYQQKSPGSA PVTLIYADTNRP SNIPSRFS GSTSGSTTTLTITGVQAED EAVYYCGSSST GDGIFGAGTTLTVL

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
173	Human	VH	278	DVQLVESGGGVVRPGESLRLSCAASGFAP SDHMSWVRQGPGEGLWVAGISLRGGV TWYADSVKGRFTISRDN SKNTLYLRLFSL RTEDTAIYYCARESWNTFFDYWGLGLVT VSS
173	Human	VL	279	EIVLTQSPGTL SLSPGETATLSCRASQNVRS NLAWYQQKPGQAPRLLIYDASSRATGIPD RFGSGSGTDFTLTISLQSEDFAVYYCQQ YGNPPLTFGGG TKVEIK
209	Human	VH	280	DVQLVESGGGAVVRPGESLRLSKASGFTF TNFAMSWVRQAPGEGLWVSGISGSDTT YYADSVKGRFTISRDNSESTLYLQMNSLR AEDTAVYYCVKDS TVSWNTFFDYWGLGT LVTVSS
209	Chicken	VL	281	ALTQPASVSANLGGTVKITCSGGYGSDDG SSSYYGWYQQKSPGSA PVILIIYDDKRPS DIPSRFSGSTSGSTTTLTITGVQAEDEAVYF CGTYDTSSGAIFGAGTTLTVL
132	Human	VH	282	DVQLVESGGGVVRPGESLRLSCAASGFSF RSYAMNWRQAPGEGLWVSRINSGGGG TDYADSVK GRFTISRDN SKNTLYLQMNSLRAEDTAVY YCAKQYDWN SF FDYWGLGALVTVSS
132	Human	VL	283	ETVLTQSPATLSVSPGERATLSCRASQTVG SKLAWHQKPGQAPRLLIYDA SNRATGIPDRFSGSGSGTDFTLTISSPQTED SAVYYCQQYYWPPYRFGG G TKVEIK
218	Human	VH	284	DVQLVESGGGVVRPGESLRLSCTASGFTFT SSTMNWRQAPGEGLDWSSISTSGVITY YADSVKGRATISRDN SKNTLYLRLFSLRA DDTAIYYCATD TFDHWGPGLTVTVSS
218	Chicken	VL	285	ALTQPASVSANPGETVKITCFGSSGNYGW FQQKSPGSA PVTVIHYNNKRPS DIPSRFSGS KSGSTGTLTITGVRAEDEAVYFCGAWETG SATFGAGTTLTVL
149	Human	VH	286	DVQLVESGGGVVRPGESLRLSCAASGFTF SNFAMNWRQAPGEGLWVSLISVTATT YYADSVKGRFTISRDN SKNTLYLQMNSLR AEDTAVYYCAKVTWNNLFDYWGLGLTV TVSS
149	Human	VL	287	EIVLTQSPGTL SLSPGERATLSCRASQPID YLAWYQQKPGQAPRLLIYNTVTRATGIPD RFGSGSGTDFTLTINRLEPEDFAVYYCQH QYDWPPYIFGGG TKVEIK
161	Human	VH	288	DVQLVESGGGVVRPGESLRLSCAASGFTF SNFAMTWVRQAPGKGP EWVSLVSVTATT YYADSVKGRFTISRDN SKSTLYLQMNSLR AEDTAVYYCAKITWNNLFDYWGLGLVT VSS
161	Human	VL	289	EIVLTQSPGTL SLSPGERATLSCRASQTVGS KLAWYQQKPGQAPRLLIYDSSSRASGIPDR FSGSGSGTDFTLTISLQSEDSAVYYCQH NDWPPYTFGGG TKVEIK
162	Human	VH	290	DVQLVESGGGLVRPGESLRLSCAASGFTFT NYAVTWVRQAPGEGLWVSLISVTGTTY YADSVKGRFTISRDN SKSTLYLQMNSLRA EDTAVYYCAKVTKNVFDYWGLGLVT VSS
162	Human	VL	291	EIVLTQSPGTL SVSPGERATFSCRASQTVGS KLAWYQQKPGQAPRLLIYDANRATGIPA RFGSGSGTEFTLTISLQSEDFAVYYCQQ HTDWPPYTFGGG TKVEIK

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
194	Human	VH	292	DVQLVESGGGVVVRPGESLRLSCAASGFTF RNYGMSWVRQAPGEGLEWVSASSSGGST YYADSVKGRFTISRDN SKNTLYLQMNSLR AEDTAIYYCAKVTWNNFFDYWGLGLVT VSS
194	Human	VL	293	EIVLAQSPDTLSVSPGERATLTCRASQDVA GKLAWYQQKPGQAPRLLIHATSSRADGIP ARFSGSGSGTEFTLTITGLQSEDFAVYYCQ QHYDWSPLTFGGGKVEIK
119 M34L	Human with liability mutation	HVR-H1	305	NFALT
135 M34L	Human with liability mutation	HVR-H1	306	IYALS
119 mut all	Human with F21L, R39K, E60A, and T76S mutations	VL	312	EIVLTQSPATLSVSPGERATLSCRASQNVK NDLAWYQQKPGQAPRLLIYAA RIRETGIPARFSGSGSGTEFTLTISLQSEDF AVYYCQQYYDWPPTFGGKVEIK
136 M34L	Human with liability mutation	HVR-H1	313	SYDLN
21/25 M34L	Human with liability mutation	HVR-H1	318	SNALS
27 M34L	Human with liability mutation	HVR-H1	319	SYALS
119_VH_MutAll_V34L	Human with germline back- mutations and liability mutation	VH	327	EVQLLES GGGVVQPGGSLRLSCAASGFSFS NFALTWVRQAPGKGLEWVSTI GSGDTYYADSVKGRFTISRDN SKNTLYLQ MNSLRAEDTAVYYCAKDS TVSWSGDFFD YWQGTLVTVSS
135_VH_MutAll_V34L	Human with germline back- mutations and liability mutation	VH	328	EVQLVES GGGVVQPGGSLRLSCAASGFSF SIYALSWVRQAPGKGLEWVSTI GADDTYYADSVKGRFTISRDN SKNTLYLQ MNSLRAEDTAVYYCAKDS TVGWSGDFFD YWQGTLVTVSS
136_VH_Mutall_V34L	Human with germline back- mutations and liability mutation	VH	329	EVQLVES GGGVVQPGRSRLRLSCAASGFTF SSYDLN WVRQAPGKGLEWVSLI SGSGEIIYYADSVKGRFTISRDN SKNTLYL QMNSLRAEDTAVYYCAKENNRYRFFDDW GQGT LVTVSS
AB21_HC_MutAll_M34L	Human with germline back- mutations and liability mutation	VH	330	EVQLVES GGGVVQPGGSLRLSCAASGFTF SSNALSWVRQAPGKGLEWVAGISAGGSD TYYPASVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCARETNH LFDYWGQGT L VTVSS
AB25_HC_MutAll_M34L	Human with germline back- mutations and liability mutation	VH	331	EVQLVES GGGVVQPGGSLRLSCAASGFTF SSNALSWVRQAPGKGLEWVAGISSGSDTY YGD SVKGRFTISRDN SKNTLYLQMNSLTA EDTAVYYCARETNH LFDYWGQGT LVT VSS
AB27_HC_MutAll_M34L	Human with germline back- mutations and liability mutation	VH	332	EVQLVES GGGVVQPGGSLRLSCAASGFRF SSYALSWVRQAPGKGLEWVSGISSGSDTY YVD SVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCARETNH LFDYWGQGT LVT VSS

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
218_Hum13 (218 VL with human IGLV2)	Humanized	VL	333	QSALTQPASVSGSPGQSITISCFGSSGNYGL VSWYQQHPGKAPKLMIIYNNKRPSGVSN RFGSKSGNTASLTISGLQAEDEADYYCG AWETGSATFGGGTKLTVL
218_Hum14 (218 VL with human IGLV3)	Humanized	VL	334	SYELTQPPSVSVSPGQTASITCFGSSGNYG WYQQKPGQSPVLVIYYNNKRPSGIPERFSG SNSGNTATLTISGTQAMDEADYYCGAWE TGSATFGGGTKLTVL
119	Human	VH	335	DVQLLESGGGVVPGGSLRLSCAASGFSF SNFAMTWVRQAPGEGLEWVSTIGSGDTY YADSVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCAKDSTVSWSGDFFDYWGLGT LVTVSS
135	Human	VH	341	DVQLVESGGGVVRPGESLRLSCAASGFSFS IYAMSWVRQAPGEGLEWVSTIGADDTYY ADSVKGRFTISRDN SKNTLYLQMNSLRAE DTAVYYCAKDSTVGWSGDFFDYWGLGTL VTVSS
16	Human	VH	294	DVQLVESGGGVVRPGESLRLSCAVSGFRFSSY AMSWVRQAPGKGLEWVSGISSGGDTYYVDSV KGRFTISRDN SKNTLYLQVNSLTAEDTAIYYCA RETWNHLFDYWGLGTLVTVSS
16	Chicken	VL	295	ALTQPASVSANLGETVKITCSGGDSSSHYWG YQQKSPGSAPVTVIYSDDERPSDIPSRFSGSASG STATLTITGVRVEDEAIYYCGAYDGSTYANTF GAGTTTLTVL
17	Human	VH	342	DVQLVESGGGAVVRPGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGGSDTYIIDS VKGRFTISRDNSENLYLQMNSLTAEDTAVYY CARETWNHLFDYWGLGTLVTVSS
17	Chicken	VL	343	ALTQPASVSANPGETVKITCSGGDWYSTYYG WYQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSA SGSAATLTITGVRVEDEAVYYCAGYDGRTYIN TFGAGTTTLTVL
22	Human	VH	344	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGGSDTYIIDS VKGRFTISRQFQEQSLSPNEPALTAEDTAVYY CARETWNHLFDYWGLGTLVTVSS
22	Chicken	VL	345	ALTQPASVSANPGETVKITCSGGDYYSTYYGW YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG SAATLTITGVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL
23	Human	VH	346	DVQLVESGGGVVRPGESLRLSCAASGFTFSSH AMSWVRQAPGKGLEWLAGISAGGSDTYIIDS VKGRFTISRDN SKSLYLRMNSLTAEDTAVYY CARETWNHLFDYWGLGTLVTVSS
23	Chicken	VL	347	ALTQPASVSANPGETVKITCSGGDYYSTYYAW YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG SAATLTITGVRVEDEAVYFCGGYDGRTYINTFG AGTTTLTVL
24	Human	VH	348	DVQLVESGGGVVRPGESLRLSCAASGFTFSSN AMSWVRQAPGKGLEWLAGISAGGSDTYYPAS VKGRFTISRDNPKNTLYLQMNTLTAEDTAVYY CARETWNHLFDYWGLGTLVTVSS
24	Chicken	VL	349	ALTQPASVSANPGETVKIACSGGSYYSYYYGW YQQKSPGSALVTVIYSDDKRPSGIPSRFSGSAS GSTATLTITGVRVEDEAVYFCGGYDYSSYTND FGAGTTTLTVL

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
26	Human	VH	350	DVQLVESGGGVVPRGESLRLSCAASGFTTFSTY AMSWVRQAPGKGLEWVSGISASGSGTYYGDS VKGRFTMSRDNSKNTLYLQMNSLTAEDTAVY YCARETNHLPDYWGLGTLTVSS
26	Chicken	VL	351	ALTQPASVSANLGGTVEITCSGGSSSYGWWYQ QKSPGSAPVTVIYSDNQRPSDIPSRFSGSASDST ATLTITGVQVEDEAIYYCGGYDSSTYANTFGA GTTTLTVL
28	Human	VH	352	DVQLVESGGGVVPRGESLRLSCAASGFSFSSN AMSWVRQAPGKGLEWVAGISASGDTYYSGSM KGRFTISRDNKNTLYLQMNSLTAEDTAVYYC ARETNHLPDYWGLGTLTVSS
28	Chicken	VL	353	ALTQPASVSANPGETVKITCSGGSDSYYYGWH QQKSPGSAPVTVIYSDDQRPPDIPSRFSGSASGS TTTLTITGVRAEDEAVYYCGGYDYSTYTNTFG AGTTLTVL
29	Human	VH	354	DVQLVESGGGVVPRGESLRLSCAVSGFRFSSY AMSWVRQAPGKGLEWVSGISSDDAYYVDSV KGRFTISRDNKNTLYLQVNSLTAEDTAVYYC ARETNHLPDYWGLGTMVTVSS
29	Chicken	VL	355	ALTQPASVSANLGETVKITCSGGDSSSHYYGW FQQKSPGSAPVTLIYSDDERPSDIPSRFSGSASG STATLTITGVRVEDEAIYFCGAYDGSTYTNTFG AGTTLTVL
30	Human	VH	356	DVQLVESGGGVVPRGESLRLSCEASGFTFSSDA MSWVRQAPGKGLEWVSGISSGSTYYGGSVK GRFTISRDNKNTLYLQMNSLTAEDTAVYYCA RETWNHLPDYWGLGTLTVSS
30	Chicken	VL	357	ALTQPASVSASPGETVEITCSGGSDSSYYGWW YQQKSPGSAPVTVIYSDNKRPSNIPSRFSGSASG STATLTITGVRVEDEAVYYCGGYDYSTYTNP GAGTTLTVL
55	Human	VH	358	DVQLVESGGGVVPRGESLRLSCAVSGFRFSSY AMSWVRQAPGKGLEWVSGISSGDTYYVDSV KGRFTISRDNKNTLYLQVNSLTAEDTAIYYCA RETWNHLPDYWGLGTLTVSS
55	Chicken	VL	359	ALTQPASVSANLGETVEITCSGGDSSSHYYGW YQQKSPGSAPVTVIYSDDERPSDIPSRFSGSASG STATLTITGVRVEDEAIYYCGAYDGSTYANTF GAGTTLTVL
56	Human	VH	360	DVQLVESGGGVVPRGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGSDTYIIDS VKGRFTISRDNKNSLYLQVNSLTAEDTAVYY CARETNHLPDYWGLGTLTVSS
56	Chicken	VL	361	ALTQPASVSANPGETVKITCSGGGYSTYYGW YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG SAATLTITGVRVEDEAVYYCAGYDGRYINTF GAGTTLTVL
59	Human	VH	362	DVQLVESGGGVVPRGESLRLSCAVSGFRFSSH AMSWVRQAPGKGLEWVSGISSGDTYYVDSV KGRFTISRDNKNTLYLQVNSLTAEDTAIYYCA RETWNHLPDYWGLGTLTVSS
59	Chicken	VL	363	ALTQPASVSANLGETVKITCSGGDSSSHYYGW YQQKSPGSAPVTVIYSDDERPSDIPSRFSGSASG STATLTITGVRVEDEAIYYCGAYDGSTYANTF GAGTTLTVL
60	Human	VH	364	DVQLVDSGGGVVPRGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGSDTYIIDS VKGRFTISRDNKNSLYLQMNSLTAEDTAVYY CARETNHLPDYWGLGTLTVSS

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
60	Chicken	VL	365	ALTQPASVSANPGETVKITCSGGDYSTYYGW YQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG SAATLTITGVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL
65	Human	VH	366	DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGGSDTYIDS VKGRFTISRDNKNSLYLQMNSLTAEDEVAVY CARETNHLPDYWGLGTLTVSS
65	Chicken	VL	367	ALTQPASVSANPGETVKITCSGGDYSTYYGW YQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG SAATLTITGVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL
69	Human	VH	368	DLQLVESGGGVVVRPGESLRLSCAASGFTFSSYA MSWVRQAPGKGLEWLAGISAGGSDTYIDSV KGRFTISRDNKNSLYLQMNSLTAEDEVAVYC ARETNHLPDYWGLGTLTVSS
69	Chicken	VL	369	ALTQPASVSANPGETVKIICSGGDYSTYYGW YQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG SAATLTITGVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL
70	Human	VH	370	DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGGSDAYIDS VKGRFTISRDNKNSLYLQMNSLTAEDEVAVY CARETNHLPDYWGLGTLTVSS
70	Chicken	VL	371	ALTQPASVSANPGETVKITCSGGDWYSTYYG WYQKSPGSAPVTVIHSDDKRPSDIPSRFSGSA SGSAATLTITGVRVEDEAVYFCAGYDGRTYIN TFGAGTTTLTVL
71	Human	VH	372	DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGGSDTYIDS VKGRFTISRDNKNSLYLQMNSLTAEDEVAVY CARETNHLPDYWGLGTLTVSS
71	Chicken	VL	373	ALTQPASVSANPGETVKITCSGGGYSTYYGW YQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG SAATLTITGVRVEDEAVYFCAGYDARTYINTF GAGTTTLTVL
73	Human	VH	374	DVQLVESGGGVVVRPGESLRLSCEASGFTFSSNA MSWARQAPGKGLEWVAGISGSDTYIGDSVK GRLTISRDNKNSLYLQMNSLTAEDEVAVYCA RETWNHLPDYWGLGTLTVSS
73	Chicken	VL	375	ALTQPASVSANPGETVKITCSGGIYSSYYYAW YQKSPGSAPVTLIYSDDKRPSNIPSRFSGSASG STATLTITGVRAEDEAVYFCGGYDQSSYTNP FAGTTTLTVL
74	Human	VH	376	DVQLVESGGGVVVRPGESLRLSCAASGFTFSSN AMSWVRQAPGKGLEWLAGISAGSDTYYPAS VKGRFTISRDNPKNTLYLQMNLTAEDEVAVY CARETNHLPDYWGLGTLTVSS
74	Chicken	VL	377	ALTQPASVSANPGETVKIACSGGSYYYSSYYGW YQKSPGSALVTVIYSDDKRPSGIPSRFSGSAS GSTATLTITGVRAEDEAVYCGGYDYSSYTND FGAGTTTLTVL
76	Human	VH	378	DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGGSDTYIDS VKGRFTISRDNKNSLYLQMNSLTAEDEVAVY CARETNHLPDYWGLGTLTVSS
76	Chicken	VL	379	ALTQPASVSANPGETVKITCSGGDWYSTYYG WYQKSPGSAPVTVIHSDDKRPSDIPSRFSGSA SGSAATLTITGVRVEDEAVYFCAGYDGRTYIN TFGAGTTTLTVL

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
201	Human	VH	380	DVQLVESGGAVVRPGETLRLSCTASGFTFSSY AMSWVRQAPGKGLEWVSGISASGSDTYADSD VKGRSTISRDNKNTLYLRMSSLTAEDTAVYY CARETNHLPDYWGLGTLTVSS
201	Chicken	VL	381	ALTQPASVSANPGETVKITCSGGGTSSYYGWY QQKSPGSAPVTLIHSDDKRPSDIPSRFSGSASGS TATLTITGVQVEDEAVYYCGGYDYTTYVNTFG AGTTTLTVL
202	Human	VH	382	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY AMSWVRQAPGKGLEWLAGISAGSDTYIDSD VKGRFTISRDNKNSLYLQMNSLTAEADTAVYY CARETNHLPDYWGLGTLTVSS
202	Chicken	VL	383	ALTQPASVSANPGETVKITCSGGGYSTYYGW YQQRSPGSAPVTVIHSDDKRPSDIPSRFSGSASGS SAATLTITGVRVEDEAVYYCAGYDGRTYLNTF GAGTTTLTVL
206	Human	VH	384	DVQLVESGGAVVRPGETLRLSCTASGFTFSSY AMSWVRQAPGKGLEWVSGISASGSDTYADSD VKGRSTISRDNKNTLYLRMSSLTAEDTAVYY CARETNHLPDYWGLGTLTVLSS
206	Chicken	VL	385	ALTQPASVSANPGETVKITCSGGGTSSYYGWY QQKSPGSAPVTLIHSDDKRPSDIPSRFSGSASGS TATLTITGVQAEDEAVYYCGGYDYTTYVNTFG AGTTTLTVL
175	Human	VH	386	DVQLVESGGGVVRPGESLRLSCAASGFTFSTD MNVWRQAPGEGLEWVSLISGSGEITYADSVK GRFTISRDNKNTLYLQMNSLRAEDTAVYYCA KENDRYRFFDYWGLGTLTVSS
175	Human	VL	387	ETVLTQSPGILSLSPGERATLTCRASQSVYTYL AWYQEKPGQAPRLLIYGASSRAAGIPDRFSGS SGTEFTLTISLQSEDFAVYYCQQYYDRPPLT FGGGTKVEIK
177	Human	VH	388	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY DMNVWRQAPGEGLEWVSLISGSGEIYYADSV KGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKENNRYRFFDYWGLGTLTVSS
177	Human	VL	389	KTVLTQSPGTLTLSPGERATLTCRADQSVYTYL AWYQERPGQAPRLLIYDASSRATGIPDRFSGSG SGTEFTLTISLQSEDFAVYYCQQYYDRPPLTF GGGTKVEIK
178	Human	VH	390	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY DMNVWRQAPGEGLEWVSLISGSGEIYYADSV KGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKENNRYRFFDYWGLGTLTVSS
178	Human	VL	391	ETVLTQSPGTLTLSPGERATLTCRASQSVYTYL AWYQEKPGQAPRLLIYGASSRATGVPDRFSGS SGTEFTLTISLQSEDFAVYYCQQYYDRPPLT FGGGTKVEIK
180	Human	VH	392	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY DMNVWRQAPGEGLEWVSLISGSGEIYYADSV KGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKENDRYRFFDYWGLGTLTVSS
180	Human	VL	393	ETVLTQSPGTLTLSPGERATLTCRASQSVYTYL AWYQEKPGQAPRLLIYGASSRATGIPDRFSGSG SGTEFTLTISLQSEDFAVYYCQQYYDRPPLTF GGGTKVEIK
184	Human	VH	394	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY DMNVWRQAPGEGLEWVSLISGSGEIYYADSV KGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKENNRYRFFDYWGLGTLTVSS

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
184	Human	VL	395	ETVLTQSPGTLSSLSPGERATLNCRASQSVYTYL AWYQEKPGQAPRLLIYDASSRATGIPDRFSGSG SGTEFTLTISLSEDFAVYYCQQYYDRPPLTF GGGTKVEIK
185	Human	VH	396	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV KGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKENNRYRFFDDWGLGLTVTVSS
185	Human	VL	397	ETVLTQSPGTLSSLSPGERATLNCRASQSVYSYL AWYQERPGQAPRLLIYGASTRATGIPDRFSGSG SGTEFTLTISLSEDDFAVYYCQQYYDRPPLTF GGGTKVEIK
189	Human	VH	398	DVQLVESGGGVVRPGESLRLSCAASGFTFSSSD MNWVRQAPGEGLEWVSLISGSGEITYYADSVK GRFTISRDNKNTLYLQMNSLRAEDTAIYYCA KENNMYRFFDYWGLGLTVTVSS
189	Human	VL	399	ETVLTQSPGTLSSLSPGERATLSCRASQSVYTYL AWYQQKPGQPPRLLIHAARNRAAGIPDRFSGS GSGTEFTLTISLQSEDFAVYYCQQYYDRPPLT FGGGTKVEIK
190	Human	VH	400	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV KGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKEDNRYRFFDYWGLGALTVTVSS
190	Human	VL	401	ETVLTQSPGTLTLSPGERTTLTCRASQSVYTYL AWYQEKPGQAPRLLIYGASSRATGIPDRFSGSG SGTEFTLTISLQSEDFAVYYCQQYYDRPPLTF GGGTKVEIK
193	Human	VH	402	DVQLVESGGGVVRPGESLRLSCAASGFTFSSY DMNWVRQAPGEGLEWVSLISGSGEITYYADSV KGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKENNRYRFFDYWGLGLTVTVSS
193	Human	VL	403	ETVLTQSPGTLSSLSPGERATLSCRASQSVYTYL AWYQEKPGQAPRLLIYAASRATGIPDRFSGSG SGTEFTLTISLQSEDFAVYYCQQYYDRPPLTF GGGTKVEIK
191	Human	VH	404	DVQLVESGGGVVRPGESLRLSCAASGFSFSSY AMNWVRQAPGEGLEWVSRINSGGGTDYAES VKGRFTISRDNSENTLYLQMNSLRAEDTAVYY CAKQYDWNNGFFDYWGLGALTVTVSS
191	Human	VL	405	ETVLTQSPATLSVSPGERATLSCRASQTVGSKL AWHQKPGQAPRLLIYD ATNTRATGIPDRFSGSGSGTDFTLTISGLQTEDSA VYYCQQYYWPPYRFGGGTKVEIK
198	Human	VH	406	DVQLVESGGGVVRPGESLRLSCAASGFSFSSH AMNWVRQAPGEGLEWVSRINSGGGTDYAES VKGRFTISRDNSENTLYLQMNSLRAEDTAVYY CAKQYDWNNGFFDYWGLGALTVTVSS
198	Human	VL	407	ETVLTQSPATLSVSPGERATLSCRASQTVGSKL AWHQKPGQAPRLLIYD ATNTRATGIPDRFSGSGSGTDFTLTISLQTEDSA VYYCQQYYWPPYRFGGGTKVEIK
163	Human	VH	408	DVQLVESGGGLVRPGESLRLSCAASGFTFTNY AVTWVRQAPGEGLEWVSLISVTGTTYADSV KGRFTISRDNKSTLYLQMNGLRAEDTAVYYC AKVTWKNVFDYWGLGLTVTVSS
163	Human	VL	409	EIVLTQSPGTLSPGERATFSCRASQTVGSKL AWYQQKPGQAPRLLIYDANTRATGIPARFSGS GSGTEFTLTISLQSEDFAVYYCQQHTDWPPYT FGGGTKVEIK

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
164	Human	VH	410	DVQLVESGGGLVPRGESLRLSCAASGFTFTNY AVTWVRQAPGEGLEWVSLISVTGTTYADSV KGRFTISRDNKSKSTLYLQMNSLRAEDTAVYYC AKVTWKNVFDYWGLGLTVTVSS
164	Human	VL	411	EIVLTQSPGTLSPGERATFSCRASQTVGSKL AWYQQKPGQAPRLLIYDANTRATGIPARFSGS RSGTEFTLTISLQSEDFAVYYCQHTDWPPYT FGGGTKVEIK
174	Human	VH	412	DVQLVESGGGVPRGESLRLSCAASGFTFSDH DMSWVRQGPGEGLWVAGISLRGGVTWYAD SVKGRFTISRDNKNTLYLRLFLRTEDTAIYY CARESWNTFFDYWGLGLTVTVSS
174	Human	VL	413	EIVLTQSPGTLSPGETATLSCRASQNVRSNL AWYQQKPGQAPRLLIYDASSRATGIPDRFSGS GSGTDFTLTISLQSEDFAVYYCQYGNPPLT FGGGTKVEIK
214	Human	VH	414	DVQLVESEGGVPRGESLRLSCEASGFTFRNSY MTWVRQAPGEGLEWVSTISDTGDTAYYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAIYYC AKSWIWWTFDYWGLGLTVTVSS
214	Chicken	VL	415	ALTQPASVSANLGGTVEITCSGGNSNHGWYQ QKSPGSAPVTLIYADTNRPSNIPSRFSGSTSGST STLTIITGVQAEDEAVYYCGSS TGDGIFGAGTT LTVL
Hum8	Humanized	VL	416	SYELTQPPSVSVSPGQTARITCSGGAYSSYYA WYQQKPGQAPVLVIYSDSKRPSGIPERFSGSS GTTVTLTISGVQAEDEADYYCGYDQSSYTNP FGGGTKLTVL
136_MutAll(Light chain)_S12T	Human with D1E, R13Q, E16R, E43K, and L111Q mutations	VL	417	EIVLTQSPGTLTLSPGERATLSCRASQSVYTYL AWYQQKPGQAPRLLIYGA SSRATGIPDRFSGSGSGIEFTLTISLQSEDFAV YYCQYYDRPPLTFGGGTKEIK
136_MutAll(Light chain)_S22T	Human with germline back- mutations	VL	418	EIVLTQSPGTLTLSPGERATLTCRASQSVYTYL AWYQQKPGQAPRLLIYGA SSRATGIPDRFSGSGSGIEFTLTISLQSEDFAV YYCQYYDRPPLTFGGGTKEIK
136_MutAll(Light chain)_Q38E	Human with germline back- mutations	VL	419	EIVLTQSPGTLTLSPGERATLSCRASQSVYTYL AWYQEKPGQAPRLLIYGA SSRATGIPDRFSGSGSGIEFTLTISLQSEDFAV YYCQYYDRPPLTFGGGTKEIK
119_wt_M34L	Human with liability mutation	VH	420	DVQLLESGGGVVPQGGSLRLSCAASGFSFSNF ALTWVRQAPGEGLEWVSTIGSGDTYYADSVK GRFTISR DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVS WSGDFFDYWGLGLTVTVSS
119_wt_M34V	Human with liability mutation	VH	421	DVQLLESGGGVVPQGGSLRLSCAASGFSFSNF AVTWVRQAPGEGLEWVSTIGSGDTYYADSVK GRFTISR DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVS WSGDFFDYWGLGLTVTVSS
135_wt_M34L	Human with liability mutation	VH	422	DVQLVESGGGVPRGESLRLSCAASGFSFSIYA LSWVRQAPGEGLEWVSTIGADTTYADSVKG RFTISR DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVG WSGDFFDYWGLGLTVTVSS
135_wt_M34V	Human with liability mutation	VH	423	DVQLVESGGGVPRGESLRLSCAASGFSFSIYA VSWVRQAPGEGLEWVSTIGADTTYADSVKG RFTISR DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVG WSGDFFDYWGLGLTVTVSS

TABLE 2-continued

Amino acid sequences of antibody clones described herein.				
Clone/ Antibody	Framework	Domain	SEQ ID NO	Sequence
136_wt_M34L	Human with liability mutation	VH	424	DVQLVESGGGVVPRGESLRLSCAASGFTFSSY DLNWRQAPGEGLEWVSLISGSGEIIYYADSV KGRFTI SRDNSKNTLYLQMNSLRAEDTAVYYCAKENN RYRFFDDWGLGTLVTVSS
136_wt_M34V	Human with liability mutation	VH	425	DVQLVESGGGVVPRGESLRLSCAASGFTFSSY DVNWRQAPGEGLEWVSLISGSGEIIYYADSV KGRFTI SRDNSKNTLYLQMNSLRAEDTAVYYCAKENN RYRFFDDWGLGTLVTVSS
119 HVR-H1	Human	HVR-H1	175	NFAMT

In some embodiments, an antibody of the present disclosure binds to a human SIRP- α polypeptide at one or more residues. It is to be understood that residues of a SIRP- α polypeptide that are bound by an antibody of the present disclosure may be described with respect to a reference SIRP- α polypeptide, but this description is not limited to a single SIRP- α polypeptide (i.e., the reference SIRP- α polypeptide). Rather, specific amino acid residues of a reference SIRP- α polypeptide are described to identify corresponding amino acid positions that can be identified on various SIRP- α polypeptides. For example, specific residues that correspond to one or more amino acid positions of SEQ ID NO:296 can be identified for various human SIRP- α polypeptides, such as v1 and/or v2. Since the amino acid sequences of SEQ ID NO:296 and SEQ ID NO:5 differ only at the N80 position (excepting a small number of C-terminal residues of SEQ ID NO:296 useful for protein production and purification), one of skill in the art will appreciate that references herein to amino acid positions with respect to the amino acid sequence of SEQ ID NO:296 will correspond to the same positions in the amino acid sequence of SEQ ID NO:5. Techniques for determining the residues of a SIRP- α polypeptide bound by an antibody are known in the art; exemplary and non-limiting descriptions are provided in Example 4.

SEQ ID NO: 296

EEELQVIQPD KSVLVAAGET ATLRCATATSL IPVGPIQWFR
GAGPGRELIY NQKEGHFPRV TTVSDLTKRN NMDFSIRIGA
ITPADAGTYY CVKFRKGSPP DVEPKSGAGT ELSVRAPKST
RHHHHHH

In some embodiments, an antibody of the present disclosure binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from I31, V33, Q52, K53, T67, R69, N70, and K96, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from L30, P32, E54, T62, N71, M72, F74, and R95, according to SEQ ID NO:296.

In some embodiments, an antibody of the present disclosure binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from 17, P9, D10, K11, S12, A42, A108, and E111, according to SEQ ID NO:296. In

some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at K11, A42, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at 17, P9, D10, K11, S12, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from L14, T26, T28, T88, Y90, S106, S113, and A116, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T88, Y90, S106, S113, and A116 of human SIRP- α v1, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T26, and T28, according to SEQ ID NO:296.

In some embodiments, an antibody of the present disclosure binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from E47, L48, P58, R59, T82, and A84, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from A17, P44, G45, 149, E54, G55, H56, F57, and P83, according to SEQ ID NO:296.

In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide with a dissociation constant (K_D) of less than 100 nM, less than 50 nM, or less than 30 nM. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide with a dissociation constant (K_D) of less than 100 nM, less than 50 nM, or less than 30 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide.

In some embodiments, an antibody of the present disclosure binds the D1 domain of a human SIRP- α polypeptide and does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide with a dissociation constant (K_D) of less than 100 nM, less than 50 nM, or less than 30 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide.

In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody. Techniques for determining whether an antibody competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody are known in the art; exemplary and non-limiting descriptions are provided in Example 5.

In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 119, 120, 121, 122, 21, 25, 27, 66, and 135. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:120 and a VL domain comprising the amino acid sequence of SEQ ID NO:97, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:121 and a VL domain comprising the amino acid sequence of SEQ ID NO:98, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:130 and a VL domain comprising the amino acid sequence of SEQ ID NO:107, (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:122 and a VL domain comprising the amino acid sequence of SEQ ID NO:99, (e) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:135 and a VL domain comprising the amino acid sequence of SEQ ID NO:136, (f) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:137 and a VL domain comprising the amino acid sequence of SEQ ID NO:138, (g) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:139 and a VL domain comprising the amino acid sequence of SEQ ID NO:140, (h) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:141 and a VL domain comprising the amino acid sequence of SEQ ID NO:142, and (i) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:127 and a VL domain comprising the amino acid sequence of SEQ ID NO:104.

In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the

D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 136 and 137. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:133 and a VL domain comprising the amino acid sequence of SEQ ID NO:134, and (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:128 and a VL domain comprising the amino acid sequence of SEQ ID NO:105.

In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 3, 213, 173, and 209. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:242 and a VL domain comprising the amino acid sequence of SEQ ID NO:243, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:275 and a VL domain comprising the amino acid sequence of SEQ ID NO:276, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:278 and a VL domain comprising the amino acid sequence of SEQ ID NO:279, and (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:280 and a VL domain comprising the amino acid sequence of SEQ ID NO:281.

In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 115, 116, 117, 118, and 132. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:116 and a VL domain comprising the amino acid sequence of SEQ ID NO:93, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:117 and a VL domain comprising the amino acid sequence of SEQ ID NO:94, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:118 and a VL domain comprising the amino acid sequence of SEQ ID NO:95, (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:119 and a VL domain comprising the amino acid sequence of SEQ ID NO:96, and (e) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:282 and a VL domain comprising the amino acid sequence of SEQ ID NO:283.

In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 218, 123, 149, 161, 162, and 194. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising

ing a VH domain comprising the amino acid sequence of SEQ ID NO:284 and a VL domain comprising the amino acid sequence of SEQ ID NO:285, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:123 and a VL domain comprising the amino acid sequence of SEQ ID NO:100, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:286 and a VL domain comprising the amino acid sequence of SEQ ID NO:287, (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:288 and a VL domain comprising the amino acid sequence of SEQ ID NO:289, (e) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:290 and a VL domain comprising the amino acid sequence of SEQ ID NO:291, and (f) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:292 and a VL domain comprising the amino acid sequence of SEQ ID NO:293.

In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with reference anti-SIRP- α antibody 45. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:244 and a VL domain comprising the amino acid sequence of SEQ ID NO:245.

The present disclosure provides multiple families of anti-SIRP- α antibodies, each family comprising multiple antibodies. As demonstrated herein, antibodies within a given family may share certain structural properties (e.g., similar or identical HVR or CDR sequences) as well as one or more functional properties, including but not limited to binding affinity to human, monkey, and/or mouse SIRP- α polypeptide(s), binding affinity to SIRP-0 polypeptides, binding affinity to SIRP- γ polypeptides, mode of binding to SIRP- α (e.g., CD47 blocking, CD47 non-blocking, or “kick off” binding), induction of phagocytosis (e.g., in an in vitro assay), activation of dendritic cells, anti-tumor efficacy, SIRP- α binding epitope residue(s) or “bin” (e.g., as determined by a binning assay), and the like (see, e.g., Tables P-T). Because of these shared properties, it is contemplated that HVR and/or VH or VL sequences of antibodies belonging to the same family can be interchanged or intermingled, such that an anti-SIRP- α antibody can comprise HVR and/or VH or VL derived from more than one specific anti-SIRP- α antibody described herein. As discussed in greater detail herein, various methodologies for determining HVR or CDR sequences of an antibody variable domain are known in the art and can be used interchangeably herein, including without limitation the Kabat, Chothia, and IMGT definitions, as well as combinations thereof.

In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104,

and 312; and/or an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; and an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328. In some embodiments, an antibody of the present disclosure comprises an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; and an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312.

In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; and/or an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385. In some embodiments, an antibody of the present disclosure

In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs:116, 117, 118, 119, 282, 404, and 406; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 116, 117, 118, 119, 282, 404, and 406; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 116, 117, 118, 119, 282, 404, and 406; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs:93, 94, 95, 96, 283, 405, and 407; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 93, 94, 95, 96, 283, 405, and 407; and/or an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 93, 94, 95, 96, 283,

In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415; and/or an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; and an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414. In some embodiments, an antibody of the present disclosure comprises an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOS: 275 and 414; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOS: 276 and 415.

In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence set forth in SEQ ID NO: 286; an HVR-H2 from a VH domain sequence set forth in SEQ ID NO: 286; an HVR-H3 from a VH domain sequence set forth in SEQ ID NO: 286; and/or an HVR-L1 from a VL domain sequence set forth in SEQ ID NO: 287; an HVR-L2 from a VL domain sequence set forth in SEQ ID NO: 287; and/or an HVR-L3 from a VL domain sequence set forth in SEQ ID NO: 287. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence set forth in SEQ ID NO: 286; an HVR-H2 from a VH domain sequence set forth in SEQ ID NO: 286; and an HVR-H3 from a VH domain sequence set forth in SEQ ID NO: 286. In some embodiments, an antibody of the present disclosure comprises an

[illegible]

In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of NFAMT (SEQ ID NO:175), NFAVT (SEQ ID NO:204), or NFALT (SEQ ID NO:305), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGSGDTYYADSVKG (SEQ ID NO:144), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVSWSGDFFDY (SEQ ID NO:145); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVKNDLA (SEQ ID NO:146), (ii) an HVR-L2 sequence comprising the amino acid sequence of AARIRET (SEQ ID NO:147), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:120, 246, 258, or 327; and/or the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:97 or 312. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:120, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:120, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; or the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:312.

In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of IYAMS (SEQ ID NO:269), IYAVS (SEQ ID NO:213), or IYALS (SEQ ID NO:306), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGADDTYYADSVKG (SEQ ID NO:150), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVGWSGDFFDY (SEQ ID NO:151); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVRSIDIA (SEQ ID NO:152), (ii) an HVR-L2 sequence comprising the amino acid sequence of AASSRDT (SEQ ID NO:153), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:341, 127, 247, 259, or 328; and/or the VL domain comprises an

amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:104 or 248. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:341, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:341, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; or the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:248.

In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of $X_1X_2DX_3N$, wherein X_1 is S or T; X_2 is Y or S; and X_3 is M, L, or V (SEQ ID NO:307); (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIX₁YYADSVKG, wherein X_1 is I or T (SEQ ID NO:308); and (iii) an HVR-H3 sequence comprising the amino acid sequence of $EX_1X_2X_3YRFFDX_4$, wherein X_1 is N or D; X_2 is N or D; X_3 is R or M; and X_4 is D or Y (SEQ ID NO:309); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of RAX_1QSVYX_2YLA , wherein X_1 is S or D; and X_2 is T or S (SEQ ID NO:310); (ii) an HVR-L2 sequence comprising the amino acid sequence of $X_1AX_2X_3RAX_4$, wherein X_1 is G, A, or D; X_2 is S or R; X_3 is S, N, or T; and X_4 is T or A (SEQ ID NO:311); and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of SYDMN (SEQ ID NO:270), SYDVN (SEQ ID NO:221), or SYDLN (SEQ ID NO:313), (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIYYADSVKG (SEQ ID NO:156), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ENNRYYRFFDD (SEQ ID NO:157); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQSVYTYLA (SEQ ID NO:158), (ii) an HVR-L2 sequence comprising the amino acid sequence of GASSRAT (SEQ ID NO:159), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least

97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:133, 260, 329, or 249; and/or the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:134, 250, 251, 417, 418, or 419. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:419; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:419; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:419; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino

acid sequence of SEQ ID NO:250; or the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:251.

In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of $X_1X_2AX_3S$, wherein X_1 is S or T; X_2 is N, Y, H, or D; and X_3 is M, L, or V (SEQ ID NO:297); (ii) an HVR-H2 sequence comprising the amino acid sequence of $GISX_1X_2X_3X_4X_5X_6YYX_7X_8SX_9KG$, wherein X_1 is A or S; X_2 is G, S, or absent; X_3 is S, D or G; X_4 is G or S; X_5 is D, S, or G; X_6 is T or A; X_7 is P, G, V, I, A, or S; X_8 is A, D, or G; and X_9 is V or M (SEQ ID NO:298); and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of $SGGX_1X_2X_3SX_4YYX_5$, wherein X_1 is D, G, S, I, or absent; X_2 is S, W, G, Y, D, or absent; X_3 is S, Y, T, or D; X_4 is H, T, S, or Y; and X_5 is G or A (SEQ ID NO:299); (ii) an HVR-L2 sequence comprising the amino acid sequence of $SDX_1X_2RPX_3$, wherein X_1 is D or N; X_2 is E, K, or Q; and X_3 is S or P (SEQ ID NO:300); and (iii) an HVR-L3 sequence comprising the amino acid sequence of $X_1X_2YDX_3X_4X_5YX_6NX_7$, wherein X_1 is G or A; X_2 is G or A; X_3 is G, Y, Q, S, or A; X_4 is S, R, or T; X_5 is T or S; X_6 is A, I, V, L, or T; and X_7 is T, A, D, or P (SEQ ID NO:301).

In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of SX_1AX_2S , wherein X_1 is N or Y; and wherein X_2 is M, L, or V (SEQ ID NO:302); (ii) an HVR-H2 sequence comprising the amino acid sequence of $GISX_1GX_2X_3DTYYX_4X_5SVKG$, wherein X_1 is A or S; X_2 is G or absent; X_3 is S or G; X_4 is P, G, or V; and X_5 is A or D (SEQ ID NO:303); and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of $SGGX_1YSSYYA$, wherein X_1 is S or A (SEQ ID NO:304); (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336); and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172).

In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GIS-AGGSDTYYPASVKG (SEQ ID NO:195), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193).

In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO: 135, 263, 264, or 330.

In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISSGSDTYYGDSVKG (SEQ ID NO:197), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193).

In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%,

at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO: 137, 265, 266, or 331. In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SYAMS (SEQ ID NO:200), SYAVS (SEQ ID NO:272), or SYALS (SEQ ID NO:319), (ii) an HVR-H2 sequence comprising the amino acid sequence of GIS-SGGDTYYVDSVKG (SEQ ID NO:201), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO: 139, 267, 268, or 332. In some embodiments, the VL domain comprises one or more human IGLV3 framework sequences. In some embodiments, the VL domain comprises the sequence FW1-HVR-L1-FW2-HVR-L2-FW3-HVR-L3-FW4 (N-terminus to C-terminus), wherein FW1 comprises the amino acid sequence SYELTQPPSVSVSPGQTARITC (SEQ ID NO:314), FW2 comprises the amino acid sequence WYQQKPGQAPVTLII (SEQ ID NO:315), FW3 comprises the amino acid sequence NIPERFSGSSSGTTVLTISGVQAEDEADYYC (SEQ ID NO:316), and FW4 comprises the amino acid sequence FGGGTKLTVL (SEQ ID NO:317). In some embodiments, the VL domain comprises (i) an HVR-L1 sequence comprising the amino acid sequence of SGGSYSSYYYA (SEQ ID NO:170), (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336), and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:252. In some embodiments, the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:416. In some embodiments, the VL domain comprises (i) an HVR-L1 sequence comprising the amino acid sequence of SGGAYSSYYYA (SEQ ID NO:261), (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336), and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:262. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO:330, and the

VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:137, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:268, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:332, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:330, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:332, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO:254; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO: 254; the VH domain comprises the amino acid sequence of SEQ ID NO:330, and the VL domain comprises the amino acid sequence of SEQ ID NO: 254; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO: 254; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO: 254; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO: 254; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO:268, and the VL domain comprises the amino acid sequence of SEQ ID NO: 254; the VH

domain comprises the amino acid sequence of SEQ ID NO:332, and the VL domain comprises the amino acid sequence of SEQ ID NO: 254; the VH domain comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:330, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:268, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:332, and the VL domain comprises the amino acid sequence of SEQ ID NO: 416; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:137, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:254; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:416; the VH domain comprises the amino acid sequence of SEQ ID NO:137, and the VL domain comprises the amino acid sequence of SEQ ID NO:254; the VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:254; the VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:416; or the VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:262.

In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:120 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:97. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:127 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:104. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:133 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:134. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:136. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain

comprising the amino acid sequence of SEQ ID NO:138. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:140. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:142.

In some embodiments, the antibody comprises (a) an HVR-H1 sequence comprising the amino acid sequence of $\text{GFSPSX}_1\text{X}_2\text{AMX}_3$, wherein X_1 is N or I; X_2 is F or Y; and X_3 is T or S (SEQ ID NO:185); (b) an HVR-H2 sequence comprising the amino acid sequence of $\text{TIGX}_4\text{X}_5\text{DTYYADSVKG}$, wherein X_4 is S or A and X_5 is G or D (SEQ ID NO:186); (c) an HVR-H3 sequence comprising the amino acid sequence of $\text{DSTVX}_6\text{WSGDFFDY}$, wherein X_6 is S or G (SEQ ID NO:187); (d) an HVR-L1 sequence comprising the amino acid sequence of $\text{RASQNVX}_7\text{X}_8\text{DX}_9\text{A}$, wherein X_7 is K or R; X_8 is N or S; and X_9 is L or I (SEQ ID NO:188); (e) an HVR-L2 sequence comprising the amino acid sequence of $\text{AAX}_{10}\text{X}_{11}\text{RX}_{12}\text{T}$, wherein X_{10} is R or S; X_{11} is I or S; and X_{12} is E or D (SEQ ID NO:189); and (f) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148).

In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:143-148 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:143-145 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:146-148). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:148-153 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:149-151 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:152, 153, and 148). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:155-160 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:155-157 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:158-160). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161-166 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:161-163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:164-166). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161-166 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:161-163 and/or one, two, or three light chain HVR sequences of a variable domain shown in Table 2). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161, 163, 168, and 170-172 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs: 161, 168, and 163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:170-172). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences

In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:143, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:144, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:145; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:146, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:147, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:149, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:150, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:151; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:152, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:153, and an HVR-L3 comprising the amino

In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166. In some

121

embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:170, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:171, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:172. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:176, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:177, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:178. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:184.

In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:161, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:162, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:161, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:168, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:173, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:174, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:179, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:162, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2

122

comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184.

In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:242; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:243. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:244; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:245.

As described supra, various techniques for delineating hypervariable regions (HVRs) or complementarity determining regions (CDRs) are known in the art and can be applied to the variable domain sequences described herein. In some embodiments, an antibody of the present disclosure comprises HVRs as defined by Chothia, Kabat, IMGT, or a combination thereof (e.g., one or more HVRs as defined by one delineation and one or more HVRs as defined by a different delineation). HVR sequences of antibodies of the present disclosure delineated using three known delineation (Chothia, Kabat, and IMGT) are provided in Table 5. As used herein, unless otherwise specified, the numbering of HVR residues is defined by Kabat numbering.

TABLE 5

HVR delineations.			
Hum 1 (Chothia)	HVR-L1	170	SGGSYSSYYA
	HVR-L2	336	SDDKRPS
	HVR-L3	172	GGYDQSSYTNP
Hum 1 (Kabat)	HVR-L1	170	SGGSYSSYYA
	HVR-L2	336	SDDKRPS
	HVR-L3	172	GGYDQSSYTNP
Hum 1 (IMGT)	HVR-L1	337	GSYSS
	HVR-L2	338	IYS
	HVR-L3	339	GGYDQSSYT
Hum 7 (Chothia)	HVR-L1	164	SGGDYSSYYG
	HVR-L2	336	SDDKRPS
	HVR-L3	166	GGDYSTYANA

123

TABLE 5-continued

HVR delineations.			
Hum 7 (Kabat)	HVR-L1 HVR-L2 HVR-L3	164 336 166	SGGDYYSYYG SDDKRPS GGYDYSTYANA
Hum 7 (IMGT)	HVR-L1 HVR-L2 HVR-L3	340 338 190	GDYYS IYS GGYDYSTYA
AB21 (Chothia)	HVR-H1 HVR-H2 HVR-H3	191 192 193	GFTFSSN SAGGSD ETWNHLFDY
AB21 (Kabat)	HVR-H1 HVR-H2 HVR-H3	194 195 193	SNAMS GISAGGSDTYYPASVKG ETWNHLFDY
AB21 (IMGT)	HVR-H1 HVR-H2 HVR-H3	161 162 163	GFTFSSNA ISAGGSDT ARETNHLFDY
AB25 (Chothia)	HVR-H1 HVR-H2 HVR-H3	191 196 193	GFTFSSN SSGSD ETWNHLFDY
AB25 (Kabat)	HVR-H1 HVR-H2 HVR-H3	194 197 193	SNAMS GISSGSDTYYGDSVKG ETWNHLFDY
AB25 (IMGT)	HVR-H1 HVR-H2 HVR-H3	161 168 163	GFTFSSNA ISSGSDT ARETNHLFDY
AB27 (Chothia)	HVR-H1 HVR-H2 HVR-H3	198 199 193	GFRFSSY SSGGD ETWNHLFDY
AB27 (Kabat)	HVR-H1 HVR-H2 HVR-H3	200 201 193	SYAMS GISSGGDTYYVDSVKG ETWNHLFDY
AB27 (IMGT)	HVR-H1 HVR-H2 HVR-H3	173 174 163	GFRFSSYA ISSGGDT ARETNHLFDY
AB119 (Chothia)	HVR-H1 HVR-H2 HVR-H3	202 203 145	GFSFSNF GSGD DSTVSWSGDFFDY
AB119 (Kabat)	HVR-H1 HVR-H2 HVR-H3	204 144 145	NFAVT TIGSGDTYYADSVKG DSTVSWSGDFFDY
AB119 (IMGT)	HVR-H1 HVR-H2 HVR-H3	205 206 207	GFSFSNFA IGSGDT AKDSTVSWSGDFFDY
AB119 (Chothia)	HVR-L1 HVR-L2 HVR-L3	146 147 148	RASQNVKNDLA AARIRET QQYYDWPPFT
AB119 (Kabat)	HVR-L1 HVR-L2 HVR-L3	146 147 148	RASQNVKNDLA AARIRET QQYYDWPPFT
AB119 (IMGT)	HVR-L1 HVR-L2 HVR-L3	208 209 210	QNVKND AAR QQYYDWP
AB135 (Chothia)	HVR-H1 HVR-H2 HVR-H3	211 212 151	GFSFSIY GADD DSTVWWSGDFFDY
AB135 (Kabat)	HVR-H1 HVR-H2 HVR-H3	213 150 151	IYAVS TIGADDTYYADSVKG DSTVWWSGDFFDY

124

TABLE 5-continued

HVR delineations.			
5 AB135 (IMGT)	HVR-H1 HVR-H2 HVR-H3	214 215 216	GFSFSIYA IGADDT AKDSTVWWSGDFFDY
10 AB135 (Chothia)	HVR-L1 HVR-L2 HVR-L3	152 153 148	RASQNVRSIDIA AASSRDT QQYYDWPPFT
15 AB135 (IMGT)	HVR-L1 HVR-L2 HVR-L3	152 153 148	RASQNVRSIDIA AASSRDT QQYYDWPPFT
20 AB136 (Chothia)	HVR-H1 HVR-H2 HVR-H3	219 220 157	GFTFSSY SGSGEI ENNRYRFFDD
25 AB136 (IMGT)	HVR-H1 HVR-H2 HVR-H3	221 156 157	SYDVN LISGSGEIIYYADSVKG ENNRYRFFDD
30 AB136 (Chothia)	HVR-L1 HVR-L2 HVR-L3	222 223 224	GFTFSSYD ISGSGEII AKENNRYRFFDD
35 AB136 (IMGT)	HVR-L1 HVR-L2 HVR-L3	158 159 160	RASQSVYTYLA GASSRAT QQYYDRPPLT
40 AB3 (CHOTHIA)	HVR-H1 HVR-H2 HVR-H3	227 228 229	DYGMN QITSGSRITYGA AVKG DFGSGVGSIDA
45 AB3 (Chothia)	HVR-L1 HVR-L2 HVR-L3	230 231 234	GFIFSDY TSGSR GSYDGSIDI
50 AB45 (Kabat)	HVR-L1 HVR-L2 HVR-L3	232 233 234	SGSRGRYG RDNQRP GSYDGSIDI
55 AB45 (CHOTHIA)	HVR-L1 HVR-L2 HVR-L3	232 233 234	SGSRGRYG RDNQRP GSYDGSIDI
60 AB45 (Chothia)	HVR-L1 HVR-L2 HVR-L3	232 233 234	SGSRGRYG RDNQRP GSYDGSIDI
65 AB45 (Kabat)	HVR-L1 HVR-L2 HVR-L3	235 236 237	SYAMG GIDDDGSTANYGPAVKG ASVTGWSAHISGRLD
	HVR-L1 HVR-L2 HVR-L3	219 238 237	GFTFSSY DDGST ASVTGWSAHISGRLD
	HVR-L1 HVR-L2 HVR-L3	239 240 241	SGGGIYYYY ENDKRPS GGYDSNTTSGI
	HVR-L1 HVR-L2 HVR-L3	239 240 241	SGGGIYYYY ENDKRPS GGYDSNTTSGI

In some embodiments, an antibody of the present disclosure comprises 1, 2, 3, 4, 5, or 6 HVRs listed in Table 5 (e.g., a VL domain comprising 1, 2, or 3 light chain HVRs listed in Table 5 and/or a VH domain comprising 1, 2, or 3 heavy chain HVRs listed in Table 5).

In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 202, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 203, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:145; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:146, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:147, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 211, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 212, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:151; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:152, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:153, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 219, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 220, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:157;

127

and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:158, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:159, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:160.

In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 204, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:144, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:145; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:146, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:147, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 213, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:150, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:151; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:152, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:153, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:156, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:157; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:158, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:159, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:160.

In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 205, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 206, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 207; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO: 208, an HVR-L2 comprising the amino acid sequence of SEQ ID NO: 209, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO: 210. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 214, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 215, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 216; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO: 217, an HVR-L2 comprising the amino acid sequence of SEQ ID NO: 218, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 222, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 223, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 224; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO: 225, an HVR-L2 comprising the amino acid sequence of SEQ ID NO: 226, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:160.

In some embodiments of any of the above embodiments, the antibody enhances phagocytosis by a macrophage

128

expressing a human SIRP- α polypeptide, enhances activation of a dendritic cell expressing a human SIRP- α polypeptide, inhibits in vivo growth of a tumor or tumor cell(s) that expresses CD47, and/or does not prevent interactions between a CD47-expressing cell and a T cell. Exemplary assays for measuring phagocytosis, dendritic cell activation, tumor growth inhibition, and interactions between CD47-expressing cells and T cells (e.g., adhesion assays) are described herein and known in the art.

Antibody Production and Other Antibody Properties

An antibody of the present disclosure may be produced by any means known in the art. Exemplary techniques for antibody production are described below; however these exemplary techniques are provided for illustrative purposes only and are not intended to be limiting. In addition, exemplary antibody properties contemplated for use with the antibodies described herein are further described.

In some embodiments, an antibody that “binds” an antigen has a dissociation constant (K_D) for the antigen that is less than or equal to 1 μ M at 25° C. In some embodiments, an antibody of the present disclosure has a dissociation constant (K_D) for human v1 and/or v2 SIRP- α polypeptides that is less than or equal to 1 μ M at 25° C., less than or equal to 500 nM at 25° C., less than or equal to 400 nM at 25° C., less than or equal to 300 nM at 25° C., less than or equal to 250 nM at 25° C., less than or equal to 200 nM at 25° C., less than or equal to 200 nM at 25° C., less than or equal to 100 nM at 25° C., or less than or equal to 50 nM at 25° C. In some embodiments, an antibody that binds a human SIRP- α polypeptide and one or more non-human SIRP- α polypeptides binds the human SIRP- α polypeptide at a higher affinity (e.g., 10-fold or 100-fold higher) than the non-human SIRP- α polypeptide, though it still considered to “bind” both polypeptides. In some embodiments, an antibody that binds a non-human SIRP- α polypeptide and one or more human SIRP- α polypeptides binds the non-human SIRP- α polypeptide at a higher affinity (e.g., 10-fold or 100-fold higher) than the human SIRP- α polypeptide, though it still considered to “bind” both polypeptides. Assays for determining binding affinity are known in the art and include without limitation surface plasmon resonance (SPR), e.g., as described herein; radiolabeled antigen binding assay (RIA), e.g., using a Fab version of an antibody and its antigen; and the like. Other exemplary binding assays are described herein.

To prepare an antigen, the antigen may be purified or otherwise obtained from a natural source, or it may be expressed using recombinant techniques. In some embodiments, the antigen may be used as a soluble protein. In some embodiments, the antigen may be conjugate to another polypeptide or other moiety, e.g., to increase its immunogenicity. For example, an antigen described herein may be coupled with an Fc region. In some embodiments, a cell expressing the antigen on its cell surface may be used as the antigen.

Polyclonal antibodies can be raised in an animal by multiple subcutaneous (sc) or intraperitoneal (ip) injections of the antigen and an adjuvant. For example, descriptions of chicken immunization are described herein. In some embodiments, the antigen is conjugated with an immunogenic protein, e.g., keyhole limpet hemocyanin, serum albumin, bovine thyroglobulin, or soybean trypsin inhibitor using a bifunctional or derivatizing agent. Exemplary methods for immunization of chickens are provided herein. Relevant methods suitable for a variety of other organisms, such as mammals, are well known in the art.

As described supra, monoclonal antibodies may be produced by a variety of methods. In some embodiments, a monoclonal antibody of the present disclosure is made using the hybridoma method first described by Kohler et al., *Nature*, 256:495 (1975), and further described in Hongo et al., *Hybridoma*, 14 (3): 253-260 (1995); Harlow et al., *Antibodies: A Laboratory Manual*, (Cold Spring Harbor Laboratory Press, 2nd ed. 1988); and Hammerling et al., in: *Monoclonal Antibodies and T-Cell Hybridomas* 563-681 (Elsevier, N.Y., 1981). Human hybridoma technology (Trioma technology) is described in Vollmers and Brandlein, *Histology and Histopathology*, 20(3):927-937 (2005) and Vollmers and Brandlein, *Methods and Findings in Experimental and Clinical Pharmacology*, 27(3):185-91 (2005). A culture medium in which hybridoma cells are grown may be screened for the presence of an antibody of interest, e.g., by in vitro binding assay, immunoprecipitation, ELISA, RIA, etc.; and the binding affinity may be determined, e.g., by Scatchard analysis. A hybridoma that produces an antibody with desired binding properties can be subcloned and grown using known culture techniques, grown in vivo as ascites tumors in an animal, and the like.

In some embodiments, a monoclonal antibody is made using a library method, such as a phage display library. See, e.g., Hoogenboom et al. in *Methods in Molecular Biology* 178:1-37 (O'Brien et al., ed., Human Press, Totowa, NJ, 2001). In some embodiments, repertoires of VH and VL genes are cloned by polymerase chain reaction (PCR) and recombined randomly in phage libraries, which are then screened for antigen-binding phage, e.g., as described in Winter et al., *Ann. Rev. Immunol.*, 12: 433-455 (1994). Phage typically display antibody fragments, either as single-chain Fv (scFv) fragments or as Fab fragments. Alternatively, the naive repertoire can be cloned (e.g., from human) to provide a single source of antibodies to a wide range of non-self and also self-antigens without any immunization as described by Griffiths et al., *EMBO J.*, 12: 725-734 (1993). Finally, naive libraries can also be made synthetically by cloning unrearranged V-gene segments from stem cells, and using PCR primers containing random sequence to encode the highly variable CDR3 regions and to accomplish rearrangement in vitro, as described by Hoogenboom and Winter, *J. Mol. Biol.*, 227: 381-388 (1992).

In some embodiments, an antibody of the present disclosure is a chicken antibody. Chicken antibodies can be produced using various techniques known in the art; see, e.g., U.S. Pat. Nos. 6,143,559; 8,592,644; and 9,380,769.

In some embodiments, an antibody of the present disclosure is a chimeric antibody. See, e.g., U.S. Pat. No. 4,816,567 and Morrison et al., *Proc. Natl. Acad. Sci. USA*, 81:6851-6855 (1984). In some embodiments, a chimeric antibody comprises a non-human variable region (e.g., a variable region derived from a chicken, mouse, rat, hamster, rabbit, or non-human primate, such as a monkey) and a human constant region. In some embodiments, a chimeric antibody is a "class switched" antibody in which the class or subclass has been changed from that of the parent antibody. Chimeric antibodies include antigen-binding fragments thereof.

In some embodiments, a chimeric antibody is a humanized antibody. A non-human antibody can be humanized to reduce immunogenicity to humans, while retaining the specificity and affinity of the parental non-human antibody. Generally, a humanized antibody comprises one or more variable domains in which HVRs, e.g., CDRs, (or portions thereof) are derived from a non-human antibody (e.g., a chicken antibody), and FRs (or portions thereof) are derived

from human antibody sequences. A humanized antibody optionally will also comprise at least a portion of a human constant region. In some embodiments, some FR residues in a humanized antibody are substituted with corresponding residues from a non-human antibody (e.g., the antibody from which the HVR residues are derived), e.g., to restore or improve antibody specificity or affinity. Humanized antibodies and methods of making them are reviewed, e.g., in Almagro and Fransson, *Front. Biosci.* 13:1619-1633 (2008). Methods of humanizing a chicken antibody have also been described, e.g., in WO2005014653.

Human framework regions useful for humanization include but are not limited to: framework regions selected using the "best-fit" method; framework regions derived from the consensus sequence of human antibodies of a particular subgroup of light or heavy chain variable regions; human somatically mutated framework regions or human germline framework regions; and framework regions derived from screening FR libraries. See, e.g., Sims et al. *J. Immunol.* 151:2296 (1993); Carter et al. *Proc. Natl. Acad. Sci. USA*, 89:4285 (1992); Presta et al. *J. Immunol.*, 151:2623 (1993); Almagro and Fransson, *Front. Biosci.* 13:1619-1633 (2008); and Baca et al., *J. Biol. Chem.* 272:10678-10684 (1997).

In some embodiments, an antibody of the present disclosure is a human antibody. Human antibodies can be produced using various techniques known in the art. In some embodiments, the human antibody is produced by a non-human animal, such as the genetically engineered chickens (see, e.g., U.S. Pat. Nos. 8,592,644; and 9,380,769) and/or mice described herein. Human antibodies are described generally in Lonberg, *Curr. Opin. Immunol.* 20:450-459 (2008).

In some embodiments, an antibody of the present disclosure is generated by or derived from a chicken, e.g., using the methods described herein.

In some embodiments, an antibody of the present disclosure is an antibody fragment, including without limitation a Fab, F(ab')₂, Fab'-SH, Fv, or scFv fragment, or a single domain, single heavy chain, or single light chain antibody. Antibody fragments can be generated, e.g., by enzymatic digestion or by recombinant techniques. In some embodiments, Proteolytic digestion of an intact antibody is used to generate an antibody fragment, e.g., as described in Morimoto et al., *Journal of Biochemical and Biophysical Methods* 24:107-117 (1992) and Brennan et al., *Science*, 229:81 (1985). In some embodiments, an antibody fragment is produced by a recombinant host cell. For example, Fab, Fv and ScFv antibody fragments are expressed by and secreted from *E. coli*. Antibody fragments can alternatively be isolated from an antibody phage library.

Fab'-SH fragments can be directly recovered from *E. coli* and chemically coupled to form F(ab')₂ fragments. See Carter et al., *Bio Technology* 10:163-167 (1992). F(ab')₂ fragments can also be isolated directly from a recombinant host cell culture. Fab and F(ab')₂ fragment with increased in vivo half-life comprising salvage receptor binding epitope residues are described in U.S. Pat. No. 5,869,046.

In some embodiments, an antibody is a single chain Fv fragment (scFv). See WO 93/16185 and U.S. Pat. Nos. 5,571,894 and 5,587,458. scFv fusion proteins can be constructed to produce a fusion of an effector protein at either the amino or the carboxy terminus of an scFv. The antibody fragment may also be a "linear antibody", e.g., as described in U.S. Pat. No. 5,641,870, for example. Such linear antibodies may be monospecific or bispecific.

In some embodiments, an antibody of the present disclosure is a multispecific antibody. Multispecific antibodies

possess binding specificities against more than one antigen (e.g., having two, three, or more binding specificities). In some embodiments, the antibody is a bispecific antibody. In some embodiments, a bispecific antibody comprises two different binding specificities for the same antigen (e.g., having different binding affinity and/or specific epitope of the same antigen). In some embodiments, a bispecific antibody comprises binding specificities for two distinct antigens. In some embodiments, the bispecific antibody is a full-length or intact antibody. In some embodiments, the bispecific antibody is an antibody fragment of the present disclosure.

Bispecific or multispecific antibodies with a variety of combinations of binding specificities are contemplated herein. In some embodiments, the bispecific antibody has a first binding specificity for one or more SIRP- α polypeptides as described herein. In some embodiments, the bispecific antibody has a second binding specificity for an antigen expressed by a cancer cell, e.g., on the cell surface. Exemplary such antigens include without limitation CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PTK7, PD-L1, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le^v, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. Without wishing

deletion of favorable interactions. A hole refers to a void that is created when an original amino acid in a protein is replaced with a different amino acid having a smaller side-chain volume. A knob refers to a bump that is created when an original amino acid in a protein is replaced with a different amino acid having a larger side-chain volume. For example, in some embodiments, an amino acid being replaced is in the CH3 antibody constant domain of an Fc domain monomer and involved in the dimerization of two Fc domain monomers. In some embodiments, a hole in one CH3 antibody constant domain is created to accommodate a knob in another CH3 antibody constant domain, such that the knob and hole amino acids act to promote or favor the heterodimerization of the two Fc domain monomers. In some embodiments, a hole in one CH3 antibody constant domain is created to better accommodate an original amino acid in another CH3 antibody constant domain. In some embodiments, a knob in one CH3 antibody constant domain is created to form additional interactions with original amino acids in another CH3 antibody constant domain.

In some embodiments, a hole is constructed by replacing amino acids having larger side chains such as tyrosine or tryptophan with amino acids having smaller side chains such as alanine, valine, or threonine, for example a Y407V mutation in the CH3 antibody constant domain. Similarly, in some embodiments, a knob is constructed by replacing amino acids having smaller side chains with amino acids having larger side chains, for example a T366W mutation in the CH3 antibody constant domain. In some embodiments, one Fc domain monomer includes the knob mutation T366W and the other Fc domain monomer includes hole mutations T366S, L358A, and Y407V. In some embodiments, a polypeptide of the disclosure including a high affinity SIRP- α D1 variant is fused to an Fc domain monomer including the knob mutation T366W to limit unwanted knob-knob homodimer formation. Examples of knob-into-hole amino acid pairs are included, without limitation, in Table 3.

TABLE 3

Knob-into-hole amino acid pairs.								
Fc domain monomer 1	Y407A	Y407A	F405A	T394S	T366S L358A Y407V	T394W Y407T	T394S Y407A	T366W T394S
Fc domain monomer 2	T366Y	T366W	T394W	F405W	T366W	T366Y F405A	T366W F405W	F405W Y407A

to be bound to theory, it is thought that combining such a binding specificity with a binding specificity against a SIRP- α is particularly advantageous, e.g., to direct FcR-expressing leukocytes to target a tumor cell with the second binding specificity while also inhibiting the responsiveness of SIRP- α expressed by the leukocyte to any CD47 expressed by the tumor cell with the first binding specificity.

Various methods are known in the art for generating and purifying a bispecific antibody. Numerous approaches have been described. One approach is the "knobs-into-holes" or "protuberance-into-cavity" approach (see, e.g., U.S. Pat. No. 5,731,168). In some embodiments, heterodimerization of Fc domain monomers is promoted by introducing different, but compatible, substitutions in the two Fc domain monomers, such as "knob-into-hole" residue pairs and charge residue pairs. The knob and hole interaction favors heterodimer formation, whereas the knob-knob and the hole-hole interaction hinder homodimer formation due to steric clash and

Another approach uses antibody variable domains with the desired binding specificities (antibody-antigen combining sites) fused to immunoglobulin constant domain sequences, e.g., with an immunoglobulin heavy chain constant domain, comprising at least part of the hinge, CH2, and CH3 regions. In some embodiments, the bispecific antibody has a hybrid immunoglobulin heavy chain with a first binding specificity in one arm and a hybrid immunoglobulin heavy chain-light chain pair (providing a second binding specificity) in the other arm. See WO 94/04690. Another approach uses cross-linking (see, e.g., U.S. Pat. No. 4,676,980) to produce a heteroconjugate antibody. In some embodiments, bispecific antibodies can be prepared using chemical linkage (see, e.g., Brennan et al., *Science*, 229: 81 (1985)) to proteolytically cleave an intact antibody into F(ab')₂ fragments that are reduced in the presence of a dithiol complexing agent and converted to thionitrobenzoate (TNB) derivatives, one of which is reconverted to the Fab'-thiol by

reduction and mixed with the other Fab'-TNB derivative to form the bispecific antibody. In some embodiments, Fab'-SH fragments are chemically coupled. In some embodiments, bispecific antibody fragments are produced in cell culture using leucine zippers, as in Kostelny et al., *J. Immunol.*, 148(5):1547-1553 (1992). For other bispecific antibody formats, see, e.g., Spiess, C. et al. (2015) *Mol. Immunol.* 67:95-106.

In some embodiments, an antibody of the present disclosure is a diabody. See, e.g., Hollinger et al., *Proc. Natl. Acad. Sci. USA*, 90:6444-6448 (1993). In a diabody, the VH and VL domains of one fragment pair with complementary VL and VH domains of another fragment, thus forming two antigen-binding sites. Another strategy for making bispecific antibody fragments by the use of single-chain Fv (sFv) dimers has also been reported. See Gruber et al., *J. Immunol.*, 152:5368 (1994).

In some embodiments, an antibody of the present disclosure is a single-domain antibody. A single-domain antibody refers to a single polypeptide chain comprising all or a portion of the heavy chain variable domain or all or a portion of the light chain variable domain of an antibody. In certain embodiments, a single-domain antibody is a human single-domain antibody (see, e.g., U.S. Pat. No. 6,248,516 B1). In one embodiment, a single-domain antibody includes all or a portion of the heavy chain variable domain of an antibody. Camelid antibodies are also known.

Antibodies can be produced using recombinant methods. For recombinant production of an anti-antigen antibody, nucleic acid encoding the antibody is isolated and inserted into a replicable vector for further cloning (amplification of the DNA) or for expression. DNA encoding the antibody may be readily isolated and sequenced using conventional procedures (e.g., by using oligonucleotide probes that are capable of binding specifically to genes encoding the heavy and light chains of the antibody). Many vectors are available. The vector components generally include, but are not limited to, one or more of the following: a signal sequence, an origin of replication, one or more marker genes, an enhancer element, a promoter, and a transcription termination sequence.

An antibody of the present disclosure can be produced recombinantly as a fusion polypeptide with a heterologous polypeptide, e.g., a signal sequence or other polypeptide having a specific cleavage site at the N-terminus of the mature protein or polypeptide. The heterologous signal sequence selected can be one that is recognized and processed (e.g., cleaved by a signal peptidase) by the host cell. For prokaryotic host cells that do not recognize and process a native antibody signal sequence, the signal sequence is substituted by a prokaryotic signal sequence selected, for example, from alkaline phosphatase, penicillinase, lpp, or heat-stable enterotoxin II leaders. For yeast secretion the native signal sequence may be substituted by, e.g., the yeast invertase leader, a factor leader (including *Saccharomyces* and *Kluyveromyces* α -factor leaders), or acid phosphatase leader, the *C. albicans* glucoamylase leader, etc. In mammalian cell expression, mammalian signal sequences as well as viral secretory leaders, for example, the herpes simplex gD signal, are available.

Both expression and cloning vectors contain a nucleic acid sequence that enables the vector to replicate in one or more selected host cells, e.g., to allow the vector to replicate independently of the host chromosomal DNA. This sequence can include origins of replication or autonomously replicating sequences. Such sequences are well known for a variety of bacteria, yeast, and viruses. Generally, the origin

of replication component is not needed for mammalian expression vectors (the SV40 origin may be used because it contains the early promoter).

Expression and cloning vectors can contain a selection gene or selectable marker. Typical selection genes encode proteins that (a) confer resistance to antibiotics or other toxins, e.g., ampicillin, neomycin, methotrexate, or tetracycline, (b) complement auxotrophic deficiencies, or (c) supply critical nutrients not available from complex media. Examples of dominant selection use the drugs neomycin, mycophenolic acid and hygromycin. Another example of suitable selectable markers for mammalian cells are those that enable the identification of cells competent to take up antibody-encoding nucleic acid, such as DHFR, glutamine synthetase (GS), thymidine kinase, metallothionein-I and -II, preferably primate metallothionein genes, adenosine deaminase, ornithine decarboxylase, and the like. For example, a Chinese hamster ovary (CHO) cell line deficient in endogenous DHFR activity transformed with the DHFR gene is identified by culturing the transformants in a culture medium containing methotrexate (Mtx), a competitive antagonist of DHFR.

Alternatively, host cells (particularly wild-type hosts that contain endogenous DHFR) transformed or co-transformed with DNA sequences encoding an antibody of interest, wild-type DHFR gene, and another selectable marker such as aminoglycoside 3'-phosphotransferase (APH) can be selected by cell growth in medium containing a selection agent for the selectable marker such as an aminoglycosidic antibiotic, e.g., kanamycin, neomycin, or G418.

Expression and cloning vectors generally contain a promoter that is recognized by the host organism and is operably linked to nucleic acid encoding an antibody. Promoters suitable for use with prokaryotic hosts include the phoA promoter, β -lactamase and lactose promoter systems, alkaline phosphatase promoter, a tryptophan (trp) promoter system, and hybrid promoters such as the tac promoter. However, other known bacterial promoters are suitable. Promoter sequences are known for eukaryotes. Yeast promoters are well known in the art and can include inducible promoters/enhancers regulated by growth conditions. Virtually all eukaryotic genes have an AT-rich region located approximately 25 to 30 bases upstream from the site where transcription is initiated. Examples include without limitation the promoters for 3-phosphoglycerate kinase or other glycolytic enzymes, such as enolase, glyceraldehyde-3-phosphate dehydrogenase, hexokinase, pyruvate decarboxylase, phosphofructokinase, glucose-6-phosphate isomerase, 3-phosphoglycerate mutase, pyruvate kinase, triosephosphate isomerase, phosphoglucose isomerase, and glucokinase. Antibody transcription from vectors in mammalian host cells can be controlled, for example, by promoters obtained from the genomes of viruses. The early and late promoters of the SV40 virus are conveniently obtained as an SV40 restriction fragment that also contains the SV40 viral origin of replication. The immediate early promoter of the human cytomegalovirus is conveniently obtained as a HindIII E restriction fragment. Alternatively, the Rous Sarcoma Virus long terminal repeat can be used as the promoter.

Transcription of a DNA encoding an antibody of this invention by higher eukaryotes is often increased by inserting an enhancer sequence into the vector. Many enhancer sequences are now known from mammalian genes (globin, elastase, albumin, α -fetoprotein, and insulin). Typically, however, one will use an enhancer from a eukaryotic cell virus.

Expression vectors used in eukaryotic host cells (yeast, fungi, insect, plant, animal, human, or nucleated cells from other multicellular organisms) will also contain sequences necessary for the termination of transcription and for stabilizing the mRNA.

Suitable host cells for cloning or expressing the DNA in the vectors herein are the prokaryote, yeast, or higher eukaryote cells described above. Suitable prokaryotes for this purpose include eubacteria, such as Gram-negative or Gram-positive organisms, for example, Enterobacteriaceae such as *Escherichia*, e.g., *E. coli*, *Enterobacter*, *Erwinia*, *Klebsiella*, *Proteus*, *Salmonella*, e.g., *Salmonella typhimurium*, *Serratia*, e.g., *Serratia marcescans*, and *Shigella*, etc. In addition to prokaryotes, eukaryotic microbes such as filamentous fungi or yeast are suitable cloning or expression hosts for antibody-encoding vectors. *Saccharomyces cerevisiae*, or common baker's yeast, is the most commonly used among lower eukaryotic host microorganisms. Certain fungi and yeast strains may be selected in which glycosylation pathways have been "humanized," resulting in the production of an antibody with a partially or fully human glycosylation pattern. See, e.g., Li et al., *Nat. Biotech.* 24:210-215 (2006).

Plant cell cultures of cotton, corn, potato, soybean, petunia, tomato, duckweed (Lemnaceae), alfalfa (*M. truncatula*), and tobacco can also be utilized as hosts.

Suitable host cells for the expression of glycosylated antibody are also derived from multicellular organisms (invertebrates and vertebrates). Examples of invertebrate cells include plant and insect cells. Numerous baculoviral strains and variants and corresponding permissive insect host cells from hosts such as *Spodoptera frugiperda* (caterpillar), *Aedes aegypti* (mosquito), *Aedes albopictus* (mosquito), *Drosophila melanogaster* (fruitfly), and *Bombyx mori* have been identified.

Vertebrate cells may be used as hosts, and propagation of vertebrate cells in culture (tissue culture) has become a routine procedure. Examples of useful mammalian host cell lines are monkey kidney CV1 line transformed by SV40 (COS-7, ATCC CRL 1651); human embryonic kidney line (293 or 293 cells subcloned for growth in suspension culture, Graham et al., *J. Gen. Virol.* 36:59 (1977)); baby hamster kidney cells (BHK, ATCC CCL 10); mouse sertoli cells (TM4, Mather, *Biol. Reprod.* 23:243-251 (1980)); monkey kidney cells (CV1 ATCC CCL 70); African green monkey kidney cells (VERO-76, ATCC CRL-1587); human cervical carcinoma cells (HELA, ATCC CCL 2); canine kidney cells (MDCK, ATCC CCL 34); buffalo rat liver cells (BRL 3A, ATCC CRL 1442); human lung cells (W138, ATCC CCL 75); human liver cells (Hep G2, HB 8065); mouse mammary tumor (MMT 060562, ATCC CCL51); TRI cells (Mather et al., *Annals N.Y. Acad. Sci.* 383:44-68 (1982)); MRC 5 cells; FS4 cells; and a human hepatoma line (Hep G2). Other useful mammalian host cell lines include Chinese hamster ovary (CHO) cells, including DHFR-CHO cells (Urlaub et al., *Proc. Natl. Acad. Sci. USA* 77:4216 (1980)); and myeloma cell lines such as NSO and Sp2/0. For a review of certain mammalian host cell lines suitable for antibody production, see, e.g., Yazaki and Wu, *Methods in Molecular Biology*, Vol. 248 (B. K. C. Lo, ed., Humana Press, Totowa, N.J., 2003), pp. 255-268.

The host cells of the present disclosure may be cultured in a variety of media. Commercially available media such as Ham's F10 (Sigma), Minimal Essential Medium ((MEM), (Sigma), RPMI-1640 (Sigma), and Dulbecco's Modified Eagle's Medium ((DMEM), Sigma) are suitable for culturing the host cells. In addition, any of the media described in Ham et al., *Meth. Enz.* 58:44 (1979), Barnes et al., *Anal. Biochem.* 102:255 (1980), U.S. Pat. Nos. 4,767,704; 4,657,866; 4,927,762; 4,560,655; or 5,122,469; WO 90/03430; WO 87/00195; or U.S. Pat. Re. 30,985 may be used as culture media for the host cells. Any of these media may be supplemented as necessary with hormones and/or other growth factors (such as insulin, transferrin, or epidermal growth factor), salts (such as sodium chloride, calcium, magnesium, and phosphate), buffers (such as HEPES), nucleotides (such as adenosine and thymidine), antibiotics (such as GENTAMYCIN™ drug), trace elements (defined as inorganic compounds usually present at final concentrations in the micromolar range), and glucose or an equivalent energy source. Any other necessary supplements may also be included at appropriate concentrations that would be known to those skilled in the art. The culture conditions, such as temperature, pH, and the like, are those previously used with the host cell selected for expression, and will be apparent to one of skill in the art.

When using recombinant techniques, the antibody can be produced intracellularly, in the periplasmic space, or directly secreted into the medium. If the antibody is produced intracellularly, as a first step, the particulate debris, either host cells or lysed fragments, are removed, for example, by centrifugation or ultrafiltration. Carter et al., *Bio Technology* 10:163-167 (1992) describe a procedure for isolating antibodies which are secreted to the periplasmic space of *E. coli*.

The antibody composition prepared from the cells can be purified using, for example, hydroxylapatite chromatography, hydrophobic interaction chromatography, gel electrophoresis, dialysis, and affinity chromatography, with affinity chromatography being among one of the typically preferred purification steps.

In some embodiments, an antibody of the present disclosure comprises a kappa or lambda light chain constant region. In some embodiments, an antibody of the present disclosure comprises a light chain constant region comprising the amino acid sequence of SEQ ID NO:325, 326, or 426. Exemplary and non-limiting light chain constant region sequences are provided in Table 6. In some embodiments, an antibody of the present disclosure comprises an IGLC3 lambda light chain constant region or an IGLC7 constant region.

In some embodiments, an antibody of the present disclosure includes an Fc region. For example, in some embodiments, the Fc region is a human Fc region, e.g., IgG1, IgG2, or IgG4 and subtypes thereof. Exemplary and non-limiting Fc regions are provided within the amino acid sequences of SEQ ID NOs:320-324 shown in Table 6. In some embodiments, an Fc region within one or more of the amino acid sequences of SEQ ID NOs:320-324 comprises one or more of the mutations described herein, e.g., infra.

TABLE 6

Exemplary constant region sequences		
Name	SEQ ID NO	Sequence
IgG1 wildtype	320	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTWNS GALTSGVHTFPAVLQSS GLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC DKHTHTCPPCPAPELLGG PSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD GVEVHNAKTKPREEQYN STYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREE MTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLD SDGSFFLYSKLTVDKSRW QQGNVFCSCVMHEALHNYHTQKSLSLSPGK
IgG1_AAA_N297A	321	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTWNS GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVN HKPSNTKVDKKVEPKSCDKHTHTCPPCPAPEAAGAPSVFLFPPK PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKAL PAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF YPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKKS RWQQGNVFCSCVMHEALHNYHTQKSLSLSPGK
IgG2	322	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTWNS GALTSGVHTFPAVLQSSG LYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKTKVERKCC VECPCCPAPPVAGPSVFL FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVQFNWYVDGVEV HNAKTKPREEQFNSTFRV VSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPR EPQVYTLPPSREEMTKNQ VSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPMLSDGSFF LYSKLTVDKSRWQQGNV FSCVMHEALHNYHTQKSLSLSPGK
IgG2Da	323	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTWNS GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVN HKPSNTKVDKTKVERKCCVECPCCPAPPVAGPSVFLFPPKPKDT LMISRTPEVTCVVVDVSHEDPEVQFNWYVDGVEVHNAKTKP REEQFNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPSSIE KTIKTKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDI AVEWESNGQPENNYKTPPMLSDGSFFLYSKLTVDKSRWQ QGNVFCSCVMHEALHNYHTQKSLSLSPGK
IgG4_S228P	324	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTWNS GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVN HKPSNTKVDKRVESKYGPCCPAPAEFLGGPSVFLFPPKPKDT LMISRTPEVTCVVVDVSDQEDPEVQFNWYVDGVEVHNAKTKP REEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIE KTIKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSD IAVEWESNGQPENNYKTPPVLDSDGSFFLYSRLTVDKSRWQ EGNVFCSCVMHEALHNYHTQKSLSLSLGK
Human Kappa	325	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKV DNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVY ACEVTHQGLSSPVTKSFNRGEC
Human Lambda IGLC1	326	GQPKANPTVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK ADGSPVKAGVETTKPSKQSNNKYAASSYLSLTPEQWKSHRSY SCQVTHEGSTVEKTVAPTECS
Human Lambda IGLC2	426	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK ADSSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHRSY SCQVTHEGSTVEKTVAPTECS

In some embodiments, the Fc region includes one or more mutations that influence one or more antibody properties, such as stability, pattern of glycosylation or other modifications, effector cell function, pharmacokinetics, and so forth. In some embodiments, an antibody of the present disclosure has reduced or minimal glycosylation. In some embodiments, an antibody of the present disclosure has ablated or reduced effector function. Exemplary Fc mutations include without limitation (i) a human IgG1 Fc region

mutations L234A, L235A, G237A, and N297A; (ii) a human IgG2 Fc region mutations A330S, P331S and N297A; and (iii) a human IgG4 Fc region mutations S228P, E233P, F234V, L235A, delG236, and N297A (EU numbering). In some embodiments, the human IgG2 Fc region comprises A330S and P331S mutations. In some embodiments, the human IgG4 Fc region comprises an S288P mutation. In some embodiments, the human IgG4 Fc region comprises S288P and L235E mutations.

Antibodies that target cell surface antigens can trigger immunostimulatory and effector functions that are associated with Fc receptor (FcR) engagement on immune cells. There are a number of Fc receptors that are specific for particular classes of antibodies, including IgG (gamma receptors), IgE (eta receptors), IgA (alpha receptors) and IgM (mu receptors). Binding of the Fc region to Fc receptors on cell surfaces can trigger a number of biological responses including phagocytosis of antibody-coated particles (antibody-dependent cell-mediated phagocytosis, or ADCP), clearance of immune complexes, lysis of antibody-coated cells by killer cells (antibody-dependent cell-mediated cytotoxicity, or ADCC) and, release of inflammatory mediators, placental transfer, and control of immunoglobulin production. Additionally, binding of the C1 component of complement to antibodies can activate the complement system. Activation of complement can be important for the lysis of cellular pathogens. However, the activation of complement can also stimulate the inflammatory response and can also be involved in autoimmune hypersensitivity or other immunological disorders. Variant Fc regions with reduced or ablated ability to bind certain Fc receptors are useful for developing therapeutic antibodies and Fc-fusion polypeptide constructs which act by targeting, activating, or neutralizing ligand functions while not damaging or destroying local cells or tissues.

In some embodiments, a Fc domain monomer refers to a polypeptide chain that includes second and third antibody constant domains (e.g., CH2 and CH3). In some embodiments, an Fc domain monomer also includes a hinge domain. In some embodiments, the Fc domain monomer is of any immunoglobulin antibody isotype, including IgG, IgE, IgM, IgA, and IgD. Additionally, in some embodiments, an Fc domain monomer is of any IgG subtype (e.g., IgG1, IgG2, IgG2a, IgG2b, IgG2c, IgG3, and IgG4). In some embodiments, Fc domain monomers include as many as ten changes from a wild-type Fc domain monomer sequence (e.g., 1-10, 1-8, 1-6, 1-4 amino acid substitutions, additions or insertions, deletions, or combinations thereof) that alter the interaction between an Fc domain and an Fc receptor.

In some embodiments, an Fc domain monomer of an immunoglobulin or a fragment of an Fc domain monomer is capable of forming an Fc domain with another Fc domain monomer. In some embodiments, an Fc domain monomer of an immunoglobulin or a fragment of an Fc domain monomer is not capable of forming an Fc domain with another Fc domain monomer. In some embodiments, an Fc domain monomer or a fragment of an Fc domain is fused to a polypeptide of the disclosure to increase serum half-life of the polypeptide. In some embodiments, an Fc domain monomer or a fragment of an Fc domain monomer fused to a polypeptide of the disclosure dimerizes with a second Fc domain monomer to form an Fc domain which binds an Fc receptor, or alternatively, an Fc domain monomer binds to an Fc receptor. In some embodiments, an Fc domain or a fragment of the Fc domain fused to a polypeptide to increase serum half-life of the polypeptide does not induce any immune system-related response. An Fc domain includes two Fc domain monomers that are dimerized by the interaction between the CH3 antibody constant domains.

A wild-type Fc domain forms the minimum structure that binds to an Fc receptor, e.g., FcγRI, FcγRIIa, FcγRIIb, FcγRIIIa, FcγRIIIb, and FcγRIV. In some embodiments, the Fc domain in an antibody of the present disclosure comprises one or more amino acid substitutions, additions or insertions, deletions, or any combinations thereof that lead

to decreased effector function such as decreased antibody-dependent cell-mediated cytotoxicity (ADCC), decreased complement-dependent cytotoxicity (CDC), decreased antibody-dependent cell-mediated phagocytosis (ADCP), or any combinations thereof. For example, an antibody of the present disclosure can exhibit decreased binding (e.g., minimal binding or absence of binding) to a human Fc receptor and decreased binding (e.g., minimal binding or absence of binding) to complement protein C1q; decreased binding (e.g., minimal binding or absence of binding) to human FcγRI, FcγRIIa, FcγRIIb, FcγRIIIa, FcγRIIIb, or any combinations thereof, and C1q; altered or reduced antibody-dependent effector function, such as ADCC, CDC, ADCP, or any combinations thereof, and so forth. Exemplary mutations include without limitation one or more amino acid substitutions at E233, L234, L235, G236, G237, D265, D270, N297, E318, K320, K322, A327, A330, P331, or P329 (numbering according to the EU index of Kabat (Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD (1991))).

In some embodiments, an antibody of the present disclosure has reduced or ablated binding to CD16a, CD32a, CD32b, CD32c, and CD64 Fcγ receptors. In some embodiments, an antibody with a non-native Fc region described herein exhibits at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in C1q binding compared to an antibody comprising a wild-type Fc region. In some embodiments, an antibody with a non-native Fc region as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in CDC compared to an antibody comprising a wild-type Fc region.

In some embodiments, the Fc variants herein are minimally glycosylated or have reduced glycosylation relative to a wild-type sequence. In some embodiments, deglycosylation is accomplished with a mutation of N297A, or by mutating N297 to any amino acid which is not N.

In some embodiments, variants of antibody IgG constant regions (e.g., Fc variants) possess a reduced capacity to specifically bind Fcγ receptors or have a reduced capacity to induce phagocytosis. In some embodiments, variants of antibody IgG constant regions (e.g., Fc variants) possess a reduced capacity to specifically bind Fcγ receptors and have a reduced capacity to induce phagocytosis. For example, in some embodiments, an Fc domain is mutated to lack effector functions, typical of a "dead" Fc domain. For example, in some embodiments, an Fc domain includes specific amino acid substitutions that are known to minimize the interaction between the Fc domain and an Fcγ receptor. In some embodiments, an Fc domain monomer is from an IgG1 antibody and includes one or more of amino acid substitutions L234A, L235A, G237A, and N297A (as designated according to the EU numbering system per Kabat et al., 1991). In some embodiments, one or more additional mutations are included in such IgG1 Fc variant. Non-limiting examples of such additional mutations for human IgG1 Fc variants include E318A and K322A. In some instances, a human IgG1 Fc variant has up to 12, 11, 10, 9, 8, 7, 6, 5 or 4 or fewer mutations in total as compared to wild-type human IgG1 sequence. In some embodiments, one or more additional deletions are included in such IgG1 Fc variant. For example, in some embodiments, the C-terminal lysine of the Fc IgG1 heavy chain constant region is deleted, for example to increase the homogeneity of the polypeptide when the polypeptide is produced in bacterial or mammalian cells. In some instances, a human IgG1 Fc variant has up to

12, 11, 10, 9, 8, 7, 6, 5 or 4 or fewer deletions in total as compared to wild-type human IgG1 sequence.

In some embodiments, an Fc domain monomer is from an IgG2 antibody and includes amino acid substitutions A330S, P331S, or both A330S and P331S. The aforementioned amino acid positions are defined according to Kabat, et al. (1991). The Kabat numbering of amino acid residues can be determined for a given antibody by alignment at regions of homology of the sequence of the antibody with a "standard" Kabat numbered sequence. In some embodiments, the Fc variant comprises a human IgG2 Fc sequence comprising one or more of A330S, P331S and N297A amino acid substitutions (as designated according to the EU numbering system per Kabat, et al. (1991)). In some embodiments, one or more additional mutations are included in such IgG2 Fc variants. Non-limiting examples of such additional mutations for human IgG2 Fc variant include V234A, G237A, P238S, V309L and H268A (as designated according to the EU numbering system per Kabat et al. (1991)). In some instances, a human IgG2 Fc variant has up to 12, 11, 10, 9, 8, 7, 6, 5, 4, 3 or fewer mutations in total as compared to wild-type human IgG2 sequence. In some embodiments, one or more additional deletions are included in such IgG2 Fc variant.

When the Fc variant is an IgG4 Fc variant, in some embodiments, such Fc variant comprises a S228P, E233P, F234V, L235A, L235E, or delG236 mutation (as designated according to Kabat, et al. (1991)). In some instances, a human IgG4 Fc variant has up to 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2 or 1 mutation(s) in total as compared to wild-type human IgG4 sequence.

In some embodiments, the Fc variant exhibits reduced binding to an Fc receptor of the subject compared to the wild-type human IgG Fc region. In some embodiments, the Fc variant exhibits ablated binding to an Fc receptor of the subject compared to the wild-type human IgG Fc region. In some embodiments, the Fc variant exhibits a reduction of phagocytosis compared to the wild-type human IgG Fc region. In some embodiments, the Fc variant exhibits ablated phagocytosis compared to the wild-type human IgG Fc region.

Antibody-dependent cell-mediated cytotoxicity, which is also referred to herein as ADCC, refers to a form of cytotoxicity in which secreted Ig bound onto Fc receptors (FcRs) present on certain cytotoxic cells (e.g., Natural Killer (NK) cells and neutrophils) enabling these cytotoxic effector cells to bind specifically to an antigen-bearing target cell and subsequently kill the target cell. Antibody-dependent cell-mediated phagocytosis, which is also referred to herein as ADCP, refers to a form of cytotoxicity in which secreted Ig bound onto Fc receptors (FcRs) present on certain phagocytic cells (e.g., macrophages) enabling these phagocytic effector cells to bind specifically to an antigen-bearing target cell and subsequently engulf and digest the target cell. Ligand-specific high-affinity IgG antibodies directed to the surface of target cells can stimulate the cytotoxic or phagocytic cells and can be used for such killing. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit reduced ADCC or ADCP as compared to a polypeptide construct comprising a wild-type Fc region. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in ADCC or ADCP compared to a polypeptide construct comprising a wild-type Fc region. In some embodiments, antibodies comprising an Fc variant as

described herein exhibit ablated ADCC or ADCP as compared to a polypeptide construct comprising a wild-type Fc region.

Complement-directed cytotoxicity, which is also referred to herein as CDC, refers to a form of cytotoxicity in which the complement cascade is activated by the complement component C1q binding to antibody Fc. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in C1q binding compared to a polypeptide construct comprising a wild-type Fc region. In some cases, polypeptide constructs comprising an Fc variant as described herein exhibit reduced CDC as compared to a polypeptide construct comprising a wild-type Fc region. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in CDC compared to a polypeptide construct comprising a wild-type Fc region. In some cases, antibodies comprising an Fc variant as described herein exhibit negligible CDC as compared to a polypeptide construct comprising a wild-type Fc region.

Fc variants herein include those that exhibit reduced binding to an Fcγ receptor compared to the wild-type human IgG Fc region. For example, in some embodiments, an Fc variant exhibits binding to an Fcγ receptor that is less than the binding exhibited by a wild-type human IgG Fc region to an Fcγ receptor. In some instances, an Fc variant has reduced binding to an Fcγ receptor by a factor of 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99% or 100% (fully ablated effector function). In some embodiments, the reduced binding is for any one or more Fcγ receptors, e.g., CD16a, CD32a, CD32b, CD32c, or CD64.

In some instances, the Fc variants disclosed herein exhibit a reduction of phagocytosis compared to its wild-type human IgG Fc region. Such Fc variants exhibit a reduction in phagocytosis compared to its wild-type human IgG Fc region, wherein the reduction of phagocytosis activity is e.g., by a factor of 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99% or 100%. In some instances, an Fc variant exhibits ablated phagocytosis compared to its wild-type human IgG Fc region.

In some embodiments, the Fc variants disclosed herein are coupled to one or more fusion partners. In some cases the fusion partner is a therapeutic moiety, such as a cytotoxic agent of the present disclosure. In some cases, the fusion partner is selected to enable targeting of an expressed protein, purification, screening, display, and the like. In some embodiments, the fusion partner also affects the degree of binding to Fc receptors or the degree of phagocytosis reduction.

In some embodiments, fusion partners are linked to the Fc variant sequence via a linker sequence. In some embodiments, the linker sequence generally comprises a small number of amino acids, such as less than ten amino acids, although longer linkers are also utilized. In some cases, the linker has a length less than 10, 9, 8, 7, 6, or 5 amino acids or shorter. In some cases, the linker has a length of at least 10, 11, 12, 13, 14, 15, 20, 25, 30, or 35 amino acids or longer. Optionally, in some embodiments, a cleavable linker is employed.

In some embodiments, a fusion partner is a targeting or signal sequence that directs an Fc variant protein and any associated fusion partners to a desired cellular location or to the extracellular media. In some embodiments, certain sig-

naling sequences target a protein to be either secreted into the growth media, or into the periplasmic space, located between the inner and outer membrane of the cell. In some embodiments, a fusion partner is a sequence that encodes a peptide or protein that enables purification or screening. Such fusion partners include, but are not limited to, polyhistidine tags (His-tags) (for example His6 and His10) or other tags for use with Immobilized Metal Affinity Chromatography (IMAC) systems (e.g., Ni+2 affinity columns), GST fusions, MBP fusions, Strep-tag, the BSP biotinylation target sequence of the bacterial enzyme BirA, and epitope tags which are targeted by antibodies (for example c-myc tags, flag-tags, and the like).

In some embodiments, such tags are useful for purification, for screening, or both. For example, in some embodiments, an Fc variant is purified using a His-tag by immobilizing it to a Ni+2 affinity column, and then after purification the same His-tag is used to immobilize the antibody to a Ni+2 coated plate to perform an ELISA or other binding assay.

Various fusion partners that enable a variety of selection methods are available. For example, by fusing the members of an Fc variant library to the gene III protein, phage display can be employed. In some embodiments, fusion partners enable Fc variants to be labeled. Alternatively, in some embodiments, a fusion partner binds to a specific sequence on the expression vector, enabling the fusion partner and associated Fc variant to be linked covalently or noncovalently with the nucleic acid that encodes them.

In some embodiments, when a fusion partner is a therapeutic moiety, the therapeutic moiety is, e.g., a cytotoxic agent, a peptide, a protein, an antibody, a siRNA, or a small molecule.

In some embodiments, an antibody of the present disclosure is bound to various carriers or labels and used to detect the presence of specific antigen expressing cells. Examples of carriers include glass, polystyrene, polypropylene, polyethylene, dextran, nylon, amylases, natural and modified celluloses, polyacrylamides, agaroses, and magnetite. The nature of the carrier can be either soluble or insoluble. Various different labels and methods of labeling are known. Examples of labels include enzymes, radioisotopes, fluorescent compounds, colloidal metals, chemiluminescent compounds, and bio-luminescent compounds. Various techniques for binding labels to antibodies disclosed herein are available. In some embodiments, the antibodies are coupled to low molecular weight haptens. These haptens are then specifically detected by means of a second reaction. For example, in some embodiments, the hapten biotin is used with avidin or the haptens dinitrophenol, pyridoxal, or fluorescein are detected with specific anti-hapten antibodies (e.g., anti-dinitrophenol antibodies, anti-pyridoxal antibodies, and anti-fluorescein antibodies respectively). In some embodiments, the antibodies described herein are utilized in vitro for binding assays, such as immune assays. For example, in some embodiments, the antibodies are utilized in liquid phase or bound to a solid phase carrier. In some embodiments, antibodies utilized for immunoassays are detectably labeled in various ways.

Methods of Identifying and/or Generating Antibodies

Certain aspects of the present disclosure relate to methods of identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide. Advantageously, the methods described herein can be used to identify antigen binding domains that block binding between human CD47 and a human SIRP- α polypeptide, do not block binding between

human CD47 and a human SIRP- α polypeptide, or reduce affinity of a human SIRP- α polypeptide for human CD47.

In some embodiments, the methods include providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide. Exemplary antigen binding domains and antibodies that bind the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide, and/or other SIRP polypeptides, are described herein, as are exemplary methods for identifying such antigen binding domains/antibodies. Exemplary methods are described in greater detail in the Examples infra.

In some embodiments, the methods include assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47. In some embodiments, "assembling" the complex includes providing a solution containing both a SIRP- α D1 variant and a polypeptide comprising an IgSF domain of CD47. In some embodiments, the SIRP- α D1 variant is a non-naturally occurring variant, e.g., that binds to human CD47 with an affinity that is at least 10-fold, at least 100-fold, or at least 100-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. Advantageously, this facilitates antibody screening, as a natural SIRP- α :CD47 interaction may be too weak for use in binding and screening assays. Exemplary variants are described in greater detail infra.

In some embodiments, the methods include contacting the antigen binding domain with the assembled complex. In some embodiments, binding, or a lack or deficiency thereof, of the antigen binding domain to the complex is detected. Various detection techniques are described herein. In some embodiments, SPR or ELISA is used. Detectable binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. A lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. In addition, SPR is used to distinguish blocking, non-blocking and kick-off antibodies. For example, in some embodiments, a non-blocking antibody increases RUs when injected on top of pre-formed SIRP- α :CD47 complex, a kick-off antibody increases K_{off} of a pre-formed SIRP- α :CD47 complex, and blocking antibody does not change the RUs or K_{off} of a pre-formed SIRP- α :CD47 complex.

In some embodiments, the IgSF domain of CD47 is a human IgSF domain. In some embodiments, the polypeptide comprising the IgSF domain of CD47 comprises a human CD47 extracellular domain. In some embodiments, the IgSF domain of CD47 comprises the amino acid sequence of QLLFNKTKSVEFTFSNDTVVIPCFVTNMEAQNTTE-VYVKWKFGRDIYTFDGLNKSTV PTFSSAKI-EVSQLLKGDASLKMDSKSDAVSHTGNYTCEVTEL-TREGETHIELKYRVVS (SEQ ID NO:16). In some embodiments, the polypeptide comprising the IgSF domain of CD47 is conjugated to another polypeptide or other moiety, e.g., an Ig Fc region.

High Affinity SIRP- α D1 Domain Variants

A variety of high affinity SIRP- α D1 variants are contemplated for use herein. For example, in certain embodiments, the SIRP- α D1 variant comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 17-52 (see Table 4). Further descriptions of SIRP- α D1 variants follow.

TABLE 4

Exemplary SIRP- α D1 variant amino acid sequences.		
SEQ ID NO	Sequence	
17	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQREG PFPRVTTVSDTTKRNNMDFSIRIGAITPADAGTYCYCKFRKGSPDDVEFKSGAG TELSVRAKPS	
18	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
19	EEELQVIQPDKSVSVAAGESAILHCTVTSLFPVGPIQWFRGAGPARELIYNQRQ GPFRVTTVSESTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGT ELSVRAKPS	
20	EEELQVIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
21	EEELQIIQPDKSVSVAAGESAILHCTVTSLFPVGPIQWFRGAGPARVLIYNQRQG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
22	EEELQIIQPDKSVSVAAGESAILHCTITSLIPVGPIQWFRGAGPARVLIYNQRQGP FPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTEL SVRAKPS	
23	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQRQG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
24	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQKQG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
25	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQREG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
26	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG HFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
27	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG PFPRVTTVSESTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTEL SVRAKPS	
28	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
29	EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQREG PFPRVTTVSESTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
30	EEELQVIQPDKSVSVAAGESAILHCTVTSLFPVGPIQWFRGAGPARELIYNQRE GPFRVTTVSESTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGT ELSVRAKPS	
31	EEELQVIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREG PFPRVTTVSESTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
32	EEELQVIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREG PFPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
33	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREGP FPRVTTVSESTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTEL SVRAKPS	
34	EEELQVIQPDKSVSVAAGESAILHCTITSLIPVGPIQWFRGAGPARELIYNQREG PFPRVTTVSESTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTE LSVRAKPS	
35	EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREGP FPRVTTVSETTKRENMDFSISISNITPADAGTYCYCKFRKGSPDTEFKSGAGTEL SVRAKPS	

TABLE 4-continued

Exemplary SIRP- α D1 variant amino acid sequences.		
SEQ ID NO	Sequence	
36	EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDLTNRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
37	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG PFPFRVTTVSDLTNRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSGAG TELSVRAKPS	
38	EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDTTRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
39	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG PFPFRVTTVSDTTRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSGAG TELSVRAKPS	
40	EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDLTNRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
41	EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDLTNRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
42	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG PFPFRVTTVSDTTRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSGAG TELSVRAKPS	
43	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG PFPFRVTTVSDTTRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGAG TELSVRAKPS	
44	EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDLTNRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
45	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG PFPFRVTTVSDTTRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGAG TELSVRAKPS	
46	EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDLTNRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
47	EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDLTNRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
48	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG PFPFRVTTVSDLTNRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGAG TELSVRAKPS	
49	EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG GPFPRVTTVSDTTRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGA GTELSVRAKPS	
50	EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPQWFRGAGPGRELIYNQKE GHFPRVTTVSDLTNRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFKSG AGTELSVRAKPS	
51	EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPQWFRGAGPGRELIYNQREG PFPFRVTTVSDTTRNMDFSIIRIGAITPADAGTYTCVKFRKGSDDVEFKSGAG TELSVRAKPS	
52	EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPQWFRGAGPGRELIYNQR QGPFPRVTTVSDLTNRNMDFSIIRIGNITPADAGTYTCVKFRKGSDDVEFK SGAGTELSVRAKPS	

In some embodiments, a SIRP- α D1 variant polypeptide or fragment thereof binds to CD47 with a K_D less than 1×10^{-8} M, less than 5×10^{-9} M, less than 1×10^{-9} M, less than 5×10^{-10} M, less than 1×10^{-10} M or less than 1×10^{-11} M. In some embodiments, a SIRP- α D1 variant polypeptide or fragment thereof binds to CD47 with a K_D between about

500 nM and 100 nM, between about 100 nM and 50 nM, between about 50 nM and 10 nM, between about 10 nM and 5 nM, between about 5 nM and 1 nM, between about 1 nM and 500 pM, between about 500 pM and 100 pM, between about 100 pM and 50 pM, or between about 50 pM and 10 pM.

In some embodiments, fragments include polypeptides of less than 10 amino acids in length, about 10 amino acids in length, about 20 amino acids in length, about 30 amino acids in length, about 40 amino acids in length, about 50 amino acids in length, about 60 amino acids in length, about 70 amino acids in length, about 80 amino acids in length, about 90 amino acids in length, about 100 amino acids in length, or more than about 100 amino acids in length. Fragments retain the ability to bind to CD47. Preferably, SIRP- α D1 variant polypeptides and fragments thereof bind to CD47 with a higher affinity than a SIRP- α polypeptide binds to CD47.

In some embodiments, the above-mentioned SIRP- α D1 variant polypeptides are attached or fused to a second polypeptide. In some embodiments, the second polypeptide includes, without limitation, an Fc polypeptide, an Fc variant, an HSA polypeptide, an albumin peptide, a PEG polymer or a fragment of the foregoing.

In some embodiments, the polypeptide includes a high affinity SIRP- α D1 domain that has at least 85% sequence identity (e.g., at least 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity) to any variant provided in Table 4.

Antibody Generation in Chicken

Certain aspects of the present disclosure relate to methods of producing an anti-SIRP- α antibody that binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides. Without wishing to be bound to theory, it is thought that the use of chicken is particularly advantageous because of the greater diversity between chicken SIRP- α and mammalian (e.g., human, monkey, mouse, etc.) SIRP- α . Further, the phylogenetic distance between chickens and mammals allows for the identification of antibodies that cross-react against, e.g., human and mouse SIRP- α polypeptides, which can be difficult to achieve by producing antibodies in mouse due to self-tolerance.

In some embodiments, the methods include immunizing a chicken with a peptide comprising at least a portion of a human SIRP- α extracellular domain (e.g., the D1 domain). An exemplary immunization schedule is described infra. Methods for chicken immunization are described, e.g., in Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16. In some embodiments, the methods include obtaining an antibody from an antibody-producing cell from the immunized chicken.

In some embodiments, the methods include detecting binding between the antibody obtained from the cell and the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides. For example, in some embodiments, human SIRP- α v1 and v2, e.g., as described herein, are used. Exemplary detection techniques are described herein and include without limitation the GEM assay (see, e.g., WO2009111014 and Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16), SPR, and ELISA.

Methods of Treatment

Certain aspects of the present disclosure relate to treating a disease or disorder using an antibody described herein. In some embodiments, the disease is cancer. In some embodiments, the disease is an autoimmune or inflammatory disease.

For example, provided herein are methods of treating or delaying progression of cancer in an individual by administering an effective amount of an antibody of the present disclosure. Without wishing to be bound to theory, it is thought that the antibodies described herein may be useful in the treatment of cancer, e.g., by abrogating the cancer's

ability to inhibit phagocytosis and immune surveillance through the CD47: SIRP- α signaling axis, or by otherwise enhancing activation of the immune system (such as by activation of dendritic cells).

In some embodiments, an antibody of the present disclosure is administered in combination with a chemotherapeutic agent.

In some embodiments, an antibody of the present disclosure is administered in combination with a second antibody, e.g., an antibody that binds an antigen expressed by the cancer. Exemplary antigens expressed by cancers are known in the art and include without limitation CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PTK7, PD-L1, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le^x, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. For example, in some embodiments, an antibody of the present disclosure is administered in combination with a monoclonal antibody that binds CD123 (also known as IL-3 receptor alpha), such as talacotuzumab (also known as CSL362 and JNJ-56022473). In some embodiments, an antibody of the present disclosure is administered in combination with a monoclonal antibody that binds EGFR (such as cetuximab). In some embodiments, the second antibody includes one or more effector functions, e.g., effector functions that are associated with Fc receptor (FcR) engagement on immune cells including without limitation ADCC or ADCP, and/or complement-dependent cytotoxicity (CDC). Without wishing to be bound to theory, it is thought that combining such an antibody with an antibody of the present disclosure is particularly advantageous, e.g., to direct FcR-expressing leukocytes to target a tumor cell to which the second antibody is bound while also inhibiting the responsiveness of SIRP- α expressed by the leukocyte to any CD47 expressed by the tumor cell with the SIRP- α antibody.

In some embodiments, an antibody of the present disclosure is administered in combination with an immunotherapeutic agent. An immunotherapeutic agent may refer to any therapeutic that targets the immune system and promotes a therapeutic redirection of the immune system, such as a modulator of a costimulatory pathway, cancer vaccine, recombinantly modified immune cell, etc. In some embodiments, the immunotherapeutic agent comprises an antibody. Exemplary antigens of immunotherapeutic antibodies are known in the art and include without limitation PD-1, PD-L1, OX40, CTLA-4, CD137/4-1BB, B7-H3, FZD7, CD27, TNFR2, CCR4, CSF1R, CSF, TIM-3, LAG-3, VISTA, ICOS, CCR2, IDO, A2R, CD39, CD73, TIGIT, CD80, CD47, arginase, TDO, and PVRIG. Immunotherapeutic agents that are approved or in late-stage clinical testing include, without limitation, ipilimumab, pembrolizumab, nivolumab, atezolizumab, avelumab, durvalumab, and the like. Without wishing to be bound to theory, it is thought that the antibodies of the present disclosure are

suitable for use with immunotherapeutic agents due to complementary mechanisms of action, e.g., in activating both macrophages and T_{effector} cells to target tumor cells. In certain embodiments, an antibody of the present disclosure is administered in combination with an inhibitor of the PD-L1/PD-1 pathway, e.g., an anti-PD-L1 or anti-PD-1 antibody. As demonstrated herein, combined administration of an anti-SIRP- α antibody of the present disclosure and an inhibitor of the PD-L1/PD-1 pathway can result in synergistic anti-tumor activity. For example, in some embodiments, a blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-1 antibody. In some embodiments, a non-blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-1 antibody. In some embodiments, a blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-L1 antibody. In some embodiments, a non-blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-L1 antibody.

Any cancer type known in the art may be included, such as but not limited to carcinoma, sarcoma, lymphoma, leukemia, lymphoma, and blastoma. More particular examples of such cancers include, but are not limited to, lung cancer, squamous cell cancer, brain tumors, glioblastoma, head and neck cancer, hepatocellular cancer, colorectal cancer (e.g., colon or rectal cancers), liver cancer, bladder cancer, gastric or stomach cancer, pancreatic cancer, cervical cancer, ovarian cancer, cancer of the urinary tract, breast cancer, peritoneal cancer, uterine cancer, salivary gland cancer, kidney or renal cancer, prostate cancer, vulval cancer, thyroid cancer, anal carcinoma, penile carcinoma, melanoma, multiple myeloma and B-cell lymphoma (including non-Hodgkin's lymphomas (NHL)); acute lymphoblastic leukemia (ALL); chronic lymphocytic leukemia (CLL); acute myeloid leukemia (AML); Merkel cell carcinoma; hairy cell leukemia; chronic myeloblastic leukemia (CML); and associated metastases.

In addition to cancer therapies, the antibodies provided herein are useful in therapies in which monoclonal antibodies are administered for the purpose of depleting cells, e.g., in the treatment of inflammatory diseases by depletion immune cells. For such purposes the an antibody provided herein is administered in combination with a second therapeutic antibody, e.g. with rituximab for depletion of B cells in inflammatory diseases and autoimmune conditions; alemtuzumab for multiple sclerosis; OKT3 for immunosuppression; others for bone marrow transplant conditioning; and the like.

Further provided herein are methods of treating or delaying progression of an autoimmune disease or an inflammatory disease in an individual by administering an effective amount of an antibody of the present disclosure. Autoimmune diseases and inflammatory diseases amenable to treatment according to the disclosure include, but are not limited to, multiple sclerosis, rheumatoid arthritis, a spondyloarthropathy, systemic lupus erythematosus, an antibody-mediated inflammatory or autoimmune disease, graft versus host disease, sepsis, diabetes, psoriasis, atherosclerosis, Sjogren's syndrome, progressive systemic sclerosis, scleroderma, acute coronary syndrome, ischemic reperfusion, Crohn's Disease, endometriosis, glomerulonephritis, myasthenia gravis, idiopathic pulmonary fibrosis, asthma, acute respiratory distress syndrome (ARDS), vasculitis, and inflammatory autoimmune myositis. In some embodiments, an antibody of the present disclosure is administered in combination with a therapeutic agent, such as an immuno-

suppressive, anti-inflammatory, or immunomodulatory agent. In some embodiments, an antibody provided herein is used in the treatment of an autoimmune disease or an inflammatory disease, e.g., multiple sclerosis, rheumatoid arthritis, a spondyloarthropathy, systemic lupus erythematosus, an antibody-mediated inflammatory or autoimmune disease, graft versus host disease, sepsis, diabetes, psoriasis, psoriatic arthritis, atherosclerosis, Sjogren's syndrome, progressive systemic sclerosis, scleroderma, acute coronary syndrome, ischemic reperfusion, Crohn's Disease, ulcerative colitis, endometriosis, glomerulonephritis, IgA nephropathy, polycystic kidney disease, myasthenia gravis, idiopathic pulmonary fibrosis, asthma, atopic dermatitis, acute respiratory distress syndrome (ARDS), vasculitis, or inflammatory autoimmune myositis.

In some embodiments, an antibody of the present disclosure is part of a pharmaceutical formulation, e.g., including the antibody and one or more pharmaceutically acceptable carriers. Pharmaceutical compositions and formulations as described herein can be prepared by mixing the active ingredients (such as an antibody or a polypeptide) having the desired degree of purity with one or more optional pharmaceutically acceptable carriers (*Remington's Pharmaceutical Sciences* 16th edition, Osol, A. Ed. (1980)), in the form of lyophilized formulations or aqueous solutions. Pharmaceutically acceptable carriers are generally nontoxic to recipients at the dosages and concentrations employed, and include, but are not limited to: buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives; low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g. Zn-protein complexes); and/or non-ionic surfactants such as polyethylene glycol (PEG). In some embodiments, an antibody of the present disclosure is lyophilized.

EXAMPLES

The present disclosure will be more fully understood by reference to the following examples. The examples should not, however, be construed as limiting the scope of the present disclosure. It is understood that the examples and embodiments described herein are for illustrative purposes only and that various modifications or changes in light thereof will be suggested to persons skilled in the art and are to be included within the spirit and purview of this application and scope of the appended claims.

Example 1: Identification of Antibodies with Novel Binding Specificities to SIRP- α Proteins

Methods

Antibody Production

The following proteins were used for immunization. Each includes a human or mouse SIRP- α peptide fused to a modified Fc region (either an S228P human IgG4 Fc or an L234A/L235A/G237A/N297A human IgG1 Fc designated as IgG1_AAA_N297A) for increased immunogenicity.

TABLE A

Immunogen sequences.		
Description	SEQ ID NO	Sequence
Human sirpa v1 (Fusion with Fc of IgG4_S228P)	1	EEELQVIQPDKSVLVAAGETATLRCTATSLIPV GPIQWFRGAGPGRELIYNQKEGHFPRVTTVSD LTKRNNMDFSIRIGNITPADAGTYTCVKFRKG SPDDVEFKSGAGTELSVRAPSESKYGPPCPP CPAPEFLGGPSVFLFPPPKDLMISRTPEVTC VVVDVSQEDPEVQFNWYVDGVEVHNAKTKP REEQFNSTYRVVSVLTVLHQDWLNGKEYKCK VSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQ EEMTKNQVSLTCLVKGFYPSDIAVEWESNGQ PENNYKTTTPVLDSDGSFFLYSRLTVDKSRWQ EGNVFSCSVMEALHNHYTQKSLSLGLGK
Human sirpa v2 (Fusion with Fc of IgG4_S228P)	2	EEELQVIQPDKSVSVAAGESAILHCTVTSLIPV GPIQWFRGAGPARELIYNQKEGHFPRVTTVSE STKRENMDFSISISNITPADAGTYTCVKFRKGS PDTEFKSGAGTELSVRAPSESKYGPPCPPCPA PEFLGGPSVFLFPPPKDLMISRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREE QFNSTYRVVSVLTVLHQDWLNGKEYKCKVS NKGLPSSIEKTISKAKGQPREPQVYTLPPSQEE MTKNQVSLTCLVKGFYPSDIAVEWESNGQPE NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQ GNVFCSCVMHEALHNHYTQKSLSLGLGK
Mouse 129 sirpa (Fusion with Fc of IgG1_AAA_N297A)	3	KELKVTQPEKSVSVAAGDSTVLNCTLTSLLPV GPIKWYRGVQSRLLIYSFTGEHFPRVTNVSD ATKRNNMDFSIRISNVTPEAGTYTCVKFQKG PSEPDTEIQSGGGTEVYVLAKPSDKTHTCPPCP APEAAGAPSVFLFPPPKDLMISRTPEVTCVV VDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYASTYRVVSVLTVLHQDWLNGKEYKCKV SNKALPAIEKTISKAKGQPREPQVYTLPPSRE EMTKNQVSLTCLVKGFYPSDIAVEWESNGQP ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ QGNVFCSCVMHEALHNHYTQKSLSLSPGK
Mouse NOD sirpa (Fusion with Fc of IgG1_AAA_N297A)	4	TEVKVIQPEKSVSVAAGDSTVLNCTLTSLLPV GPIRWYRGVQSRQUYSFTEHFPRVTNVSD ATKRNNLDFSIRISNVTPEAGTYTCVKFQRG SPDTEIQSGGGTEVYVLAKDKTHTCPPCPAPE AAGAPSVFLFPPPKDLMISRTPEVTCVVVD VSHEDPEVKFNWYVDGVEVHNAKTKPREEQ YASTYRVVSVLTVLHQDWLNGKEYKCKVSN KALPAIEKTISKAKGQPREPQVYTLPPSREEM TKNQVSLTCLVKGFYPSDIAVEWESNGQPEN NYKTTTPVLDSDGSFFLYSKLTVDKSRWQQG NVFSCSVMEALHNHYTQKSLSLSPGK

45

The above proteins were used to immunize wild-type chickens, SynVH chickens which are transgenic chickens containing VH from human and VL from chicken, or chickens with fully human "HuMAB" immunoglobulin loci (Crystal Bioscience; see, e.g., WO2012162422, WO2011019844, and WO2013059159). Chickens were immunized with varied schedules having alternating doses of antigen. An exemplary immunization schedule is as follows: initial immunization with 100 µg dose of antigen having the sequence of SEQ ID NO:1 at week 1, boost of 100 µg of antigen having the sequence of SEQ ID NO: 2 at week 3, draw at week 4, boost with 50 µg dose of antigen having the sequence of SEQ ID NO:1 at week 5, draw at week 6, boost with 50 µg of antigen having the sequence of SEQ ID NO: 2 at week 7, and draw at week 8. Additional descriptions of chicken immunization may be found, e.g., in Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16. Screening and Expression of Antibody Clones

Clones were generated and screened according to the GEM assay (see Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16). Clones were tested in a scFv-Fc format in which the light chain is fused via linker with the heavy

chain. The scFv was fused to the N-terminus of human Fc-IgG1. Clones were expressed in FreeStyle™ 293-FS cells (Thermo Fisher) and secreted media was used for ELISA and SPR binding characterization.

50 SPR

Binding of the antibody clones to various SIRP proteins was determined using surface plasmon resonance (SPR) detection on a ProteOn XPR36 instrument (Bio-Rad, Hercules, CA) using phosphate buffered saline (PBS, pH 7.4) supplemented with 0.01% Tween-20 (PBST) as running buffer. The pre-filtered media containing the secreted antibodies was used directly for the assay. First, anti-human IgG Fc (BR-1008-39, GE Healthcare) was amine-coupled onto a GLC sensor chip to generate the capture surfaces for the antibodies. About 4000 RU per flow cell of immobilized anti-human IgG Fc is achieved. Each clone is screened using the same method as follows. The SIRP analytes used for the screen are listed in Table C.

- (1) ~5-10 uL of pre-filtered media in 10 mM sodium acetate buffer (pH4.5) was injected for 2 mins at 30 ul/min;
- (2) buffer flow for 1 min at 100 uL/min;

155

- (3) SIRP analyte (100 nM) injected for 1 min at 100 uL/min, followed by a dissociation cycle of 10 mins;
- (4) regeneration of chip surface by flowing 3M Magnesium Chloride for 1 min at 25 uL/min in both orientation; and
- (5) buffer flow for 1 min at 100 uL/min.

Biosensor data were double-referenced by subtracting the interspot data (containing no immobilized anti-human IgG Fc) from the reaction spot data (immobilized anti-human IgG Fc) and then subtracting the response of a buffer "blank" analyte injection from that of an analyte injection. Binding was fitted using a 1:1 Langmuir and K_{off} (1/S) values calculated. All SPR assays were performed at 25° C.

ELISA

ELISA assays were carried out to screen binding of antibody clones to SIRP analytes and SIRP- α :CD47 complex. Briefly, 96-well flat-bottom, high binding plates (Greiner Bio-One #655061) were coated with the following proteins in separate ELISA experiments: Avidin (Sigma A9275) (2 ug/ml) followed by biotinylated human SIRP α V1 (0.5 ug/ml), Avidin (2 ug/ml) followed by human SIRP α V2 biotin (0.5 ug/ml), mouse NOD SIRP α (2 ug/ml), human SIRP γ (2 ug/ml), CD47 (2 ug/ml) followed by high affinity human SIRP α V1 and V2 (at 2 ug/ml each) or anti-hFc (2 ug/ml Rockland 609-4103). Plates were blocked with PBSTM (Phosphate Buffered Saline pH 7.4, 0.05% Tween®20 polysorbate, 3% milk), 50 ul of supernatant containing the secreted scFv-Fc are added for 1 hr at room temp. Plates were washed with PBST. 50 ul of anti-hFc-HRP (1:5000 Rockland 609-4303) added for 1 hr at room temp. Plates were washed with PBST. TMB was developed for 5 min and stopped with 1N HCl. ELISA results were read using BioTek Synergy H1 Hybrid Reader. The corresponding proteins used for the assay and described here are: human SIRP α V1 (SEQ ID NO: 5); human SIRP α V2 (SEQ ID NO:6); cynomolgus SIRP α (SEQ ID NO:11); mouse NOD SIRP α (SEQ ID NO: 8); human SIRP γ (SEQ ID NO: 15); CD47 (SEQ ID NO: 16); high affinity SIRP α V1 (SEQ ID NO: 42); high affinity SIRP α V2 (SEQ ID NO:17).

Results

SIRP- α is a highly polymorphic protein in humans, monkeys, and mice. For example, 20 amino acid differences have been identified between SIRP- α proteins in the NOD and C57BL/6 mouse strains, and these polymorphisms lead to functional consequences related to CD47 binding and engraftment of human hematopoietic stem cells in these mouse strains. In humans, at least 10 distinct alleles of SIRP α have been identified, and amino acid variations that distinguish the alleles are found in predicted CD47-binding residues (Takenaka, K. et al. (2007) *Nat. Immunol.* 8:1313-23). An alignment of 10 human variant SIRP- α protein sequences is provided in FIG. 1A. The identification of antibodies having different binding specificities with intra- and/or inter-species cross-reactivity is of great interest for development of clinical candidates that are effective across human populations and the characterization of these candidates in various animal models.

The D1 domains of SIRP- α proteins from human (v1 and v2 variants), cynomolgus monkey, and mouse 129 were aligned to identify conserved amino acids (FIG. 1B). While the cynomolgus monkey and mouse 129 sequences are much more divergent from human v1 and v2 than human variants v3-v10 as shown in FIG. 1A, these alignments demonstrate some degree of conservation among all four proteins, suggesting that cross-reactive antibodies could perhaps be iden-

156

tified. However, each protein also shows unique polymorphisms, suggesting that specific antibodies were also possible.

As shown in FIG. 1C, SIRP- α protein sequences representing multiple human variants and mouse strains (e.g., 129, NOD, C57BL/6, and BALB/c) were also aligned. R1, R2, R3 delineate residues located around binding sites of SIRP α to CD47. These alignments demonstrated that the R2 and R3 are considerably more divergent among and between human and mouse sequences than the R1. Without wishing to be bound to theory, it is thought that anti-SIRP- α antibodies that bind to specific murine SIRP- α proteins may be useful, e.g., for pharmacokinetic studies (such as those using CD-1 mice), development of transgenic mice (such as those in a C57BL/6 background), and/or characterization in SCID (e.g., in a NOD background) or syngeneic (e.g., in a BALB/c or C57BL/6 background) mouse models.

Mammalian SIRP- α D1 domains described above were also aligned with the D1 domain of chicken SIRP- α comprising the sequence DFKLQQPQSSVVVKGDTLTLNC-TASGSGPIGAVKWVKGWGSNDQTVYEHKGSFPRVM RAVPDPNTNDFTRISNVSLDAGTYYCVKLRK-GIVDDVVFTTRGGGTEVSVHA (SEQ ID NO:84) (FIG. 2). Compared with the mammalian SIRP- α sequences, the sequence of chicken SIRP α was found to be significantly more divergent. Pairwise comparisons of the percentage of sequence identity between various SIRP- α proteins is shown in Table B.

TABLE B

Pairwise sequence identities (%) between SIRP- α proteins.					
	Human v1 (SEQ ID NO: 5)	Human v2 (SEQ ID NO: 6)	Mouse (SEQ ID NO: 7)	Cyno (SEQ ID NO: 11)	Chicken (SEQ ID NO: 84)
Human v1 (SEQ ID NO: 5)	—	89	66	91	46
Human v2 (SEQ ID NO: 6)	89	—	68	88	41
Mouse (SEQ ID NO: 7)	66	68	—	72	47
Cyno (SEQ ID NO: 11)	91	88	72	—	47
Chicken (SEQ ID NO: 84)	46	41	47	47	—

Without wishing to be bound to theory, it was thought that this divergence would provide unique opportunities to generate antibodies that cross-react across multiple mammalian SIRP- α proteins. For example, it may be difficult to generate anti-SIRP- α antibodies that cross-react with the murine sequence from a mouse host due to immune tolerance. Moreover, the greater divergence between the chicken and mammalian immune systems may lead to a greater diversity in antibody production.

In order to identify antibodies with novel binding specificities to SIRP- α proteins, antibody clones were characterized in the scFv-Fc format as described above. Each antibody clone was tested at concentrations between 0.008 and 1.0 μ g/mL for binding using ELISA to the following targets: the D1 domain of human SIRP- α v1 (sequence according to SEQ ID NO:5), the D1 domain of human SIRP- α v2 (sequence according to SEQ ID NO:6), the D1 domain of a cynomolgus SIRP- α variant (sequence according to SEQ ID

157

NO:11), the D1 domain of mouse SIRP- α (sequence according to SEQ ID NO:7), and a human SIRP γ isoform (sequence according to SEQ ID NO:15). As used hereinafter, antibody clones are referred to by clone ID number. In addition, the notation "S[clone number]" refers to an sc-Fv-Fc format; the notation "AB[clone number]" refers to a full IgG antibody format; the notation "AB[clone number]a" refers to a human IgG1 with L234A, L235A, G237A, and N297A mutations; "AB[clone number]b" refers to a mouse IgG1 N297A format; "AB[clone number]c" refers to a mouse IgG2a format; and "[clone number] Fab" refers to a Fab fragment format.

In addition, the binding of each antibody clone to a pre-complex prepared using a 1:1 mix of two high affinity SIRP- α variants (SEQ ID NO:42 and SEQ ID NO:18) bound with the IgSF domain of CD47 (sequence according to SEQ ID NO:16) was characterized. Advantageously, since the affinity between the wild-type SIRP- α D1 domain and the IgSF domain of CD47 is relatively low, the use of a complex comprising a high affinity SIRP- α variant allows the identification of antibodies that bind to SIRP- α while it is complexed with CD47 (e.g., a non-blocking antibody). It also allows the identification of antibodies that are unable to bind SIRP- α while it is complexed with CD47, suggesting that the antibody and CD47 compete for the same binding interface on SIRP- α (e.g., a blocking antibody). The two SIRP α variants used (SEQ ID NO:17 and SEQ ID NO:19) correspond to high affinity SIRP α D1 domain engineered using human SIRP α polypeptide variant 1 and variant 2, respectively.

FIG. 3A shows the ELISA binding curve for clone S130 (SEQ ID NO:71). This clone demonstrated cross-reactivity across mammalian SIRP- α proteins, with binding to multiple human variants as well as cynomolgus and murine proteins (FIG. 3B). However, its binding was also isoform-specific, since no binding was observed with human SIRP γ . This clone was also identified as a blocker of the interaction between SIRP- α and CD47, as no binding to a pre-formed complex containing CD47 bound to a high-affinity SIRP- α variant was detected.

FIG. 4A shows the ELISA binding curve for clone S121 (SEQ ID NO:75). This clone also demonstrated cross-reactivity across mammalian SIRP- α proteins, but it also bound to human SIRP γ , indicating pan-isoformic binding (FIG. 4B). This clone was also identified as a blocker of the interaction between SIRP- α and CD47.

FIG. 5A shows the ELISA binding curve for clone S137 (SEQ ID NO:73). Like the clone shown in FIGS. 3A & 3B, this clone demonstrated cross-reactivity across mammalian SIRP- α proteins with isoform-specific binding, as it did not bind to human SIRP γ (FIG. 5B). However, this clone was identified as a non-blocker of the interaction between SIRP- α and CD47, since it bound to the pre-formed complex.

FIG. 6A shows the ELISA binding curve for clone S128 (SEQ ID NO:70). This clone demonstrated isoform-specific cross-reactivity across primate SIRP- α proteins, but it did not bind to the murine protein (FIG. 6B). This clone was also identified as a blocker of the interaction between SIRP- α and CD47.

FIG. 7A shows the ELISA binding curve for clone S135 (SEQ ID NO:72). Similar to the clone shown in FIGS. 6A & 6B, this blocking clone demonstrated cross-reactivity across primate SIRP- α proteins, but it did not bind to the murine protein (FIG. 7B). Unlike clone S128, this clone was found to cross-react with human SIRP γ .

158

FIG. 8A shows the ELISA binding curve for clone S126 (SEQ ID NO:69). This clone was found to be human-specific, binding to both human SIRP- α variants but none of the other peptides (FIG. 8B). This clone was identified as a blocker of the interaction between SIRP- α and CD47.

FIG. 9A shows the ELISA binding curve for clone S138 (SEQ ID NO:74). This clone only bound human SIRP- α variant 1, demonstrating a high degree of intra- and inter-species binding specificity (FIG. 9B). This clone was identified as a blocker of the interaction between SIRP- α and CD47.

In conclusion, novel antibodies with a number of unique binding specificities to SIRP- α proteins were characterized. FIGS. 10A-10C provide an alignment showing scFv-Fc variable domain sequences obtained from a chicken that produces chicken antibodies (CDRs are indicated). FIGS. 10D-10F provide an alignment showing scFv-Fc variable domain sequences obtained from a chicken that produces human antibodies (CDRs are indicated). These studies highlight the utility of the chicken as a source of antibody diversity for the identification of anti-SIRP- α antibodies.

To assess if antibodies can block binding of SIRP α to CD47, an SPR screen was also carried out, in addition to screening by ELISA. Antibody capture was carried out using an anti-human IgG-Fc immobilized GLC surface prepared as described above. A SIRP α variant (SEQ ID NO:17) engineered to bind CD47 (SEQ ID NO:16) with high nM affinity was used for the screen rather than a wildtype SIRP- α . This is because the wildtype SIRP- α variant has low uM binding affinity to CD47, which does not allow stable complex interaction to assess sandwich formation. Phosphate buffered saline (PBS, pH 7.4) supplemented with 0.01% Tween-20 (PBST) as running buffer.

First, approximately 5-10 uL of pre-filtered media containing the antibodies in 10 mM sodium acetate buffer (pH4.5) was injected for 2 mins at 30 uL/min and captured over the anti-human IgG-Fc immobilized GLC surface followed by a brief buffer flow of 1 min at 100 uL/min. Next, 100 nM of a high affinity SIRP- α variant (SEQ ID NO:17) pre-mixed with CD47 (SEQ ID NO:16) at different concentrations of 0, 20, 55, 500, or 1500 nM was injected separately for a minute at 100 uL/min with a dissociation time for 10 mins. For the description below, SIRP- α refers to SEQ ID NO:17.

For an antibody that does not block binding of SIRP- α to CD47, it would be expected to bind to SIRP- α /CD47 complex and form a sandwich. Therefore, at increasing concentration of CD47, the resonance increased accordingly due to increased sandwich formation. Antibody clone S123 was found to match this profile. An example of the profile for S123 is shown in FIG. 12A.

For an antibody that blocks binding of SIRP- α to CD47, the profile will be different than a non-blocking antibody. At increasing concentration of CD47, one would expect fewer molecules of SIRP- α to be available to bind to the antibody since the antibody competes for the same binding site as CD47 and most/all SIRP- α is complexed with CD47. Therefore, one would expect the resonance (RU) to decrease with increasing concentration of CD47 in the mixture. Antibody clone S119 was found to match this profile. An example of the profile for S119 is shown in FIG. 12B.

In addition to blockers and non-blockers, a third category of antibodies was isolated that have a "kick off" profile. Antibody clone S118 was found to match this profile. An example of the profile for S118 is shown in FIG. 12C. At higher concentration of CD47 (e.g 500 nM and 1500 nM CD47), a transient sandwich was formed between the anti-

159

body, SIRP- α and CD47 as indicated by the higher resonance of 300 RU which then decayed over time. This decay of resonance units indicates that the antibody was able to bind to SIRP- α in the complex and “kick SIRP- α off” from binding to CD47.

The k_{off} rates of binding of each clone to each SIRP analyte were determined using SPR (Table C). The SPR screening conditions have been described herein, and the K_{off} values were determined using Langmuir kinetic fit-

160

tings. The CD47 blocking properties (block, non-block, kick-off) are described in the last column of the Table C. Each antibody is identified according to its corresponding SEQ ID NO. SIRP analyte sequences are as follows: CV1-3, SEQ ID NO:18; v1, SEQ ID NO:5; v2, SEQ ID NO:6; cyno1, SEQ ID NO:11; cyno2, SEQ ID NO:12; m129, SEQ ID NO:7; NOD, SEQ ID NO:8; BL6, SEQ ID NO: 9; sirpb1, SEQ ID NO:13; sirpg, SEQ ID NO:15.

TABLE C

Langmuir kinetic fittings (k_{off} 1/s)												
Clone	SEQ ID NO	CV1-3	v1	v2	cyno1	cyno2	m129	NOD	BL6	sirpb1	sirpg	cd47 blocking
S1	53	2.19E-03	2.33E-03	NB	3.48E-02	4.06E-03	2.12E-03	7.89E-03	1.70E-03	NB	2.19E-03	Block
S2	54	1.49E-01	2.52E-03	NB	LB	5.59E-03	1.42E-02	NB	8.85E-03	—	—	Block
S8	55	LB	1.14E-04	1.42E-04	4.15E-03	6.12E-04	1.29E-02	1.79E-03	6.96E-03	9.30E-04	3.23E-04	Block
S9	56	1.68E-04	1.29E-04	1.40E-04	3.38E-04	2.70E-04	NB	1.81E-02	NB	1.03E-04	7.10E-03	Block
S11	57	2.69E-03	3.35E-05	1.86E-04	1.11E-03	3.92E-03	NB	LB	LB	1.59E-04	3.86E-04	Block
S12	58	5.15E-02	1.01E-04	1.68E-04	1.17E-03	5.24E-04	NB	2.60E-02	1.98E-03	1.73E-04	3.07E-04	Block
S13	59	LB	2.07E-04	2.91E-04	1.84E-03	5.01E-04	4.93E-02	3.83E-03	1.03E-02	1.35E-04	4.02E-04	Block
S14	60	3.33E-03	5.42E-05	7.64E-05	6.49E-04	4.52E-04	3.73E-02	4.98E-03	1.81E-02	6.01E-05	1.26E-04	Block
S115	61	4.84E-04	7.86E-06	1.95E-05	2.22E-05	6.91E-05	NB	NB	NB	—	3.34E-05	Kick off
S116	62	4.80E-04	2.84E-05	5.07E-05	7.01E-05	1.20E-04	NB	NB	NB	—	8.25E-05	Kick off
S117	63	2.75E-04	1.20E-05	3.40E-05	1.83E-05	5.92E-05	NB	NB	NB	—	4.30E-06	Kick off
S118	64	2.47E-04	9.17E-06	4.12E-05	6.97E-05	8.06E-05	NB	NB	NB	1.31E-05	1.13E-05	Kick off
S119	65	5.65E-04	4.15E-04	1.48E-04	2.34E-04	3.10E-04	NB	LB	NB	4.28E-04	3.95E-04	Block
S120	66	6.26E-04	4.04E-04	1.49E-04	2.36E-04	3.14E-04	NB	LB	NB	4.25E-04	3.93E-04	Block
S122	67	6.63E-04	4.98E-04	2.26E-04	2.66E-04	3.27E-04	NB	LB	NB	3.52E-04	3.61E-04	Block
S123	68	1.31E-03	1.55E-03	NB	1.57E-03	1.54E-03	NB	NB	NB	NB/LB	1.43E-03	Non-block
S126	69	NB	NB	NB	NB	NB	NB	NB	NB	NB	NB	—
S128	70	NB	4.59E-03	2.11E-03	1.52E-02	1.47E-02	NB	NB	NB	4.21E-03	NB	—
S130	71	3.73E-03	2.82E-03	2.08E-02	LB	LB	LB	NB	NB	LB	5.06E-03	Block
S135	72	3.63E-03	6.58E-04	8.78E-05	4.73E-04	3.88E-04	NB	NB	NB	5.21E-04	1.46E-03	Block
S137	73	1.37E-03	1.65E-03	1.88E-03	2.04E-03	1.79E-03	5.15E-04	4.43E-04	4.63E-03	3.76E-03	1.48E-02	Non-block
S138	74	LB	NB	NB	NB	NB	NB	NB	NB	NB	NB	—

NB = No binding

LB = very low binding (K_{off} cannot be calculated by Langmuir fittings)

— = not screened

These data indicate that antibody clones with a variety of binding specificities for SIRP- α proteins across different species and intra-species variants were identified. The binding specificities for selected antibodies were further characterized using ELISA as described above by generating binding curves against the human v1 (SEQ ID NO: 5), human v2 (SEQ ID NO: 6), NOD mouse (SEQ ID NO:8), and cynomolgus SIRP- α D1 domains (SEQ ID NO:11), as well as human SIRP γ (SEQ ID NO: 15) and a pre-formed complex of a 1:1 mixture of two high-affinity SIRP- α variants (SEQ ID NOs:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO: 16).

Sequences of scFv-Fc clones tested are provided in Table D.

TABLE D

scFv-Fc sequences.			
Clone	Framework/ CDR	SEQ ID NO	Sequence
S1	Chicken	53	ALTQPASVSNPGETVEITCSGGGSNNAYGWFOQ KSPGSAPLTVIYDNGKRPSDIPSRFSGSKSDSTGLT ITRVQAEDAEVYYCGSADNSGAGVFGAGTTTLTVL GQSSRSSGGGSGGGGMAAVTLDESGLLQTP GGALSLVCKGSGFTFSSHAMNWRVQAPKGLEW VAGISSDGRFTYYGAHVQGRATISRDNQSTVRLQ LNNLRAEDTATYYCTKNGCGSGGDLDCIDAWG HGTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGPSV FLPPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKF NRYVDGVEVHNATKPREQYNSTYRVVSVLTV

TABLE D-continued

scFv-Fc sequences.			
Clone	Framework/ CDR	SEQ ID NO	Sequence
			LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ PREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIA VEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK V
S2	Chicken	54	AVTQPASVSANPGETVTRITCSGDSSTYYSWHQQKS PGSAPVSVIYNTDRPSDIPSRFSGSASGSTATLTIT GVQAEDEAVYFCGAYDSSSDSIFGAGTTLTVLGQ SSRSSGGGGSSGGGSMMAAVTLDES GGLQTPGG GLSLVCKASGFTFSNFMWVRQAPGKLEFVAA EISDTGTPYYGSAVQGRATISRDNQSTVRLQLN NLRAEDTGTYFCTRNFGSSVSSIDAWGHGTEVIVS SSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPK DTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGV EVHNAKTKPREEQYNSTYRVVSVLTVTLHQDWLN GKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT LPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQ GNVFSCSVMHEALHNHYTQKSLSLSPGKV
S8	Chicken	55	AVTQPSSVSANPGETVEITCSGSSYYGWYQKSP GSAPVTVIYDNDKRPSDIPSRFSGSKSGSTHTLIITG VQVEDEAVYFCGNEDNNYVAIFGAGTTLTVLGQS SSRSSGGGGSSGGGSMMAAVTLDES GGLQTPGGA LSLVCKASGFTFSYNMGWVRQAPGKLEFVAGI YASGSTDTDTYGPVAGRATISRDNQSTVRLQ LNNLRAEDTGTYCAKAAGCSTHTCTAYIADSID AWGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLG GPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPE VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYP SDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSK LTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSL SPGKV
S9	Chicken	56	ALTQPSSVSANPGETVKITCSGDSNAHYYYGWYQ QKSPGSAPVTVIYNDKRPSGIPSRFSGSASGSTAT LIITGVQVEDEAVYFCGSADSNPAIFGAGTTLTVL GQSSRSSGGGGSSGGGSMMAAVTLDES GGLQTP GRALSLVCRGSGFSISSYNMGWVRQAPGKLEFIA SIGSDGSSTHYAPAVKGRATITRDVGQSTVRLQLN NLRAEDTGTYFCAKDAYQCSYATCNDYLDITDAW GHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGPS VFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVK FNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLT VLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDI AVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD DKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG KV
S11	Chicken	57	AVTQPASVSANPGETVKITCSGSSSGSYGWYQKKS PGSAPVTLIYETNKRPSNIPSRFSGSKSGSTATLTIT GVQADDEAVYCGSEDSSTYLSIFGAGTTLTVLGQ SSRSSGGGGSSGGGSMMAAVTLDES GGLQTPGG ALSLVCKASGFTFSNFMWVRQAPGKLEFVAA IYSGNSAEYGAAVQGRATISRDNQSTVRLQLN RAEDTGTYFCAKDAGSGCYSGVCACTSSIDAWGH GTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNAKTKPREEQYNSTYRVVSVLTVTLHQ DWLNGEYKCKVSNKALPAPIEKTISKAKGQPREP QVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEW ESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSR WQQGNVFSCSVMHEALHNHYTQKSLSLSPGKV
S12	Chicken	58	AVTQPASVSANPGETVKITCSGDSSTYYGWYQKKS PGSAPVTVIYDNDKRPSNIPSRFSGSKSGSTGTLTIT GVQADDEAVYFCGNEDNSYVAIFGAGTTLTVLGQ SSRSSGGGGSSGGGSMMAAVTLDES GGLQTPGG ALSLVCKASGFTFSYNMGWVRQAPGKLEFVAG IYIASGDLGTYGAAVQGRATISRDNQSTVRLQL NNLRAEDTGTYFCAKSAGGCSAHSCTYIADSIDA

TABLE D-continued

scFv-Fc sequences.			
Clone	Framework/ CDR	SEQ ID NO	Sequence
			WGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGP SVFLFPPPKPD TLMISRTPEVTCVVVDVSHEDPEV KFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAK GQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPS DIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKL TVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLS PGKV
S13	Chicken	59	AVTQPASVSANPGETVKITCSGSSSYGYRQKSP GSAPVTLIYDNDKRPSGIPSRFSGSKSGSTNTLTITG VQADDEAVYYCGNEDNSYVGIFGAGTTLTVLGQS SRSSGGGGSSGGGGMAAVTLDES GGLQTPGGA LSLVCKASGFTFSSYNMGWVRQAPDKGLEFVAGI YTGSDAGLSTTYGAAVQGRATISRDNGQSTVRLQ LNNLGAEDTGIYFCKSAGGCSYNDAYIADSID AWGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLG GPSVFLFPPPKPD TLMISRTPEVTCVVVDVSHEDPE VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAK KGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPS SDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSK LTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSL SPGKV
S14	Chicken	60	AVTQPASVSANLGETVKITCSGDSYGYWYQQA PGSAPVTLIYDNDKRPSNIPSRFSGSKSGSTATLTIT GVQADDEAVYYCGNEDMNYVGIFGAGTTLTVLG QSSRSSGGGGSSGGGGMAAVTLDES GGLQTPG GALSLVCKASGFTFNSYNMGWVRQAPDKGLEFV AGIYSAGGDTSTTYGAAVNGRATISRDNGQSTVRL QLNLRADTGIYFCAKAAGGCTAHNCDAYIADSI DAWGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELL GGPSVFLFPPPKPD TLMISRTPEVTCVVVDVSHEDP EVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV VLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAK AKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPS PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS KLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSL SPGKV
S115	Human	61	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW HQQKPGQAPRLLIYDATNRATGISDRFSGSGSGTD FTLTISSLQTEDSAVYYCQYYWPPYRFGGGTKV EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRP GESLRRLSCAASGFSSSYAMNWVRQAPGEGLEWV SRINSGGGTDYAESVKGRFTISRDNSNTLYLQM NSLRAEDTAVYYCAKQYDWSFFDYWGLGALVT VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFCSCVMHEALHNHYTQKSLSLSPGKV
S116	Human	62	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW HQQKPGQAPRLLIYDATNRATGISDRFSGSGSGTD FTLTISSLQTEDSAVYYCQYYWPPYRFGGGTKV EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRP GESLRRLSCAASGFSSSYAMNWVRQAPGEGLEWV SRINSGGGTDYAESVKGRFTISRDNSNTLYLQM NSLRAEDTAVYYCAKQYDWSFFDYWGLGALVT VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFCSCVMHEALHNHYTQKSLSLSPGKV
S117	Human	63	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW HQQKPGQAPRLLIYDATNRATGIPDRFSGSGSGTD FTLTISSLQTEDSAVYYCQYYWPPYRFGGGTKV EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRP

TABLE D-continued

scFv-Fc sequences.			
Clone	Framework/ CDR	SEQ ID NO	Sequence
			GESLRRLSCAASGFSFSSYAMNWVRQAPGEGLEWV SRINSGGGGTDYAESVKGRFTISRDNSENTLYLQM NSLRAEDTAVYYCAKQYDWNQFDYWGLGALVT VSSSLDPKSSDKTHTCPPCPVPELLGGPSVFLFPPKP KDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVMEALHNHYTQKSLSLSPGKV
S118	Human	64	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW HQQKPGQAPRLLIYDASRRATGIPDRFSGSGSTDF LTITSLQSEDFAVYYCQYYWPPYRFGGKTKV EIKQSSRSSGGGSSGGGSDVQLVESGGGVVVRP GESLRRLSCAASGFSFSSYAMNWVRQAPGEGLEWV SRINSGGGGTDYAESVKGRFTISRDNSENTLYLQM NSLRAEDTAVYYCAKQYDWNQFDYWGLGALVT VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVMEALHNHYTQKSLSLSPGKV
S119	Human	65	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT LTITSLQSEDFAVYYCQYYDWPPFTFGGKTKVEI KGQSSRSSGGGSSGGGSDVQLLESGGGVVQPG GSLRLSCAASGFSFNFAMTWVRQAPGEGLEWVS TIGSGDTYYADSVKGRFTISRDNKNLTYLQMNSL RAEDTAVYYCAKDSVSWSGDFDYWGLGLVT VSSSLDPKSSDKPHTCPPCPAPELLGGPSVFLFPPKP KDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVMEALHNHYTQKSLSLSPGKV
S120	Human	66	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT LTITSLQSEDFAVYYCQYYDWPPFTFGGKTKVEI KGQSSRSSGGGSSGGGSDVQLVESGGGVVQPG GSLRLSCAASGFSFNFAMTWVRQAPGEGLEWVS TIGSGDTYYADSVKGRFTISRDNKNLTYLQMNSL RAEDTAVYYCAKDSVSWSGDFDYWGLGTRVT VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVMEALHNHYTQKSLSLSPGKV
S122	Human	67	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT LTITSLQSEDFAVYYCQYYDWPPFTFGGKTKVEI KGQSSRSSGGGSSGGGSDVQLVESGGGVVVRP ESLRLSCAASGFRFSNFAMTWVRQAPGEGLEWVS TIGSGDTYYADSVKGRFTISRDNKNLTYLQMNSL RAEDTAVYYCAKDSVSWSGDFDYWGLGLVT VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVMEALHNHYTQKSLSLSPGKV
S123	Human	68	EIVLTQSPGTLSPGERVTLTLCRASQGIAGKIAWY QQKPGQAPRLLIYDASSRATGIPGRFSGSGSGTEFT LTITSLQSEDFAVYYCQHYDWSPLTFGGKTKVEI KGQSSRSSGGGSSGGGSDVQLVESGGGLVQPG

TABLE D-continued

scFv-Fc sequences.			
Clone	Framework/ CDR	SEQ ID NO	Sequence
			GSLRLSCTASGFTFRNYGMSWVRQAPGEGLEWVS ASSGSGSTYYTDSVKGRFTISRDNKNTLYLQMNS LRAEDTAIYYCAKVWNNFFDYWGLGLTVTVSSS LDPKSSDKTHTCPPCPAPPELLGGPSVFLFPPKPKDT LMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEV HNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPP SRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPE NNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHYTQKSLSLSPGKV
S126	Human	69	EIVLTQSPGTLTLSPGERATLSCRASQSIGSSYLAW YQQKPGQAPRLLIYDATNRATGIPDRFSGSGSGTD FTLTISSLQTEDSAVYYCQQYYWPPYRFGGGTKV EIKGQSSRSSGGGSSGGGSDVQLVESGGGVVRP GESLRLSCAASGFTFSNYDMTWVRQAPGEGLEWV SGISGNGSTYYADSVKGRFTISRDNKNTLYLQM NSLRAEDTAIYYCAMNRWWFDYWGLGLTVTVSS SLDPKSSDKTHTCPPCPAPPELLGGPSVFLFPPKPKD TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE VHNKTKPREEQYNSTYRVVSVLTVLHQDWLNG KEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL PPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ PENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQG NVFSCSVMHEALHNHYTQKSLSLSPGKV
S128	Human	70	ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW HQKPGQAPRLLIYDASNATGIPDRFSGSGSGTD FTLTISSLQTEDSAVYYCQQYYWPPYRFGGGTKV EIKGQSSRSSGGGSSGGGSDVQLVESGGGVVRP GESLRLSCAASGFTFSYAMNWRQAPGEGLEWV SRIDSGGGTDYADSVKGRFTISRDNKNTLYLQM NSLRAEDTAIYYCAKQYDWSFFDYWGLGAPVT VSSSLDPKSSDKTHTCPPCPAPPELLGGPSVFLFPPKPK KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPSPRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ QGNVFCSCSVMHEALHNHYTQKSLSLSPGKV
S130	Human	71	EIVLTQSPGTLTVSPGERATLSCRASQNVRSDLAW YQKLGQAPRLLIYDANRATDIPDRFSGSGSGTE FTLTISSLQSEDFAVYYCQHYDWPPTFGGGTKV EIKGQSSRSSGGGSSGGGSDVQLVESGGGVVRP GESLRLSCAASGFTFSNYAMSWVRQAPGEGLEWV SLITNGDGAYYADSVKGRFTISRDNKNTLYLQM NSLRAEDTAIYYCAKDGAAHYDIFFDYWGLGTP VTSSSLDPKSSDKTHTCPPCPAPPELLGGPSVFLFPP KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYV DGVEVHNKTKPREEQYNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQ VYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWE SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRW QQGNVFCSCSVMHEALHNHYTQKSLSLSPGKV
S135	Human	72	EIVLTQSPATLSVSPGERVTFSCRASQNVRSIDIAWY QQKPGQAPRLLIYAASSRDGIPDRFSGSGSGTDFT LTISSLQSEDFGVYYCQQYYDWPPTFGGGTKVEI KGQSSRSSGGGSSGGGSDVQLVESGGGVVRPG ESLRLSCAASGFTFSIYAMSWVRQAPGEGLEWVST IGADDTYYADSVKGRFTISRDNKNTLYLQMNSLR AEDTAIYYCAKDSVWGSDFFDYWGLGLTVTV SSSSLDPKSSDKTHTCPPCPAPPELLGGPSVFLFPPKPK DTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG EVHNKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPSPRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ GNVFSCSVMHEALHNHYTQKSLSLSPGKV
S137	Human	73	ETVLTQSPGTLTLSPGERATLTCRASQSVYTYLAW YQKPGQAPRLLIYGASSRATGIPDRFSGSGSGTVF TLTISSLQSEDFAVYYCQQYYDRPPLTFGGGTKVEI KGQSSRSSGGGSSGGGSDVQLVESGGGVVRPG

TABLE D-continued

scFv-Fc sequences.			
Clone	Framework/ CDR	SEQ ID NO	Sequence
			ESLRLSCAASGFTTFSSYDMNWRQAPGEGLEWVS LISGSGEIIYYADSVKGRFTISRDNKNTLYLQMNS LRAEDTAVVYCAKENNRYRFFDDWGLGTLVTVSS SLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKD TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNG KEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL PPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ PENNYKTPPVLDSDGSFFLYSLKTVDKSRWQQG NVFSCSVMHEALHNHYTQKSLSLSPGKV
S138	Human	74	EIVLTQSPGTLSPGGERVILTCRASQSVDTYNLAW YQQKPGQAPRLLIYDLSTRATGIPDRFSGSGSGTEF TLTINSLEPEDFAVYYCHQYDWPPTFGGGTKVE IKGQSSRSSGGGSSGGGSDVQLVESGGGVVRPG ESLRLSCAASGFTTFSSYAMNWRQAPGEGLEWVS GISGRGGDTYYADSVKGRFTISRDNKNTLYLQM NSLRAEDTAIYYCAKGTWNYGSFDYWGGLTLVT VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPK KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG V
S121	Human	75	EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT LTITSLQSEDFAVYYCQYYDWPPTFGGGTKVEI KGQSSRSSGGGSSGGGSDVQLVESGGGVVQPG GSLRLSCAASGFSFNFAMTWVRQAPGEGLEWVS TIGSGDTYYADSVKGRFTISRDNKNTLYLQMNSL RAEDTAVYYCAKDSVSWSGDFFDYWGGLTLVT VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPK KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTPPVLDSDGSFFLYSLKTVDKSRWQ QGNVFCSCVMHEALHNHYTQKSLSLSPGKV

Without wishing to be bound to theory, it is thought that antibodies with cross-reactive binding among human, cynomolgus, and/or murine proteins may allow for characterization of antibodies in both animal models and clinical testing. Antibodies with isoform- and/or variant-specific binding may be useful for personalized medicine approaches to specific human populations and/or studies on specific variants of interest. In addition to profiling intra- and inter-species binding specificities, the methods described herein also allow for identification of antibodies that block or do not block binding between SIRP- α and CD47, as well as antibodies that “kick SIRP- α -off” from binding to CD47.

Example 2: Identification of Additional Antibodies with Novel Binding Specificities to SIRP- α Proteins

Methods

Determination of K_D

The interactions of anti-SIRP α antibodies with SIRP α from various species (human v1, human v2, cynomolgus, mouse 129, BL6, BALBc, NOD), SIRP β and SIRP γ were analyzed using two methods, direct immobilization of the antibodies (via GLC chip) or capture via biotinylated Protein A (via NLC chip), according to the following protocols. All experiments were performed at 25° C. using a SPR-based ProteOn XPR36 biosensor (BioRad, Inc, Hercules, CA) equipped with GLC or NLC sensor chips. Antibodies were expressed using FreeStyle™ 293-FS cells (Thermo Fisher) as described in Example 1. Purification was carried out by

standard Protein A affinity column chromatography and eluted antibodies were stored in PBS buffer.

The running buffer was PBS pH 7.4 with 0.01% Tween-20 (PBST+). All analytes were used at their nominal concentrations as determined by A280 Absorbance and using their molar calculated extinction coefficient. Analytes were injected in a “one-shot” kinetic mode as described elsewhere (see, e.g., Bravman, T. et al. (2006) *Anal. Biochem.* 358: 281-288).

For the method using GLC chip, the analytes were injected and flowed over anti-SIRP α antibodies immobilized (~1000 RUs) on GLC chips using Proteon™ Amine Coupling Kit. For the immobilization step, GLC chip was activated with EDAC/Sulpho-NHS 1:1 (Biorad) diluted 1/100 for 300 s at 25 μ L/min. Anti-SIRP α antibodies were diluted to 80 nM concentration in 10 mM sodium acetate buffer pH 4.5 and immobilized to the chip at 30 μ L/min for 50 s. Chip was inactivated with ethanolamine for 300 s at 25 μ L/min. The analytes (e.g., SIRP- α from different species, SIRP- β , SIRP- γ) were injected in a “one-shot” kinetic mode at nominal concentrations of 100, 33, 11, 3.7, 1.2 and 0 nM. Association times were monitored for 90 s at 100 μ L/min, and dissociation times were monitored for 1200 s. The surfaces were regenerated with a 2:1 v/v blend of Pierce IgG elution buffer/4M NaCl.

Alternatively, K_D determination was performed using antibody capture via an NLC chip. In this case, 15 μ g/mL biotinylated protein A (ThermoFisher) was injected at 30 μ L/min for 120 s over the NLC chip to obtain an immobilization response of ~1000-1200RUs. Next, anti-SIRP α

171

antibodies (~160 nM) were injected for 80 s at 30 $\mu\text{L}/\text{min}$. The analytes (SIRP- α from different species, SIRP- β and SIRP- γ) were subsequently injected in a “one-shot” kinetic mode at nominal concentrations of 100, 33, 11, 3.7, 1.2 and 0 nM. Association times were monitored for 60 s at 25 $\mu\text{L}/\text{min}$, and dissociation times were monitored for 120 s. The surfaces were regenerated with a 2:1 v/v blend of Pierce IgG elution buffer/4M NaCl.

Biosensor data were double-referenced by subtracting the interspot data (containing no immobilized protein) from the reaction spot data (immobilized protein) and then subtracting the response of a buffer “blank” analyte injection from that of an analyte injection. Double-referenced data were fit globally to a simple Langmuir model and the K_D value was calculated from the ratio of the apparent kinetic rate constants ($K_D = k_d/k_a$).

Results

Three representative antibodies (a blocking, non-blocking, and “kick off” antibody) were further characterized for binding to various SIRP- α , - β , - γ proteins as described above. The three antibodies contain human sequences and were derived from the HuMab chicken immunization experiments. These antibodies were tested as full-length human IgG1 antibodies with L234A, L235A, G237A, and N297A mutations. SIRP- α proteins examined in these experiments corresponded to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), human SIRP β isoform 1 (SEQ ID NO:13), human SIRP β isoform 2 (SEQ ID NO:14), and human SIRP γ isoform 1 (SEQ ID NO:15). Human SIRP β isoform 1 (SEQ ID NO:13) is also known in the art as SIRP β 1 isoform 1.

K_D values obtained for the three antibodies are shown in Table E below.

TABLE E

Summary of K_D values (M)							
		v1	v2	cyno	beta isoform 1	beta isoform 2	gamma isoform 1
Kick-off	AB132a	4.26×10^{-10}	1.86×10^{-9}	2.41×10^{-9}	1.31×10^{-10}	1.64×10^{-10}	5.19×10^{-10}
Blocker	AB119a	1.98×10^{-10}	7.99×10^{-11}	1.45×10^{-10}	3.42×10^{-10}	3.76×10^{-10}	8.86×10^{-10}
Non-blocker	AB136a	6.88×10^{-10}	3.22×10^{-9}	2.65×10^{-9}	4.35×10^{-9}	2.27×10^{-9}	1.20×10^{-7}

Next, the binding kinetics of various antibody clones to selected mouse SIRP- α proteins were determined as described above. Mouse proteins that were tested include BALBc (SEQ ID NO:10), BL6 (SEQ ID NO:9), NOD (SEQ ID NO:8), and m129 (SEQ ID NO:7) SIRP- α proteins. The results are summarized in Tables F-I below. Antibody clones labeled as “c” were tested as full-length mouse IgG2a antibodies; antibody clones labeled as “a” were tested as full-length human IgG1 antibodies with L234A, L235A, G237A, and N297A mutations.

TABLE F

Summary of kinetics for binding of selected antibodies to BALBc mouse SIRP- α protein (SEQ ID NO: 10)			
Antibody	K_{on} (1/Ms)	K_{off} (1/s)	K_D (M)
AB136a	8.27×10^5	2.73×10^{-4}	3.31×10^{-10}
AB3c	7.76×10^5	4.12×10^{-3}	5.31×10^{-9}
AB21c	4.62×10^5	6.18×10^{-4}	1.34×10^{-9}
AB25c	3.03×10^5	2.92×10^{-3}	9.64×10^{-9}

172

TABLE F-continued

Summary of kinetics for binding of selected antibodies to BALBc mouse SIRP- α protein (SEQ ID NO: 10)			
Antibody	K_{on} (1/Ms)	K_{off} (1/s)	K_D (M)
AB27c	1.50×10^5	2.26×10^{-3}	1.50×10^{-8}
AB66c	2.24×10^5	7.62×10^{-4}	3.41×10^{-9}

TABLE G

Summary of kinetics for binding of selected antibodies to BL6 mouse SIRP- α protein (SEQ ID NO: 9)			
Antibody	K_{on} (1/Ms)	K_{off} (1/s)	K_D (M)
AB136a	6.24×10^5	5.85×10^{-3}	9.37×10^{-9}
AB3c	4.12×10^5	6.19×10^{-3}	1.50×10^{-8}
AB21c	2.76×10^5	2.41×10^{-4}	8.76×10^{-10}
AB25c	1.42×10^5	3.99×10^{-4}	2.81×10^{-9}
AB27c	1.47×10^5	1.40×10^{-3}	9.49×10^{-9}
AB66c	1.07×10^5	6.19×10^{-4}	5.80×10^{-9}

TABLE H

Summary of kinetics for binding of selected antibodies to NOD mouse SIRP- α protein (SEQ ID NO: 8)			
Antibody	K_{on} (1/Ms)	K_{off} (1/s)	K_D (M)
AB136a	8.41×10^5	4.48×10^{-4}	5.33×10^{-10}
AB3c	9.99×10^5	1.43×10^{-3}	1.43×10^{-9}
AB21c	7.49×10^5	4.79×10^{-4}	6.40×10^{-10}
AB25c	3.66×10^5	1.43×10^{-3}	3.90×10^{-9}

TABLE H-continued

Summary of kinetics for binding of selected antibodies to NOD mouse SIRP- α protein (SEQ ID NO: 8)			
Antibody	K_{on} (1/Ms)	K_{off} (1/s)	K_D (M)
AB27c	2.96×10^5	1.01×10^{-3}	3.42×10^{-9}
AB66c	4.12×10^5	2.32×10^{-4}	5.64×10^{-10}

TABLE I

Summary of kinetics for binding of selected antibodies to m129 mouse SIRP- α protein (SEQ ID NO: 7)			
Antibody	K_{on} (1/Ms)	K_{off} (1/s)	K_D (M)
AB136a	6.88×10^5	5.97×10^{-4}	8.67×10^{-10}
AB3c	5.34×10^5	1.18×10^{-2}	2.20×10^{-8}
AB21c	5.63×10^5	3.31×10^{-5}	5.88×10^{-11}
AB25c	3.52×10^5	2.07×10^{-5}	5.87×10^{-11}

TABLE I-continued

Summary of kinetics for binding of selected antibodies to m129 mouse SIRP- α protein (SEQ ID NO: 7)			
Antibody	K_{on} (1/Ms)	K_{off} (1/s)	K_D (M)
AB27c	2.07×10^5	1.01×10^{-4}	4.87×10^{-10}
AB66c	2.07×10^5	1.98×10^{-5}	9.55×10^{-11}

For the above antibodies, VH and VL domain sequences (respectively) were as follows: AB136a: SEQ ID NO: 133 and 134; AB3c: SEQ ID NO: 242 and 243; AB21c: SEQ ID NO: 135 and 136; AB25c: SEQ ID NO: 137 and 138; AB27c: SEQ ID NO: 139 and 140; AB66c: SEQ ID NO: 141 and 142.

These results demonstrate that multiple antibody clones bound to all four mouse SIRP- α proteins, making these antibodies suitable for characterization in in vivo mouse models.

Example 3: Functional Properties of Anti-SIRP- α Antibodies

The previous Examples describe the identification and characterization of anti-SIRP- α antibodies with a diverse array of properties, such as binding specificity for various human, cynomolgus, and/or murine SIRP- α , SIRP- β , and SIRP- γ proteins, and whether or not the antibodies block binding between SIRP- α and CD47. Antibodies representing particular categories of interest were next examined in a variety of in vitro and in vivo functional assays aimed at characterizing their effects on SIRP- α 's biological functions. In particular, "blocking" (i.e., antibodies that block binding between SIRP- α and CD47) and "non-blocking" (i.e., antibodies that do not block binding between SIRP- α and CD47) were characterized for activities in various in vitro and in vivo assays.

As noted above, antibody clones labeled as "a" were tested as full-length human IgG1 antibodies with L234A, L235A, G237A, and N297A mutations. Antibody clones labeled as "b" were tested as full-length mouse IgG1 antibodies with an N297A mutation. Antibody clones labeled as "c" were tested as full-length mouse IgG2a antibodies.

Methods

Tumor Cell Line Culturing

DLD-1 (human colorectal adenocarcinoma) and OE19 (human esophageal carcinoma) cells were maintained in growth medium comprised of RPMI (Gibco) supplemented with 10 percent heat-inactivated Fetal Bovine Serum (Gibco), one percent penicillin/streptomycin (Gibco), and one percent GlutamaxTM glutamine supplement (Gibco). Derivation and Culture of Human Monocyte-Derived Macrophages

Trima residuals were received from Blood Centers of the Pacific and diluted 1:4 with Phosphate Buffered Saline (PBS, Gibco). Diluted blood was split into four tubes and underlaid with 20 ml Ficoll-Paque[®] Plus (GE Healthcare) lymphocyte medium. Tubes were centrifuged for 30 minutes at 400 $\times$ g. PBMCs were collected from the interface and resuspended in FACS buffer (PBS with 0.5 percent Bovine Serum Albumin (Gibco)). CD14⁺ monocytes were purified by negative selection using the Monocyte Isolation Kit II (Miltenyi Biotec) and LS columns (Miltenyi Biotec) according to the manufacturer's protocol.

For nonpolarized macrophages, CD14⁺ monocytes were seeded into 15 cm tissue culture plates (Corning) at 10 million cells per dish in 25 ml IMDM (Gibco) supplemented

with 10 percent human AB serum (Corning), one percent penicillin/streptomycin, and one percent GlutamaxTM glutamine supplement. Cells were cultured for seven to ten days.

For M2 polarized macrophages, CD14⁺ monocytes were seeded into 15 cm tissue culture plates (Corning) at 6 million cells per dish in 25 ml RPMI (Gibco) supplemented with 10 fetal bovine serum (Thermo Fisher), one percent penicillin/streptomycin, and one percent Glutamax, and 50 ng/ml M-CSF (Miltenyi). Cells were cultured for seven to ten days.

In Vitro Phagocytosis Assays

DLD-1 and OE19 cells were detached from culture plates by washing twice with 20 ml PBS and incubation in 10 ml TrypLETM Select (Gibco) adherent cell dissociation enzyme for 10 minutes at 37° C. Cells were centrifuged, washed in PBS, and resuspended in medium. Cells were labeled with the CelltraceTM CFSE Cell Proliferation kit (Thermo Fisher) according to the manufacturer's instructions and resuspended in IMDM. Macrophages were detached from culture plates by washing twice with 20 ml PBS and incubation in 10 ml TrypLETM Select adherent cell dissociation enzyme for 20 minutes at 37° C. Cells were removed with a cell scraper (Corning), washed in PBS, and resuspended in IMDM.

Phagocytosis assays were assembled in ultra-low attachment U-bottom 96 well plates (Corning) containing 100,000 DLD-1 or OE19 cells, 50,000 macrophages, five-fold serial dilutions of anti-SIRP- α antibody from 100 nM to 6.4 pM and cetuximab (Absolute Antibody) at 1 or 0.01 ug/ml, trastuzumab at 0.01 ug/ml, or control antibody of the same isotype (Southern Biotech). All anti-SIRP- α antibodies tested had a human IgG1 with L234A, L235A, G237A, and N297A mutations except AB136c, which had a mouse IgG2a. Plates were incubated two hours at 37° C. in a humidified incubator with 5 percent carbon dioxide. Cells were pelleted by centrifugation for five minutes at 400 $\times$ g and washed in 250 μ l FACS buffer. Macrophages were stained on ice for 15 minutes in 50 μ l FACS buffer containing 10 μ l human FcR Blocking Reagent (Miltenyi Biotec), 0.5 μ l anti-CD33 BV421 (Biolegend), and 0.5 μ l anti-CD206 APC-Cy7 (Biolegend). Cells were washed in 200 μ l FACS buffer, washed in 250 μ l PBS, and stained on ice for 30 minutes in 50 μ l Fixable Viability Dye eFluorTM 506 (ebioscience) dye diluted 1:1000 in PBS. Cells were washed twice in 250 μ l FACS buffer and fixed for 30 minutes on ice in 75 μ l CytotfixTM (BD Biosciences) fixation solution. Cells were washed in 175 μ l FACS buffer and resuspended in 75 μ l FACS buffer. Cells were analyzed on a FACS CantoTM II (BD Biosciences) flow cytometer, with subsequent data analysis by Flowjo 10.7 (Treestar). Dead cells were excluded by gating on the e506-negative population. Macrophages that had phagocytosed tumor cells were identified as cells positive for CD33, CD206, and CFSE.

In Vivo Dendritic Cell Activation

Balb/c mice (n=3/group) were intravenously injected with a control rat anti-mouse anti-SIRP- α antagonistic antibody (clone p84), rat IgG control, AB136b (AB136 with a mouse IgG1 Fc region bearing an N297A mutation), mouse IgG control, or vehicle (PBS) at 10 mg/kg. Five hours post injection, spleens were harvested and processed into single cell suspension by mechanical dissociation through a 70 μ m cell strainer. Cells were washed two with PBS and red blood cell lysis was performed. Subsequently, cells were washed for an additional two times with PBS/10% FBS and prepared for cell staining. Cells were stained with fluorochrome conjugated MHC-II, CD8, 33D1, CD4, CD11c, CD86, CCR7 and viability dye at 4 degrees for one hour. Cells were

washed two times and analyzed using a BD Canto™ II flow cytometer and data were processed using Flowjo.

In Vivo Anti-Tumor Activity

For the CT26 syngeneic mouse colon carcinoma model, CT26 cells were implanted subcutaneously in BALB/c mice and randomized into groups (8-9 mice/group). Treatment groups included vehicle (PBS) and AB136b. AB136 anti-SIRP- α had a mouse IgG1 Fc region bearing an N297A mutation. Treatment was initiated when tumors were an average of 75-80 mm³, day 7 or 8 post implant. Mice were dosed intraperitoneally (IP) at 3 mg/kg twice a week for three weeks with AB136b. Animals were sacrificed when tumors reached a volume of ~2000 mm³.

For the MC38 syngeneic mouse colon carcinoma model, MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8-10 mice/group). Treatment groups included vehicle (PBS), AB25b, AB25c, AB27b, AB3b, and AB136b. All anti-SIRP α antibodies had a mouse IgG1 Fc region bearing an N297A mutation except for AB25c, which had a mouse IgG2a. Treatment was initiated when tumors were an average of 60-65 mm³, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks for anti-SIRP α . Animals were sacrificed when tumors reached a volume of ~2000 mm³.

Results

Various anti-SIRP- α antibodies were examined for induction of phagocytosis in in vitro phagocytosis assays using polarized and non-polarized macrophages.

Additional anti-SIRP- α antibodies AB3 and AB45 were tested in the experiments described below. Both are non-blocking antibodies. The k_{off} rates of binding of each clone (in scFv-Fc format) to each SIRP analyte were determined using SPR (Table J1). The SPR screening conditions have been described herein, and the Koff values were determined using Langmuir kinetic fittings. SIRP analyte sequences are as follows: CV1-3, SEQ ID NO:18; v1, SEQ ID NO:5; v2, SEQ ID NO:6; cyno1, SEQ ID NO: 11; cyno2, SEQ ID NO:12; m129, SEQ ID NO:7; NOD, SEQ ID NO:8; BL6, SEQ ID NO:9; sirpb1, SEQ ID NO: 13; sirpg, SEQ ID NO:15.

TABLE J1

Langmuir kinetic fittings K_D (M) for AB3 and AB45										
SEQ ID NO:										
	5 V1	6 v2	11 cyno1	12 Cyno2	8 NOD	9 BL6	10 BALBc	15 sirp-g	13 Sirpb1	14 Sirpb2
AB3*	1.62E-10	7.67E-11	2.29E-09	2.85E-09	1.63E-09	3.65E-09	1.16E-09	8.36E-08	1.63E-09	6.78E-11
AB45**	6.63E-11	1.34E-10	NB	NB	NB	NB	NB	2.71E-08	1.06E-08	2.53E-11

NB—No binding.

*See SEQ ID NOs: 242 and 243 for VH and VL domains, respectively.

**See SEQ ID NOs: 244 and 245 for VH and VL domains, respectively.

As shown in FIGS. 13A & 13B, non-blocking anti-SIRP- α antibody AB3a was found to induce phagocytosis of DLD-1 tumor cells in M2 polarized macrophages. In particular, treatment of macrophages with cetuximab (anti-EGFR antibody) and AB3a led to robust induction of tumor cell phagocytosis. Similarly, non-blocking anti-SIRP- α antibody AB45a was found to induce phagocytosis of DLD-1 tumor cells M2 polarized macrophages, and treatment of macrophages with cetuximab (anti-EGFR antibody) and AB45a led to robust induction of tumor cell phagocytosis (FIGS. 13C & 13D). Blocking anti-SIRP- α antibodies AB119a (FIG. 13E) and AB135a (FIG. 13F) were also found

to induce phagocytosis of OE19 tumor cells when co-administered with trastuzumab (anti-HER2 antibody). Non-blocking anti-SIRP- α antibody AB136c was also found to induce phagocytosis of DLD-1 tumor cells (FIG. 13G).

A dendritic cell activation assay was used to characterize the in vivo effects of non-blocking anti-SIRP- α antibody AB136b (SEQ ID NOs:133 and 134 for VH and VL domain sequences, respectively). Failure to engage mouse SIRP- α receptor on splenic dendritic cells via CD47 binding leads to splenic dendritic cell activation. Control anti-SIRP- α antagonist antibody p84 activated splenic dendritic cells when injected intravenously into mice (FIG. 14). Non-blocking anti-SIRP- α antibody AB136b was tested in vivo to determine if it leads to dendritic cell activation. Interestingly, AB136b treatment led to activation of splenic dendritic cells at a similar level as p84 (FIG. 14).

Next, the in vivo anti-tumor effects of various anti-SIRP- α antibodies were assayed in two syngeneic mouse colon carcinoma models. The anti-tumor activities of blocking anti-SIRP- α antibodies AB25b, AB25c and AB27b and non-blocking anti-SIRP- α antibodies AB3b and AB136b were examined in an MC38 syngeneic mouse colon carcinoma model to assess their single agent activities. Both blocking (AB25b, AB25c and AB27b) and non-blocking (AB3b and AB136b) anti-SIRP- α antibodies delayed tumor formation at 10 mg/kg as compared to vehicle alone (FIG. 15) in the MC38 syngeneic mouse model. On day 25, groups treated with anti-SIRP α antibodies had three mice below 600 mm³ for AB25b and AB27b, four mice below 600 mm³ for AB25c and AB3b and five mice below 600 mm³ for AB136b, while the vehicle-treated group had only two mice below 600 mm³.

The anti-tumor activity of non-blocking AB136b was next evaluated in a CT26 syngeneic mouse colon carcinoma model to assess single agent activity. These results confirmed that AB136b treatment delayed tumor formation at 3 mg/kg as compared to vehicle alone in the CT26 syngeneic mouse model (FIG. 16).

Taken together, these results demonstrate the efficacy of anti-SIRP- α antibody treatment in inducing tumor cell

phagocytosis by macrophages, activating dendritic cells, and inhibiting tumor growth in vivo. Multiple blocking anti-SIRP- α antibodies were found to promote tumor cell phagocytosis, activate dendritic cells and block in vivo tumor growth. Surprisingly, however, treatment with non-blocking anti-SIRP- α antibody was also found to activate splenic dendritic cells and inhibit tumor growth in vivo in two different syngeneic mouse tumor models. The finding that non-blocking anti-SIRP- α antibodies were able to increase phagocytosis and block in vivo tumor growth was both surprising and unexpected. Previous work has suggested that

only blocking anti-SIRP- α antibodies would be able to inhibit in vivo tumor growth (see Yanagita, T. et al. (2017) *JCI Insight* 2:e89140).

Example 4: Structural Analysis of Anti-SIRP- α Antibody Epitopes on SIRP- α

As described in Examples 1-3 above, anti-SIRP- α antibodies have been generated with a variety of specificities and modes of binding to SIRP- α , e.g., antibodies that block CD47 binding to SIRP- α , antibodies that do not block CD47 binding to SIRP- α , and antibodies that bind to SIRP- α and reduce its affinity for binding CD47 ("kick off" antibodies). Structural analyses were undertaken in order to understand how these types of antibodies bind to the D1 domain of SIRP- α as compared to CD47 and characterize the epitopes of selected anti-SIRP- α antibodies.

Methods

Crystallography and Structural Analysis

Expression of the Fabs and SIRP α was similar to previously established protocols and involved traditional methods of affinity chromatography and size exclusion for protein purification. A human SIRP- α v1 mutant bearing an N80A mutation as compared to SEQ ID NO:5 was used for ease of protein production in Expi293 (SEQ ID NO:296). The goal of using an N80A SIRP α was to produce a homogenous, non-glycosylated form of SIRP α that would be most amenable for crystallization. The final purification buffer is minimal with only 10 mM Tris pH 7.5 and 50 mM NaCl. The purified complex sample is stable at 4° C. and was concentrated to 10-12 mg/mL in preparation for crystallization experiments and eventual structure determination.

The methods for the Fab:SIRP α complex project followed well established principles or crystallography starting with the sparse matrix technique, which led to custom optimized conditions, establishing a routine protocol. Crystallization was carried out utilizing the sitting drop vapor diffusion technique. For the initial sparse matrix screening, condition kits commercially available through Qiagen® were utilized (Table J2). These crystallization experiments were set in drops with varying ratios of protein to crystallant condition. The ratios set were in the range of 1:1, 2:1, and 3:1 protein to condition in total volume of 1 μ L in the subwells of the 96 well (8 $\times$ 12) tray. 100 μ L of crystallization condition was placed in the well reservoir. These drops were set by utilizing the Mosquito drop setting robot and the completed plates were sealed and stored in a 12° C. incubator. The experiment was monitored by viewing the plates/drops under the microscope to see developments that included drop precipitation, aggregation, phase separation, amorphous formation, as well as initial protein crystals.

TABLE J2

Kits used for initial sparse matrix screening.	
Kit	Cat. No./ID
Classics Suite	130901
Classics II Suite	130923
Classics L Suite	130902
PEGs Suite	130904
PEGs II Suite	130916
PACT Suite	130918
ProComplex Suite	130915
AmSO ₄ Suite	130905

Crystallization Summary and Crystal Harvesting

Crystallization of the 4 complexes was achieved with derivatives of two main conditions as shown in Table J3). Crystal harvesting was done on optimal crystal forms so that manipulation and cryo-freezing in liquid nitrogen would not jeopardize the integrity of the crystal prior to X-ray diffraction screening and possible data collection. To prevent icing, a cryo-protectant was implemented during freezing. The typical cryo-protectant included an addition of 20% glycerol to the crystallization condition that formed the crystal. When crystals formed in conditions with 30% PEG 4000 or above, the addition of glycerol was not necessary. The high percentage of PEG 4000 behaved as a viable cryo-protectant. Crystals of complexes were manipulated with cryo loops that are either nylon or Mitigen® crystal mount style. Single crystals were isolated and excised out of the drop in which they formed and transferred into cryo-protectant for a short period before plunging into liquid nitrogen to flash freeze.

TABLE J3

Crystallization conditions for forming Fab:SIRP- α complex crystals.			
Fab Antibody	Buffer	Salt	Precipitant
119	0.1M Tris-HCl pH 8.5	0.2M MgCl ₂	15% (w/v) PEG 4000
136	0.1M Tris-HCl pH 8.5	0.2M MgCl ₂	30% (w/v) PEG 4000
3	0.1M Sodium Acetate pH 4.6	0.45M Ammonium Sulfate	35% (w/v) PEG 4000
115	0.1M Tris pH 7.2	0.2M MgCl ₂	18% (w/v) PEG 4000

Data Collection and Processing

Data were collected on crystals flash frozen in liquid nitrogen as described above. These samples remained in the cryo stream when screened for protein lattice diffraction prior to subsequent dataset collection via the oscillation method. Collection occurred at either the Advanced Photon Source or Diamond Light Source. Datasets were reduced using the xia2 suite. xia2 is a wrapper script that allows for automated reduction of macromolecular crystallographic data. The program is able to utilize multiple data reducing programs such as XDS, DIALLS, Mosflm, and Aimless. These programs allow for the diffraction data to be indexed into the appropriate space group and unit cell, integration of intensities, and scaling to produce an estimate of intensity of each unique reflection. For the initial Complex 1 dataset, phases were calculated via molecular replacement (MR) using Phaser MR of the CCP4 Suite utilizing homolog models of SIRP α , as well as Fab (see PDB CODE: 2UV3 and PDB CODE: 4NM4 respectively). For subsequent datasets of Complex 2, 3, and 4, the completed structure model coordinates of Complex 1 was used as the search model for the phase calculation and the initial model build.

Structure Building and Refinement

Model building utilized the Coot program. As an example, Complex 1 was calculated to have 4 molecules in the ASU, therefore, 4 pairs of AB119f complexed with SIRP α are to be built to complete the structure model. The strategy of structure building was to build amino acid residues into electron density following the known sequence of the target proteins. Quality of data directly correlates to the fit of the structure model into the observed crystal dataset. Therefore, building the tertiary structure of the Complex followed established protocol. Initially, the peptide bone of the residue was built. This was followed by the placement of the

correct residue side chain if the electron density map permits. In the instance an original amino acid in the protein sequence was not modeled, it was due to lack of density for the respective side chain, thus only the peptide backbone of the amino acid can be modeled and an alanine residue side chain is built in place. Stretches of the sequence may be a disordered region and cannot be built due to lack of density even for placement of the amino acid backbone. The builds were followed by subsequent rounds of refinement through the Refmac program, a part of the CCP4 Suite. The refinement program is utilized to minimize coordinate parameters through the Maximum Likelihood residual. Refinement was completed when the model parameters fell into favorable statistical tolerances for parameters including: the Rvalue, bond length and angles, and Ramachandran fit. In addition, to check physical tolerances of the model, Molprobit was also utilized to ensure complete and proper structure determination.

Epitope Mapping and Superimposition Analysis

Buried surface area of the antigen for the epitope was calculated as the difference between the solvent-accessible surface area of the antigen alone and antigen in complex with Fab fragment of the antibody. Conversely, buried surface area of the antibody heavy and light chains for the paratope was calculated as the difference between the solvent-accessible surface area of the fab fragment alone and in complex with its antigen. The surface accessible area was calculated by the rolling ball method with probe radius of 1.4 Å. Buried surface area is reported in Å². All antibody: SIRPα complexes were superimposed by selecting SIRPα from each structure and superimposing its carbon atoms. CD47: SIRPα, complex (V1 variant) structure used for the analysis is PDB:4CMM. The superimposition and the RMSD calculation was performed using PYMOL.

Results

First, the structure of blocking antibody 119 Fab bound to SIRP-α was determined and compared to CD47 binding to SIRP-α. As shown in FIG. 17A, antibody 119 and CD47 bound to a similar epitope of the SIRP-α D1 domain. 56% of the SIRP-α residues in the antibody 119 epitope were also found in the CD47 epitope, while 75% of the SIRP-α residues in the CD47 epitope were also found in the antibody 119 epitope. The SIRP-α residues participating in the interaction with antibody 119 (as determined by buried surface area changes, described supra) are shaded and shown as space-filled models in FIG. 17B. Key residues from the antibody 119 Fab paratope are also shown (black sticks), including heavy chain residues N31, S53, D55, Y57, T99, S101, and W102 (N31 is not visible in the structure orientation shown in FIG. 17B) and light chain residues Y92 and W94 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:335 and 97, respectively).

FIG. 18A shows a comparison between CD47 binding to the SIRP-α D1 domain and non-blocking antibody 136 Fab binding to the SIRP-α D1 domain. The SIRP-α binding epitopes of antibody 136 and CD47 were found to be completely non-overlapping. The SIRP-α residues partici-

pating in the interaction with antibody 136 are shaded and shown as space-filled models in FIG. 18B. Key residues from the antibody 136 Fab paratope are also shown (black sticks), including heavy chain residues E56, Y59, and R102, and light chain residues Y92 and R94 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:133 and 134, respectively).

FIG. 19A shows a comparison between CD47 binding to the SIRP-α D1 domain and non-blocking antibody 3 Fab binding to the SIRP-α D1 domain. The SIRP-α binding epitopes of antibody 3 and CD47 were also found to be completely non-overlapping. The SIRP-α residues participating in the interaction with antibody 3 are shaded and shown as space-filled models in FIG. 19B. Key residues from the antibody 3 Fab paratope are also shown (black sticks), including heavy chain residues R56 and G100 and light chain residues R24, R26, Y86, and G88 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:242 and 243, respectively).

FIG. 19C shows a comparison between CD47 binding to the SIRP-α D1 domain and "kick off" antibody 115 Fab binding to the SIRP-α D1 domain. The SIRP-α binding epitope of antibody 115 was found to be adjacent to the epitope of CD47. While mAb 115 was found to bind an epitope of the SIRP-α D1 domain adjacent to that of CD47, parts of the 115 epitope likely overlap with CD47 itself. Comparing both epitopes, there are 2 identical residues, suggesting that the interactions from the non-overlapping portions of each epitope allow the 115 Fab and CD47 to bind SIRP-α simultaneously. Kinetic analyses described herein demonstrate that antibody 115 forms a transient complex with SIRP-α and CD47 but reduces the affinity of CD47 for the SIRP-α D1 domain. The SIRP-α residues participating in the interaction with antibody 115 are shaded in FIG. 19D. Key residues from the antibody 115 Fab paratope are also shown, including heavy chain residues S31, Y32, N52, Y100, and W102 and light chain residues K32 and Y92 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:273 and 274, respectively).

FIG. 20A illustrates the binding of antibodies 119, 136, 3, and 115 to SIRP-α, as compared to CD47 binding to SIRP-α. Blocking antibody 119 was found to bind a completely non-overlapping SIRP-α epitope, as compared with the SIRP-α epitopes of both non-blocking antibodies 136 and 3. SIRP-α and antibody residues that participate in these interactions are summarized in Table K1. SIRP-α residues determined to interact with CD47 were determined as follows: 29, 30, 31, 33, 34, 35, 36, 37, 50, 51, 52, 53, 54, 66, 67, 68, 69, 74, 93, 96, 97, 98, 99, and 101 (according to SEQ ID NO:5). FIGS. 20B-20E summarize the epitopes for CD47 and anti-SIRP-α Fabs 119, 136, 3, and 115 binding to SIRP-α (residues according to SEQ ID NO:296) and lists buried surface area changes. As described herein, the numbering of amino acid residues of an antibody used to indicate the paratope is based on numbering according to the amino acid sequence of the heavy or light chain, and not, e.g., the Kabat or Chothia numbering of antibody residues.

TABLE K1

SIRP-α epitopes and antibody paratopes for selected anti-SIRP-α antibodies.			
Antibody clone	SIRP-α epitope residues	Heavy chain antibody paratope residues	Light chain antibody paratope residues
119	3, 28, 29, 30, 31, 32, 33, 35, 36, 38, 50, 52, 53, 54, 62, 64, 65, 66,	30, 31, 32, 33, 50, 52, 53, 54, 55, 56, 57, 96, 98, 99,	27, 29, 30, 32, 49, 53, 55, 56, 91, 92, 94, and 95

TABLE K1-continued

SIRP- α epitopes and antibody paratopes for selected anti-SIRP- α antibodies.			
Antibody clone	SIRP- α epitope residues	Heavy chain antibody paratope residues	Light chain antibody paratope residues
136	67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 95, 96, 97, and 98 (SEQ ID NO: 296)	100, 101, 102, 103, 104, 105, 106, and 108 (SEQ ID NO: 335)	(SEQ ID NO: 97)
	6, 8, 9, 10, 11, 12, 14, 42, 43, 44, 46, 87, 88, 90, 103, 104, 105, 106, 107, 108, 109, 111, 112, 113, 115, and 116 (SEQ ID NO: 296)	33, 52, 54, 56, 57, 59, 99, 100, 101, 102, 103, and 104 (SEQ ID NO: 133)	27, 30, 32, 92, 93, 94, and 95 (SEQ ID NO: 134)
	5, 6, 7, 8, 9, 10, 11, 12, 14, 24, 26, 28, 72, 88, 90, 108, 109, 111, 113, and 115 (SEQ ID NO: 296)	31, 50, 52, 53, 54, 56, 58, 99, 100, 101, and 102 (SEQ ID NO: 242)	24, 25, 26, 45, 86, 87, 88, and 89 (SEQ ID NO: 243)
115	17, 39, 40, 44, 45, 46, 47, 48, 49, 51, 54, 55, 56, 57, 58, 59, 80, 82, 83, 84, 85, 89, 100 (SEQ ID NO: 296)	28, 30, 31, 32, 33, 50, 52, 53, 54, 56, 57, 75, 100, 101, 102 (SEQ ID NO: 273)	32, 91, 92, 93, and 94 (SEQ ID NO: 274)

TABLE K2

Anti-SIRP- α antibody paratopes and buried surface area calculations.		
Residue name	Residue number	Change in buried area ($\text{\AA}^2$)
119 Heavy Chain		
TRP	102	-228
SER	101	-97.2
THR	99	-70.4
SER	53	-60.9
ASP	55	-59.2
ASN	31	-58.2
TYR	57	-53.5
PHE	32	-43.1
SER	103	-30.7
SER	98	-24.7
SER	30	-19.3
GLY	52	-18.4
VAL	100	-18.4
ASP	105	-17.2
ALA	33	-16.3
GLY	54	-7.8
ASP	108	-6.2
PHE	106	-5.7
GLY	104	-5.6
LYS	96	-5.3
THR	56	-2.9
THR	50	-1.7
119 Light Chain		
TYR	92	-78
TRP	94	-57.5
ILE	53	-31.5
TYR	49	-30.2
PRO	95	-20.4
GLU	55	-16.6
VAL	29	-12.3
ALA	30	-8.8
ASP	32	-8
TYR	91	-3.5
THR	56	-1.5
GLN	27	-3.4
136 Heavy Chain		
ARG	102	-114.9
GLU	56	-67.6
TYR	59	-57.3
ARG	104	-46.7
ILE	57	-30.8
ASN	101	-12.6
ASP	33	-12
TYR	103	-10.9

TABLE K2-continued

Anti-SIRP- α antibody paratopes and buried surface area calculations.		
Residue name	Residue number	Change in buried area ($\text{\AA}^2$)
25		
SER	52	-5.1
SER	54	-4
GLU	99	-3.3
ASN	100	-1.1
30		
136 Light Chain		
ARG	94	-97.6
TYR	92	-95.2
TYR	30	-89
GLN	27	-46.6
TYR	32	-46.5
ASP	93	-19.1
PRO	95	-7.3
35		
3 Heavy Chain		
ARG	56	-107.9
GLY	100	-62
ASP	31	-34.9
SER	53	-27.1
THR	52	-25.2
SER	101	-22.7
GLY	54	-15.1
TYR	58	-12.2
GLN	50	-7.3
GLY	102	-5.1
PHE	99	-4.5
40		
3 Light Chain		
ARG	24	-123.6
TYR	86	-62.5
ARG	26	-56.4
GLY	88	-49.1
ASP	87	-22.1
GLY	25	-13.1
ARG	45	-4.8
SER	89	-1.7
45		
115 Light Chain		
LYS	32	-70.2
TYR	91	-2.9
TYR	92	-85
TYR	93	-22.3
TRP	94	-21.4
50		
115 Heavy Chain		
SER	28	-35.3
SER	30	-17.2
SER	31	-70.6
55		
60		
65		

TABLE K2-continued

Anti-SIRP- α antibody paratopes and buried surface area calculations.		
Residue name	Residue number	Change in buried area ($\text{\AA}^2$)
TYR	32	-50.2
ALA	33	-1.3
ARG	50	-9.5
ASN	52	-54.4
SER	53	-39.8
GLY	54	-1.4
GLY	56	-34
GLY	57	-11.2
SER	75	-13.3
TYR	100	-53.6
ASP	101	-45.8
TRP	102	-136.4

These results elucidate the binding interfaces between the SIRP- α D1 domain and various anti-SIRP- α antibodies, as well as the binding interface between CD47 and SIRP- α . They also provide a structural basis for the observations that antibody 119 blocks binding between CD47 and SIRP- α (since they share overlapping SIRP- α epitopes), that antibodies 136 and 3 do not block binding (as their epitopes do not overlap with that of CD47), and that antibody 115 “kicks CD47 off” from binding SIRP- α (since they share only slightly overlapping SIRP- α epitopes). These studies further highlight specific antibody and SIRP- α residues that participate in each interaction.

With the SIRP α binding epitopes of non-blockers (3, 136), blockers (119) and kick-off (115) antibodies defined by crystallography, the binding locations for additional anti-SIRP α antibodies can be characterized by carrying out epitope binning described in Example 5. For instance, for any anti-SIRP α antibodies that compete with non-blocker 136 for binding to SIRP α , we can predict that these antibodies would share similar binding epitopes as 136 as defined by crystallography.

Example 5: Epitope Binning of Anti-SIRP- α Antibodies

Anti-SIRP- α antibodies described above were next assayed in epitope binning experiments with SIRP- α in order to categorize antibodies having shared and distinct epitopes.

Methods

Epitope Binning

Briefly, epitope binning was conducted as shown in FIG. 21A. A first anti-SIRP- α antibody was immobilized on a chip, then human SIRP- α v1 (SEQ ID NO:5) was injected. A second anti-SIRP- α antibody was then injected. If the second antibody was able to bind the complex formed between the first anti-SIRP- α antibody and SIRP- α , the first and second antibodies were determined to bind different epitopes. If the second antibody was not able to bind, the first and second antibodies were determined to share an epitope.

Exemplary results of a binning assay using anti-SIRP- α antibody (A) immobilized to the chip and injecting anti-SIRP- α antibodies (B-F) are shown in FIG. 21B. Anti-SIRP α antibody (B), (E) and (F) form sandwiches with the complex, and this is indicated by the increasing RU at time 60 s (upon injection of respective anti-SIRP α antibodies). As such, they are determined to bind different epitopes than anti-SIRP α (A) and are indicated by white boxes in the binning plot (FIGS. 22A & 22B). For anti-SIRP α antibody

(C) and (D), they did not form a sandwich, and this is indicated by a steady RU after injection of respective anti-SIRP α antibodies. As such, they are determined to bind the same epitope as anti-SIRP α (A) and are shaded gray in the binning plot (FIGS. 22A & 22B). The clone number for the ligand (anti-SIRP α) bound to the chip is indicated as rows, and the clone number for the analytes (anti-SIRP α) injected over the chip is indicated as columns in FIGS. 22A & 22B. “X” indicates scenarios where the data from one orientation disagreed with the other.

First, anti-SIRP α antibodies were immobilized on GLC chips using Proteon™ Amine Coupling Kit as described before. Briefly, for the immobilization step, GLC chip was activated with EDAC/Sulpho-NHS 1:1 (Biorad) diluted 1/80 for 300 s at 25 $\mu\text{L}/\text{min}$. Anti-SIRP α antibodies were diluted to 80 nM concentration in 10 mM sodium acetate buffer pH 4.5 and immobilized to the chip at 30 $\mu\text{L}/\text{min}$ for 50 s. Chip was inactivated with ethanolamine for 300 s at 25 $\mu\text{L}/\text{min}$. After which, SIRP- α v1 (SEQ ID NO:5) (100 nM) was first injected at 100 $\mu\text{L}/\text{min}$ for 60 s followed by injecting the anti-SIRP α antibodies in testing (100-150 nM) at 100 $\mu\text{L}/\text{min}$ for 60 s. The surfaces were regenerated with a 2:1 v/v blend of Pierce IgG elution buffer/4M NaCl. The resultant sensograms were used to score and group the antibodies into different bins according to their binding profiles.

Results

The following anti-SIRP- α antibody clones were binned: 3, 21, 25, 27, 45, 66, 115, 116, 117, 118, 119, 120, 121, 122, 123, 132, 135, 136, 137, 149, 161, 162, 173, 194, 209, 213, and 218. As shown in FIGS. 22A & 22B, the results demonstrated that each mode (B: blocking, NB: non-blocking, and KO: “kick off”) of anti-SIRP- α antibody binding to human SIRP- α v1 binned separately. For the non-blockers, the anti-SIRP α antibodies could be further separated into 4 bins based on their binding profiles. For instance, anti-SIRP α antibodies 123, 149, 161, 162, 194 and 218 were grouped as Bin 5. Anti-SIRP α antibodies 3, 173, 209 and 213 were grouped as Bin 4. Anti-SIRP α antibodies 136 and 137 were grouped as Bin 2. Anti-SIRP α antibody 45 has a unique binding profile and is grouped as Bin 6.

Combined with the structural analyses described in Example 4, these results led to the model of antibody and CD47 binding to the SIRP- α D1 domain shown in FIG. 23. For instance, various anti-SIRP α antibodies in Bins 1, 2, 3 and 4 are postulated to share similar binding epitopes with anti-SIRP α 119, 136, 115 and 3 which were categorized in the respective bins. The binding epitopes for anti-SIRP α 119, 136, 115 and 3 have been obtained from crystal solutions as described in Example 4.

In addition, anti-SIRP α antibodies in Bin 5 are postulated to bind adjacent to anti-SIRP α 136 (Bin 2) and 115 (Bin 3) respectively and share overlapping epitopes. This is based on binning data showing anti-SIRP α antibodies in Bin 5 competed with antibodies in Bins 2 and 3 for binding to SIRP α V1, and they did not compete with antibodies in Bins 1, 4 and 6 (FIG. 22B).

Anti-SIRP α antibody 45 (Bin 6) is postulated to bind adjacent to anti-SIRP α 3 (Bin 4) and 136 (Bin 2) respectively and share overlapping epitopes. This is based on binning data showing anti-SIRP α 45 competed with antibodies in Bins 2 and 4 for binding to SIRP α V1, and anti-SIRP α 45 did not compete with antibodies in Bins 1, 3 and 5 (FIG. 22B).

The model in FIG. 23 illustrates the epitope of each antibody bin based on structural and binning analyses, and demonstrates the overlap between various antibody bins using exemplary antibody clones. Additional anti-SIRP- α

antibodies isolated herein are grouped into families and bins based on their epitope mapping profiles, homologies in their VH/VL domain sequences, binding specificities to SIRP proteins and the sources of the antibodies. For instance, anti-SIRP- α antibodies in family 1/Bin 1 are CD47 blockers that share high sequence homologies in their VH and VL domains (Table P and FIGS. 11O & 11P). The VH and VL are fully human sequences. Anti-SIRP- α antibodies in family 2/Bin 1 are also CD47 blockers. Their VH and VL share high homologies and are from human and chicken sources respectively (Table P, FIGS. 11A-11D). The anti-SIRP- α antibodies in Family 3/Bin 2 include CD47 non-blockers, and their highly homologous VH/VL are fully human sequences (Table P, FIGS. 11E-11F). The anti-SIRP- α antibodies in Family 4/Bin3 include kick-off antibodies, and their highly homologous VH/VL are fully human sequences (Table P, FIGS. 11G & 11H). Family 5 includes CD47 non-blockers that are separately mapped into Bins 4, 5, and 6. The sequence alignments of these antibodies and their binding profiles are shown in FIGS. 11I-11N and Table P, respectively. The K_{off} binding values to various SIRP- α , SIRP- β , and SIRP- γ polypeptides of the corresponding anti-SIRP α antibodies in Families 1-5 are presented in Table T.

Example 6: Germline and Liability Mutation Variants of Anti-SIRP- α Antibodies

Some anti-SIRP- α antibodies described above are fully human antibodies generated in a chicken (e.g., antibodies 119, 135, and 136). As such, some of these antibodies may contain mutations in the variable domain framework sequences, as compared to wild-type human germline sequence, by virtue of generating these antibodies in chicken B cells. Therefore, it is desirable to “back-mutate” these respective residues to match that of human germline sequence with the goal of limiting immunogenicity when these anti-SIRP α antibodies are tested in humans as potential therapeutics. In addition to germline back-mutations, the CDRs and framework region of antibodies 119, 135 and 136 were analyzed for liability hot spots. These analyses identified sites where engineering may be desired to limit risk due to modifications that may occur during manufacturing, storage and/or drug development of anti-SIRP α antibodies.

Methods

Germline/Liability Mutations

The K_D for respective germline and liability mutants binding to SIRP were determined using direct immobilization using GLC chip as described supra.

The wildtype and mutant antibodies were expressed in Expi293 and purified by Protein A affinity column chromatography as described earlier. All antibodies were expressed as human L234A/L235A/G237A/N297A IgG1 Fc antibodies. Mutagenesis was carried out using QuikChange Lightning Site Directed Mutagenesis kit according to manufacturer's instructions (Agilent Catalog #210518).

Results

Antibodies 119, 135, and 136 were examined. Selected antibody sequences were aligned with available human germline sequences using IgBlast (NCBI). For instance, a total of 7 sites on the heavy and light chains of 119 were identified. As shown in FIG. 24A, residues that were not commonly occurring in human germline sequence of 119 VH (e.g., D1, E43, and L112) were back-mutated to match human germline sequence (e.g., D1E, E43K, and L112Q) while keeping CDR sequences intact. As for 119 VL (FIG. 24B), residues that were not commonly occurring in human

germline sequence of 119 VL (e.g. F21, R39, E60, and T76) were back-mutated to match human germline sequence (e.g. F21L, R39K, E60A, and T76S) while keeping CDR sequences intact. As used herein, the terms “all mut” and “mut” refer to variable domains containing all of the germline mutations described herein for a particular antibody variable domain. The amino acid numberings used to describe the germline and liability mutations are based on sequential numbering accordingly to respective SEQ IDs.

In addition, antibody sequences were also analyzed for “liability” hot spots, including residues that may be susceptible to oxidation, deamidation, isomerization, hydrolysis, and N-linked glycosylation. Potential hot spots are shown in Table L. In particular, M34V and M34L variants of HVR-H1 were generated for the VH domain of multiple antibodies.

TABLE L

Potential and known hot spots	
Oxidation	M
Deamidation	NG NS NT
Isomerization	DG DS DT
Hydrolysis	DP
N-linked glycosylation site(s)	NXS/T

Variants of antibody 119 were generated using heavy and/or light chain variable domains bearing germline and liability back-mutations. A 119 mutant (“mut”) VH domain was generated with the germline back mutations D1E, E43K, and L112Q, as well as the M34V mutations in CDR-H1 that remove a methionine residue that could potentially be oxidized (see SEQ ID NOs:246 for VH sequences). Another variant was generated with the germline back mutations D1E, E43K, and L112Q (see SEQ ID NO:258 for VH sequence). Alignments between the parental and variant sequences are shown in FIG. 24A. A 119 mutant (“mut”) light chain was also generated with the germline back mutations F21L, R39K, E60A, and T76S; an alignment between the parental and variant sequence is shown in FIG. 24B.

Antibody 119 variants with the mutant heavy and/or light chain were compared with the parental 119 antibody with an IgG1 Fc region bearing L234A, L235A, G237A, and N297A mutations (EU numbering), and a parental 119 antibody with an IgG4 Fc region bearing an S228P mutation (EU numbering), for binding affinity to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), and cynomolgus SIRP- α (SEQ ID NO:11). The 119 mutant heavy and light chains were both found to cause slight reductions in binding affinity to all three SIRP- α proteins. However, the 119 antibody variant with mutated heavy and light chains still displayed strong binding to both human SIRP- α proteins, with a K_D of approximately 30 nM (Table M). Compared with the parental antibody, yield of the 119 antibody variant with mutated heavy and light chains also decreased by approximately 4.5-fold (Table M).

To investigate effect of methionine in HVR-H1 on 119 SIRP α binding, M34V and M34L single mutations were generated in the 119 VH wildtype background and combined with 119 wildtype light chain. Both 119 wt/wt_M34V and 119 wt/wt_M34L had comparable affinities (K_D , M) to human SIRP α v1 and v2 as compared with 119 wt/wt. This indicates that residue M34 is not critical for SIRP- α binding

187

and can be substituted with M34L or M34V mutations. The corresponding VH sequences for M34V and M34L single mutations generated in the 119 VH wildtype background are SEQ ID NO: 421 and 420, respectively.

TABLE M

Binding affinities (K_D , M) of 119 variant antibodies to human and cyno SIRP- α proteins					
Antibody	VL/VH	K_D for human v1	K_D for human v2	K_D for cyno	Yield (mg/ml)
119	Mut/mut	3.17E-10	8.75E-11	1.95E-10	1.575
119	wt/mut	2.54E-10	6.94E-11	1.55E-10	5.377
119	mut/wt	2.14E-10	8.64E-11	1.38E-10	3.527
119	wt/wt	1.83E-10	6.82E-11	1.12E-10	7.192
119	Mut/	2.15E-10	6.88E-11	1.18E-10	NT
	mut_V34M				
AB119	Wt/wt with hIgG4	1.74E-10	5.98E-11	1.16E-10	0.659

NT = not tested

Next, similar variants of antibody 135 were generated. A 135 mutant (mut) heavy chain was generated with the germline back mutations D1E, R13Q, E16G, E43K, and L12Q, as well as the M34V mutation in CDR-H1 that removes a methionine residue that could potentially be oxidized (see SEQ ID NO:247 for VH sequence). A similar variant was constructed without the M34V mutation in CDR-H1 (see SEQ ID NO:259 for VH sequence). A 135 mutant (mut) light chain was generated with the germline back mutations F21L and D60A (see SEQ ID NO:248 for VL sequence). Alignments between the parental and variant sequences are shown in FIGS. 25A & 25B.

Antibody 135 variants with the mutant heavy and/or light chain were compared with the parental 135 antibody for binding affinity to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), and human SIRP- γ v1 (SEQ ID NO:15). The 135 mutant heavy and light chains had comparable binding affinity to all four SIRP- α proteins, as well as comparable yields (Table N1).

To investigate the effect of methionine in HVR-H1 of 135 on SIRP- α binding, M34V and M34L single mutations were generated in the 135 VH wildtype background and combined with a 135 wildtype light chain. Both 135 wt/wt_M34V and 135 wt/wt_M34L had comparable affinities (K_D , M) to human SIRP- α v1 and v2 as compared with 135 wt/wt. This indicates that residue M34 is not critical for SIRP- α binding and can be substituted with an M34L or M34V mutation. The corresponding VH sequences for M34V and M34L single mutations generated in the 135 VH wildtype background are SEQ ID NO: 423 and 422, respectively.

TABLE N1

Binding affinities (K_D , M) of 135 variant antibodies to human and cyno SIRP- α proteins and human SIRP- γ protein						
Anti-body	VL/ VH	K_D for human α v1	K_D for human α v2	K_D for cyno	K_D for human γ v1	Yield (mg/mL)
135	mut/ mut	1.92E-10	1.75E-11	1.27E-10	7.33E-10	3.189
135	wt/ mut	1.84E-10	1.97E-11	1.25E-10	7.79E-10	2.885
135	mut/ wt	1.49E-10	2.66E-11	1.02E-10	5.22E-10	2.689

188

TABLE N1-continued

Binding affinities (K_D , M) of 135 variant antibodies to human and cyno SIRP- α proteins and human SIRP- γ protein						
Anti-body	VL/VH	K_D for human α v1	K_D for human α v2	K_D for cyno	K_D for human γ v1	Yield (mg/mL)
135	wt/	1.51E-10	2.90E-11	9.69E-11	5.39E-10	3.264
	wt					
135	wt/	1.50E-10	1.61E-11	7.94E-11	NT	NT
	mut V34M					

NT = not tested

Similar variants of antibody 136 were generated. A 136 mutant (mut) heavy chain was generated with the germline back mutations D1E, R13Q, E16R, E43K, and L111Q, as well as the M34V mutation in CDR-H1 that removes a methionine residue that could potentially be oxidized (see SEQ ID NO:249 for VH sequence). A similar variant was constructed without the M34V mutation in CDR-H1 (see SEQ ID NO:260 for VH sequence). A 136 mutant (mut) light chain was generated with the germline back mutations T2I, T12S, T22S, and E38Q (see SEQ ID NO:250 for VL). Alignments between the parental and variant sequences are shown in FIGS. 26A & 26B.

As shown in FIG. 27A, antibody 136 variants with the mutant heavy and/or light chain were compared with the parental ("wt") 136 antibody as IgG1_AAA_N297A for binding affinity to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), NOD mouse SIRP- α (SEQ ID NO:8), BL/6 mouse SIRP- α (SEQ ID NO:9), and BALB/c mouse SIRP- α (SEQ ID NO:10). In FIG. 27A, the Y-axis shows the ratio of K_D mut/ K_D wt binding to various SIRP. If ratio=1 indicated no change in K_D compare to parental antibody binding. Ratio=>1 and <1 indicated decrease and increase affinity binding to SIRP- α (s) compare to parental antibody. While the 136 wt (VL)/mut (VH) antibody behaved similar to wt (VL)/wt(VH), both variants with the mutated light chain had poorer binding affinities, indicating that some VL mutations were not tolerated.

In order to analyze the effect of each light chain back mutation on binding affinity, additional antibody 136 variants were constructed and characterized for binding affinity to BL/6 mouse SIRP- α (SEQ ID NO:9), NOD mouse SIRP- α (SEQ ID NO:8), BALB/c mouse SIRP- α (SEQ ID NO:10), human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), and human SIRP- γ v1 (SEQ ID NO:15). Starting with the "all mutant" background, each individual mutation was reversed. I2T, S12T, S22T, and Q38E mutations were individually tested in otherwise "all mutant" light chains, as shown in FIG. 27B. The I2T mutation in an otherwise all mut background showed consistently similar binding affinity, as compared with the parental wt/wt antibody (see SEQ ID NO:251 for I2T in all mut background and FIG. 27A for alignment with parental and mutant 136 antibodies). However, the other three reverse mutations (S12T, S22T, and Q38E) consistently showed binding affinities more similar to 136 mut/mut, indicating that the T2I mutation is responsible for reduced binding affinity to various SIRP proteins. Additional data from these experiments are provided in Table Q infra.

TABLE N2

Binding affinities (K_D , M) of 136 variant antibodies to human, mouse, and cyno SIRP- α proteins and human SIRP- γ protein								
Light chain/ Heavy chain	V1 SEQ ID NO: 5	V2 SEQ ID NO: 6	NOD SEQ ID NO: 8	BL6 SEQ ID NO: 9	BALBc SEQ ID NO: 10	Cyno SEQ ID NO: 11	Sirpg SEQ ID NO: 15	
136 MutRv_I2T/mut	7.17E-10	1.86E-09	6.80E-10	1.53E-08	4.22E-10	2.33E-09	4.00E-08	
136 MutRv_S12T/mt	4.88E-09	1.05E-08	2.85E-09	1.63E-08	2.09E-09	7.99E-09	6.67E-08	
136 MutRv_S22T/mut	4.99E-09	8.28E-09	2.44E-09	1.33E-08	1.64E-09	6.84E-09	6.17E-08	
136 MutRv_Q38E/mut	6.18E-09	1.25E-08	5.98E-09	3.31E-08	2.98E-09	1.04E-08	1.93E-08	
136 MutRv_I2T/mut_V34M	5.51E-10	1.74E-09	5.96E-10	1.38E-08	3.61E-10	2.11E-09	3.35E-08	
136 Mut/mut	7.32E-09	1.27E-08	5.42E-09	2.16E-08	2.67E-09	9.48E-09	2.88E-08	
136 Wt/wt	9.72E-10	2.54E-09	8.21E-10	1.64E-08	4.58E-10	2.83E-09	3.36E-08	

15

Example 7: Humanization of Anti-SIRP- α Antibodies

The Examples above describe the generation of anti-SIRP- α antibodies having a fully human heavy chain and a chicken light chain. In order to humanize the chicken-derived light chains, chicken HVRs of these antibodies were grafted onto various human lambda light chain frameworks.

Methods

Humanization

Antibodies were humanized using standard techniques. For measuring production yield, equal volume of Expi293 cultures expressing anti-SIRP α antibodies were purified by Protein A affinity chromatography. After buffer exchange into PBS, the protein concentration was determined by A280 and expressed in mg/mL.

Results

In order to design humanized light chains, each chicken light chain sequence was aligned to the closest human

germline framework by IgBLAST (NCBI). Using this analysis, the closest match to the chicken lambda light chain framework is human IGLV3 (see SEQ ID NOs:314-317).

In another approach, a literature search was undertaken to determine the optimal human lambda light chain framework sequences to pair with a human VH3 sequences (the human heavy chain used for these antibodies). Based on these analyses, it was thought that human VH3 would pair well with human IGLV1 and IGLV2. See Glanville, J. et al. (2009) *Proc. Natl. Acad. Sci.* 106:20216-20221; Lloyd, C. et al. (2009) *Protein Eng. Des. Sel.* 22:159-168; and Jayaram, N. et al. (2012) *Protein Eng. Des. Sel.* 25:523-529.

Therefore, six humanized light chains were created: Hum1 (AB25 HVRs+human IGLV3 framework), Hum2 (AB25 HVRs+human IGLV1 framework), Hum3 (AB66 HVRs+human IGLV3 framework), Hum4 (AB66 HVRs+human IGLV1 framework), Hum5 (AB25 HVRs+human IGLV2 framework), and Hum6 (AB21 HVRs+human IGLV1 framework). Sequences of the resulting light chain variable domains are provided in Table O1.

TABLE O1

Sequences of humanized variable light chain domains tested. HVR sequences are bolded and underlined.			
Name	LC HVRs	Human LC framework	Sequence
Hum1	AB25	IGLV3	SYELTQPPSVSVSPGQTARITC <u>SGGSYSSYYAWYQ</u> QKPGQAPVTLIY <u>SDDKRPS</u> NIPIERFSGSSSGTTVTLTII SGVQAEDEADYYC <u>GGYDQSSYTN</u> PFGGGTKLTVL (SEQ ID NO: 252)
Hum2	AB25	IGLV1	QSVLTQPPSVSAAPGQKVTISCS <u>SGGSYSSYYAWYQ</u> QLPGTAPKTLIY <u>SDDKRPS</u> NIPIERFSGSKSGTSATLGI TGLQTGDEADYYC <u>GGYDQSSYTN</u> PFGTGKTVTL (SEQ ID NO: 253)
Hum3	AB66	IGLV3	SYELTQPPSVSVSPGQTARITC <u>SGGDYYSTYYAWYQ</u> QKPGQAPVTVIH <u>SDDKRPS</u> DIPERFSGSSSGTTVTLTII SGVQAEDEADYYC <u>GGYDGRTYINT</u> PFGGGTKLTVL (SEQ ID NO: 254)
Hum4	AB66	IGLV1	QSVLTQPPSVSAAPGQKVTISCS <u>GGDYYSTYYAWY</u> QQLPGTAPKTVIH <u>SDDKRPS</u> DIPDRFSGSKSGTSATL GITGLQTGDEADYYC <u>GGYDGRTYINT</u> FGTGKTVTL (SEQ ID NO: 255)
Hum5	AB25	IGLV2	QSALTQPASVSGSPGQSIITISCTGTSTSDV <u>GSYSSYY</u> <u>AWYQQHPGKAPKTLIY</u> <u>SDDKRPS</u> NIPIERFSGSKSG NTASLTISGLQAEDEADYYC <u>GGYDQSSYTN</u> PFGGG TKLTVL (SEQ ID NO: 256)
Hum6	AB21	IGLV1	QSVLTQPPSVSAAPGQKVTISCS <u>SGGDYYSSYYGWY</u> QQLPGTAPKTVIY <u>SDDKRPS</u> DIPDRFSGSKSGTSATL GITGLQTGDEADYYC <u>GGYDYSTYANA</u> FGTGKTVTL (SEQ ID NO: 257)

Each of the 6 humanized light chains was paired with each of four heavy chains (derived from A1B21, A1B25, AB327, and AB366), generating 24 unique antibodies. Antibodies were expressed as described above. Surprisingly, human IGLV1 framework sequences resulted in decreased antibody expression regardless of the heavy chain. This refers to all the heavy chain pairings with Hum2, Hum4 and Hum6 (except when pairing was carried out with heavy chain from AB366). The results are summarized in FIG. 28 as “protein yield” (row 1). In contrast, antibodies with light chains including human IGLV2 and IGLV3 frameworks (Hum 1, Hum3, Hum5) showed higher levels of expression regardless of the heavy chain.

Selected antibodies were next characterized for binding to a variety of SIRP proteins (e.g., to human SIRP- α v1, human SIRP- α v2, cynomolgus SIRP- α , mouse BALB/c SIRP- α , and human SIRP-7). These data are also summarized in FIG. 28. Selected humanized light chains caused a decrease in binding to one or more antigens. For instance, the human IGLV3 framework (represented by Hum1 and Hum3) was found to allow for superior levels of antibody production without perturbing binding affinity. For example, light chain variable domains with the IGLV3 frameworks and either the antibody 25 or antibody 66 HVR sequences (represented by Hum1 and Hum 3 respectively) combined well with a variety of heavy chains (e.g., heavy variable domains from antibodies 21, 25, 27, and 66) and showed similar binding to different SIRP- α and SIRP- γ proteins. In contrast, IGLV1 and IGLV2 frameworks (represented by Hum2, Hum4, Hum5 and Hum6) were found to either lower expression and/or decrease binding to SIRP when paired with heavy chains from antibodies 21, 25, and 27 and 66. Additional binding data from these experiments are provided in Table R infra. The human IGLV3 framework was selected for further testing.

Another goal for humanization of these antibodies was antibody sequences having greater than or equal to 85% identity to human germline light chain/heavy chain sequences. Additional VL domain Hum9 and Hum8 was generated based on the Hum1 VL domain. Compared to Hum1, Hum9 contains 4 amino acid substitutions near or in HVR-L1 and -L2 that increase the humanness of the light chain to greater than or equal to 85% identity to human light chain sequence (FIG. 29). Compared to Hum1, Hum8 contains 5 amino acid substitutions respectively near or in HVR-L1 and -L2 that increase the humanness of the light chain to greater than or equal to 85% identity to human light chain sequence (FIG. 29). Hum1, Hum8 and Hum9 VLs when paired with heavy chain all_mut_AB21 (carrying germline mutations) produced anti-SIRP α antibodies that bind to human v1 with affinity equal or better than 10 pM (Table S). Similarly, when Hum1, Hum8 and Hum9 VLs were paired with heavy chain all_mut_AB25 (carrying germline mutations), the anti-SIRP α antibodies bind to human v1 with affinity equal or better than 10 pM. Additional binding data from these experiments are provided in Table S infra. These light chains can be combined interchangeably with antibody VH domains 21, 25, and 27 (as well as variants thereof, which were modified as described supra for antibodies 119, 135, and 136; FIG. 30). Without wishing to be bound to theory, it is thought that the humanization process described above can be applied to the light chain of any antibody of family 2 (bin 1).

Example 8: Induction of Phagocytosis and Dendritic Cell Activation by Anti-SIRP- α Antibodies

Various anti-SIRP- α antibodies representing different modes of binding to SIRP- α (e.g., blocking, non-blocking, and “kick off” antibodies) were next examined in phagocytosis assays.

CD47-Fc Binding Assays

CD47-Fc was conjugated with Alexa Fluor 647 (AF647) using the Alexa Fluor 647 Microscale Protein Labeling Kit (Thermo Fisher Scientific). In 96 well polypropylene plates (Corning), 100,000 PBMCs were suspended in 100 μ l 0.25 μ M AF647-labeled CD47-Fc and 1 μ l anti-CD14 PE antibody (Biolegend) in FACS buffer. Cells were incubated on ice for 30 minutes, washed in FACS buffer, and incubated in 10-fold serial dilutions of anti-SIRP- α antibody from 1.25 μ M to 25 pM. Cells were incubated on ice for 30 minutes, washed in FACS buffer, and fixed in 75 μ l of 0.5 percent formaldehyde. Cells were analyzed on a FACS Canto™ II (BD Biosciences) flow cytometer, with subsequent data analysis by Flowjo 10.7 (Treestar). Geometric mean fluorescence intensity of the CD47-Fc signal was determined in the CD14-positive monocyte population.

Dendritic Cell Activation Assays

The dendritic cell activation assays were performed as described elsewhere herein. Briefly, Balb/c mice (n=3/group) were intravenously injected with a control rat anti-mouse anti-SIRP- α antagonistic antibody (clone p84), human IgG1 control, various anti-SIRP- α antibodies, mouse IgG control, or vehicle (PBS) at 10 mg/kg. Five hours post injection, spleens were harvested and processed into single cell suspension by mechanical dissociation. Activation marker CD86, MHCII and CCR7 level on CD4+ splenic dendritic cells was measured by flow cytometry.

Cell Adhesion Assays

The adhesion assay was performed using isogenic cells lacking or expressing human CD47. Hamster CD47 knock-out CHO cells (CHO^{CD47 KO}) were generated using CRISPR technology. Isogenic cells expressing human CD47 were generated by transiently transfecting human CD47 into CHO^{CD47 KO} 24 hours post transfection, human CD47 transfected CHO^{CD47 KO} cells (4×10^5) were re-plated onto 24 well tissue culture treated plates and allowed to reattach and reach confluency overnight at 37° C.

Human peripheral blood mononuclear cells (PBMCs) were separated from peripheral blood of healthy donors by Ficoll gradient centrifugation. CD3+ and CD14+ cells were isolated from PBMCs by negative selection using magnetic beads. Isolated CD3+ and CD14+ cells (5×10^6) were pre-incubated with Hum1/AB21mutall, Hum9/AB21mutall or 136 wt/mut antibodies containing mutated human IgG1 (L234A L235A G237A, and N297A) (20 μ g/mL) at 37° C. for 20 minutes, before plating onto CHO^{CD47 KO} and CHO^{hCD47+} cells and allowed to adhere for 1 hour at 37° C. Non-adherent cells were removed by five gentle washes with PBS. Adherent cells were detached with trypsin and neutralized with 10% FBS. Cells were transferred to 96 well plate and washed 2 times with PBS+0.5% BSA followed by cell surface labeling with fluorochrome conjugated human CD3, CD4, CD8 and CD14 antibodies. Counting beads were added and cell adhesion was quantified using a BD Canto™ II flow cytometer. Analysis of flow cytometry data was done using Flowjo.

Results

Five antibodies that “kick off” CD47 from binding SIRP- α were examined for their effects on phagocytosis of

HER2(+) OE19 cells by M2 macrophages in combination with the anti-HER2 antibody trastuzumab (FIG. 31). All 5 “kick off” antibodies were found to enhance trastuzumab-induced phagocytosis.

Five antibodies that do not block CD47 from binding SIRP- α were next examined for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 32). All non-blocking antibodies were found to enhance cetuximab-induced phagocytosis.

Next, humanized antibodies described above were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 33A). All humanized antibodies were found to enhance cetuximab-induced phagocytosis. Additional variants of antibody 136 (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 33B). All variants of antibody 136 enhanced cetuximab-induced phagocytosis, but to varying degrees. Additional humanized antibodies described above (antibody 25 and 27 heavy chain variants combined with Hum1 or Hum9 light chain) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 33C). All humanized antibodies were found to enhance cetuximab-induced phagocytosis.

To investigate the effect of anti-SIRP- α antibody Fc regions on phagocytosis, two non-blocking antibodies were tested either as full-length antibodies or F(ab)₂ fragments for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 34). These results indicated that non-blocking antibodies lose the ability to induce phagocytosis as F(ab)₂ fragments, suggesting that the Fc region of this class of antibody is required for induction of phagocytosis.

Three variants of blocking antibody 119 (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 35). The results demonstrated that all three variants enhanced cetuximab-induced phagocytosis.

Two variants of blocking antibody 135 (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 36). The results demonstrated that both variants enhanced cetuximab-induced phagocytosis.

Additional non-blocking antibodies (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 37). All non-blocking antibodies were found to enhance cetuximab-induced phagocytosis.

Next, various anti-SIRP- α antibodies were examined for their effects on in vivo dendritic cell activation (FIGS. 38A-38B), including known anti-SIRP- α antibody p84 (see, e.g., Tangsheng, Y. et al (2015) *Immunity* 43:1-12). Failure to engage mouse SIRP- α receptor on splenic dendritic cells via CD47 binding leads to splenic dendritic cell activation. Control anti-SIRP- α antagonist antibody p84 activated splenic dendritic cells when injected intravenously into mice. Non-blocking anti-SIRP- α antibodies (AB136b, AB3b and AB136 wt/mut) and blocking anti-SIRP- α antibodies (Hum1/AB21mutall, Hum8/AB21mutall, and Hum9/AB21mutall) were tested in vivo to determine if it leads to

dendritic cell activation. As determined by CD86 and MHCII expression, both SIRP- α blockers and non-blockers induce activation of dendritic cells. These results suggest that blocking and non-blocking anti-SIRP- α antibodies induce activation of dendritic cells.

Two variants of antibody 218, 218-Hum13/VH_wt and 218-Hum14/VH_wt, were generated by expressing humanized light chains (Hum13 and Hum14 respectively) with the wild-type heavy chain of antibody 218. Hum13 used the human IGLV2 framework, whereas Hum14 used the human IGLV3 framework (see SEQ ID NOs:333 and 334 for VL sequences of Hum13 and Hum14, respectively). Both clones showed lower affinity binding to v1 and v2 (K_D is ~29.3 to 53.1 nM), as shown in Table O2.

TABLE O2

Binding affinity (K_D , M) of antibody 218 and 218 variants for human SIRP- α v1 and v2.		
K_D (M)	v1 ALX135	v2 ALX269
AB218a	1.60E-10	3.23E-10
218-Hum13/VH_wt	2.93E-08	3.38E-08
218-Hum14/VH_wt	3.45E-08	5.31E-08

Both antibodies were tested in a phagocytosis assay in combination with cetuximab in DLD-1 cells. Interestingly, neither humanized variant was able to enhance phagocytosis over the parental AB218a antibody (FIG. 39A). Without wishing to be bound to theory, these results suggest that antibodies with K_D at or below an approximate range of 30-50 nM may be more effective in inducing phagocytosis than antibodies that bind with weaker affinity.

Exemplary blocking, non-blocking, and kick off anti-SIRP- α antibodies were next tested for induction of phagocytosis as single agents. First, three antibodies that block CD47 from binding SIRP- α , AB119a, AB120a, and AB122a, were examined for their effects as single agents on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages (as described above). All blocking antibodies were found to induce phagocytosis as single agents (FIG. 39B). Next, two antibodies that do not block CD47 from binding SIRP- α , AB136a and AB137a, were examined for their effects as single agents on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages. All non-blocking antibodies were found to induce phagocytosis as single agents (FIG. 39C). Finally, five “kick-off” anti-SIRP- α antibodies, AB115a, AB116a, AB117a, AB118a, and AB132a, were examined for their effects as single agents on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages. All “kick-off” antibodies were found to induce phagocytosis as single agents (FIG. 39D).

Example 9: Synergistic Anti-Tumor Effects of Combining Blocking or Non-Blocking Anti-SIRP- α Antibodies with Inhibition of the PD-L1/PD-1 Pathway

Methods

In Vivo Anti-Tumor Activity

For the CT26 syngeneic mouse colon carcinoma model, CT26 cells were implanted subcutaneously in BALB/c mice and randomized into groups (8-9 mice/group). Treatment groups included vehicle (PBS), AB25b, anti-PD-L1, and AB25b/anti-PD-L1. Anti-PD-L1 is generated by fusing the VH and VL domain of Atezolizumab with mouse IgG1 Fc region bearing an N297A mutation. All anti-SIRP- α anti-

bodies also have a mouse IgG1 Fc region bearing an N297A mutation. Treatment was initiated when tumors were an average of 75-80 mm³, day 7 or 8 post implant. Mice were dosed intraperitoneally (IP) at 3 mg/kg or 10 mg/kg twice a week for three weeks for anti-SIRP α antibodies and three doses at 3 mg/kg, five days apart for anti-PD-L1. Animals were sacrificed when tumors reached a volume of ~2000 mm³.

For the MC38 syngeneic mouse colon carcinoma model, MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8-10 mice/group). Treatment groups included vehicle (PBS), AB25b, AB136b, anti-PD1 (clone RMP1-14, BioXCell), AB136b/anti-PD1, and AB25b/anti-PD1. All anti-SIRP α antibodies had a murine IgG1 Fc region bearing an N297A mutation except for AB25c. Treatment was initiated when tumors were an average of 60-65 mm³, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks for anti-SIRP α and three doses at 2 mg/kg for anti-PD1. Animals were sacrificed when tumors reached a volume of ~2000 mm³.

Results

Anti-tumor activity of the blocking AB25b anti-SIRP- α antibody was tested alone and in combination with an anti-PD-L1 antibody in the CT26 syngeneic mouse colon carcinoma model. As shown in FIG. 40, administration of AB25b at 10 mg/kg in combination with anti-PD-L1 at 3 mg/kg delayed tumor formation when compared to treatment with each single agent or vehicle control. On day 27 of

the study, the combination treatment group had six mice with tumors below 600 mm³ in size, as compared to two, two, and two mice with tumors below 600 mm³ in size in the vehicle, anti-PD-L1 single agent, and anti-SIRP- α single agent treatment groups, respectively.

Next, the anti-tumor activities of the blocking AB25b anti-SIRP- α antibody and non-blocking AB136b anti-SIRP- α antibody were tested alone and in combination with an anti-PD-1 antibody in the MC38 syngeneic mouse colon carcinoma model. As shown in FIG. 41, combining either AB25b or AB136b (at 10 mg/kg) with anti-PD-1 at 5 mg/kg delayed tumor formation when compared to treatment with each single agent or vehicle control. On day 27 of the study, the AB25b/PD-1 combination treatment group had seven mice with tumors below 600 mm³ in size, and the AB136b/PD-1 combination treatment group had six mice with tumors below 600 mm³ in size, as compared to one, five, two, and one mice with tumors below 600 mm³ in size in the vehicle, anti-PD-1 single agent, AB25b single agent, and AB136b single agent treatment groups, respectively.

A summary of antibodies described herein and their properties is provided in Table P. Additional binding data are provided in Table T.

Although the foregoing disclosure has been described in some detail by way of illustration and example for purposes of clarity of understanding, the descriptions and examples should not be construed as limiting the scope of the present disclosure. The disclosures of all patent and scientific literature cited herein are expressly incorporated in their entirety by reference.

TABLE P

Anti-SIRP- α antibody summary.													
Anti-body	Type of	In	In	Species Binding (+/-) (Koff)		Human Isoforms (+/-) (Koff)		K _D		Heavy	Light	chain	Chain
	Binding (NB/B/ KO)	Bin No.	vitro phago (+/-)	vivo mouse (+/-)	Human v1 (SEQ ID NO: 5)	Cyno (SEQ ID NO: 1)	Mouse 129 (SEQ ID NO: 7)	Beta (SEQ ID NO: 13)	Gamma (SEQ ID NO: 15)	Human V1 (SEQ ID NO: 5)	Human V2 (SEQ ID NO: 6)	(Human/ Chicken)	(Human/ Chicken)
Family 1													
119	B	1	+	N.A.	+	+	-	+	+	1.83E-10	7.99E-11	Human	Human
120	B	1	+	N.A.	+	+	-	+	+	2.14E-10	8.56E-11	Human	Human
121	B	1	+	N.A.	+	+	-	+	+	1.57E-10	5.01E-11	Human	Human
122	B	1		N.A.	+	+	-	+	+	2.14E-10	7.78E-11	Human	Human
135	b	1	+	N.A.	+	+	-	+	+	1.51E-10	2.90E-11	Human	Human
Family 2													
16	B	1	+		+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken
17	B	1	+		+	+	+	+	+			Human	Chicken
21	B	1	+	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken
22	B	1			+	+	+	+	+			Human	Chicken
23	B	1			+	+	+	+	+			Human	Chicken
24	B	1			+	+	+	+	+			Human	Chicken
25	B	1	+	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken
26	B	1	+	NT	+	+	+	+	+			Human	Chicken
27	B	1	+	+	+	+	+	+	+	2.15E-11		Human	Chicken
28	B	1	NT	NT	+	+	+	+	+			Human	Chicken
29	B	1	NT	NT	+	+	+	+	+			Human	Chicken
30	B	1	+	NT	+	+	+	+	+			Human	Chicken
55	B	1	NT	NT	+	+	+	+	+			Human	Chicken
56	B	1	NT	NT	+	+	+	+	+			Human	Chicken
59	B	1	NT	NT	+	+	+	+	+			Human	Chicken
60	B	1	NT	NT	+	+	+	+	+			Human	Chicken
65	B	1	NT	NT	+	+	+	+	+			Human	Chicken
66	B	1	NT	+	+	+	+	+	+	<1.0E-12		Human	Chicken
69	B	1	NT	NT	+	+	+	+	+			Human	Chicken
70	B	1	NT	NT	+	+	+	+	+			Human	Chicken
71	B	1	NT	NT	+	+	+	+	+			Human	Chicken
73	B	1	NT	NT	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken
74	B	1	NT	NT	+	+	+	+	+			Human	Chicken
76	B	1	NT	NT	+	+	+	+	+			Human	Chicken

TABLE P-continued

Anti-SIRP- α antibody summary.													
Anti-body	Type of	In	In	Species Binding (+/-) (Koff)			Human Isoforms (+/-) (Koff)		K_D	K_D	Heavy	Light	
	Binding (NB/B/ KO)	Bin No.	vitro phago (+/-)	vivo mouse (+/-)	Human v1 (SEQ ID NO: 5)	Cyno (SEQ ID NO: 1)	Mouse 129 (SEQ ID NO: 7)	Beta (SEQ ID NO: 13)	Gamma (SEQ ID NO: 15)	Human V1 (SEQ ID NO: 5)	Human V2 (SEQ ID NO: 6)	chain (Human/ Chicken)	Chain (Human/ Chicken)
201	B	1	+	NT	+	+	+	+	+	<1.0E-12	4.87E-12	Human	Chicken
202	B	1	+	NT	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken
206	B	1	NT	NT	+	+	+	+	+			Human	Chicken
Family 3													
136	NB	2	+	+	+	+	+	+	-	4.58E-10	3.22E-09	Human	Human
137	NB	2	+	NT	+	+	+	+	-	7.74E-10	3.14E-09	Human	Human
175	NB	2	+	NT	+	+	+	+	NT	1.37E-10	6.62E-10	Human	Human
177	NB	2	NT	NT	+	+	+	+	NT			Human	Human
178	NB	2	NT	NT	+	+	+	+	NT			Human	Human
180	NB	2	NT	NT	+	+	+	+	NT			Human	Human
184	NB	2	NT	NT	+	+	+	+	NT			Human	Human
185	NB	2	NT	NT	+	+	+	+	NT			Human	Human
189	NB	2	+	NT	+	+	+	+	NT	3.10E-10	1.90E-09	Human	Human
190	NB	2	NT	NT	+	+	+	+	NT			Human	Human
193	NB	2	+	NT	+	+	+	+	NT	7.79E-10	5.23E-09	Human	Human
Family 4													
115	KO	3	+	N.A.	+	+	-	+	+	1.77E-11	1.50E-11	Human	Human
116	KO	3	+	N.A.	+	+	-	+	+	1.10E-10	2.99E-10	Human	Human
117	KO	3	+	N.A.	+	+	-	+	+	3.12E-11		Human	Human
118	KO	3	+	N.A.	+	+	-	+	+	4.03E-11		Human	Human
132	KO	3	+	N.A.	+	+	-	+	+	4.26E-10	1.86E-09	Human	Human
191	KO	3	NT	N.A.	+	+	-	+	NT			Human	Human
198	KO	3	NT	N.A.	+	+	-	+	NT			Human	Human
Family 5 (additional non-blockers)													
3	NB	4	+	+	+	+	+	+	+	1.62E-10	7.67E-11	Chicken	Chicken
173	NB	4	+	NT	+	-	-	-	-	9.37E-10	9.28E-09	Human	Human
174	NB	4	NT	NT	+	-	-	-	-			Human	Human
209	NB	4	+	NT	+	-	-	+	-	1.71E-10	5.01E-09	Human	Chicken
213	NB	4	+	NT	-	+	-	-	NT	6.05E-09	1.69E-09	Human	Chicken
214	NB	4	NT	NT	-	+	-	-	NT			Human	Chicken
123	NB	5	+	NT	+	+	-	-	+	6.05E-10	NLB	Human	Human
149	NB	5	+	NT	+	+	-	+	-	8.73E-10	2.38E-10	Human	Human
161	NB	5	+	NT	+	+	-	+	-	1.03E-9	1.27E-10	Human	Human
162	NB	5	+	NT	+	+	-	+	+	4.50E-10	1.57E-08	Human	Human
163	NB	5	NT	NT	+	+	-	+	+			Human	Human
164	NB	5	NT	NT	+	+	-	+	+			Human	Human
194	NB	5	+	NT	+	+	-	-	NT	4.97E-10	NLB	Human	Human
218	NB	5	+	NT	+	+	-	NT	+	1.23E-10	2.76E-10	Chicken	Human
S45	NB	6	+	NT	+	-	-	-	-	6.63E-11	1.34E-10	Chicken	Chicken

B = blocker; NB = non-blocker; KO = kick off.

NT or blank = not tested; NA = not applicable (antibodies do not cross-react).

NLB = no binding

TABLE Q

Anti-SIRP- α antibody germline/liability mutation summary.														
Anti-body	Hu-man Light Chain	Hu-man Heavy Chain	ing (NB/ B/ KO)	Type of Bind- Bin No.	K_D (M)								In vitro phago (+/-)	In vivo mouse (+/-)
					SEQ ID NO: 5 Human v1	SEQ ID NO: 6 Human v2	SEQ ID NO: 11 Cyno	SEQ ID NO: 8 NOD	SEQ ID NO: 9 BL6	SEQ ID NO: 10 BALBc	SEQ ID NO: 13 Human SIRPb	SEQ ID NO: 15 Human SIRPg		
119	wt	wt	B	1	1.83E-10	6.82E-11	1.12E-10	NLB	NLB	NLB	3.42E-10	2.67E-10	+	N.A.
119	Mut	wt	B	1	2.14E-10	8.64E-11	1.38E-10	NLB	NLB	NLB	NT	2.33E-10	NT	N.A.
119	wt	Mut	B	1	2.54E-10	6.94E-11	1.55E-10	NLB	NLB	NLB	NT	NT	+	N.A.
119	Mut	Mut	B	1	3.17E-10	8.75E-11	1.95E-10	NLB	NLB	NLB	4.60E-10	3.36E-10	+	N.A.
119	Mut	Mut_ V34M	B	1	2.15E-10	6.88E-11	1.18E-10	NLB	NLB	NLB	3.37E-10	2.63E-10	NT	N.A.

TABLE Q-continued

Anti-SIRP- α antibody germline/liability mutation summary.														
Anti-body	Hu-man Light Chain	Hu-man Heavy Chain	Type of Bind- ing (NB/ B/ KO)	Bin No.	K_D (M)								In vitro phago (+/-)	In vivo mouse (+/-)
					SEQ ID NO: 5 Human v1	SEQ ID NO: 6 Human v2	SEQ ID NO: 11 Cyno	SEQ ID NO: 8 NOD	SEQ ID NO: 9 BL6	SEQ ID NO: 10 BALBc	SEQ ID NO: 13 Human SIRPb	SEQ ID NO: 15 Human SIRPg		
135	wt	wt	B	1	1.51E-10	2.90E-11	9.69E-11	NLB	NLB	NLB		5.39E-10	+	N.A.
135	Mut	wt	B	1	1.49E-10	2.66E-11	1.02E-10	NLB	NLB	NLB		5.22E-10	NT	N.A.
135	Wt	Mut	B	1	1.84E-10	1.97E-11	1.25E-10	NLB	NLB	NLB	7.79E-10	7.33E-10	NT	N.A.
135	Mut	Mut	B	1	1.92E-10	1.75E-11	1.27E-10	NLB	NLB	NLB	1.88E-10	7.33E-10	+	N.A.
135	wt	Mut_	B	1	1.50E-10	1.61E-11	7.94E-11	NLB	NLB	NLB	1.60E-10	5.33E-10	NT	N.A.
		V34M												
136	wt	wt	NB	2	4.58E-10	1.63E-09	2.15E-09	5.54E-10	1.27E-08	3.50E-10	4.35E-9	2.39E-08	+	+
136	Mut	wt	NB	2	7.28E-09	1.74E-08	1.13E-08	4.12E-09	3.26E-08	2.80E-09		1.96E-08	-	
136	wt	Mut	NB	2	5.58E-10	1.74E-09	2.26E-09	6.78E-10	2.31E-08	4.16E-10	3.54E-09	1.65E-08	NT	
136	Mut	Mut	NB	2	7.29E-09	1.95E-08	1.30E-08	5.21E-09	3.17E-08	3.14E-09		1.68E-09	-	
136	Mut_	Mut	NB	2	7.17E-10	1.86E-09	2.33E-09	6.80E-10	1.53E-08	4.22E-10	3.16E-09	4.00E-08	+	
	I2T													
136	Mut_	Mut	NB	2	4.88E-09	1.05E-08	7.99E-09	2.85E-09	1.63E-08	2.09E-09		6.67E-08	NT	
	S12T													
136	Mut_	Mut	NB	2	4.99E-09	8.28E-09	6.84E-09	2.44E-09	1.33E-08	1.64E-09		6.17E-08	NT	
	S22T													
136	Mut_	Mut	NB	2	6.18E-09	1.25E-08	1.04E-08	5.98E-09	3.31E-08	2.98E-09		1.93E-08	NT	
	Q38E													
136	Mut_	Mut_	NB	2	5.51E-10	1.74E-09	2.11E-09	5.96E-10	1.38E-08	3.61E-10	2.22E-09	3.35E-08	NT	
	I2T	V34M												

B = blocker; NB = non-blocker.

NT or blank = not tested; NA = not applicable (antibodies do not cross-react); NLB = no binding

119 heavy chain mut = D1E, E43K, L112Q, M34V

119 light chain mut = F21L, R39K, E60A, T76S

135 heavy chain mut = D1E, R13Q, E16G, M34V, E43K, L112Q

TABLE R

Anti-SIRP- α antibody humanization summary (round 1).										
Antibody Designation	VL	VH	K _{off} (1/s)					In vitro phago (+/-)	In vivo mouse (+/-)	
			SEQ ID NO: 5	SEQ ID NO: 6	SEQ ID NO: 11	SEQ ID NO: 10	SEQ ID NO: 15			
			Human V1	Human V2	Cyno	BALBc	SIRPg			
Parental Antibodies										
AB21	Chicken (AB21_LC_wt)SEQ ID NO: 136	Human (AB21_HC_wt)SEQ ID NO: 135)	7.07E-04	1.92E-03	2.29E-03	2.41E-03	9.02E-04	NT	+	
AB25	Chicken (AB25_LC_wt)SEQ ID NO: 138	Human (AB25_HC_wt)SEQ ID NO: 137	1.63E-04	3.53E-04	3.94E-04	1.78E-03	2.03E-04	+	+	
AB27	Chicken (AB27_LC_wt)SEQ ID NO: 140	Human (AB27_HC_wt)SEQ ID NO: 139	3.15E-04	5.79E-04	6.83E-04	5.00E-03	4.07E-04	+	+	
AB66	Chicken (AB66_LC_wt)SEQ ID NO: 142	Human (AB66_HC_wt)SEQ ID NO: 141	9.49E-04	2.81E-03	2.69E-03	3.46E-03	9.97E-04	NT	+	
Humanization of chicken light chain of AB25, AB66 - replaced with human IGLV3 framework										
Hum1/AB21_HC_wt	Hum1_Humanized (AB25_IGLV3)SEQ ID NO: 252	Human (AB21_HC_wt)SEQ ID NO: 135	1.93E-04	3.03E-04	3.95E-04	2.91E-03	1.88E-04	+		
Hum1/AB25_HC_wt	Hum1_Humanized (AB25_IGLV3)SEQ ID NO: 252	Human (AB25_HC_wt)SEQ ID NO: 137	1.33E-04	2.67E-04	3.30E-04	3.74E-03	2.03E-04	+		
Hum1/AB27_HC_wt	Hum1_Humanized (AB25_IGLV3)SEQ ID NO: 252	Human (AB27_HC_wt)SEQ ID NO: 139	1.92E-04	2.92E-04	3.79E-04	3.70E-03	1.78E-04	+		
Hum1/AB66_HC_wt	Hum1_Humanized (AB25_IGLV3)SEQ ID NO: 252	Human (AB66_HC_wt)SEQ ID NO: 141	1.24E-04	2.39E-04	3.03E-04	2.46E-03	1.36E-04	NT		
Hum3/	Hum3_Humanized	Human	1.37E-04	4.38E-04	4.32E-04	2.31E-03	2.10E-04	NT		

TABLE R-continued

Anti-SIRP- α antibody humanization summary (round 1).									
Antibody Designation	VL	VH	Koff (1/s)					In vitro phago (+/-)	In vivo mouse (+/-)
			SEQ ID NO: 5 Human V1	SEQ ID NO: 6 Human V2	SEQ ID NO: 11 Cyno	SEQ ID NO: 10 BALBc	SEQ ID NO: 15 SIRPg		
AB21_HC_wt	(AB66_IGLV3) SEQ ID NO: 254	(AB21_HC_wt) SEQ ID NO: 135							
Hum3/ AB25_HC_wt	Hum3_Humanized (AB66_IGLV3) SEQ ID NO: 254	Human (AB25_HC_wt) SEQ ID NO: 137	5.94E-05	3.53E-04	3.75E-04	2.27E-04	1.69E-04	NT	
Hum3/ AB27_HC_wt	Hum3_Humanized (AB66_IGLV3) SEQ ID NO: 254	Human (AB27_HC_wt) SEQ ID NO: 139	2.16E-04	3.96E-04	4.64E-04	2.02E-04	2.07E-04	NT	
Hum3/ AB66 HC wt	Hum3_Humanized (AB66_IGLV3) SEQ ID NO: 254	Human (AB66_HC_wt) SEQ ID NO: 141	1.24E-04	2.87E-04	3.06E-04	2.05E-03	1.35E-04	NT	

NT or blank = not tested.

20

TABLE S

Anti-SIRP- α antibody humanization summary (round 2).											
			KD (M)								
Antibody Designation	VL	VH	SEQ ID NO: 5 Human v1	SEQ ID NO: 6 Human V2	SEQ ID NO: 11 Cyno	SEQ ID NO: 8 NOD	SEQ ID NO: 9 BL6	SEQ ID NO: 10 BALBc	SEQ ID NO: 13 SIRPb	SEQ ID NO: 15 SIRPg	In vitro phago (+/-)
Pairing of humanized light chain with heavy chain (Germline mut)											
Hum1/ AB21_HC_ Mutall	Hum1_ Humanized (AB25_IGL V3) SEQ ID NO: 252	Human (AB21_HC_ Mutall) SEQ ID NO: 263	5.32E-12	4.60E-12	2.91E-11	3.70E-09	9.50E-09	7.91E-09	6.7E-12	>1.0E-12	+
Hum1/ AB25_HC_ Mutall	Hum1_ Humanized (AB25_IGL V3) SEQ ID NO: 252	Human (AB25_HC_ Mutall) SEQ ID NO: 265	5.19E-12			4.83E-09	1.03E-08	7.25E-09			+
Hum1/ AB27_HC_ Mutall	Hum1_ Humanized (AB25_IGL V3) SEQ ID NO: 252	Human (AB27_HC_ MutAll) SEQ ID NO: 267	4.37E-12			2.92E-09	9.03E-09	5.77E-09			+
Mutation of humanized light chain to increase % humaness											
Hum8/ AB21_HC_ Mutall	Hum8_ Humanized (AB25_IGL V3) + 5aa in CDR SEQ ID NO: 416	Human (AB21_HC_ Mutall) SEQ ID NO: 263	2.01E-11			2.78E-08	4.15E-04	7.12E-08			+
Hum8/ AB25_HC_ Mutall	Hum8_ Humanized (AB25_IGL V3) + 5aa in CDR SEQ ID NO: 416	Human (AB25_HC_ Mutall) SEQ ID NO: 265	2.62E-11			2.05E-08	4.97E-04	5.99E-08			NT
Hum9/ AB21_HC_ Mutall	Hum9_ Humanized (AB25_IGL V3) + 4aa in CDR SEQ ID NO: 262	Human (AB21_HC_ Mutall) SEQ ID NO: 263	1.19E-11	1.19E-10	2.22E-10	2.41E-08	5.33E-04	1.36E-07	5.69E-11	3.45E-11	+
Hum9/ AB25_HC_ Mutall	Hum9_ Humanized (AB25_IGL V3) + 4aa in	Human (AB25_HC_ Mutall) SEQ ID	1.29E-11			6.12E-08	1.15E-08	3.83E-08			+

TABLE S-continued

Anti-SIRP- α antibody humanization summary (round 2).											
Antibody Designation	VL	VH	KD (M)								In vitro phago (+/-)
			SEQ ID NO: 5	SEQ ID NO: 6	SEQ ID NO: 11	SEQ ID NO: 8	SEQ ID NO: 9	SEQ ID NO: 10	SEQ ID NO: 13	SEQ ID NO: 15	
			Human v1	Human V2	Cyno	NOD	BL6	BALBc	SIRPb	SIRPg	
	CDR SEQ ID NO: 262	NO: 265									

NT or blank = not tested.

TABLE T

Anti-SIRP- α antibody binding data summary. Values indicated are tested by SPR (K_{off} 1/s).											
Antibody	CV1-3 SEQ ID NO: 18	v1 SEQ ID NO: 5	v2 SEQ ID NO: 6	cyno1 SEQ ID NO: 11	cyno2 SEQ ID NO: 12	m129 SEQ ID NO: 7	NOD SEQ ID NO: 8	BL6 SEQ ID NO: 9	SIRPb SEQ ID NO: 13	SIRPg SEQ ID NO: 15	CD47 blocking
Family 1											
S119	5.65E-04	4.15E-04	1.48E-04	2.34E-04	3.10E-04	NLB	NLB	NLB	4.28E-04	3.95E-04	block
S120	6.26E-04	4.04E-04	1.49E-04	2.36E-04	3.14E-04	NLB	NLB	NLB	4.25E-04	3.93E-04	block
S121	NT	4.79E-04	1.12E-04	3.02E-04	3.19E-04	NLB	NLB	NLB	5.64E-04	4.82E-04	block
S122	6.63E-04	4.98E-04	2.26E-04	2.66E-04	3.27E-04	NLB	NLB	NLB	3.52E-04	3.61E-04	block
S135	3.63E-03	6.58E-04	8.78E-05	4.73E-04	3.88E-04	NLB	NLB	NLB	5.21E-04	1.46E-03	block
Family 2											
S16	1.00E-04	6.22E-05	1.11E-04	1.32E-04	1.38E-04	1.76E-04	1.28E-03	2.35E-03	1.13E-04	8.17E-05	block
S17	1.54E-04	1.24E-04	1.97E-04	1.93E-04	2.07E-04	2.34E-04	6.12E-04	1.13E-03	1.25E-04	1.02E-04	block
S21	1.95E-04	1.80E-04	2.07E-04	2.33E-04	2.52E-04	2.81E-04	2.64E-03	8.06E-04	1.90E-04	1.84E-04	block
S22	1.35E-04	1.21E-04	1.46E-04	1.77E-04	1.77E-04	1.53E-04	5.83E-04	7.36E-04	1.38E-04	1.08E-04	block
S23	1.06E-04	8.35E-05	1.35E-04	1.60E-04	1.83E-04	1.34E-04	8.23E-04	1.32E-03	1.14E-04	1.04E-04	block
S24	2.16E-04	2.57E-04	3.17E-04	3.82E-04	3.34E-04	4.02E-04	3.86E-03	1.26E-03	1.94E-04	1.97E-04	block
S25	1.40E-04	1.12E-04	2.09E-04	2.19E-04	2.12E-04	1.33E-04	7.79E-04	2.90E-04	1.71E-04	1.41E-04	block
S26	4.74E-05	5.81E-05	1.11E-04	9.67E-04	7.43E-04	1.08E-03	7.64E-04	NLB	1.29E-04	8.35E-05	block
S27	8.94E-05	5.81E-05	9.97E-05	1.38E-04	1.61E-04	1.36E-04	1.30E-03	1.84E-03	8.97E-05	9.46E-05	block
S28	6.59E-05	2.11E-05	1.47E-05	1.15E-04	1.32E-04	3.06E-04	6.16E-04	1.86E-03	6.29E-05	3.27E-05	block
S29	3.43E-04	9.61E-05	1.78E-04	3.87E-04	3.69E-04	1.14E-03	NLB	1.86E-03	2.79E-04	2.16E-04	block
S30	1.42E-04	1.32E-04	1.88E-04	3.35E-04	3.66E-04	4.76E-04	NLB	2.44E-03	2.37E-04	1.93E-04	block
S55	6.00E-05	4.18E-05	7.16E-05	9.63E-05	1.16E-04	1.43E-04	1.19E-03	1.97E-03	6.48E-05	4.69E-05	block
S56	2.03E-04	2.28E-04	2.86E-04	2.95E-04	2.80E-04	4.06E-04	1.06E-03	4.29E-03	3.62E-04	2.15E-04	block
S59	8.19E-05	4.98E-05	8.63E-05	1.06E-04	1.13E-04	2.09E-04	1.35E-03	2.39E-03	1.07E-04	4.28E-05	block
S60	1.45E-04	1.40E-04	1.86E-04	1.92E-04	1.75E-04	2.21E-04	7.26E-04	1.50E-03	1.53E-04	1.00E-04	block
S65	1.47E-04	1.45E-04	1.72E-04	1.88E-04	1.90E-04	1.59E-04	6.07E-04	7.86E-04	1.54E-04	1.34E-04	block
S66	1.10E-04	1.18E-04	2.06E-04	2.50E-04	2.55E-04	1.27E-04	4.90E-04	1.02E-03	1.38E-04	1.26E-04	block
S69	2.33E-04	2.05E-04	2.67E-04	2.55E-04	2.16E-04	2.46E-04	7.09E-04	7.65E-04	2.29E-04	1.99E-04	block
S70	2.60E-04	2.25E-04	2.90E-04	2.96E-04	2.45E-04	2.70E-04	6.72E-04	9.87E-04	2.25E-04	2.22E-04	block
S71	3.79E-04	3.35E-04	4.03E-04	3.64E-04	3.13E-04	4.17E-04	1.12E-03	3.41E-03	3.88E-04	3.16E-04	block
S73	5.28E-05	1.90E-05	7.49E-05	1.27E-04	1.19E-04	1.64E-04	1.48E-03	3.73E-04	1.08E-04	7.30E-05	block
S74	1.66E-04	1.61E-04	2.78E-04	3.37E-04	3.20E-04	4.12E-04	NLB	1.37E-03	2.66E-04	2.25E-04	block
S76	1.55E-04	1.40E-04	2.09E-04	2.77E-04	2.69E-04	1.79E-04	5.76E-04	9.98E-04	1.87E-04	1.77E-04	block
S201	8.32E-05	3.46E-05	4.27E-05	6.36E-04	5.22E-04	8.18E-04	2.15E-03	1.11E-03	4.10E-03	1.85E-04	block
S202	6.86E-05	3.75E-05	5.40E-05	1.13E-04	1.20E-04	4.58E-05	5.09E-04	6.30E-04	2.05E-03	4.44E-05	block
S206	8.22E-05	3.50E-05	3.60E-05	6.24E-04	5.18E-04	9.01E-04	2.23E-03	1.57E-03	4.37E-03	1.92E-04	block
Family 3											
S136	1.59E-03	8.45E-04	1.61E-03	1.79E-03	1.32E-03	4.59E-04	3.86E-04	4.78E-03	3.60E-03	NLB	Non-block
S137	1.37E-03	1.65E-03	1.88E-03	2.04E-03	1.79E-03	5.15E-04	4.43E-04	4.63E-03	3.76E-03	NLB	Non-block
S175	1.65E-03	8.39E-04	1.83E-03	1.77E-03	1.07E-03	6.05E-04	4.17E-04	3.20E-03	2.99E-03	NT	Non-block
S177	NLB	3.32E-03	5.37E-03	4.47E-03	2.69E-03	1.04E-03	7.36E-04	NLB	NLB	NT	Non-block
S178	4.26E-03	2.23E-03	3.17E-03	3.44E-03	1.77E-03	1.12E-03	7.62E-04	NLB	4.73E-03	NT	Non-block
S180	3.07E-03	1.60E-03	2.27E-03	2.50E-03	1.38E-03	8.06E-04	5.76E-04	4.66E-03	3.75E-03	NT	Non-block
S184	5.14E-03	2.53E-03	3.91E-03	3.90E-03	2.21E-03	8.21E-04	6.38E-04	4.22E-03	NLB	NT	Non-block
S185	2.39E-03	1.22E-03	2.05E-03	1.78E-03	1.03E-03	7.17E-04	5.73E-04	5.14E-03	3.85E-03	NT	Non-block
S189	2.28E-03	1.06E-03	2.81E-03	3.31E-03	1.87E-03	8.37E-04	5.04E-04	3.90E-03	2.78E-03	NT	Non-block
S190	3.08E-03	1.56E-03	1.99E-03	2.17E-03	1.24E-03	7.73E-04	5.63E-04	NLB	2.56E-03	NT	Non-block
S193	NLB	3.08E-03	5.17E-03	NLB	4.28E-03	2.09E-03	1.57E-03	3.89E-03	NLB	NT	Non-block
Family 4											
S115	4.84E-04	7.86E-06	1.95E-05	2.22E-05	6.91E-05	NLB	NLB	NLB	3.69E-05	3.34E-05	Kick-off
S116	4.80E-04	2.84E-05	5.07E-05	7.01E-05	1.20E-04	NLB	NLB	NLB	3.09E-05	8.25E-05	Kick-off
S117	2.75E-04	1.20E-05	3.40E-05	1.83E-05	5.92E-05	NLB	NLB	NLB	3.54E-05	4.30E-06	Kick-off
S118	2.47E-04	9.17E-06	4.12E-05	6.97E-05	8.06E-05	NLB	NLB	NLB	1.31E-05	1.13E-05	Kick-off
S132	7.95E-05	3.34E-05	2.06E-05	4.28E-05	7.12E-05	NLB	NLB	NLB	1.01E-05	1.05E-05	Kick-off
S191	2.95E-04	6.67E-05	3.53E-05	5.54E-05	1.38E-05	NLB	NLB	NLB	1.01E-05	NT	Kick-off

TABLE T-continued

Anti-SIRP- α antibody binding data summary. Values indicated are tested by SPR (K_{off} 1/s).											
Antibody	CV1-3 SEQ ID NO: 18	v1 SEQ ID NO: 5	v2 SEQ ID NO: 6	cyno1 SEQ ID NO: 11	cyno2 SEQ ID NO: 12	m129 SEQ ID NO: 7	NOD SEQ ID NO: 8	BL6 SEQ ID NO: 9	SIRPb SEQ ID NO: 13	SIRPg SEQ ID NO: 15	CD47 blocking
S198 Family 5	7.87E-04	3.67E-05	4.95E-05	8.34E-05	7.73E-05	NLB	NLB	NLB	4.72E-05	NT	kick-off
S3	3.94E-04	4.10E-04	2.76E-04	2.18E-03	1.74E-03	6.55E-04	1.19E-03	1.97E-03	1.92E-03	1.62E-04	Non-block
S45	4.76E-05	7.24E-05	1.44E-05	NLB	NLB	NLB	NLB	NLB	NLB	NLB	Non-block
S123	1.31E-03	1.55E-03	NLB	1.57E-03	1.54E-03	NLB	NLB	NLB	NLB	1.43E-03	Non-block
S149	NLB	4.15E-03	5.44E-04	NLB	2.58E-03	NLB	NLB	NLB	3.31E-04	NLB	Non-block
S161	NLB	3.37E-03	3.84E-04	NLB	2.15E-03	NLB	NLB	NLB	3.21E-03	NLB	Non-block
S162	1.52E-03	6.84E-04	NLB	NLB	3.18E-04	NLB	NLB	NLB	1.98E-03	4.66E-03	Non-block
S163	1.74E-03	5.35E-04	6.64E-03	NLB	2.42E-04	NLB	NLB	NLB	1.70E-03	3.15E-03	Non-block
S164	1.34E-03	5.95E-04	9.89E-03	NLB	4.08E-04	NLB	NLB	NLB	1.67E-03	4.26E-03	Non-block
S173	1.37E-03	1.12E-03	5.04E-03	NLB	NLB	NLB	NLB	NLB	NLB	NLB	Non-block
S174	1.89E-03	1.43E-03	3.99E-03	NLB	NLB	NLB	NLB	NLB	NLB	NLB	Non-block
S194	1.11E-03	1.17E-03	NLB	1.31E-03	1.11E-03	NT	NLB	NLB	NLB	NT	Non-block
S209	2.45E-03	6.59E-04	4.01E-03	NLB	NLB	NT	NLB	NLB	6.54E-05	NLB	Non-block
S213	4.82E-03	NLB	6.85E-04	2.60E-04	3.13E-03	NT	NLB	NLB	NLB	NT	Non-block
S214	4.53E-03	NLB	6.77E-04	3.51E-03	2.71E-03	NT	NLB	NLB	NLB	NT	Non-block
S218	NT	7.06E-05	5.47E-05	2.49E-05	NT	NT	NLB	NLB	NT	2.12E-05	Non-block

NLB = $K_{off} > 5 \times 10^3$ and no binding; NT = not tested.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 426

<210> SEQ ID NO 1

<211> LENGTH: 348

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 1

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15

Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser Glu Ser Lys Tyr Gly Pro Pro Cys Pro
115 120 125

Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe
130 135 140

Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val
145 150 155 160

Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe
165 170 175

Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro
180 185 190

-continued

Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr
 195 200 205
 Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val
 210 215 220
 Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala
 225 230 235 240
 Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln
 245 250 255
 Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly
 260 265 270
 Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro
 275 280 285
 Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser
 290 295 300
 Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu
 305 310 315 320
 Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His
 325 330 335
 Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys
 340 345

<210> SEQ ID NO 2
 <211> LENGTH: 347
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 2

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
 1 5 10 15
 Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Ile Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
 100 105 110
 Val Arg Ala Lys Pro Ser Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro
 115 120 125
 Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro
 130 135 140
 Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr
 145 150 155 160
 Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn
 165 170 175
 Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg
 180 185 190
 Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val
 195 200 205

-continued

```

Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser
 210          215          220

Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys
 225          230          235          240

Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu
          245          250          255

Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe
          260          265          270

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
          275          280          285

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
          290          295          300

Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly
 305          310          315          320

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
          325          330          335

Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys
          340          345

```

```

<210> SEQ ID NO 3
<211> LENGTH: 346
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 3

```

```

Lys Glu Leu Lys Val Thr Gln Pro Glu Lys Ser Val Ser Val Ala Ala
 1          5          10          15

Gly Asp Ser Thr Val Leu Asn Cys Thr Leu Thr Ser Leu Leu Pro Val
          20          25          30

Gly Pro Ile Lys Trp Tyr Arg Gly Val Gly Gln Ser Arg Leu Leu Ile
          35          40          45

Tyr Ser Phe Thr Gly Glu His Phe Pro Arg Val Thr Asn Val Ser Asp
          50          55          60

Ala Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn Val
          65          70          75          80

Thr Pro Glu Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Gln Lys Gly
          85          90          95

Pro Ser Glu Pro Asp Thr Glu Ile Gln Ser Gly Gly Gly Thr Glu Val
          100          105          110

Tyr Val Leu Ala Lys Pro Ser Asp Lys Thr His Thr Cys Pro Pro Cys
          115          120          125

Pro Ala Pro Glu Ala Ala Gly Ala Pro Ser Val Phe Leu Phe Pro Pro
          130          135          140

Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys
          145          150          155          160

Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp
          165          170          175

Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu
          180          185          190

Glu Gln Tyr Ala Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu
          195          200          205

His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn
          210          215          220

```

-continued

Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly
 225 230 235 240
 Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu
 245 250 255
 Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr
 260 265 270
 Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn
 275 280 285
 Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe
 290 295 300
 Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn
 305 310 315 320
 Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
 325 330 335
 Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 340 345

<210> SEQ ID NO 4
 <211> LENGTH: 342
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 4

Thr Glu Val Lys Val Ile Gln Pro Glu Lys Ser Val Ser Val Ala Ala
 1 5 10 15
 Gly Asp Ser Thr Val Leu Asn Cys Thr Leu Thr Ser Leu Leu Pro Val
 20 25 30
 Gly Pro Ile Arg Trp Tyr Arg Gly Val Gly Gln Ser Arg Gln Leu Ile
 35 40 45
 Tyr Ser Phe Thr Thr Glu His Phe Pro Arg Val Thr Asn Val Ser Asp
 50 55 60
 Ala Thr Lys Arg Ser Asn Leu Asp Phe Ser Ile Arg Ile Ser Asn Val
 65 70 75 80
 Thr Pro Glu Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Gln Arg Gly
 85 90 95
 Ser Pro Asp Thr Glu Ile Gln Ser Gly Gly Gly Thr Glu Val Tyr Val
 100 105 110
 Leu Ala Lys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu
 115 120 125
 Ala Ala Gly Ala Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp
 130 135 140
 Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp
 145 150 155 160
 Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly
 165 170 175
 Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Ala
 180 185 190
 Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp
 195 200 205
 Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro
 210 215 220
 Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu
 225 230 235 240

-continued

Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
245 250 255

Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
260 265 270

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
275 280 285

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
290 295 300

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
305 310 315 320

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
325 330 335

Ser Leu Ser Pro Gly Lys
340

<210> SEQ ID NO 5
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15

Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 6
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 6

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1 5 10 15

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Ile Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

-continued

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
 100 105 110

Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 7
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 7

Lys Glu Leu Lys Val Thr Gln Pro Glu Lys Ser Val Ser Val Ala Ala
 1 5 10 15

Gly Asp Ser Thr Val Leu Asn Cys Thr Leu Thr Ser Leu Leu Pro Val
 20 25 30

Gly Pro Ile Lys Trp Tyr Arg Gly Val Gly Gln Ser Arg Leu Leu Ile
 35 40 45

Tyr Ser Phe Thr Gly Glu His Phe Pro Arg Val Thr Asn Val Ser Asp
 50 55 60

Ala Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn Val
 65 70 75 80

Thr Pro Glu Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Gln Lys Gly
 85 90 95

Pro Ser Glu Pro Asp Thr Glu Ile Gln Ser Gly Gly Gly Thr Glu Val
 100 105 110

Tyr Val Leu Ala Lys Pro Ser
 115

<210> SEQ ID NO 8
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 8

Thr Glu Val Lys Val Ile Gln Pro Glu Lys Ser Val Ser Val Ala Ala
 1 5 10 15

Gly Asp Ser Thr Val Leu Asn Cys Thr Leu Thr Ser Leu Leu Pro Val
 20 25 30

Gly Pro Ile Arg Trp Tyr Arg Gly Val Gly Gln Ser Arg Gln Leu Ile
 35 40 45

Tyr Ser Phe Thr Thr Glu His Phe Pro Arg Val Thr Asn Val Ser Asp
 50 55 60

Ala Thr Lys Arg Ser Asn Leu Asp Phe Ser Ile Arg Ile Ser Asn Val
 65 70 75 80

Thr Pro Glu Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Gln Arg Gly
 85 90 95

Ser Pro Asp Thr Glu Ile Gln Ser Gly Gly Gly Thr Glu Val Tyr Val
 100 105 110

Leu Ala Lys
 115

<210> SEQ ID NO 9
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 9

Lys Glu Leu Lys Val Thr Gln Pro Glu Lys Ser Val Ser Val Ala Ala

-continued

1	5	10	15
Gly Asp Ser Thr Val Leu Asn Cys Thr Leu Thr Ser Leu Leu Pro Val	20	25	30
Gly Pro Ile Arg Trp Tyr Arg Gly Val Gly Pro Ser Arg Leu Leu Ile	35	40	45
Tyr Ser Phe Ala Gly Glu Tyr Val Pro Arg Ile Arg Asn Val Ser Asp	50	55	60
Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn Val	65	70	75
Thr Pro Ala Asp Ala Gly Ile Tyr Tyr Cys Val Lys Phe Gln Lys Gly	85	90	95
Ser Ser Glu Pro Asp Thr Glu Ile Gln Ser Gly Gly Gly Thr Glu Val	100	105	110
Tyr Val Leu Ala Lys	115		

<210> SEQ ID NO 10
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 10

Thr Glu Val Lys Val Thr Gln Pro Glu Lys Ser Val Ser Val Ala Ala	1	5	10	15
Gly Asp Ser Thr Ile Leu Asn Cys Thr Val Thr Ser Leu Leu Pro Val	20	25	30	
Gly Pro Ile Arg Trp Tyr Arg Gly Val Gly Gln Ser Arg Leu Leu Ile	35	40	45	
Tyr Ser Phe Thr Gly Glu His Phe Pro Arg Ile Arg Asn Val Ser Asp	50	55	60	
Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn Val	65	70	75	80
Thr Pro Glu Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Gln Arg Gly	85	90	95	
Ser Ser Glu Pro Asp Thr Glu Ile Gln Ser Gly Gly Gly Thr Glu Val	100	105	110	
Tyr Val Leu Ala Lys	115			

<210> SEQ ID NO 11
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Macaca fascicularis

<400> SEQUENCE: 11

Glu Glu Glu Leu Gln Val Ile Gln Pro Glu Lys Ser Val Ser Val Ala	1	5	10	15
Ala Gly Glu Ser Ala Thr Leu Asn Cys Thr Ala Thr Ser Leu Ile Pro	20	25	30	
Val Gly Pro Ile Gln Trp Phe Arg Gly Val Gly Pro Gly Arg Glu Leu	35	40	45	
Ile Tyr His Gln Lys Glu Gly His Phe Pro Arg Val Thr Pro Val Ser	50	55	60	
Asp Pro Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn	65	70	75	80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys				

-continued

85	90	95
Gly Ser Pro Asp Val Glu Leu Lys Ser Gly Ala Gly Thr Glu Leu Ser		
100	105	110
Val Arg Ala Lys Pro Ser		
115		

<210> SEQ ID NO 12
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: *Macaca fascicularis*

<400> SEQUENCE: 12

Glu Glu Glu Leu Gln Val Ile Gln Pro Glu Lys Ser Val Ser Val Ala		
1	5	10
Ala Gly Asp Ser Ala Thr Leu Asn Cys Thr Val Ser Ser Leu Ile Pro		
20	25	30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu		
35	40	45
Ile Tyr Asn Leu Lys Glu Gly His Phe Pro Arg Val Thr Ala Val Ser		
50	55	60
Asp Pro Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn		
65	70	75
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys		
85	90	95
Gly Ser Pro Asp Val Glu Leu Lys Ser Gly Ala Gly Thr Glu Leu Ser		
100	105	110
Val Arg Ala Lys Pro Ser		
115		

<210> SEQ ID NO 13
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: *Homo sapiens*

<400> SEQUENCE: 13

Glu Asp Glu Leu Gln Val Ile Gln Pro Glu Lys Ser Val Ser Val Ala		
1	5	10
Ala Gly Glu Ser Ala Thr Leu Arg Cys Ala Met Thr Ser Leu Ile Pro		
20	25	30
Val Gly Pro Ile Met Trp Phe Arg Gly Ala Gly Ala Gly Arg Glu Leu		
35	40	45
Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser		
50	55	60
Glu Leu Thr Lys Arg Asn Asn Leu Asp Phe Ser Ile Ser Ile Ser Asn		
65	70	75
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys		
85	90	95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu		
100	105	110
Ser Val Arg Ala Lys Pro Ser		
115		

<210> SEQ ID NO 14
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: *Homo sapiens*

<400> SEQUENCE: 14

-continued

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Ile Ser Val Ala
 1 5 10 15
 Ala Gly Glu Ser Ala Thr Leu His Cys Thr Val Thr Ser Leu Ile Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp His Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 15
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 15

Glu Glu Glu Leu Gln Met Ile Gln Pro Glu Lys Leu Leu Val Thr
 1 5 10 15
 Val Gly Lys Thr Ala Thr Leu His Cys Thr Val Thr Ser Leu Leu Pro
 20 25 30
 Val Gly Pro Val Leu Trp Phe Arg Gly Val Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Ser
 65 70 75 80
 Ile Thr Pro Ala Asp Val Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Glu Asn Val Glu Phe Lys Ser Gly Pro Gly Thr Glu Met
 100 105 110
 Ala Leu Gly Ala Lys Pro Ser
 115

<210> SEQ ID NO 16
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 16

Gln Leu Leu Phe Asn Lys Thr Lys Ser Val Glu Phe Thr Phe Ser Asn
 1 5 10 15
 Asp Thr Val Val Ile Pro Cys Phe Val Thr Asn Met Glu Ala Gln Asn
 20 25 30
 Thr Thr Glu Val Tyr Val Lys Trp Lys Phe Lys Gly Arg Asp Ile Tyr
 35 40 45
 Thr Phe Asp Gly Ala Leu Asn Lys Ser Thr Val Pro Thr Asp Phe Ser
 50 55 60
 Ser Ala Lys Ile Glu Val Ser Gln Leu Leu Lys Gly Asp Ala Ser Leu
 65 70 75 80

-continued

Lys Met Asp Lys Ser Asp Ala Val Ser His Thr Gly Asn Tyr Thr Cys
85 90 95

Glu Val Thr Glu Leu Thr Arg Glu Gly Glu Thr Ile Ile Glu Leu Lys
100 105 110

Tyr Arg Val Val Ser
115

<210> SEQ ID NO 17
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 17

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15

Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 18
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 18

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1 5 10 15

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu
35 40 45

Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110

Val Arg Ala Lys Pro Ser
115

-continued

```

<210> SEQ ID NO 19
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 19

```

```

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1          5          10          15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Phe Pro
20          25          30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
35          40          45
Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50          55          60
Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65          70          75          80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85          90          95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100         105         110
Val Arg Ala Lys Pro Ser
115

```

```

<210> SEQ ID NO 20
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 20

```

```

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1          5          10          15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
20          25          30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu
35          40          45
Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50          55          60
Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65          70          75          80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
85          90          95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100         105         110
Val Arg Ala Lys Pro Ser
115

```

```

<210> SEQ ID NO 21
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 21

```

```

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1          5          10          15

```

-continued

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Phe Pro
 20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu
 35 40 45

Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60

Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
 65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
 85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
 100 105 110

Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 22
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 22

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
 1 5 10 15

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Ile Pro
 20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu
 35 40 45

Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60

Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
 65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
 85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
 100 105 110

Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 23
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 23

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
 1 5 10 15

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
 20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
 35 40 45

Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60

Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn

-continued

65	70	75	80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys			
	85	90	95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser			
	100	105	110
Val Arg Ala Lys Pro Ser			
	115		

<210> SEQ ID NO 24
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 24

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala			
1	5	10	15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro			
	20	25	30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu			
	35	40	45
Ile Tyr Asn Gln Lys Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser			
	50	55	60
Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn			
65	70	75	80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys			
	85	90	95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser			
	100	105	110
Val Arg Ala Lys Pro Ser			
	115		

<210> SEQ ID NO 25
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 25

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala			
1	5	10	15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro			
	20	25	30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu			
	35	40	45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser			
	50	55	60
Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn			
65	70	75	80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys			
	85	90	95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser			
	100	105	110
Val Arg Ala Lys Pro Ser			
	115		

-continued

<210> SEQ ID NO 26
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 26

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1 5 10 15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu
35 40 45
Ile Tyr Asn Gln Arg Gln Gly His Phe Pro Arg Val Thr Thr Val Ser
50 55 60
Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
85 90 95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110
Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 27
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 27

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1 5 10 15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu
35 40 45
Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60
Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
85 90 95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110
Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 28
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 28

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala

-continued

1	5	10	15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro	20	25	30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Val Leu	35	40	45
Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser	50	55	60
Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn	65	70	75
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys	85	90	95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser	100	105	110
Val Arg Ala Lys Pro Ser	115		

<210> SEQ ID NO 29
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 29

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala	1	5	10	15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Ile Pro	20	25	30	
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu	35	40	45	
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser	50	55	60	
Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn	65	70	75	80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys	85	90	95	
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser	100	105	110	
Val Arg Ala Lys Pro Ser	115			

<210> SEQ ID NO 30
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 30

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala	1	5	10	15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Phe Pro	20	25	30	
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu	35	40	45	
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser	50	55	60	

-continued

Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110

Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 31
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 31

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1 5 10 15

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110

Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 32
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 32

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1 5 10 15

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110

Val Arg Ala Lys Pro Ser
115

-continued

```

<210> SEQ ID NO 33
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 33

```

```

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1      5      10      15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
20     25     30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
35     40     45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50     55     60
Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65     70     75     80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85     90     95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100    105    110
Val Arg Ala Lys Pro Ser
115

```

```

<210> SEQ ID NO 34
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 34

```

```

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1      5      10      15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Ile Pro
20     25     30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
35     40     45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50     55     60
Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65     70     75     80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85     90     95
Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100    105    110
Val Arg Ala Lys Pro Ser
115

```

```

<210> SEQ ID NO 35
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 35

```

-continued

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Ser Val Ala
 1 5 10 15
 Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ile Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Glu Thr Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
 100 105 110
 Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 36
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 36

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 37
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 37

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60

-continued

Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 38
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 38

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15

Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 39
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 39

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15

Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser

-continued

115

<210> SEQ ID NO 40
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 40

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15
Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
65 70 75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110
Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 41
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 41

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15
Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Phe Pro
20 25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
65 70 75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110
Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 42
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 42

-continued

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Val Leu
 35 40 45
 Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 43
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 43

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Val Leu
 35 40 45
 Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 44
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 44

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser

-continued

50	55	60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala		
65	70	75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys		
	85	90 95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu		
	100	105 110
Ser Val Arg Ala Lys Pro Ser		
115		

<210> SEQ ID NO 45
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 45

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala		
1	5	10 15
Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro		
	20	25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Val Leu		
	35	40 45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser		
	50	55 60
Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala		
65	70	75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Ile Lys Phe Arg Lys		
	85	90 95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu		
	100	105 110
Ser Val Arg Ala Lys Pro Ser		
115		

<210> SEQ ID NO 46
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 46

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala		
1	5	10 15
Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Phe Pro		
	20	25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu		
	35	40 45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser		
	50	55 60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala		
65	70	75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys		
	85	90 95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu		
	100	105 110

-continued

Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 47
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 47

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15
Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala
65 70 75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110
Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 48
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 48

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15
Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
20 25 30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45
Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
50 55 60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala
65 70 75 80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110
Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 49
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 49

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 50

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 50

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 51

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 51

Glu Glu Glu Leu Gln Ile Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ile Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Arg Glu Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60

-continued

Asp Thr Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 52
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 52

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Phe Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Arg Gln Gly Pro Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn
 65 70 75 80
 Ile Thr Val Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 53
 <211> LENGTH: 484
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 53

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15
 Glu Ile Thr Cys Ser Gly Gly Gly Ser Asn Asn Ala Tyr Gly Trp Phe
 20 25 30
 Gln Gln Lys Ser Pro Gly Ser Ala Pro Leu Thr Val Ile Tyr Asp Asn
 35 40 45
 Gly Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser
 50 55 60
 Asp Ser Thr Gly Thr Leu Thr Ile Thr Arg Val Gln Ala Glu Asp Glu
 65 70 75 80
 Ala Val Tyr Tyr Cys Gly Ser Ala Asp Asn Ser Gly Ala Gly Val Phe
 85 90 95
 Gly Ala Gly Thr Thr Leu Thr Val Leu Gly Gln Ser Ser Arg Ser Ser
 100 105 110
 Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Met Ala Ala Val Thr

-continued

115	120	125
Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly Ala Leu Ser		
130	135	140
Leu Val Cys Lys Gly Ser Gly Phe Thr Phe Ser Ser His Ala Met Asn		
145	150	155
Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ala Gly Ile		
165	170	175
Ser Ser Asp Gly Arg Phe Thr Tyr Tyr Gly Ala Ala Val Gln Gly Arg		
180	185	190
Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr Val Arg Leu Gln Leu		
195	200	205
Asn Asn Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys Thr Lys Asn		
210	215	220
Gly Gly Cys Gly Ser Gly Gly Asp Leu Asp Cys Ile Asp Ala Trp Gly		
225	230	235
His Gly Thr Glu Val Ile Val Ser Ser Ser Leu Asp Pro Lys Ser Ser		
245	250	255
Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly		
260	265	270
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met		
275	280	285
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His		
290	295	300
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val		
305	310	315
His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr		
325	330	335
Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly		
340	345	350
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile		
355	360	365
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val		
370	375	380
Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser		
385	390	395
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu		
405	410	415
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro		
420	425	430
Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val		
435	440	445
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met		
450	455	460
His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser		
465	470	475
Pro Gly Lys Val		

<210> SEQ ID NO 54

<211> LENGTH: 480

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 54

Ala 1	Val 2	Thr 3	Gln 4	Pro 5	Ala 6	Ser 7	Val 8	Ser 9	Ala 10	Asn 11	Pro 12	Gly 13	Glu 14	Thr 15	Val 16
Arg 17	Ile 18	Thr 19	Cys 20	Ser 21	Gly 22	Asp 23	Ser 24	Ser 25	Ser 26	Tyr 27	Tyr 28	Ser 29	Trp 30	His 31	Gln 32
Gln 33	Lys 34	Ser 35	Pro 36	Gly 37	Ser 38	Ala 39	Pro 40	Val 41	Ser 42	Val 43	Ile 44	Tyr 45	Ser 46	Asn 47	Thr 48
Asp 49	Arg 50	Pro 51	Ser 52	Asp 53	Ile 54	Pro 55	Ser 56	Arg 57	Phe 58	Ser 59	Gly 60	Ser 61	Ala 62	Ser 63	Gly 64
Ser 65	Thr 66	Ala 67	Thr 68	Leu 69	Thr 70	Ile 71	Thr 72	Gly 73	Val 74	Gln 75	Ala 76	Glu 77	Asp 78	Glu 79	Ala 80
Val 81	Tyr 82	Phe 83	Cys 84	Gly 85	Ala 86	Tyr 87	Asp 88	Ser 89	Ser 90	Ser 91	Asp 92	Ser 93	Asp 94	Ile 95	Phe 96
Gly 97	Ala 98	Gly 99	Thr 100	Thr 101	Leu 102	Thr 103	Val 104	Leu 105	Gly 106	Gln 107	Ser 108	Ser 109	Arg 110	Ser 111	Ser 112
Gly 113	Gly 114	Gly 115	Gly 116	Ser 117	Ser 118	Gly 119	Gly 120	Gly 121	Gly 122	Ser 123	Met 124	Ala 125	Ala 126	Val 127	Thr 128
Leu 129	Asp 130	Glu 131	Ser 132	Gly 133	Gly 134	Gly 135	Leu 136	Gln 137	Thr 138	Pro 139	Gly 140	Gly 141	Gly 142	Leu 143	Ser 144
Leu 145	Val 146	Cys 147	Lys 148	Ala 149	Ser 150	Gly 151	Phe 152	Asp 153	Phe 154	Ser 155	Asn 156	Phe 157	Asn 158	Met 159	Ala 160
Trp 161	Val 162	Arg 163	Gln 164	Gly 165	Pro 166	Gly 167	Lys 168	Gly 169	Leu 170	Glu 171	Tyr 172	Val 173	Ala 174	Glu 175	Ile 176
Ser 177	Asp 178	Thr 179	Gly 180	Ser 181	Thr 182	Pro 183	Tyr 184	Tyr 185	Gly 186	Ser 187	Ala 188	Val 189	Gln 190	Gly 191	Arg 192
Ala 193	Thr 194	Ile 195	Ser 196	Arg 197	Asp 198	Asn 199	Gly 200	Gln 201	Ser 202	Thr 203	Val 204	Arg 205	Leu 206	Gln 207	Leu 208
Asn 209	Asn 210	Leu 211	Arg 212	Ala 213	Glu 214	Asp 215	Thr 216	Gly 217	Thr 218	Tyr 219	Phe 220	Cys 221	Thr 222	Arg 223	Asn 224
Phe 225	Gly 226	Ser 227	Ser 228	Val 229	Ser 230	Ser 231	Ile 232	Asp 233	Ala 234	Trp 235	Gly 236	His 237	Gly 238	Thr 239	Glu 240
Val 241	Ile 242	Val 243	Ser 244	Ser 245	Ser 246	Leu 247	Asp 248	Pro 249	Lys 250	Ser 251	Ser 252	Asp 253	Lys 254	Thr 255	His 256
Thr 257	Cys 258	Pro 259	Pro 260	Cys 261	Pro 262	Ala 263	Pro 264	Glu 265	Leu 266	Leu 267	Gly 268	Gly 269	Pro 270	Ser 271	Val 272
Phe 273	Leu 274	Phe 275	Pro 276	Pro 277	Lys 278	Pro 279	Lys 280	Asp 281	Thr 282	Leu 283	Met 284	Ile 285	Ser 286	Arg 287	Thr 288
Pro 289	Glu 290	Val 291	Thr 292	Cys 293	Val 294	Val 295	Val 296	Asp 297	Val 298	Ser 299	His 300	Glu 301	Asp 302	Pro 303	Glu 304
Val 305	Lys 306	Phe 307	Asn 308	Trp 309	Tyr 310	Val 311	Asp 312	Gly 313	Val 314	Glu 315	Val 316	His 317	Asn 318	Ala 319	Lys 320
Thr 321	Lys 322	Pro 323	Arg 324	Glu 325	Glu 326	Gln 327	Tyr 328	Asn 329	Ser 330	Thr 331	Tyr 332	Arg 333	Val 334	Val 335	Ser 336
Val 337	Leu 338	Thr 339	Val 340	Leu 341	His 342	Gln 343	Asp 344	Trp 345	Leu 346	Asn 347	Gly 348	Lys 349	Glu 350	Tyr 351	Lys 352
Cys 353	Lys 354	Val 355	Ser 356	Asn 357	Lys 358	Ala 359	Leu 360	Pro 361	Ala 362	Pro 363	Ile 364	Glu 365	Lys 366	Thr 367	Ile 368
Ser 369	Lys 370	Ala 371	Lys 372	Gly 373	Gln 374	Pro 375	Arg 376	Glu 377	Pro 378	Gln 379	Val 380	Tyr 381	Thr 382	Leu 383	Pro 384
Pro 385	Ser 386	Arg 387	Asp 388	Glu 389	Leu 390	Thr 391	Lys 392	Asn 393	Gln 394	Val 395	Ser 396	Leu 397	Thr 398	Cys 399	Leu 400
Val 401	L														

-continued

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
 420 425 430

Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
 435 440 445

Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
 450 455 460

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Val
 465 470 475 480

<210> SEQ ID NO 55
 <211> LENGTH: 491
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 55

Ala Val Thr Gln Pro Ser Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15

Glu Ile Thr Cys Ser Gly Ser Ser Thr Tyr Tyr Gly Trp Tyr Gln Gln
 20 25 30

Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Asp Asn Asp Lys
 35 40 45

Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser
 50 55 60

Thr His Thr Leu Ile Ile Thr Gly Val Gln Val Glu Asp Glu Ala Val
 65 70 75 80

Tyr Phe Cys Gly Asn Glu Asp Asn Asn Tyr Val Ala Ile Phe Gly Ala
 85 90 95

Gly Thr Thr Leu Thr Val Leu Gly Gln Ser Ser Arg Ser Ser Gly Gly
 100 105 110

Gly Gly Ser Ser Gly Gly Gly Gly Ser Met Ala Ala Val Thr Leu Asp
 115 120 125

Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly Ala Leu Ser Leu Val
 130 135 140

Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Tyr Asn Met Gly Trp Val
 145 150 155 160

Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Val Ala Gly Ile Tyr Ala
 165 170 175

Ser Gly Ser Ser Thr Asp Thr Asp Thr Thr Tyr Gly Pro Ala Val Ala
 180 185 190

Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr Val Arg Leu
 195 200 205

Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Thr Tyr Tyr Cys Ala
 210 215 220

Lys Ala Ala Gly Gly Cys Ser Thr His Thr Cys Thr Ala Tyr Ile Ala
 225 230 235 240

Asp Ser Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile Val Ser Ser
 245 250 255

Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys
 260 265 270

Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro
 275 280 285

Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys
 290 295 300

-continued

Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp
305					310					315					320
Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu
			325						330					335	
Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu
			340					345					350		
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn
		355					360					365			
Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly
	370					375					380				
Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu
385					390					395					400
Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr
			405					410						415	
Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn
		420						425					430		
Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe
		435					440					445			
Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn
	450					455					460				
Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr
465					470					475					480
Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Val					
			485					490							

<210> SEQ ID NO 56

<211> LENGTH: 489

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 56

Ala	Leu	Thr	Gln	Pro	Ser	Ser	Val	Ser	Ala	Asn	Pro	Gly	Glu	Thr	Val
1				5					10					15	
Lys	Ile	Thr	Cys	Ser	Gly	Asp	Asn	Ser	Ala	His	Tyr	Tyr	Tyr	Gly	Trp
			20				25						30		
Tyr	Gln	Gln	Lys	Ser	Pro	Gly	Ser	Ala	Pro	Val	Thr	Val	Ile	Tyr	Tyr
		35				40					45				
Asn	Asp	Lys	Arg	Pro	Ser	Gly	Ile	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Ala
	50					55				60					
Ser	Gly	Ser	Thr	Ala	Thr	Leu	Ile	Ile	Thr	Gly	Val	Gln	Val	Glu	Asp
65				70					75					80	
Glu	Ala	Val	Tyr	Phe	Cys	Gly	Ser	Ala	Asp	Ser	Ser	Asn	Pro	Ala	Ile
			85					90						95	
Phe	Gly	Ala	Gly	Thr	Thr	Leu	Thr	Val	Leu	Gly	Gln	Ser	Ser	Arg	Ser
		100						105					110		
Ser	Gly	Gly	Gly	Gly	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Met	Ala	Ala	Val
		115					120					125			
Thr	Leu	Asp	Glu	Ser	Gly	Gly	Gly	Leu	Gln	Thr	Pro	Gly	Arg	Ala	Leu
	130				135					140					
Ser	Leu	Val	Cys	Arg	Gly	Ser	Gly	Phe	Ser	Ile	Ser	Ser	Tyr	Asn	Met
145				150					155						160
Gly	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Phe	Ile	Ala	Ser
			165				170							175	

-continued

```

Ile Gly Ser Asp Gly Ser Ser Thr His Tyr Ala Pro Ala Val Lys Gly
    180                      185                      190

Arg Ala Thr Ile Thr Arg Asp Val Gly Gln Ser Thr Val Arg Leu Gln
    195                      200                      205

Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Thr Tyr Phe Cys Ala Lys
    210                      215                      220

Asp Ala Tyr Gln Cys Ser Tyr Ala Thr Cys Asn Asp Tyr Leu Asp Thr
    225                      230                      235                      240

Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile Val Ser Ser Ser Leu
    245                      250                      255

Asp Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
    260                      265                      270

Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
    275                      280                      285

Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
    290                      295                      300

Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val
    305                      310                      315                      320

Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln
    325                      330                      335

Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln
    340                      345                      350

Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala
    355                      360                      365

Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro
    370                      375                      380

Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr
    385                      390                      395                      400

Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser
    405                      410                      415

Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr
    420                      425                      430

Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr
    435                      440                      445

Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe
    450                      455                      460

Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
    465                      470                      475                      480

Ser Leu Ser Leu Ser Pro Gly Lys Val
    485

```

```

<210> SEQ ID NO 57
<211> LENGTH: 487
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 57

```

```

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1          5          10          15

Lys Ile Thr Cys Ser Gly Ser Ser Ser Gly Ser Tyr Gly Trp Tyr Gln
20          25          30

Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Glu Thr Asn
35          40          45

```

Lys 50	Arg	Pro	Ser	Asn	Ile	Pro 55	Ser	Arg	Phe	Ser	Gly 60	Ser	Lys	Ser	Gly
Ser 65	Thr	Ala	Thr	Leu	Thr 70	Ile	Thr	Gly	Val	Gln 75	Ala	Asp	Asp	Glu	Ala 80
Val	Tyr	Tyr	Cys	Gly 85	Ser	Glu	Asp	Ser	Ser 90	Thr	Tyr	Leu	Ser	Ile 95	Phe
Gly	Ala	Gly	Thr	Thr 100	Leu	Thr	Val	Leu 105	Gly	Gln	Ser	Ser	Arg 110	Ser	Ser
Gly	Gly	Gly	Gly	Ser	Ser	Gly 120	Gly	Gly	Gly	Ser	Met	Ala 125	Ala	Val	Thr
Leu	Asp 130	Glu	Ser	Gly	Gly 135	Gly	Leu	Gln	Thr	Pro 140	Gly	Gly	Ala	Leu	Ser
Leu 145	Val	Cys	Lys	Ala 150	Ser	Gly	Phe	Thr	Phe 155	Ser	Ser	Phe	Asn	Met	Gly 160
Trp	Val	Arg	Gln	Ala 165	Pro	Gly	Lys	Gly	Leu 170	Glu	Phe	Val	Ala 175	Ala	Ile
Tyr	Ser	Gly	Asn 180	Ser	Ala	Glu	Tyr	Gly 185	Ala	Ala	Val	Gln 190	Gly	Arg	Ala
Thr	Ile 195	Ser	Arg	Asp	Asn	Gly 200	Gln	Ser	Thr	Val	Arg 205	Leu	Gln	Leu	Asn
Asn 210	Leu	Arg	Ala	Glu	Asp 215	Thr	Gly	Ile	Tyr	Phe 220	Cys	Ala	Lys	Asp	Ala
Gly 225	Ser	Gly	Cys	Tyr 230	Ser	Gly	Val	Cys	Ala 235	Gly	Thr	Ser	Ser	Ile	Asp 240
Ala	Trp	Gly	His	Gly 245	Thr	Glu	Val	Ile	Val 250	Ser	Ser	Ser	Leu	Asp 255	Pro
Lys	Ser	Ser	Asp 260	Lys	Thr	His	Thr	Cys 265	Pro	Pro	Cys	Pro 270	Ala	Pro	Glu
Leu 275	Leu	Gly	Gly	Pro	Ser	Val	Phe 280	Leu	Phe	Pro	Pro	Lys 285	Pro	Lys	Asp
Thr 290	Leu	Met	Ile	Ser	Arg 295	Thr	Pro	Glu	Val	Thr	Cys 300	Val	Val	Val	Asp
Val 305	Ser	His	Glu	Asp 310	Pro	Glu	Val	Lys	Phe 315	Asn	Trp	Tyr	Val	Asp	Gly 320
Val	Glu	Val	His	Asn 325	Ala	Lys	Thr	Lys	Pro 330	Arg	Glu	Glu	Gln 335	Tyr	Asn
Ser	Thr	Tyr	Arg 340	Val	Val	Ser	Val	Leu 345	Thr	Val	Leu	His 350	Gln	Asp	Trp
Leu	Asn 355	Gly	Glu	Glu	Tyr	Lys	Cys 360	Lys	Val	Ser	Asn 365	Lys	Ala	Leu	Pro
Ala 370	Pro	Ile	Glu	Lys	Thr 375	Ile	Ser	Lys	Ala 380	Lys	Gly	Gln	Pro	Arg	Glu
Pro 385	Gln	Val	Tyr	Thr 390	Leu	Pro	Pro	Ser	Arg 395	Asp	Glu	Leu	Thr	Lys	Asn 400
Gln	Val	Ser	Leu 405	Thr	Cys	Leu	Val	Lys 410	Gly	Phe	Tyr	Pro	Ser	Asp 415	Ile
Ala	Val	Glu	Trp 420	Glu	Ser	Asn	Gly 425	Gln	Pro 430	Glu	Asn	Asn 435	Tyr	Lys	Thr
Thr	Pro 435	Pro	Val	Leu	Asp 440	Ser	Asp	Gly	Ser 445	Phe	Phe	Leu 450	Tyr	Ser	Lys
Leu 455	Thr	Val	Asp 460	Lys	Ser	Arg 465	Trp	Gln	Gln 470	Gly	Asn 475	Val	Phe	Ser	Cys
Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu

-continued

465	470	475	480
-----	-----	-----	-----

Ser Leu Ser Pro Gly Lys Val
 485

<210> SEQ ID NO 58
 <211> LENGTH: 489
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 58

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15

Lys Ile Thr Cys Ser Gly Asp Ser Ser Tyr Tyr Gly Trp Tyr Gln Gln
 20 25 30

Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Asp Asp Asn Lys
 35 40 45

Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser
 50 55 60

Thr Gly Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ala Val
 65 70 75 80

Tyr Phe Cys Gly Asn Glu Asp Asn Ser Tyr Val Ala Ile Phe Gly Ala
 85 90 95

Gly Thr Thr Leu Thr Val Leu Gly Gln Ser Ser Arg Ser Ser Gly Gly
 100 105 110

Gly Gly Ser Ser Gly Gly Gly Gly Ser Met Ala Ala Val Thr Leu Asp
 115 120 125

Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly Ala Leu Ser Leu Val
 130 135 140

Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Tyr Asn Met Gly Trp Val
 145 150 155 160

Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Val Ala Gly Ile Tyr Ile
 165 170 175

Ala Ser Gly Asp Leu Gly Thr Thr Tyr Gly Ala Ala Val Gln Gly Arg
 180 185 190

Ala Thr Ile Ser Arg Asp Asp Gly Gln Ser Thr Val Arg Leu Gln Leu
 195 200 205

Asn Asn Leu Arg Ala Glu Asp Thr Gly Thr Tyr Phe Cys Ala Lys Ser
 210 215 220

Ala Gly Gly Cys Ser Ala His Ser Cys Asp Thr Tyr Ile Ala Asp Ser
 225 230 235 240

Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile Val Ser Ser Ser Leu
 245 250 255

Asp Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
 260 265 270

Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
 275 280 285

Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
 290 295 300

Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val
 305 310 315 320

Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln
 325 330 335

Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln

-continued

340	345	350
Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala		
355	360	365
Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro		
370	375	380
Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr		
385	390	395
Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser		
405	410	415
Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr		
420	425	430
Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr		
435	440	445
Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe		
450	455	460
Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys		
465	470	475
Ser Leu Ser Leu Ser Pro Gly Lys Val		
485		

<210> SEQ ID NO 59
 <211> LENGTH: 490
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 59

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val	
1	15
Lys Ile Thr Cys Ser Gly Ser Ser Ser Tyr Tyr Gly Trp Tyr Arg Gln	
20	30
Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Asp Asn Asp Lys	
35	45
Arg Pro Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser	
50	60
Thr Asn Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ala Val	
65	80
Tyr Tyr Cys Gly Asn Glu Asp Asn Ser Tyr Val Gly Ile Phe Gly Ala	
85	95
Gly Thr Thr Leu Thr Val Leu Gly Gln Ser Ser Arg Ser Ser Gly Gly	
100	110
Gly Gly Ser Ser Gly Gly Gly Gly Ser Met Ala Ala Val Thr Leu Asp	
115	125
Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly Ala Leu Ser Leu Val	
130	140
Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Tyr Asn Met Gly Trp Val	
145	160
Arg Gln Ala Pro Asp Lys Gly Leu Glu Phe Val Ala Gly Ile Tyr Thr	
165	175
Gly Ser Asp Ala Gly Leu Ser Thr Thr Tyr Gly Ala Ala Val Gln Gly	
180	190
Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr Val Arg Leu Gln	
195	205
Leu Asn Asn Leu Gly Ala Glu Asp Thr Gly Ile Tyr Phe Cys Thr Lys	

-continued

210	215	220
Ser Ala Gly Gly Cys Ser Asp Tyr Asn Cys Asp Ala Tyr Ile Ala Asp		
225	230	235 240
Ser Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile Val Ser Ser Ser		
	245	250 255
Leu Asp Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro		
	260	265 270
Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys		
	275	280 285
Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val		
	290	295 300
Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr		
305	310	315 320
Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu		
	325	330 335
Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His		
	340	345 350
Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys		
	355	360 365
Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln		
	370	375 380
Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu		
385	390	395 400
Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro		
	405	410 415
Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn		
	420	425 430
Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu		
	435	440 445
Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val		
	450	455 460
Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln		
465	470	475 480
Lys Ser Leu Ser Leu Ser Pro Gly Lys Val		
	485	490

<210> SEQ ID NO 60

<211> LENGTH: 489

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 60

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Glu Thr Val		
1	5	10 15
Lys Ile Thr Cys Ser Gly Asp Ser Ser Tyr Tyr Gly Trp Tyr Gln Gln		
	20	25 30
Lys Ala Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Asp Asn Asp Lys		
	35	40 45
Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser		
	50	55 60
Thr Ala Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ala Val		
65	70	75 80
Tyr Tyr Cys Gly Asn Glu Asp Met Asn Tyr Val Gly Ile Phe Gly Ala		

-continued

85							90					95				
Gly	Thr	Thr	Leu	Thr	Val	Leu	Gly	Gln	Ser	Ser	Arg	Ser	Ser	Gly	Gly	
			100					105					110			
Gly	Gly	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Met	Ala	Ala	Val	Thr	Leu	Asp	
		115					120					125				
Glu	Ser	Gly	Gly	Gly	Leu	Gln	Thr	Pro	Gly	Gly	Ala	Leu	Ser	Leu	Val	
	130					135					140					
Cys	Lys	Ala	Ser	Gly	Phe	Thr	Phe	Asn	Ser	Tyr	Asn	Met	Gly	Trp	Val	
145					150					155					160	
Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Phe	Val	Ala	Gly	Ile	Tyr	Ser	
				165					170					175		
Ala	Gly	Gly	Asp	Thr	Ser	Thr	Thr	Tyr	Gly	Ala	Ala	Val	Asn	Gly	Arg	
			180					185					190			
Ala	Thr	Ile	Ser	Arg	Asp	Asn	Gly	Gln	Ser	Thr	Val	Arg	Leu	Gln	Leu	
	195						200					205				
Asn	Asn	Leu	Arg	Ala	Glu	Asp	Thr	Gly	Ile	Tyr	Phe	Cys	Ala	Lys	Ala	
	210					215					220					
Ala	Gly	Gly	Cys	Thr	Ala	His	Asn	Cys	Asp	Ala	Tyr	Ile	Ala	Asp	Ser	
225					230					235					240	
Ile	Asp	Ala	Trp	Gly	His	Gly	Thr	Glu	Val	Ile	Val	Ser	Ser	Ser	Leu	
				245					250					255		
Asp	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	
			260					265					270			
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	
	275						280					285				
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	
	290					295					300					
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	
305					310					315					320	
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	
				325					330					335		
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	
			340					345					350			
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	
	355						360					365				
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	
	370					375					380					
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	
385					390					395					400	
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	
				405					410					415		
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	
			420					425					430			
Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	
	435						440					445				
Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	
	450					455					460					
Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	
465					470					475					480	
Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Val								
				485												

-continued

```

<211> LENGTH: 480
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 61
Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1      5      10      15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
20     25     30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35     40     45
Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Ser Asp Arg Phe Ser Gly
50     55     60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
65     70     75     80
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Trp Pro Pro
85     90     95
Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser
100    105    110
Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val
115    120    125
Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser Leu
130    135    140
Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr Ala Met
145    150    155    160
Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Arg
165    170    175
Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val Lys Gly
180    185    190
Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr Leu Gln
195    200    205
Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
210    215    220
Gln Tyr Asp Trp Asn Ser Phe Phe Asp Tyr Trp Gly Leu Gly Ala Leu
225    230    235    240
Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr His
245    250    255
Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val
260    265    270
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
275    280    285
Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
290    295    300
Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
305    310    315    320
Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
325    330    335
Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
340    345    350
Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile
355    360    365
Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
370    375    380

```


-continued

```

Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
385                390                395                400

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
                405                410                415

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
                420                425                430

Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
                435                440                445

Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
                450                455                460

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Val
465                470                475                480

```

```

<210> SEQ ID NO 62
<211> LENGTH: 480
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 62

```

```

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1          5          10          15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
20        25        30

Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35        40        45

Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Ser Asp Arg Phe Ser Gly
50        55        60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
65        70        75        80

Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
85        90        95

Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser
100       105       110

Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val
115       120       125

Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser Leu
130       135       140

Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr Ala Met
145       150       155       160

Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Arg
165       170       175

Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val Lys Gly
180       185       190

Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr Leu Gln
195       200       205

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
210       215       220

Gln Tyr Asp Trp Asn Ser Phe Phe Asp Tyr Trp Gly Leu Gly Ala Leu
225       230       235       240

Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr His
245       250       255

Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val
260       265       270

```

-continued

```

Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
  275                               280               285

Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
  290                               295               300

Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
  305                               310               315               320

Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
  325                               330               335

Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
  340                               345               350

Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile
  355                               360               365

Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
  370                               375               380

Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
  385                               390               395               400

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
  405                               410               415

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
  420                               425               430

Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
  435                               440               445

Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
  450                               455               460

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Val
  465                               470               475               480

```

```

<210> SEQ ID NO 63
<211> LENGTH: 480
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 63

```

```

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
  1      5      10      15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
  20     25     30

Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
  35     40     45

Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
  50     55     60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
  65     70     75     80

Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
  85     90     95

Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser
  100    105    110

Arg Ser Ser Gly Gly Gly Gly Ser Ser Arg Gly Gly Gly Ser Asp Val
  115    120    125

Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser Leu
  130    135    140

Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr Ala Met
  145    150    155    160

```

-continued

```

Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Arg
      165                      170                      175
Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val Lys Gly
      180                      185                      190
Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr Leu Gln
      195                      200                      205
Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
      210                      215                      220
Gln Tyr Asp Trp Asn Gly Phe Phe Asp Tyr Trp Gly Leu Gly Ala Leu
      225                      230                      235                      240
Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr His
      245                      250                      255
Thr Cys Pro Pro Cys Pro Val Pro Glu Leu Leu Gly Gly Pro Ser Val
      260                      265                      270
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
      275                      280                      285
Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
      290                      295                      300
Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
      305                      310                      315                      320
Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
      325                      330                      335
Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
      340                      345                      350
Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile
      355                      360                      365
Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
      370                      375                      380
Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
      385                      390                      395                      400
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
      405                      410                      415
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
      420                      425                      430
Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
      435                      440                      445
Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
      450                      455                      460
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Val
      465                      470                      475                      480

```

<210> SEQ ID NO 64

<211> LENGTH: 480

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 64

```

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1          5          10          15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
20          25          30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35          40          45

```

Tyr	Asp	Ala	Ser	Arg	Arg	Ala	Thr	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	
50						55				60						
Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Thr	
65					70					75					80	
Glu	Asp	Ser	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Tyr	Tyr	Trp	Pro	Pro	
				85					90					95		
Tyr	Arg	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gln	Ser	Ser	
			100					105					110			
Arg	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Val	
		115					120					125				
Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Val	Val	Arg	Pro	Gly	Glu	Ser	Leu	
		130				135					140					
Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Ser	Phe	Ser	Ser	Tyr	Ala	Met	
145					150					155					160	
Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Glu	Gly	Leu	Glu	Trp	Val	Ser	Arg	
				165					170					175		
Ile	Asn	Ser	Gly	Gly	Gly	Gly	Thr	Asp	Tyr	Ala	Glu	Ser	Val	Lys	Gly	
			180					185					190			
Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Glu	Asn	Thr	Leu	Tyr	Leu	Gln	
		195					200					205				
Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Lys	
						215					220					
Gln	Tyr	Asp	Trp	Asn	Gly	Phe	Phe	Asp	Tyr	Trp	Gly	Leu	Gly	Ala	Leu	
225					230					235					240	
Val	Thr	Val	Ser	Ser	Ser	Leu	Asp	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	
				245					250					255		
Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	
			260					265					270			
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	
		275					280					285				
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	
		290				295					300					
Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	
305					310					315					320	
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	
				325				330						335		
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	
			340					345				350				
Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	
		355					360					365				
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	
		370				375					380					
Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val						

-continued

His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Val
465					470					475					480

<210> SEQ ID NO 65
 <211> LENGTH: 481
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 65

Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala	Thr	Leu	Ser	Val	Ser	Pro	Gly
1				5					10					15	
Glu	Arg	Ala	Thr	Phe	Ser	Cys	Arg	Ala	Ser	Gln	Asn	Val	Lys	Asn	Asp
	20						25						30		
Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile
	35						40					45			
Tyr	Ala	Ala	Arg	Ile	Arg	Glu	Thr	Gly	Ile	Pro	Glu	Arg	Phe	Ser	Gly
	50					55				60					
Ser	Gly	Ser	Gly	Thr	Glu	Phe	Thr	Leu	Thr	Ile	Thr	Ser	Leu	Gln	Ser
65				70					75					80	
Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Tyr	Asp	Trp	Pro	Pro
		85						90					95		
Phe	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gln	Ser	Ser
	100						105						110		
Arg	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Val
	115						120					125			
Gln	Leu	Leu	Glu	Ser	Gly	Gly	Gly	Val	Val	Gln	Pro	Gly	Gly	Ser	Leu
	130				135					140					
Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Ser	Phe	Ser	Asn	Phe	Ala	Met
145				150					155					160	
Thr	Trp	Val	Arg	Gln	Ala	Pro	Gly	Glu	Gly	Leu	Glu	Trp	Val	Ser	Thr
		165					170						175		
Ile	Gly	Ser	Gly	Asp	Thr	Tyr	Tyr	Ala	Asp	Ser	Val	Lys	Gly	Arg	Phe
	180						185						190		
Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn
	195				200						205				
Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Lys	Asp	Ser
	210				215						220				
Thr	Val	Ser	Trp	Ser	Gly	Asp	Phe	Phe	Asp	Tyr	Trp	Gly	Leu	Gly	Thr
225				230					235					240	
Leu	Val	Thr	Val	Ser	Ser	Ser	Leu	Asp	Pro	Lys	Ser	Ser	Asp	Lys	Pro
		245						250					255		
His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser
	260						265						270		
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
	275					280					285				
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro
	290				295						300				
Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
305				310					315					320	
Lys	Thr	Lys	Pro	Arg	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	
		325					330						335		
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
	340						345						350		

-continued

Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr
 355 360 365
 Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu
 370 375 380
 Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys
 385 390 395 400
 Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser
 405 410 415
 Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp
 420 425 430
 Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser
 435 440 445
 Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala
 450 455 460
 Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 465 470 475 480
 Val

<210> SEQ ID NO 66
 <211> LENGTH: 481
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 66

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp
 20 25 30
 Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Glu Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
 85 90 95
 Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser
 100 105 110
 Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val
 115 120 125
 Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly Ser Leu
 130 135 140
 Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe Ala Met
 145 150 155 160
 Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Thr
 165 170 175
 Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe
 180 185 190
 Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn
 195 200 205
 Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp Ser
 210 215 220
 Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly Thr

-continued

225	230	235	240
Arg Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr	245	250	255
His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser	260	265	270
Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg	275	280	285
Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro	290	295	300
Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala	305	310	315
Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val	325	330	335
Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr	340	345	350
Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr	355	360	365
Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu	370	375	380
Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys	385	390	395
Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser	405	410	415
Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp	420	425	430
Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser	435	440	445
Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala	450	455	460
Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys	465	470	475
Val			480

<210> SEQ ID NO 67

<211> LENGTH: 481

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 67

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly	1	5	10	15
Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp	20	25	30	
Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile	35	40	45	
Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Glu Arg Phe Ser Gly	50	55	60	
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser	65	70	75	80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro	85	90	95	
Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser	100	105	110	

-continued

```

Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val
    115                                120                                125
Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser Leu
    130                                135                                140
Arg Leu Ser Cys Ala Ala Ser Gly Phe Arg Phe Ser Asn Phe Ala Met
    145                                150                                155                                160
Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Thr
    165                                170                                175
Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe
    180                                185                                190
Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn
    195                                200                                205
Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp Ser
    210                                215                                220
Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly Thr
    225                                230                                235                                240
Leu Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr
    245                                250                                255
His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser
    260                                265                                270
Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg
    275                                280                                285
Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro
    290                                295                                300
Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala
    305                                310                                315                                320
Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val
    325                                330                                335
Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr
    340                                345                                350
Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr
    355                                360                                365
Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu
    370                                375                                380
Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys
    385                                390                                395                                400
Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser
    405                                410                                415
Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp
    420                                425                                430
Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser
    435                                440                                445
Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala
    450                                455                                460
Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
    465                                470                                475                                480
Val

```

<210> SEQ ID NO 68

<211> LENGTH: 478

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 68

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Val Ser Pro Gly
 1 5 10 15
 Glu Arg Val Thr Leu Thr Cys Arg Ala Ser Gln Gly Ile Ala Gly Lys
 20 25 30
 Ile Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Asp Ala Ser Ser Arg Ala Thr Gly Ile Pro Gly Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln His Tyr Asp Trp Ser Pro
 85 90 95
 Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser
 100 105 110
 Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val
 115 120 125
 Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly Ser Leu
 130 135 140
 Arg Leu Ser Cys Thr Ala Ser Gly Phe Thr Phe Arg Asn Tyr Gly Met
 145 150 155 160
 Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Ala
 165 170 175
 Ser Ser Gly Ser Gly Ser Thr Tyr Tyr Thr Asp Ser Val Lys Gly Arg
 180 185 190
 Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met
 195 200 205
 Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala Lys Val
 210 215 220
 Thr Trp Asn Asn Phe Phe Asp Tyr Trp Gly Leu Gly Thr Leu Val Thr
 225 230 235 240
 Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr His Thr Cys
 245 250 255
 Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu
 260 265 270
 Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu
 275 280 285
 Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys
 290 295 300
 Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys
 305 310 315 320
 Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu
 325 330 335
 Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys
 340 345 350
 Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys
 355 360 365
 Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser
 370 375 380
 Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys
 385 390 395 400
 Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln

-continued

405	410	415
Pro Glu Asn Asn Tyr Lys Thr Thr	Pro Pro Val Leu Asp Ser Asp Gly	
420	425	430
Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln		
435	440	445
Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn		
450	455	460
His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Val		
465	470	475
 <210> SEQ ID NO 69		
<211> LENGTH: 478		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Synthetic Construct		
 <400> SEQUENCE: 69		
Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly		
1	5	10
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Ser Ser		
20	25	30
Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu		
35	40	45
Ile Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser		
50	55	60
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln		
65	70	75
Thr Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro		
85	90	95
Pro Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser		
100	105	110
Ser Arg Ser Ser Gly Gly Gly Ser Ser Ser Gly Gly Gly Gly Ser Asp		
115	120	125
Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser		
130	135	140
Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr Asp		
145	150	155
Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser		
165	170	175
Gly Ile Ser Gly Asn Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val Lys		
180	185	190
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu		
195	200	205
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
210	215	220
Met Asn Arg Trp Trp Phe Asp Tyr Trp Gly Leu Gly Thr Leu Val Thr		
225	230	235
Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr His Thr Cys		
245	250	255
Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu		
260	265	270
Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu		
275	280	285
Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys		

-continued

290	295	300
Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys 305 310 315 320		
Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu 325 330 335		
Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys 340 345 350		
Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys 355 360 365		
Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser 370 375 380		
Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys 385 390 395 400		
Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln 405 410 415		
Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly 420 425 430		
Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln 435 440 445		
Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn 450 455 460		
His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Val 465 470 475		

<210> SEQ ID NO 70

<211> LENGTH: 480

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 70

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly 1 5 10 15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys 20 25 30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile 35 40 45
Tyr Asp Ala Ser Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly 50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr 65 70 75 80
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro 85 90 95
Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser 100 105 110
Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val 115 120 125
Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser Leu 130 135 140
Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Arg Ser Tyr Ala Met 145 150 155 160
Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Arg 165 170 175
Ile Asp Ser Gly Gly Gly Gly Thr Asp Tyr Ala Asp Ser Val Lys Gly

-continued

180					185					190					
Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln
	195					200						205			
Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Lys
	210					215					220				
Gln	Tyr	Asp	Trp	Asn	Ser	Phe	Phe	Asp	Tyr	Trp	Gly	Leu	Gly	Ala	Pro
	225					230					235				240
Val	Thr	Val	Ser	Ser	Ser	Leu	Asp	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His
			245						250					255	
Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val
			260					265					270		
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
		275					280					285			
Pro	Glu	Val	Thr	Cys	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	
	290					295					300				
Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
	305					310					315				320
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser
			325					330						335	
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
			340					345					350		
Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile
		355					360					365			
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
	370					375					380				
Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu
	385					390					395				400
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
			405					410						415	
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser
			420					425					430		
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg
		435					440					445			
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
	450					455					460				
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Val
	465					470					475				480

<210> SEQ ID NO 71

<211> LENGTH: 483

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 71

Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Val	Ser	Pro	Gly
1				5					10					15	
Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Asn	Val	Arg	Ser	Asp
		20					25						30		
Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Leu	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile
		35				40						45			
Tyr	Asp	Ala	Asn	Thr	Arg	Ala	Thr	Asp	Ile	Pro	Asp	Arg	Phe	Ser	Gly
	50				55				60						
Ser	Gly	Ser	Gly	Thr	Glu	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Ser

[illegible]

-continued

```

<210> SEQ ID NO 72
<211> LENGTH: 481
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 72

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1             5             10            15

Glu Arg Val Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Arg Ser Asp
20            25            30

Ile Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35            40            45

Tyr Ala Ala Ser Ser Arg Asp Thr Gly Ile Pro Asp Arg Phe Ser Gly
50            55            60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65            70            75            80

Glu Asp Phe Gly Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
85            90            95

Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser
100           105           110

Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val
115           120           125

Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser Leu
130           135           140

Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr Ala Met
145           150           155           160

Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Thr
165           170           175

Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe
180           185           190

Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn
195           200           205

Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp Ser
210           215           220

Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly Thr
225           230           235           240

Leu Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr
245           250           255

His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser
260           265           270

Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg
275           280           285

Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro
290           295           300

Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala
305           310           315           320

Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val
325           330           335

Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr
340           345           350

Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr
355           360           365

```

-continued

```

Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu
 370                               375                               380

Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys
385                               390                               395                               400

Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser
                               405                               410                               415

Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp
                               420                               425                               430

Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser
                               435                               440                               445

Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala
                               450                               455                               460

Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
465                               470                               475                               480

Val

```

```

<210> SEQ ID NO 73
<211> LENGTH: 480
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 73

```

```

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly
1                               5                               10                               15

Glu Arg Ala Thr Leu Thr Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
20                               25                               30

Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35                               40                               45

Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
50                               55                               60

Ser Gly Ser Gly Thr Val Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65                               70                               75                               80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
85                               90                               95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser
100                              105                              110

Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Gly Ser Asp Val
115                              120                              125

Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser Leu
130                              135                              140

Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr Asp Met
145                              150                              155                              160

Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Leu
165                              170                              175

Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val Lys Gly
180                              185                              190

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
195                              200                              205

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
210                              215                              220

Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Leu Gly Thr Leu
225                              230                              235                              240

Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr His

```

-continued

245					250					255					
Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val
			260					265					270		
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
		275					280					285			
Pro	Glu	Val	Thr	Cys	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	
		290					295					300			
Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
305					310					315					320
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser
				325					330						335
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
			340					345					350		
Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile
		355					360						365		
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
		370					375					380			
Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu
385							390					395			400
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
				405					410						415
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser
			420					425					430		
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg
		435						440					445		
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
		450					455						460		
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Val
465						470					475				480

<210> SEQ ID NO 74

<211> LENGTH: 315

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 74

Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Val	Ser	Pro	Gly
1				5					10					15	
Glu	Arg	Val	Ile	Leu	Thr	Cys	Arg	Ala	Ser	Gln	Ser	Val	Asp	Thr	Tyr
			20					25					30		
Asn	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu
		35					40					45			
Ile	Tyr	Asp	Leu	Ser	Thr	Arg	Ala	Thr	Gly	Ile	Pro	Asp	Arg	Phe	Ser
	50					55					60				
Gly	Ser	Gly	Ser	Gly	Thr	Glu	Phe	Thr	Leu	Thr	Ile	Asn	Ser	Leu	Glu
65					70				75					80	
Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	His	Gln	Tyr	Tyr	Asp	Trp	Pro
			85					90						95	
Pro	Tyr	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gln	Ser
			100				105						110		
Ser	Arg	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Asp
		115					120					125			
Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Val	Val	Arg	Pro	Gly	Glu	Ser

-continued

130	135	140
Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr Ala		
145	150	155 160
Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser		
	165	170 175
Gly Ile Ser Gly Arg Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys		
	180	185 190
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu		
	195	200 205
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala		
	210	215 220
Lys Gly Thr Trp Asn Tyr Gly Ser Phe Asp Tyr Trp Gly Leu Gly Thr		
	225	230 235 240
Leu Val Thr Val Ser Ser Ser Leu Asp Pro Lys Ser Ser Asp Lys Thr		
	245	250 255
Asp Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser		
	260	265 270
Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg		
	275	280 285
Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro		
	290	295 300
Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val		
305	310	315

<210> SEQ ID NO 75

<211> LENGTH: 481

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 75

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly		
1	5	10 15
Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp		
	20	25 30
Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile		
	35	40 45
Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Glu Arg Phe Ser Gly		
	50	55 60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser		
	65	70 75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro		
	85	90 95
Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gln Ser Ser		
	100	105 110
Arg Ser Ser Gly Gly Gly Gly Ser Ser Gly Gly Gly Ser Asp Val		
	115	120 125
Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly Ser Leu		
	130	135 140
Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe Ala Met		
	145	150 155 160
Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser Thr		
	165	170 175
Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe		

-continued

180					185					190					
Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn
	195						200					205			
Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Lys	Asp	Ser
	210					215					220				
Thr	Val	Ser	Trp	Ser	Gly	Asp	Phe	Phe	Asp	Tyr	Trp	Gly	Leu	Gly	Thr
	225				230					235					240
Leu	Val	Thr	Val	Ser	Ser	Ser	Leu	Asp	Pro	Lys	Ser	Ser	Asp	Lys	Thr
				245					250					255	
His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser
			260					265					270		
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
		275						280				285			
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro
	290					295					300				
Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
	305				310					315					320
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val
			325						330					335	
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
		340						345					350		
Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr
		355					360					365			
Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu
	370					375					380				
Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys
	385				390					395					400
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser
			405						410					415	
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp
		420						425					430		
Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser
		435						440				445			
Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala
	450					455					460				
Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys
	465				470				475					480	
Val															

<210> SEQ ID NO 76

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 76

Glu	Glu	Glu	Leu	Gln	Val	Ile	Gln	Pro	Asp	Lys	Ser	Val	Ser	Val	Ala
1			5					10						15	
Ala	Gly	Glu	Ser	Ala	Ile	Leu	Leu	Cys	Thr	Val	Thr	Ser	Leu	Ile	Pro
		20						25					30		
Val	Gly	Pro	Ile	Gln	Trp	Phe	Arg	Gly	Ala	Gly	Pro	Ala	Arg	Glu	Leu
		35					40					45			
Ile	Tyr	Asn	Gln	Lys	Glu	Gly	His	Phe	Pro	Arg	Val	Thr	Thr	Val	Ser
	50					55					60				
Glu	Ser	Thr	Lys	Arg	Glu	Asn	Met	Asp	Phe	Ser	Ile	Ser	Ile	Ser	Asn

-continued

65	70	75	80
Ile Thr Pro Ala Asp	Ala Gly Thr Tyr	Tyr Cys Val Lys Phe Arg	Lys
	85	90	95
Gly Ser Pro Asp Thr	Glu Phe Lys Ser	Gly Ala Gly Thr	Glu Leu Ser
	100	105	110
Val Arg Ala Lys Pro Ser			
	115		

<210> SEQ ID NO 77
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 77

Glu Glu Gly Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala	
1	15
Ala Gly Glu Ser Ala Ile Leu His Cys Thr Ala Thr Ser Leu Ile Pro	
20	30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu	
35	45
Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser	
50	60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn	
65	80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys	
85	95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu	
100	110
Ser Val Arg Ala Lys Pro Ser	
115	

<210> SEQ ID NO 78
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 78

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Phe Val Leu Val Ala	
1	15
Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro	
20	30
Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu	
35	45
Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser	
50	60
Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Asn	
65	80
Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys	
85	95
Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu	
100	110
Ser Val Arg Ala Lys Pro Ser	
115	

<210> SEQ ID NO 79
 <211> LENGTH: 119
 <212> TYPE: PRT

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 79

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Pro Ile Arg Ile Gly Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
 100 105 110
 Ser Val Arg Ala Lys Pro Ser
 115

<210> SEQ ID NO 80

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 80

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
 1 5 10 15
 Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Ile Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
 50 55 60
 Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
 65 70 75 80
 Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
 85 90 95
 Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
 100 105 110
 Val Arg Gly Lys Pro Ser
 115

<210> SEQ ID NO 81

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 81

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
 1 5 10 15
 Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro
 20 25 30
 Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
 35 40 45
 Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
 50 55 60

-continued

Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110

Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 82
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 82

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15

Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 83
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 83

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Ser Val Ala
1 5 10 15

Ala Gly Glu Ser Ala Ile Leu His Cys Thr Val Thr Ser Leu Ile Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Ala Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Glu Ser Thr Lys Arg Glu Asn Met Asp Phe Ser Ile Ser Ile Ser Asn
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Thr Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu Ser
100 105 110

Val Arg Ala Lys Pro Ser
115

<210> SEQ ID NO 84

-continued

```

<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Gallus gallus

<400> SEQUENCE: 84

Asp Phe Lys Leu Gln Gln Pro Gln Ser Ser Val Val Val Ile Lys Gly
1           5           10           15

Asp Thr Leu Thr Leu Asn Cys Thr Ala Ser Gly Ser Gly Pro Ile Gly
          20           25           30

Ala Val Lys Trp Val Lys Gly Trp Gly Ser Asp Asn Gln Thr Val Tyr
          35           40           45

Glu His Lys Gly Ser Phe Pro Arg Val Met Arg Ala Val Pro Asp Pro
          50           55           60

Thr Asn Asp Phe Thr Ile Arg Ile Ser Asn Val Ser Leu Glu Asp Ala
65           70           75           80

Gly Thr Tyr Tyr Cys Val Lys Leu Arg Lys Gly Ile Val Asp Asp Val
          85           90           95

Val Phe Thr Arg Gly Gly Gly Thr Glu Val Ser Val His Ala
          100          105          110

```

```

<210> SEQ ID NO 85
<211> LENGTH: 105
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 85

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15

Glu Ile Thr Cys Ser Gly Gly Gly Ser Asn Asn Ala Tyr Gly Trp Phe
          20           25           30

Gln Gln Lys Ser Pro Gly Ser Ala Pro Leu Thr Val Ile Tyr Asp Asn
          35           40           45

Gly Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser
          50           55           60

Asp Ser Thr Gly Thr Leu Thr Ile Thr Arg Val Gln Ala Glu Asp Glu
65           70           75           80

Ala Val Tyr Tyr Cys Gly Ser Ala Asp Asn Ser Gly Ala Gly Val Phe
          85           90           95

Gly Ala Gly Thr Thr Leu Thr Val Leu
          100          105

```

```

<210> SEQ ID NO 86
<211> LENGTH: 105
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 86

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15

Arg Ile Thr Cys Ser Gly Asp Ser Ser Ser Tyr Tyr Ser Trp His Gln
          20           25           30

Gln Lys Ser Pro Gly Ser Ala Pro Val Ser Val Ile Tyr Ser Asn Thr
          35           40           45

Asp Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala Ser Gly
          50           55           60

```

-continued

Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Gln Ala Glu Asp Glu Ala
65 70 75 80

Val Tyr Phe Cys Gly Ala Tyr Asp Ser Ser Ser Asp Ser Asp Ile Phe
85 90 95

Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 87
<211> LENGTH: 103
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 87

Ala Val Thr Gln Pro Ser Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1 5 10 15

Glu Ile Thr Cys Ser Gly Ser Ser Thr Tyr Tyr Gly Trp Tyr Gln Gln
20 25 30

Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Asp Asn Asp Lys
35 40 45

Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser
50 55 60

Thr His Thr Leu Ile Ile Thr Gly Val Gln Val Glu Asp Glu Ala Val
65 70 75 80

Tyr Phe Cys Gly Asn Glu Asp Asn Asn Tyr Val Ala Ile Phe Gly Ala
85 90 95

Gly Thr Thr Leu Thr Val Leu
100

<210> SEQ ID NO 88
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 88

Ala Leu Thr Gln Pro Ser Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1 5 10 15

Lys Ile Thr Cys Ser Gly Asp Asn Ser Ala His Tyr Tyr Tyr Gly Trp
20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Tyr
35 40 45

Asn Asp Lys Arg Pro Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser Ala
50 55 60

Ser Gly Ser Thr Ala Thr Leu Ile Ile Thr Gly Val Gln Val Glu Asp
65 70 75 80

Glu Ala Val Tyr Phe Cys Gly Ser Ala Asp Ser Ser Asn Pro Ala Ile
85 90 95

Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 89
<211> LENGTH: 105
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 89

```

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15

Lys Ile Thr Cys Ser Gly Ser Ser Ser Gly Ser Tyr Gly Trp Tyr Gln
          20           25           30

Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Glu Thr Asn
          35           40           45

Lys Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly
          50           55           60

Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ala
65           70           75           80

Val Tyr Tyr Cys Gly Ser Glu Asp Ser Ser Thr Tyr Leu Ser Ile Phe
          85           90           95

Gly Ala Gly Thr Thr Leu Thr Val Leu
          100           105

```

<210> SEQ ID NO 90

<211> LENGTH: 103

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 90

```

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15

Lys Ile Thr Cys Ser Gly Asp Ser Ser Tyr Tyr Gly Trp Tyr Gln Gln
          20           25           30

Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Asp Asp Asn Lys
          35           40           45

Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser
          50           55           60

Thr Gly Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ala Val
65           70           75           80

Tyr Phe Cys Gly Asn Glu Asp Asn Ser Tyr Val Ala Ile Phe Gly Ala
          85           90           95

Gly Thr Thr Leu Thr Val Leu
          100

```

<210> SEQ ID NO 91

<211> LENGTH: 103

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 91

```

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15

Lys Ile Thr Cys Ser Gly Ser Ser Ser Tyr Tyr Gly Trp Tyr Arg Gln
          20           25           30

Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Asp Asn Asp Lys
          35           40           45

Arg Pro Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser
          50           55           60

Thr Asn Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ala Val
65           70           75           80

```


-continued

Tyr Tyr Cys Gly Asn Glu Asp Asn Ser Tyr Val Gly Ile Phe Gly Ala
 85 90 95

Gly Thr Thr Leu Thr Val Leu
 100

<210> SEQ ID NO 92
 <211> LENGTH: 103
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 92

Ala Val Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Glu Thr Val
 1 5 10 15

Lys Ile Thr Cys Ser Gly Asp Ser Ser Tyr Tyr Gly Trp Tyr Gln Gln
 20 25 30

Lys Ala Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Asp Asn Asp Lys
 35 40 45

Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser
 50 55 60

Thr Ala Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ala Val
 65 70 75 80

Tyr Tyr Cys Gly Asn Glu Asp Met Asn Tyr Val Gly Ile Phe Gly Ala
 85 90 95

Gly Thr Thr Leu Thr Val Leu
 100

<210> SEQ ID NO 93
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 93

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
 20 25 30

Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Ser Asp Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
 65 70 75 80

Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
 85 90 95

Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 94
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 94

-continued

```

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1           5           10          15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
           20           25           30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
           35           40           45
Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Ser Asp Arg Phe Ser Gly
           50           55           60
Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
65           70           75           80
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
           85           90           95
Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
           100          105

```

```

<210> SEQ ID NO 95
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 95

```

```

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1           5           10          15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
           20           25           30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
           35           40           45
Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
           50           55           60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
65           70           75           80
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
           85           90           95
Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
           100          105

```

```

<210> SEQ ID NO 96
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 96

```

```

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1           5           10          15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
           20           25           30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
           35           40           45
Tyr Asp Ala Ser Arg Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
           50           55           60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
65           70           75           80
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
           85           90           95

```

-continued

Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 97
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 97

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp
 20 25 30
 Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Glu Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
 85 90 95
 Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 98
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 98

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp
 20 25 30
 Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Glu Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
 85 90 95
 Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 99
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 99

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15

-continued

Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp
 20 25 30

Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Glu Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
 85 90 95

Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 100
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 100

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Val Ser Pro Gly
 1 5 10 15

Glu Arg Val Thr Leu Thr Cys Arg Ala Ser Gln Gly Ile Ala Gly Lys
 20 25 30

Ile Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ala Ser Ser Arg Ala Thr Gly Ile Pro Gly Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln His Tyr Asp Trp Ser Pro
 85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 101
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 101

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Ser Ser
 20 25 30

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45

Ile Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln
 65 70 75 80

Thr Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro
 85 90 95

Pro Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

-continued

<210> SEQ ID NO 102
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 102

```

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
           20           25           30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
           35           40           45
Tyr Asp Ala Ser Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
           50           55           60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr
65           70           75           80
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
           85           90           95
Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
           100           105

```

<210> SEQ ID NO 103
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 103

```

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Val Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Val Arg Ser Asp
           20           25           30
Leu Ala Trp Tyr Gln Gln Lys Leu Gly Gln Ala Pro Arg Leu Leu Ile
           35           40           45
Tyr Asp Ala Asn Thr Arg Ala Thr Asp Ile Pro Asp Arg Phe Ser Gly
           50           55           60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65           70           75           80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln His Tyr Tyr Asp Trp Pro Pro
           85           90           95
Val Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
           100           105

```

<210> SEQ ID NO 104
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 104

```

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1           5           10           15
Glu Arg Val Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Arg Ser Asp
           20           25           30

```

-continued

```

Ile Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
   35                               40               45

Tyr Ala Ala Ser Ser Arg Asp Thr Gly Ile Pro Asp Arg Phe Ser Gly
   50                               55               60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
   65                               70               75               80

Glu Asp Phe Gly Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
           85                               90               95

Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
   100                               105

```

```

<210> SEQ ID NO 105
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 105

```

```

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly
 1           5               10               15

Glu Arg Ala Thr Leu Thr Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
           20               25               30

Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
   35                               40               45

Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
   50                               55               60

Ser Gly Ser Gly Thr Val Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
   65                               70               75               80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
           85                               90               95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
   100                               105

```

```

<210> SEQ ID NO 106
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 106

```

```

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Val Ser Pro Gly
 1           5               10               15

Glu Arg Val Ile Leu Thr Cys Arg Ala Ser Gln Ser Val Asp Thr Tyr
   20               25               30

Asn Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
   35                               40               45

Ile Tyr Asp Leu Ser Thr Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
   50                               55               60

Gly Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Asn Ser Leu Glu
   65                               70               75               80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys His Gln Tyr Tyr Asp Trp Pro
           85                               90               95

Pro Tyr Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
   100                               105

```

```

<210> SEQ ID NO 107

```

-continued

```

<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 107
Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1         5             10             15
Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp
20        25        30
Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
35        40        45
Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Glu Arg Phe Ser Gly
50        55        60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Ser Leu Gln Ser
65        70        75        80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
85        90        95
Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100       105

```

```

<210> SEQ ID NO 108
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 108
Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
1         5             10             15
Ala Leu Ser Leu Val Cys Lys Gly Ser Gly Phe Thr Phe Ser Ser His
20        25        30
Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35        40        45
Ala Gly Ile Ser Ser Asp Gly Arg Phe Thr Tyr Tyr Gly Ala Ala Val
50        55        60
Gln Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr Val Arg
65        70        75        80
Leu Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys
85        90        95
Thr Lys Asn Gly Gly Cys Gly Ser Gly Gly Asp Leu Asp Cys Ile Asp
100       105       110
Ala Trp Gly His Gly Thr Glu Val Ile Val Ser Ser
115       120

```

```

<210> SEQ ID NO 109
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 109
Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
1         5             10             15
Gly Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Asp Phe Ser Asn Phe
20        25        30

```

-continued

```

Asn Met Ala Trp Val Arg Gln Gly Pro Gly Lys Gly Leu Glu Tyr Val
    35          40          45

Ala Glu Ile Ser Asp Thr Gly Ser Thr Pro Tyr Tyr Gly Ser Ala Val
    50          55          60

Gln Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr Val Arg
    65          70          75          80

Leu Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Thr Tyr Phe Cys
    85          90          95

Thr Arg Asn Phe Gly Ser Ser Val Ser Ser Ile Asp Ala Trp Gly His
    100         105         110

Gly Thr Glu Val Ile Val Ser Ser
    115         120

```

```

<210> SEQ ID NO 110
<211> LENGTH: 133
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 110

```

```

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
 1          5          10          15

Ala Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20         25         30

Asn Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Val
 35          40          45

Ala Gly Ile Tyr Ala Ser Gly Ser Ser Thr Asp Thr Asp Thr Thr Tyr
 50          55          60

Gly Pro Ala Val Ala Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln
 65          70          75          80

Ser Thr Val Arg Leu Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly
 85          90          95

Thr Tyr Tyr Cys Ala Lys Ala Ala Gly Gly Cys Ser Thr His Thr Cys
 100         105         110

Thr Ala Tyr Ile Ala Asp Ser Ile Asp Ala Trp Gly His Gly Thr Glu
 115         120         125

Val Ile Val Ser Ser
 130

```

```

<210> SEQ ID NO 111
<211> LENGTH: 128
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 111

```

```

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Arg
 1          5          10          15

Ala Leu Ser Leu Val Cys Arg Gly Ser Gly Phe Ser Ile Ser Ser Tyr
 20         25         30

Asn Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Ile
 35          40          45

Ala Ser Ile Gly Ser Asp Gly Ser Ser Thr His Tyr Ala Pro Ala Val
 50          55          60

Lys Gly Arg Ala Thr Ile Thr Arg Asp Val Gly Gln Ser Thr Val Arg
 65          70          75          80

```


-continued

Leu Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Thr Tyr Phe Cys
 85 90 95

Ala Lys Asp Ala Tyr Gln Cys Ser Tyr Ala Thr Cys Asn Asp Tyr Leu
 100 105 110

Asp Thr Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile Val Ser Ser
 115 120 125

<210> SEQ ID NO 112
 <211> LENGTH: 127
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 112

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
 1 5 10 15

Ala Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Asn Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Val
 35 40 45

Ala Ala Ile Tyr Ser Gly Asn Ser Ala Glu Tyr Gly Ala Ala Val Gln
 50 55 60

Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr Val Arg Leu
 65 70 75 80

Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Ile Tyr Phe Cys Ala
 85 90 95

Lys Asp Ala Gly Ser Gly Cys Tyr Ser Gly Val Cys Ala Gly Thr Ser
 100 105 110

Ser Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile Val Ser Ser
 115 120 125

<210> SEQ ID NO 113
 <211> LENGTH: 131
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 113

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
 1 5 10 15

Ala Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Asn Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Val
 35 40 45

Ala Gly Ile Tyr Ile Ala Ser Gly Asp Leu Gly Thr Thr Tyr Gly Ala
 50 55 60

Ala Val Gln Gly Arg Ala Thr Ile Ser Arg Asp Asp Gly Gln Ser Thr
 65 70 75 80

Val Arg Leu Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Thr Tyr
 85 90 95

Phe Cys Ala Lys Ser Ala Gly Gly Cys Ser Ala His Ser Cys Asp Thr
 100 105 110

Tyr Ile Ala Asp Ser Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile
 115 120 125

Val Ser Ser

-continued

130

<210> SEQ ID NO 114
 <211> LENGTH: 132
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 114

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
 1 5 10 15
 Ala Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Asn Met Gly Trp Val Arg Gln Ala Pro Asp Lys Gly Leu Glu Phe Val
 35 40 45
 Ala Gly Ile Tyr Thr Gly Ser Asp Ala Gly Leu Ser Thr Thr Tyr Gly
 50 55 60
 Ala Ala Val Gln Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser
 65 70 75 80
 Thr Val Arg Leu Gln Leu Asn Asn Leu Gly Ala Glu Asp Thr Gly Ile
 85 90 95
 Tyr Phe Cys Thr Lys Ser Ala Gly Gly Cys Ser Asp Tyr Asn Cys Asp
 100 105 110
 Ala Tyr Ile Ala Asp Ser Ile Asp Ala Trp Gly His Gly Thr Glu Val
 115 120 125
 Ile Val Ser Ser
 130

<210> SEQ ID NO 115
 <211> LENGTH: 131
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 115

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
 1 5 10 15
 Ala Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Thr Phe Asn Ser Tyr
 20 25 30
 Asn Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Val
 35 40 45
 Ala Gly Ile Tyr Ser Ala Gly Gly Asp Thr Ser Thr Thr Tyr Gly Ala
 50 55 60
 Ala Val Asn Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr
 65 70 75 80
 Val Arg Leu Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Ile Tyr
 85 90 95
 Phe Cys Ala Lys Ala Ala Gly Gly Cys Thr Ala His Asn Cys Asp Ala
 100 105 110
 Tyr Ile Ala Asp Ser Ile Asp Ala Trp Gly His Gly Thr Glu Val Ile
 115 120 125
 Val Ser Ser
 130

<210> SEQ ID NO 116
 <211> LENGTH: 118

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 116

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
1 5 10 15

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr Ala
20 25 30

Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
35 40 45

Arg Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Lys Gln Tyr Asp Trp Asn Ser Phe Phe Asp Tyr Trp Gly Leu Gly Ala
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 117
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 117

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
1 5 10 15

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr Ala
20 25 30

Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
35 40 45

Arg Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Lys Gln Tyr Asp Trp Asn Ser Phe Phe Asp Tyr Trp Gly Leu Gly Ala
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 118
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 118

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
1 5 10 15

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr Ala

-continued

20	25	30
Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser		
35	40	45
Arg Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val Lys		
50	55	60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr Leu		
65	70	75
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
85	90	95
Lys Gln Tyr Asp Trp Asn Gly Phe Phe Asp Tyr Trp Gly Leu Gly Ala		
100	105	110
Leu Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 119
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 119

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser		
1	5	10
Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr Ala		
20	25	30
Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser		
35	40	45
Arg Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val Lys		
50	55	60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr Leu		
65	70	75
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
85	90	95
Lys Gln Tyr Asp Trp Asn Gly Phe Phe Asp Tyr Trp Gly Leu Gly Ala		
100	105	110
Leu Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 120
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 120

Val Gln Leu Leu Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly Ser		
1	5	10
Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe Ala		
20	25	30
Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser		
35	40	45
Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg		
50	55	60
Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met		
65	70	75
		80

-continued

Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp
 85 90 95

Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 121
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 121

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly Ser
 1 5 10 15

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe Ala
 20 25 30

Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
 35 40 45

Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg
 50 55 60

Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met
 65 70 75 80

Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp
 85 90 95

Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110

Thr Arg Val Thr Val Ser Ser
 115

<210> SEQ ID NO 122
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 122

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
 1 5 10 15

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Arg Phe Ser Asn Phe Ala
 20 25 30

Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
 35 40 45

Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg
 50 55 60

Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met
 65 70 75 80

Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp
 85 90 95

Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 123

-continued

<211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 123

```

Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly Ser
1           5           10           15
Leu Arg Leu Ser Cys Thr Ala Ser Gly Phe Thr Phe Arg Asn Tyr Gly
20          25          30
Met Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
35          40          45
Ala Ser Ser Gly Ser Gly Ser Thr Tyr Tyr Thr Asp Ser Val Lys Gly
50          55          60
Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
65          70          75          80
Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala Lys
85          90          95
Val Thr Trp Asn Asn Phe Phe Asp Tyr Trp Gly Leu Gly Thr Leu Val
100         105         110
Thr Val Ser Ser
115

```

<210> SEQ ID NO 124
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 124

```

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
1           5           10           15
Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr Asp
20          25          30
Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
35          40          45
Gly Ile Ser Gly Asn Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Met Asn Arg Trp Trp Phe Asp Tyr Trp Gly Leu Gly Thr Leu Val Thr
100         105         110
Val Ser Ser
115

```

<210> SEQ ID NO 125
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 125

```

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
1           5           10           15

```

-continued

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Arg Ser Tyr Ala
 20 25 30
 Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
 35 40 45
 Arg Ile Asp Ser Gly Gly Gly Gly Thr Asp Tyr Ala Asp Ser Val Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80
 Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95
 Lys Gln Tyr Asp Trp Asn Ser Phe Phe Asp Tyr Trp Gly Leu Gly Ala
 100 105 110
 Pro Val Thr Val Ser Ser
 115

<210> SEQ ID NO 126
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 126

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
 1 5 10 15
 Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr Ala
 20 25 30
 Met Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
 35 40 45
 Leu Ile Thr Thr Asn Gly Asp Gly Ala Tyr Tyr Ala Asp Ser Val Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80
 Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
 85 90 95
 Lys Asp Gly Ala Ala His Tyr Tyr Asp Ile Phe Phe Asp Tyr Trp Gly
 100 105 110
 Leu Gly Thr Pro Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 127
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 127

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
 1 5 10 15
 Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr Ala
 20 25 30
 Met Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
 35 40 45
 Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg
 50 55 60
 Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met
 65 70 75 80

-continued

Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp
85 90 95

Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 128

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 128

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
1 5 10 15

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr Asp
20 25 30

Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
35 40 45

Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Leu Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 129

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 129

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu Ser
1 5 10 15

Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr Ala
20 25 30

Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
35 40 45

Gly Ile Ser Gly Arg Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
85 90 95

Lys Gly Thr Trp Asn Tyr Gly Ser Phe Asp Tyr Trp Gly Leu Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

-continued

```

<210> SEQ ID NO 130
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 130

```

```

Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly Ser
1          5          10          15
Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe Ala
20        25        30
Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val Ser
35        40        45
Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg
50        55        60
Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met
65        70        75        80
Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Asp
85        90        95
Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu Gly
100       105       110
Thr Leu Val Thr Val Ser Ser
115

```

```

<210> SEQ ID NO 131
<211> LENGTH: 102
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 131

```

```

Ala Pro Val Val Ser Gly Pro Ala Ala Arg Ala Thr Pro Gln His Thr
1          5          10          15
Val Ser Phe Thr Cys Glu Ser His Gly Phe Ser Pro Arg Asp Ile Thr
20        25        30
Leu Lys Trp Phe Lys Asn Gly Asn Glu Leu Ser Asp Phe Gln Thr Asn
35        40        45
Val Asp Pro Val Gly Glu Ser Val Ser Tyr Ser Ile His Ser Thr Ala
50        55        60
Lys Val Val Leu Thr Arg Glu Asp Val His Ser Gln Val Ile Cys Glu
65        70        75        80
Val Ala His Val Thr Leu Gln Gly Asp Pro Leu Arg Gly Thr Ala Asn
85        90        95
Leu Ser Glu Thr Ile Arg
100

```

```

<210> SEQ ID NO 132
<211> LENGTH: 99
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 132

```

```

Val Pro Pro Thr Leu Glu Val Thr Gln Gln Pro Val Arg Ala Glu Asn
1          5          10          15
Gln Val Asn Val Thr Cys Gln Val Arg Lys Phe Tyr Pro Gln Arg Leu
20        25        30
Gln Leu Thr Trp Leu Glu Asn Gly Asn Val Ser Arg Thr Glu Thr Ala
35        40        45

```

-continued

Ser Thr Val Thr Glu Asn Lys Asp Gly Thr Tyr Asn Trp Met Ser Trp
50 55 60

Leu Leu Val Asn Val Ser Ala His Arg Asp Asp Val Lys Leu Thr Cys
65 70 75 80

Gln Val Glu His Asp Gly Gln Pro Ala Val Ser Lys Ser His Asp Leu
85 90 95

Lys Val Ser

<210> SEQ ID NO 133
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 133

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30

Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35 40 45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Leu Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 134
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 134

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Thr Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
20 25 30

Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 135
<211> LENGTH: 118

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 135

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
          20           25           30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
          35           40           45
Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Pro Ala Ser Val
          50           55           60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
          65           70           75           80
Leu Gln Met Asn Thr Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          85           90           95
Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
          100          105          110
Leu Val Thr Val Ser Ser
          115

```

```

<210> SEQ ID NO 136
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 136

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15
Lys Ile Ala Cys Ser Gly Gly Asp Tyr Tyr Ser Tyr Tyr Tyr Gly Trp
          20           25           30
Tyr Gln Gln Lys Ala Pro Gly Ser Ala Leu Val Thr Val Ile Tyr Ser
          35           40           45
Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
          50           55           60
Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Ala Glu Asp
          65           70           75           80
Glu Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Tyr Ser Thr Tyr Ala Asn
          85           90           95
Ala Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
          100          105

```

```

<210> SEQ ID NO 137
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 137

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Glu Ala Ser Gly Phe Thr Phe Ser Ser Asn
          20           25           30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val

```

-continued

35	40	45
Ala Gly Ile Ser Ser Gly Ser Asp Thr Tyr Tyr Gly Asp Ser Val Lys		
50	55	60
Gly Arg Leu Thr Ile Ser Arg Asp Asn Ser Lys Asn Ile Leu Tyr Leu		
65	70	75 80
Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
	85	90 95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu		
	100	105 110
Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 138
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 138

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1 5 10 15
Glu Ile Thr Cys Ser Gly Gly Ser Tyr Ser Ser Tyr Tyr Tyr Ala Trp
20 25 30
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Ser
35 40 45
Asp Asp Lys Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Ala
50 55 60
Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Ala Glu Asp
65 70 75 80
Glu Ala Val Tyr Phe Cys Gly Gly Tyr Asp Gln Ser Ser Tyr Thr Asn
85 90 95
Pro Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 139
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 139

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Arg Phe Ser Ser Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Val Asn Ser Leu Thr Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
85 90 95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
100 105 110

-continued

Val Thr Val Ser Ser
115

<210> SEQ ID NO 140
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 140

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asp Leu Gly Glu Thr Val
 1 5 10 15
 Lys Ile Thr Cys Ser Gly Gly Asp Ser Ser Ser His Tyr Tyr Gly Trp
 20 25 30
 Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Ser
 35 40 45
 Asp Asp Glu Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60
 Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
 65 70 75 80
 Glu Ala Ile Tyr Tyr Cys Gly Ala Tyr Asp Gly Ser Thr Tyr Ala Asn
 85 90 95
 Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 141
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 141

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45
 Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Pro Lys Asn Ser Leu Tyr
 65 70 75 80
 Leu Gln Met Ser Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 142
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 142

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val

-continued

1	5	10	15
Lys Ile Thr Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr Ala Trp			
	20	25	30
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser			
	35	40	45
Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala			
	50	55	60
Ser Gly Ser Ala Ala Thr Leu Ile Ile Thr Gly Val Arg Val Glu Asp			
	65	70	75
Glu Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Gly Arg Thr Tyr Ile Asn			
	85	90	95
Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu			
	100	105	

<210> SEQ ID NO 143
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 143

Gly Phe Ser Phe Ser Asn Phe Ala Met Thr
1 5 10

<210> SEQ ID NO 144
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 144

Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 145
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 145

Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr
1 5 10

<210> SEQ ID NO 146
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 146

Arg Ala Ser Gln Asn Val Lys Asn Asp Leu Ala
1 5 10

<210> SEQ ID NO 147
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 147

Ala Ala Arg Ile Arg Glu Thr
1 5

<210> SEQ ID NO 148

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 148

Gln Gln Tyr Tyr Asp Trp Pro Pro Phe Thr
1 5 10

<210> SEQ ID NO 149

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 149

Gly Phe Ser Phe Ser Ile Tyr Ala Met Ser
1 5 10

<210> SEQ ID NO 150

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 150

Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 151

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 151

Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr
1 5 10

<210> SEQ ID NO 152

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 152

Arg Ala Ser Gln Asn Val Arg Ser Asp Ile Ala
1 5 10

<210> SEQ ID NO 153

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 153

Ala Ala Ser Ser Arg Asp Thr
1 5

<210> SEQ ID NO 154

<400> SEQUENCE: 154

000

<210> SEQ ID NO 155

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 155

Gly Phe Thr Phe Ser Ser Tyr Asp Met Asn
1 5 10

<210> SEQ ID NO 156

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 156

Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 157

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 157

Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp
1 5 10

<210> SEQ ID NO 158

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 158

Arg Ala Ser Gln Ser Val Tyr Thr Tyr Leu Ala
1 5 10

<210> SEQ ID NO 159

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 159

Gly Ala Ser Ser Arg Ala Thr
1 5

-continued

<210> SEQ ID NO 160
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 160

Gln Gln Tyr Tyr Asp Arg Pro Pro Leu Thr
1 5 10

<210> SEQ ID NO 161
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 161

Gly Phe Thr Phe Ser Ser Asn Ala
1 5

<210> SEQ ID NO 162
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 162

Ile Ser Ala Gly Gly Ser Asp Thr
1 5

<210> SEQ ID NO 163
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 163

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr
1 5 10

<210> SEQ ID NO 164
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 164

Ser Gly Gly Asp Tyr Tyr Ser Tyr Tyr Tyr Gly
1 5 10

<210> SEQ ID NO 165
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 165

Thr Val Ile Tyr Ser Asp Asp Lys Arg Pro Ser Asp
1 5 10

-continued

<210> SEQ ID NO 166
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 166

Gly Gly Tyr Asp Tyr Ser Thr Tyr Ala Asn Ala
1 5 10

<210> SEQ ID NO 167

<400> SEQUENCE: 167

000

<210> SEQ ID NO 168
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 168

Ile Ser Ser Gly Ser Asp Thr
1 5

<210> SEQ ID NO 169

<400> SEQUENCE: 169

000

<210> SEQ ID NO 170
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 170

Ser Gly Gly Ser Tyr Ser Ser Tyr Tyr Tyr Ala
1 5 10

<210> SEQ ID NO 171
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 171

Thr Leu Ile Tyr Ser Asp Asp Lys Arg Pro Ser Asn
1 5 10

<210> SEQ ID NO 172
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 172

Gly Gly Tyr Asp Gln Ser Ser Tyr Thr Asn Pro
1 5 10

-continued

<210> SEQ ID NO 173
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 173

Gly Phe Arg Phe Ser Ser Tyr Ala
1 5

<210> SEQ ID NO 174
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 174

Ile Ser Ser Gly Gly Asp Thr
1 5

<210> SEQ ID NO 175
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 175

Asn Phe Ala Met Thr
1 5

<210> SEQ ID NO 176
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 176

Ser Gly Gly Asp Ser Ser Ser His Tyr Tyr Gly
1 5 10

<210> SEQ ID NO 177
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 177

Thr Val Ile Tyr Ser Asp Asp Glu Arg Pro Ser Asp
1 5 10

<210> SEQ ID NO 178
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 178

Gly Ala Tyr Asp Gly Ser Thr Tyr Ala Asn Thr
1 5 10

<210> SEQ ID NO 179

-continued

<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 179

Gly Phe Thr Phe Ser Ser Tyr Ala
1 5

<210> SEQ ID NO 180

<400> SEQUENCE: 180

000

<210> SEQ ID NO 181

<400> SEQUENCE: 181

000

<210> SEQ ID NO 182
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 182

Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr Ala
1 5 10

<210> SEQ ID NO 183
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 183

Thr Val Ile His Ser Asp Asp Lys Arg Pro Ser Asp
1 5 10

<210> SEQ ID NO 184
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 184

Gly Gly Tyr Asp Gly Arg Thr Tyr Ile Asn Thr
1 5 10

<210> SEQ ID NO 185
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 6
<223> OTHER INFORMATION: Xaa = Asn or Ile
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 7
<223> OTHER INFORMATION: Xaa = Phe or Tyr

-continued

<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 10
<223> OTHER INFORMATION: Xaa = Thr or Ser

<400> SEQUENCE: 185

Gly Phe Ser Phe Ser Xaa Xaa Ala Met Xaa
1 5 10

<210> SEQ ID NO 186
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Ser or Ala
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 5
<223> OTHER INFORMATION: Xaa = Gly or Asp

<400> SEQUENCE: 186

Thr Ile Gly Xaa Xaa Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 187
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 5
<223> OTHER INFORMATION: Xaa = Ser or Gly

<400> SEQUENCE: 187

Asp Ser Thr Val Xaa Trp Ser Gly Asp Phe Phe Asp Tyr
1 5 10

<210> SEQ ID NO 188
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 7
<223> OTHER INFORMATION: Xaa = Lys or Arg
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 8
<223> OTHER INFORMATION: Xaa = Asn or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 10
<223> OTHER INFORMATION: Xaa = Leu or Ile

<400> SEQUENCE: 188

Arg Ala Ser Gln Asn Val Xaa Xaa Asp Xaa Ala
1 5 10

<210> SEQ ID NO 189
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 3
<223> OTHER INFORMATION: Xaa = Arg or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Ile or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 6
<223> OTHER INFORMATION: Xaa = Glu or Asp

<400> SEQUENCE: 189

Ala Ala Xaa Xaa Arg Xaa Thr
1 5

<210> SEQ ID NO 190
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 190

Gly Gly Tyr Asp Tyr Ser Thr Tyr Ala
1 5

<210> SEQ ID NO 191
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 191

Gly Phe Thr Phe Ser Ser Asn
1 5

<210> SEQ ID NO 192
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 192

Ser Ala Gly Gly Ser Asp
1 5

<210> SEQ ID NO 193
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 193

Glu Thr Trp Asn His Leu Phe Asp Tyr
1 5

<210> SEQ ID NO 194
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 194

Ser Asn Ala Met Ser
1 5

<210> SEQ ID NO 195

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 195

Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Pro Ala Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 196

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 196

Ser Ser Gly Ser Asp
1 5

<210> SEQ ID NO 197

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 197

Gly Ile Ser Ser Gly Ser Asp Thr Tyr Tyr Gly Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 198

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 198

Gly Phe Arg Phe Ser Ser Tyr
1 5

<210> SEQ ID NO 199

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 199

Ser Ser Gly Gly Asp
1 5

<210> SEQ ID NO 200

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 200

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 201

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 201

Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 202

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 202

Gly Phe Ser Phe Ser Asn Phe
1 5

<210> SEQ ID NO 203

<211> LENGTH: 4

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 203

Gly Ser Gly Asp
1

<210> SEQ ID NO 204

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 204

Asn Phe Ala Val Thr
1 5

<210> SEQ ID NO 205

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 205

Gly Phe Ser Phe Ser Asn Phe Ala
1 5

<210> SEQ ID NO 206

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 206

Ile Gly Ser Gly Asp Thr
1 5

<210> SEQ ID NO 207

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 207

Ala Lys Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr
1 5 10 15

<210> SEQ ID NO 208

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 208

Gln Asn Val Lys Asn Asp
1 5

<210> SEQ ID NO 209

<211> LENGTH: 3

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 209

Ala Ala Arg
1

<210> SEQ ID NO 210

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 210

Gln Gln Tyr Tyr Asp Trp Pro
1 5

<210> SEQ ID NO 211

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 211

Gly Phe Ser Phe Ser Ile Tyr
1 5

<210> SEQ ID NO 212

<211> LENGTH: 4

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 212

-continued

Gly Ala Asp Asp
1

<210> SEQ ID NO 213
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 213

Ile Tyr Ala Val Ser
1 5

<210> SEQ ID NO 214
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 214

Gly Phe Ser Phe Ser Ile Tyr Ala
1 5

<210> SEQ ID NO 215
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 215

Ile Gly Ala Asp Asp Thr
1 5

<210> SEQ ID NO 216
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 216

Ala Lys Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr
1 5 10 15

<210> SEQ ID NO 217
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 217

Gln Asn Val Arg Ser Asp
1 5

<210> SEQ ID NO 218
<211> LENGTH: 3
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 218

-continued

Ala Ala Ser
1

<210> SEQ ID NO 219
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 219

Gly Phe Thr Phe Ser Ser Tyr
1 5

<210> SEQ ID NO 220
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 220

Ser Gly Ser Gly Glu Ile
1 5

<210> SEQ ID NO 221
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 221

Ser Tyr Asp Val Asn
1 5

<210> SEQ ID NO 222
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 222

Gly Phe Thr Phe Ser Ser Tyr Asp
1 5

<210> SEQ ID NO 223
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 223

Ile Ser Gly Ser Gly Glu Ile Ile
1 5

<210> SEQ ID NO 224
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 224

Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp

-continued

1 5 10

<210> SEQ ID NO 225
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 225

Gln Ser Val Tyr Thr Tyr
 1 5

<210> SEQ ID NO 226
 <211> LENGTH: 3
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 226

Gly Ala Ser
 1

<210> SEQ ID NO 227
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 227

Asp Tyr Gly Met Asn
 1 5

<210> SEQ ID NO 228
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 228

Gln Ile Thr Ser Gly Ser Arg Thr Tyr Tyr Gly Ala Ala Val Lys Gly
 1 5 10 15

<210> SEQ ID NO 229
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 229

Asp Phe Gly Ser Gly Val Gly Ser Ile Asp Ala
 1 5 10

<210> SEQ ID NO 230
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 230

Gly Phe Ile Phe Ser Asp Tyr
 1 5

-continued

<210> SEQ ID NO 231
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 231

Thr Ser Gly Ser Arg
1 5

<210> SEQ ID NO 232
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 232

Ser Gly Ser Arg Gly Arg Tyr Gly
1 5

<210> SEQ ID NO 233
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 233

Arg Asp Asn Gln Arg Pro Ser
1 5

<210> SEQ ID NO 234
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 234

Gly Ser Tyr Asp Gly Ser Ile Asp Ile
1 5

<210> SEQ ID NO 235
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 235

Ser Tyr Ala Met Gly
1 5

<210> SEQ ID NO 236
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 236

Gly Ile Asp Asp Asp Gly Ser Thr Ala Asn Tyr Gly Pro Ala Val Lys
1 5 10 15

-continued

Gly

<210> SEQ ID NO 237
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 237

Ala Ser Val Thr Gly Trp Ser Ala His Ile Ser Gly Arg Leu Asp Thr
1 5 10 15

<210> SEQ ID NO 238
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 238

Asp Asp Gly Ser Thr
1 5

<210> SEQ ID NO 239
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 239

Ser Gly Gly Gly Ile Tyr Tyr Tyr Gly
1 5

<210> SEQ ID NO 240
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 240

Glu Asn Asp Lys Arg Pro Ser
1 5

<210> SEQ ID NO 241
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 241

Gly Gly Tyr Asp Ser Asn Thr Thr Ser Gly Ile
1 5 10

<210> SEQ ID NO 242
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 242

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
1 5 10 15

-continued

Ala Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Ile Phe Ser Asp Tyr
 20 25 30

Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Phe Val
 35 40 45

Ala Gln Ile Thr Ser Gly Ser Arg Thr Tyr Tyr Gly Ala Ala Val Lys
 50 55 60

Gly Arg Ala Thr Ile Ser Arg Asp Asn Arg Gln Ser Thr Val Lys Leu
 65 70 75 80

Gln Leu Asn Asn Leu Arg Ala Glu Asp Thr Gly Ile Tyr Phe Cys Ala
 85 90 95

Arg Asp Phe Gly Ser Gly Val Gly Ser Ile Asp Ala Trp Gly Asn Gly
 100 105 110

Thr Glu Val Ile Val Ser Ser
 115

<210> SEQ ID NO 243
 <211> LENGTH: 102
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 243

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Gly Thr Val
 1 5 10 15

Lys Ile Thr Cys Ser Gly Ser Arg Gly Arg Tyr Gly Trp Tyr Gln Gln
 20 25 30

Arg Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Arg Asp Asn Gln
 35 40 45

Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Ser Ser Thr Ser Gly Ser
 50 55 60

Thr Ser Thr Leu Thr Ile Thr Gly Val Gln Ala Asp Asp Glu Ser Val
 65 70 75 80

Tyr Phe Cys Gly Ser Tyr Asp Gly Ser Ile Asp Ile Phe Gly Ala Gly
 85 90 95

Thr Thr Leu Thr Val Leu
 100

<210> SEQ ID NO 244
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 244

Ala Val Thr Leu Asp Glu Ser Gly Gly Gly Leu Gln Thr Pro Gly Gly
 1 5 10 15

Ala Leu Ser Leu Val Cys Lys Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Ala Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Gly Ile Asp Asp Asp Gly Ser Thr Ala Asn Tyr Gly Pro Ala Val
 50 55 60

Lys Gly Arg Ala Thr Ile Ser Arg Asp Asn Gly Gln Ser Thr Val Arg
 65 70 75 80

Leu Gln Leu Asn Asn Pro Arg Ala Glu Asp Ser Gly Thr Tyr Phe Cys

-continued

85	90	95
Ala Lys Ala Ser Val Thr Gly Trp Ser Ala His Ile Ser Gly Arg Leu		
100	105	110
Asp Thr Trp Gly His Gly Thr Glu Val Ile Val Ser Ser		
115	120	125

<210> SEQ ID NO 245
 <211> LENGTH: 105
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 245

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1 5 10 15
Lys Ile Thr Cys Ser Gly Gly Gly Ile Tyr Tyr Tyr Gly Trp Tyr Gln
20 25 30
Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Glu Asn Asp
35 40 45
Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Thr Ser Gly
50 55 60
Ser Thr Asn Thr Phe Thr Ile Thr Gly Val Gln Ala Glu Asp Glu Ala
65 70 75 80
Val Tyr Tyr Cys Gly Gly Tyr Asp Ser Asn Thr Thr Ser Gly Ile Phe
85 90 95
Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 246
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 246

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe
20 25 30
Ala Val Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
50 55 60
Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
65 70 75 80
Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
85 90 95
Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Gln
100 105 110
Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 247
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 247

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr
 20 25 30

Ala Val Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 50 55 60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
 65 70 75 80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
 85 90 95

Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 248

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 248

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15

Glu Arg Val Thr Leu Ser Cys Arg Ala Ser Gln Asn Val Arg Ser Asp
 20 25 30

Ile Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Ala Ala Ser Ser Arg Asp Thr Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Gly Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
 85 90 95

Phe Thr Phe Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 249

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 249

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Asp Val Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val

-continued

50	55	60	
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr			
65	70	75	80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys			
	85	90	95
Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Gln Gly			
	100	105	110
Thr Leu Val Thr Val Ser Ser			
	115		

<210> SEQ ID NO 250
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 250

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly		
1	5	10 15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr		
	20	25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile		
	35	40 45
Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly		
	50	55 60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser		
65	70	75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro		
	85	90 95
Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys		
	100	105

<210> SEQ ID NO 251
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 251

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly		
1	5	10 15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr		
	20	25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile		
	35	40 45
Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly		
	50	55 60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser		
65	70	75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro		
	85	90 95
Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys		
	100	105

<210> SEQ ID NO 252
 <211> LENGTH: 108

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 252

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1           5           10          15

Thr Ala Arg Ile Thr Cys Ser Gly Gly Ser Tyr Ser Ser Tyr Tyr Tyr
20          25          30

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Thr Leu Ile Tyr
35          40          45

Ser Asp Asp Lys Arg Pro Ser Asn Ile Pro Glu Arg Phe Ser Gly Ser
50          55          60

Ser Ser Gly Thr Thr Val Thr Leu Thr Ile Ser Gly Val Gln Ala Glu
65          70          75          80

Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp Gln Ser Ser Tyr Thr
85          90          95

Asn Pro Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100         105

```

```

<210> SEQ ID NO 253
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 253

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala Ala Pro Gly Gln
1           5           10          15

Lys Val Thr Ile Ser Cys Ser Gly Gly Ser Tyr Ser Ser Tyr Tyr Tyr
20          25          30

Ala Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Thr Leu Ile Tyr
35          40          45

Ser Asp Asp Lys Arg Pro Ser Asn Ile Pro Asp Arg Phe Ser Gly Ser
50          55          60

Lys Ser Gly Thr Ser Ala Thr Leu Gly Ile Thr Gly Leu Gln Thr Gly
65          70          75          80

Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp Gln Ser Ser Tyr Thr
85          90          95

Asn Pro Phe Gly Thr Gly Thr Lys Val Thr Val Leu
100         105

```

```

<210> SEQ ID NO 254
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 254

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1           5           10          15

Thr Ala Arg Ile Thr Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr
20          25          30

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Thr Val Ile His
35          40          45

Ser Asp Asp Lys Arg Pro Ser Asp Ile Pro Glu Arg Phe Ser Gly Ser

```

-continued

50	55	60
Ser Ser Gly Thr Thr Val Thr Leu Thr Ile Ser Gly Val Gln Ala Glu		
65	70	75 80
Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp Gly Arg Thr Tyr Ile		
	85	90 95
Asn Thr Phe Gly Gly Gly Thr Lys Leu Thr Val Leu		
	100	105

<210> SEQ ID NO 255
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 255

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala Ala Pro Gly Gln
1 5 10 15
Lys Val Thr Ile Ser Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr
20 25 30
Ala Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Thr Val Ile His
35 40 45
Ser Asp Asp Lys Arg Pro Ser Asp Ile Pro Asp Arg Phe Ser Gly Ser
50 55 60
Lys Ser Gly Thr Ser Ala Thr Leu Gly Ile Thr Gly Leu Gln Thr Gly
65 70 75 80
Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp Gly Arg Thr Tyr Ile
85 90 95
Asn Thr Phe Gly Thr Gly Thr Lys Val Thr Val Leu
100 105

<210> SEQ ID NO 256
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 256

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
1 5 10 15
Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Ser Tyr
20 25 30
Ser Ser Tyr Tyr Tyr Ala Trp Tyr Gln Gln His Pro Gly Lys Ala Pro
35 40 45
Lys Thr Leu Ile Tyr Ser Asp Asp Lys Arg Pro Ser Asn Val Ser Asn
50 55 60
Arg Phe Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser
65 70 75 80
Gly Leu Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp
85 90 95
Gln Ser Ser Tyr Thr Asn Pro Phe Gly Gly Gly Thr Lys Leu Thr Val
100 105 110

Leu

<210> SEQ ID NO 257
 <211> LENGTH: 108
 <212> TYPE: PRT

-continued

```

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 257

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala Ala Pro Gly Gln
1          5          10         15

Lys Val Thr Ile Ser Cys Ser Gly Gly Asp Tyr Tyr Ser Tyr Tyr Tyr
20         25         30

Gly Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Thr Val Ile Tyr
35         40         45

Ser Asp Asp Lys Arg Pro Ser Asp Ile Pro Asp Arg Phe Ser Gly Ser
50         55         60

Lys Ser Gly Thr Ser Ala Thr Leu Gly Ile Thr Gly Leu Gln Thr Gly
65         70         75         80

Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp Tyr Ser Thr Tyr Ala
85         90         95

Asn Ala Phe Gly Thr Gly Thr Lys Val Thr Val Leu
100        105

```

```

<210> SEQ ID NO 258
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 258

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1          5          10         15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe
20         25         30

Ala Met Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35         40         45

Ser Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
50         55         60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
65         70         75         80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
85         90         95

Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Gln
100        105        110

Gly Thr Leu Val Thr Val Ser Ser
115        120

```

```

<210> SEQ ID NO 259
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 259

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1          5          10         15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr
20         25         30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35         40         45

```

-continued

Ser Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 50 55 60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
 65 70 75 80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
 85 90 95

Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 260
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 260

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Asp Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 261
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 261

Ser Gly Gly Ala Tyr Ser Ser Tyr Tyr Tyr Ala
 1 5 10

<210> SEQ ID NO 262
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 262

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
 1 5 10 15

Thr Ala Arg Ile Thr Cys Ser Gly Gly Ala Tyr Ser Ser Tyr Tyr Tyr
 20 25 30

-continued

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
 35 40 45

Ser Asp Asp Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
 50 55 60

Ser Ser Gly Thr Thr Val Thr Leu Thr Ile Ser Gly Val Gln Ala Glu
 65 70 75 80

Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp Gln Ser Ser Tyr Thr
 85 90 95

Asn Pro Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 263
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 263

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Pro Ala Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 264
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 264

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
 20 25 30

Ala Val Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Pro Ala Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr
 100 105 110

-continued

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 265
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 265

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ala Gly Ile Ser Ser Gly Ser Asp Thr Tyr Tyr Gly Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr Leu
100 105 110
Val Thr Val Ser Ser
115

<210> SEQ ID NO 266
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 266

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
20 25 30
Ala Val Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ala Gly Ile Ser Ser Gly Ser Asp Thr Tyr Tyr Gly Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr Leu
100 105 110
Val Thr Val Ser Ser
115

<210> SEQ ID NO 267
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 267

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Arg Phe Ser Ser Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr Leu
100 105 110
Val Thr Val Ser Ser
115

<210> SEQ ID NO 268

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 268

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Arg Phe Ser Ser Tyr
20 25 30
Ala Val Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr Leu
100 105 110
Val Thr Val Ser Ser
115

<210> SEQ ID NO 269

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 269

Ile Tyr Ala Met Ser
1 5

<210> SEQ ID NO 270

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 270

Ser Tyr Asp Met Asn
1 5

<210> SEQ ID NO 271
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 271

Ser Asn Ala Val Ser
1 5

<210> SEQ ID NO 272
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 272

Ser Tyr Ala Val Ser
1 5

<210> SEQ ID NO 273
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 273

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr
20 25 30

Ala Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35 40 45

Ser Arg Ile Asn Ser Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Gln Tyr Asp Trp Asn Ser Phe Phe Asp Tyr Trp Gly Leu Gly
100 105 110

Ala Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 274
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 274

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly

-continued

1	5	10	15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys	20	25	30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile	35	40	45
Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Ser Asp Arg Phe Ser Gly	50	55	60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr	65	70	75
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro	85	90	95
Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys	100	105	

<210> SEQ ID NO 275
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 275

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu	1	5	10	15
Ser Leu Arg Leu Ser Cys Glu Ala Ser Gly Phe Thr Phe Arg Asn Tyr	20	25	30	
Tyr Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val	35	40	45	
Ser Thr Ile Ser Asp Thr Gly Asp Thr Ala Tyr Tyr Ala Asp Ser Val	50	55	60	
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr	65	70	75	80
Leu Gln Met Asn Ser Leu Arg Ala Asp Asp Thr Ala Ile Tyr Tyr Cys	85	90	95	
Ala Lys Ser Trp Ile Trp Trp Thr Phe Phe Asp Tyr Trp Gly Leu Gly	100	105	110	
Thr Leu Val Thr Val Ser Ser	115			

<210> SEQ ID NO 276
 <211> LENGTH: 103
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 276

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Gly Thr Val	1	5	10	15
Glu Ile Thr Cys Ser Gly Gly Asn Ser Asn His Tyr Gly Trp Tyr Gln	20	25	30	
Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Ala Asp Thr	35	40	45	
Asn Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Thr Ser Gly	50	55	60	
Ser Thr Thr Thr Leu Thr Ile Thr Gly Val Gln Ala Glu Asp Glu Ala	65	70	75	80

-continued

Val Tyr Tyr Cys Gly Gly Ser Ser Thr Gly Asp Gly Ile Phe Gly Ala
85 90 95

Gly Thr Thr Leu Thr Val Leu
100

<210> SEQ ID NO 277

<400> SEQUENCE: 277

000

<210> SEQ ID NO 278

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 278

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ala Phe Ser Asp His
20 25 30

Asp Met Ser Trp Val Arg Gln Gly Pro Gly Glu Gly Leu Glu Trp Val
35 40 45

Ala Gly Ile Ser Leu Arg Gly Gly Val Thr Trp Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Arg Leu Phe Ser Leu Arg Thr Glu Asp Thr Ala Ile Tyr Tyr Cys
85 90 95

Ala Arg Glu Ser Trp Asn Thr Phe Phe Asp Tyr Trp Gly Leu Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 279

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 279

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Thr Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Val Arg Ser Asn
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Asp Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Asn Gly Pro Pro
85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

-continued

```

<210> SEQ ID NO 280
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 280

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Ala Val Val Arg Pro Gly Glu
1             5             10             15

Ser Leu Arg Leu Ser Cys Lys Ala Ser Gly Phe Thr Phe Thr Asn Phe
                20             25             30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
            35             40             45

Ser Gly Ile Ser Gly Ser Asp Asp Thr Thr Tyr Tyr Ala Asp Ser Val
            50             55             60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Ser Thr Leu Tyr
65             70             75             80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
            85             90             95

Val Lys Asp Ser Thr Val Ser Trp Asn Thr Phe Phe Asp Tyr Trp Gly
            100            105            110

Leu Gly Thr Leu Val Thr Val Ser Ser
            115            120

```

```

<210> SEQ ID NO 281
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 281

```

```

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Gly Thr Val
1             5             10             15

Lys Ile Thr Cys Ser Gly Gly Tyr Gly Ser Asp Asp Gly Ser Ser Ser
            20             25             30

Tyr Tyr Gly Trp Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Ile
            35             40             45

Leu Ile Tyr Trp Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe
            50             55             60

Ser Gly Ser Thr Ser Gly Ser Thr Thr Thr Leu Thr Ile Thr Gly Val
65             70             75             80

Gln Ala Glu Asp Glu Ala Val Tyr Phe Cys Gly Thr Tyr Asp Thr Ser
            85             90             95

Ser Gly Ala Ile Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
            100            105            110

```

```

<210> SEQ ID NO 282
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 282

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1             5             10             15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Arg Ser Tyr
            20             25             30

```

-continued

Ala Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45

Ser Arg Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Gln Tyr Asp Trp Asn Ser Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110

Ala Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 283
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 283

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
 20 25 30

Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ala Ser Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Pro Gln Thr
 65 70 75 80

Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro
 85 90 95

Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 284
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 284

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Thr Leu Ser Cys Thr Ala Ser Gly Phe Thr Phe Thr Ser Ser
 20 25 30

Thr Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Asp Trp Val
 35 40 45

Ser Ser Ile Ser Thr Ser Gly Val Ile Thr Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Ala Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Arg Leu Phe Ser Leu Arg Ala Asp Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95

Ala Thr Asp Thr Phe Asp His Trp Gly Pro Gly Thr Leu Val Thr Val

-continued

100	105	110
Ser Ser		
<210> SEQ ID NO 285 <211> LENGTH: 102 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic Construct <400> SEQUENCE: 285		
Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val		
1 5 10 15		
Lys Ile Thr Cys Phe Gly Ser Ser Gly Asn Tyr Gly Trp Phe Gln Gln		
20 25 30		
Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Tyr Asn Asn Lys		
35 40 45		
Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Lys Ser Gly Ser		
50 55 60		
Thr Gly Thr Leu Thr Ile Thr Gly Val Arg Ala Glu Asp Glu Ala Val		
65 70 75 80		
Tyr Phe Cys Gly Ala Trp Glu Thr Gly Ser Ala Thr Phe Gly Ala Gly		
85 90 95		
Thr Thr Leu Thr Val Leu		
100		
<210> SEQ ID NO 286 <211> LENGTH: 117 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic Construct <400> SEQUENCE: 286		
Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu		
1 5 10 15		
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Phe		
20 25 30		
Ala Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val		
35 40 45		
Ser Leu Ile Ser Val Thr Ala Thr Thr Tyr Tyr Ala Asp Ser Val Lys		
50 55 60		
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu		
65 70 75 80		
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
85 90 95		
Lys Val Thr Trp Asn Asn Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu		
100 105 110		
Val Thr Val Ser Ser		
115		
<210> SEQ ID NO 287 <211> LENGTH: 108 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic Construct <400> SEQUENCE: 287		

-continued

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Pro Ile Asp Ser Tyr
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Asn Thr Val Thr Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn Arg Leu Glu Pro
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln His Gln Tyr Asp Trp Pro Pro
 85 90 95
 Tyr Ile Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 288
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 288

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Phe
 20 25 30
 Ala Met Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Pro Glu Trp Val
 35 40 45
 Ser Leu Val Ser Val Thr Ala Thr Thr Tyr Tyr Ala Asp Ser Val Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Ser Thr Leu Tyr Leu
 65 70 75 80
 Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95
 Lys Ile Thr Trp Asn Asn Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
 100 105 110
 Val Thr Val Ser Ser
 115

<210> SEQ ID NO 289
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 289

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Asp Ser Ser Ser Arg Ala Ser Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80

-continued

Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln His Asn Asp Trp Pro Pro
 85 90 95

Tyr Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 290
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 290

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Thr Asn Tyr
 20 25 30

Ala Val Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45

Ser Leu Ile Ser Val Thr Gly Thr Thr Tyr Tyr Ala Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Ser Thr Leu Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Lys Val Thr Trp Lys Asn Val Phe Asp Tyr Trp Gly Leu Gly Thr Leu
 100 105 110

Val Thr Val Ser Ser
 115

<210> SEQ ID NO 291
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 291

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Val Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ala Asn Thr Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln His Thr Asp Trp Pro Pro
 85 90 95

Tyr Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 292
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 292

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Asn Tyr
           20           25           30
Gly Met Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
           35           40           45
Ser Ala Ser Ser Gly Ser Gly Ser Thr Tyr Tyr Ala Asp Ser Val Lys
           50           55           60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
           65           70           75           80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
           85           90           95
Lys Val Thr Trp Asn Asn Phe Phe Asp Tyr Trp Gly Leu Gly Thr Leu
           100          105          110
Val Thr Val Ser Ser
           115

```

<210> SEQ ID NO 293

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 293

```

Glu Ile Val Leu Ala Gln Ser Pro Asp Thr Leu Ser Val Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Thr Cys Arg Ala Ser Gln Asp Val Ala Gly Lys
           20           25           30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
           35           40           45
His Ala Thr Ser Ser Arg Ala Asp Gly Ile Pro Ala Arg Phe Ser Gly
           50           55           60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Thr Gly Leu Gln Ser
           65           70           75           80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln His Tyr Asp Trp Ser Pro
           85           90           95
Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
           100          105

```

<210> SEQ ID NO 294

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 294

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Arg Phe Ser Ser Tyr
           20           25           30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
           35           40           45
Ser Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys
           50           55           60

```

-continued

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Val Asn Ser Leu Thr Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
85 90 95

Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
100 105 110

Val Thr Val Ser Ser
115

<210> SEQ ID NO 295
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 295

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Glu Thr Val
1 5 10 15

Lys Ile Thr Cys Ser Gly Gly Asp Ser Ser Ser His Tyr Tyr Gly Trp
20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Ser
35 40 45

Asp Asp Glu Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
50 55 60

Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
65 70 75 80

Glu Ala Ile Tyr Tyr Cys Gly Ala Tyr Asp Gly Ser Thr Tyr Ala Asn
85 90 95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 296
 <211> LENGTH: 127
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 296

Glu Glu Glu Leu Gln Val Ile Gln Pro Asp Lys Ser Val Leu Val Ala
1 5 10 15

Ala Gly Glu Thr Ala Thr Leu Arg Cys Thr Ala Thr Ser Leu Ile Pro
20 25 30

Val Gly Pro Ile Gln Trp Phe Arg Gly Ala Gly Pro Gly Arg Glu Leu
35 40 45

Ile Tyr Asn Gln Lys Glu Gly His Phe Pro Arg Val Thr Thr Val Ser
50 55 60

Asp Leu Thr Lys Arg Asn Asn Met Asp Phe Ser Ile Arg Ile Gly Ala
65 70 75 80

Ile Thr Pro Ala Asp Ala Gly Thr Tyr Tyr Cys Val Lys Phe Arg Lys
85 90 95

Gly Ser Pro Asp Asp Val Glu Phe Lys Ser Gly Ala Gly Thr Glu Leu
100 105 110

Ser Val Arg Ala Lys Pro Ser Thr Arg His His His His His His
115 120 125

-continued

<210> SEQ ID NO 297
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 1
<223> OTHER INFORMATION: Xaa = Ser or Thr
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 2
<223> OTHER INFORMATION: Xaa = Asn or Tyr or His or Asp
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Met or Leu or Val

<400> SEQUENCE: 297

Xaa Xaa Ala Xaa Ser
1 5

<210> SEQ ID NO 298
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Ala or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 5
<223> OTHER INFORMATION: Xaa = Gly or Ser or absent
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 6
<223> OTHER INFORMATION: Xaa = Ser or Asp or Gly
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 7
<223> OTHER INFORMATION: Xaa = Gly or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 8
<223> OTHER INFORMATION: Xaa = Asp or Ser or Gly
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 9
<223> OTHER INFORMATION: Xaa = Thr or Ala
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 12
<223> OTHER INFORMATION: Xaa = Pro or Gly or Val or Ile or Ala or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 13
<223> OTHER INFORMATION: Xaa = Ala or Asp or Gly
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 15
<223> OTHER INFORMATION: Xaa = Val or Met

<400> SEQUENCE: 298

Gly Ile Ser Xaa Xaa Xaa Xaa Xaa Tyr Tyr Xaa Xaa Ser Xaa Lys
1 5 10 15

Gly

<210> SEQ ID NO 299
<211> LENGTH: 11
<212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Asp or Gly or Ser or Ile or absent
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 5
<223> OTHER INFORMATION: Xaa = Ser or Trp or Gly or Tyr or Asp
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 6
<223> OTHER INFORMATION: Xaa = Ser or Tyr or Thr or Asp
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 8
<223> OTHER INFORMATION: Xaa = His or Thr or Ser or Tyr
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 11
<223> OTHER INFORMATION: Xaa = Gly or Ala

<400> SEQUENCE: 299

Ser Gly Gly Xaa Xaa Xaa Ser Xaa Tyr Tyr Xaa
1 5 10

<210> SEQ ID NO 300
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 3
<223> OTHER INFORMATION: Xaa = Asp or Asn
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Glu or Lys or Gln
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 7
<223> OTHER INFORMATION: Xaa = Ser or Pro

<400> SEQUENCE: 300

Ser Asp Xaa Xaa Arg Pro Xaa
1 5

<210> SEQ ID NO 301
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 1
<223> OTHER INFORMATION: Xaa = Gly or Ala
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 2
<223> OTHER INFORMATION: Xaa = Gly or Ala
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 5
<223> OTHER INFORMATION: Xaa = Gly or Tyr or Gln or Ser or Ala
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 6
<223> OTHER INFORMATION: Xaa = Ser or Arg or Thr
<220> FEATURE:
<221> NAME/KEY: VARIANT

-continued

<222> LOCATION: 7
<223> OTHER INFORMATION: Xaa = Thr or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 9
<223> OTHER INFORMATION: Xaa = Ala or Ile or Val or Leu or Thr
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 11
<223> OTHER INFORMATION: Xaa = Thr or Ala or Asp or Pro

<400> SEQUENCE: 301

Xaa Xaa Tyr Asp Xaa Xaa Xaa Tyr Xaa Asn Xaa
1 5 10

<210> SEQ ID NO 302
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 2
<223> OTHER INFORMATION: Xaa = Asn or Tyr
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Met or Leu or Val

<400> SEQUENCE: 302

Ser Xaa Ala Xaa Ser
1 5

<210> SEQ ID NO 303
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Ala or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 6
<223> OTHER INFORMATION: Xaa = Gly or absent
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 7
<223> OTHER INFORMATION: Xaa = Ser or Gly
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 12
<223> OTHER INFORMATION: Xaa = Pro or Gly or Val
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 13
<223> OTHER INFORMATION: Xaa = Ala or Asp

<400> SEQUENCE: 303

Gly Ile Ser Xaa Gly Xaa Xaa Asp Thr Tyr Tyr Xaa Xaa Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 304
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

-continued

<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Ser or Ala

<400> SEQUENCE: 304

Ser Gly Gly Xaa Tyr Ser Ser Tyr Tyr Tyr Ala
1 5 10

<210> SEQ ID NO 305
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 305

Asn Phe Ala Leu Thr
1 5

<210> SEQ ID NO 306
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 306

Ile Tyr Ala Leu Ser
1 5

<210> SEQ ID NO 307
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 1
<223> OTHER INFORMATION: Xaa = Ser or Thr
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 2
<223> OTHER INFORMATION: Xaa = Tyr or Ser
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Met or Leu or Val

<400> SEQUENCE: 307

Xaa Xaa Asp Xaa Asn
1 5

<210> SEQ ID NO 308
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 9
<223> OTHER INFORMATION: Xaa = Ile or Thr

<400> SEQUENCE: 308

Leu Ile Ser Gly Ser Gly Glu Ile Xaa Tyr Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

-continued

<210> SEQ ID NO 309
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 2
<223> OTHER INFORMATION: Xaa = Asn or Asp
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 3
<223> OTHER INFORMATION: Xaa = Asn or Asp
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Arg or Met
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 10
<223> OTHER INFORMATION: Xaa = Asp or Tyr

<400> SEQUENCE: 309

Glu Xaa Xaa Xaa Tyr Arg Phe Phe Asp Xaa
1 5 10

<210> SEQ ID NO 310
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 3
<223> OTHER INFORMATION: Xaa = Ser or Asp
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 8
<223> OTHER INFORMATION: Xaa = Thr or Ser

<400> SEQUENCE: 310

Arg Ala Xaa Gln Ser Val Tyr Xaa Tyr Leu Ala
1 5 10

<210> SEQ ID NO 311
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 1
<223> OTHER INFORMATION: Xaa = Gly or Ala or Asp
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 3
<223> OTHER INFORMATION: Xaa = Ser or Arg
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 4
<223> OTHER INFORMATION: Xaa = Ser or Asn or Thr
<220> FEATURE:
<221> NAME/KEY: VARIANT
<222> LOCATION: 7
<223> OTHER INFORMATION: Xaa = Thr or Ala

<400> SEQUENCE: 311

Xaa Ala Xaa Xaa Arg Ala Xaa
1 5

-continued

<210> SEQ ID NO 312
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 312

```

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Val Lys Asn Asp
          20           25           30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
          35           40           45
Tyr Ala Ala Arg Ile Arg Glu Thr Gly Ile Pro Ala Arg Phe Ser Gly
          50           55           60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65           70           75           80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Trp Pro Pro
          85           90           95
Phe Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
          100          105

```

<210> SEQ ID NO 313
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 313

```

Ser Tyr Asp Leu Asn
1           5

```

<210> SEQ ID NO 314
 <211> LENGTH: 22
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 314

```

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1           5           10           15
Thr Ala Arg Ile Thr Cys
          20

```

<210> SEQ ID NO 315
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 315

```

Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Thr Leu Ile Tyr
1           5           10           15

```

<210> SEQ ID NO 316
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 316

```

Asn Ile Pro Glu Arg Phe Ser Gly Ser Ser Ser Gly Thr Thr Val Thr

```

-continued

1	5	10	15
---	---	----	----

Leu Thr Ile Ser Gly Val Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys
 20 25 30

<210> SEQ ID NO 317
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

 <400> SEQUENCE: 317

Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 1 5 10

<210> SEQ ID NO 318
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

 <400> SEQUENCE: 318

Ser Asn Ala Leu Ser
 1 5

<210> SEQ ID NO 319
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

 <400> SEQUENCE: 319

Ser Tyr Ala Leu Ser
 1 5

<210> SEQ ID NO 320
 <211> LENGTH: 330
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

 <400> SEQUENCE: 320

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
 1 5 10 15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20 25 30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35 40 45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50 55 60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
 65 70 75 80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
 85 90 95

Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys
 100 105 110

Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro
 115 120 125

Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys
 130 135 140

Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp
 145 150 155 160

Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu
			165						170					175	
Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu
			180					185					190		
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn
			195				200					205			
Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly
	210					215					220				
Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu
225					230					235				240	
Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr
			245						250					255	
Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn
			260					265					270		
Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe
		275					280					285			
Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn
	290					295					300				
Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr
305					310					315					320
Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys						
			325					330							

Ala 1	Ser	Thr	Lys 5	Gly	Pro	Ser	Val	Phe 10	Pro	Leu	Ala	Pro	Ser	Ser 15	Lys
Ser	Thr	Ser	Gly 20	Gly	Thr	Ala	Ala	Leu 25	Gly	Cys	Leu	Val	Lys 30	Asp	Tyr
Phe	Pro	Glu 35	Pro	Val	Thr	Val	Ser 40	Trp	Asn	Ser	Gly	Ala 45	Leu	Thr	Ser
Gly	Val	His 50	Thr	Phe	Pro	Ala 55	Val	Leu	Gln	Ser	Ser 60	Gly	Leu	Tyr	Ser
Leu 65	Ser	Ser	Val	Val	Thr 70	Val	Pro	Ser	Ser	Ser 75	Leu	Gly	Thr	Gln	Thr 80
Tyr	Ile	Cys	Asn 85	Val	Asn	His	Lys	Pro	Ser 90	Asn	Thr	Lys	Val	Asp 95	Lys
Lys	Val	Glu	Pro 100	Lys	Ser	Cys	Asp	Lys 105	Thr	His	Thr	Cys	Pro 110	Pro	Cys
Pro	Ala	Pro	Glu 115	Ala	Ala	Gly	Ala 120	Pro	Ser	Val	Phe	Leu 125	Phe	Pro	Pro
Lys	Pro 130	Lys	Asp	Thr	Leu	Met 135	Ile	Ser	Arg	Thr	Pro 140	Glu	Val	Thr	Cys
Val 145	Val	Val	Asp	Val	Ser 150	His	Glu	Asp	Pro	Glu 155	Val	Lys	Phe	Asn	Trp 160
Tyr	Val	Asp	Gly 165	Val	Glu	Val	His	Asn 170	Ala	Lys	Thr	Lys	Pro	Arg 175	Glu
Glu	Gln	Tyr	Ala 180	Ser	Thr	Tyr	Arg	Val 185	Val	Ser	Val	Leu 190	Thr	Val	Leu

-continued

```

His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn
   195                               200               205

Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly
   210                               215               220

Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu
  225                               230               235               240

Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr
   245                               250               255

Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn
   260                               265               270

Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe
   275                               280               285

Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn
   290                               295               300

Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
  305                               310               315               320

Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
   325                               330

```

<210> SEQ ID NO 322

<211> LENGTH: 326

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 322

```

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
  1                               5               10               15

Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
  20                               25               30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
  35                               40               45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
  50                               55               60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr
  65                               70               75               80

Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
  85                               90               95

Thr Val Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro
  100                              105               110

Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp
  115                              120               125

Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp
  130                              135               140

Val Ser His Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly
  145                              150               155               160

Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn
  165                              170               175

Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Val His Gln Asp Trp
  180                              185               190

Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro
  195                              200               205

Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu
  210                              215               220

Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn

```

-continued

225	230	235	240
Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile			
	245	250	255
Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr			
	260	265	270
Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys			
	275	280	285
Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys			
	290	295	300
Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu			
305	310	315	320
Ser Leu Ser Pro Gly Lys			
	325		

<210> SEQ ID NO 323

<211> LENGTH: 326

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 323

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg			
1	5	10	15
Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr			
	20	25	30
Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser			
	35	40	45
Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser			
	50	55	60
Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr			
65	70	75	80
Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys			
	85	90	95
Thr Val Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro			
	100	105	110
Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp			
	115	120	125
Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp			
	130	135	140
Val Ser His Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly			
145	150	155	160
Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn			
	165	170	175
Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Val His Gln Asp Trp			
	180	185	190
Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro			
	195	200	205
Ser Ser Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu			
	210	215	220
Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn			
225	230	235	240
Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile			
	245	250	255
Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr			

-continued

260	265	270
Thr Pro Pro Met Leu Asp Ser	Asp Gly Ser Phe Phe	Leu Tyr Ser Lys
275	280	285
Leu Thr Val Asp Lys Ser Arg	Trp Gln Gln Gly Asn Val Phe Ser Cys	
290	295	300
Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu		
305	310	315 320
Ser Leu Ser Pro Gly Lys		
325		

<210> SEQ ID NO 324
 <211> LENGTH: 327
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

 <400> SEQUENCE: 324

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg	
1	15
Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr	
20	30
Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser	
35	45
Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser	
50	60
Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr	
65	80
Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys	
85	95
Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys Pro Ala Pro	
100	110
Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys	
115	125
Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val	
130	140
Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp	
145	160
Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe	
165	175
Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp	
180	190
Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu	
195	205
Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg	
210	220
Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met Thr Lys	
225	240
Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp	
245	255
Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys	
260	270
Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser	
275	285
Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val Phe Ser	

-continued

290	295	300
-----	-----	-----

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
 305 310 315 320

Leu Ser Leu Ser Leu Gly Lys
 325

<210> SEQ ID NO 325
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 325

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
1 5 10 15
Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
20 25 30
Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
35 40 45
Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
50 55 60
Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
65 70 75 80
Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
85 90 95
Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
100 105

<210> SEQ ID NO 326
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 326

Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser Ser
1 5 10 15
Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp
20 25 30
Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser Pro
35 40 45
Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn Asn
50 55 60
Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys
65 70 75 80
Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val
85 90 95
Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
100 105

<210> SEQ ID NO 327
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 327

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1 5 10 15

-continued

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe
 20 25 30

Ala Leu Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 50 55 60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
 65 70 75 80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
 85 90 95

Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 328
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 328

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr
 20 25 30

Ala Leu Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 50 55 60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
 65 70 75 80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
 85 90 95

Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 329
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 329

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Asp Leu Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

-continued

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 330
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 330

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
 20 25 30

Ala Leu Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Pro Ala Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 331
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 331

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
 20 25 30

Ala Leu Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Gly Ile Ser Ser Gly Ser Asp Thr Tyr Tyr Gly Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr Leu
 100 105 110

Val Thr Val Ser Ser
 115

-continued

```

<210> SEQ ID NO 332
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 332

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1             5             10             15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Arg Phe Ser Ser Tyr
                20             25             30

Ala Leu Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
            35             40             45

Ser Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys
            50             55             60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65             70             75             80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
            85             90             95

Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Gln Gly Thr Leu
            100            105            110

Val Thr Val Ser Ser
            115

```

```

<210> SEQ ID NO 333
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 333

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
1             5             10             15

Ser Ile Thr Ile Ser Cys Phe Gly Ser Ser Gly Asn Tyr Gly Leu Val
            20             25             30

Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu Met Ile Tyr
            35             40             45

Tyr Asn Asn Lys Arg Pro Ser Gly Val Ser Asn Arg Phe Ser Gly Ser
            50             55             60

Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser Gly Leu Gln Ala Glu
65             70             75             80

Asp Glu Ala Asp Tyr Tyr Cys Gly Ala Trp Glu Thr Gly Ser Ala Thr
            85             90             95

Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
            100            105

```

```

<210> SEQ ID NO 334
<211> LENGTH: 103
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 334

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1             5             10             15

Thr Ala Ser Ile Thr Cys Phe Gly Ser Ser Gly Asn Tyr Gly Trp Tyr
            20             25             30

```

-continued

Gln Gln Lys Pro Gly Gln Ser Pro Val Leu Val Ile Tyr Tyr Asn Asn
 35 40 45

Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser Asn Ser Gly
 50 55 60

Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Met Asp Glu Ala
 65 70 75 80

Asp Tyr Tyr Cys Gly Ala Trp Glu Thr Gly Ser Ala Thr Phe Gly Gly
 85 90 95

Gly Thr Lys Leu Thr Val Leu
 100

<210> SEQ ID NO 335
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 335

Asp Val Gln Leu Leu Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe
 20 25 30

Ala Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45

Ser Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 50 55 60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
 65 70 75 80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
 85 90 95

Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 336
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 336

Ser Asp Asp Lys Arg Pro Ser
 1 5

<210> SEQ ID NO 337
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 337

Gly Ser Tyr Ser Ser
 1 5

<210> SEQ ID NO 338
 <211> LENGTH: 3

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 338

```

```

Ile Tyr Ser
1

```

```

<210> SEQ ID NO 339
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 339

```

```

Gly Gly Tyr Asp Gln Ser Ser Tyr Thr
1          5

```

```

<210> SEQ ID NO 340
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 340

```

```

Gly Asp Tyr Tyr Ser
1          5

```

```

<210> SEQ ID NO 341
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 341

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10          15

```

```

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr
          20          25          30

```

```

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
          35          40          45

```

```

Ser Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
          50          55          60

```

```

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
          65          70          75          80

```

```

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
          85          90          95

```

```

Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu
          100          105          110

```

```

Gly Thr Leu Val Thr Val Ser Ser
          115          120

```

```

<210> SEQ ID NO 342
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 342

```

-continued

```

Asp Val Gln Leu Val Glu Ser Gly Gly Ala Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
           20           25           30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
           35           40           45
Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
           50           55           60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Ser Leu Tyr
65           70           75           80
Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
           85           90           95
Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
           100          105          110
Leu Val Thr Val Ser Ser
           115

```

```

<210> SEQ ID NO 343
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 343

```

```

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15
Lys Ile Thr Cys Ser Gly Gly Asp Trp Tyr Ser Thr Tyr Tyr Gly Trp
           20           25           30
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
           35           40           45
Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
           50           55           60
Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
65           70           75           80
Glu Ala Val Tyr Tyr Cys Ala Gly Tyr Asp Gly Arg Thr Tyr Ile Asn
           85           90           95
Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
           100          105

```

```

<210> SEQ ID NO 344
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 344

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
           20           25           30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
           35           40           45
Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
           50           55           60
Lys Gly Arg Phe Thr Ile Ser Arg Arg Gln Phe Gln Glu Gln Ser Leu

```

-continued

65	70	75	80
Ser Pro Asn Glu Pro Ala Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr	85	90	95
Cys Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly	100	105	110
Thr Leu Val Thr Val Ser Ser	115		

<210> SEQ ID NO 345
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 345

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val	1	5	10	15
Lys Ile Thr Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr Gly Trp	20	25	30	
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser	35	40	45	
Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala	50	55	60	
Ser Gly Ser Ala Ala Thr Leu Thr Ile Ala Gly Val Arg Val Glu Asp	65	70	75	80
Glu Ala Val Tyr Phe Cys Gly Ala Tyr Asp Gly Arg Thr Tyr Ile Asn	85	90	95	
Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu	100	105		

<210> SEQ ID NO 346
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 346

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu	1	5	10	15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser His	20	25	30	
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu	35	40	45	
Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val	50	55	60	
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Ser Ser Leu Tyr	65	70	75	80
Leu Arg Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys	85	90	95	
Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr	100	105	110	
Leu Val Thr Val Ser Ser	115			

<210> SEQ ID NO 347
 <211> LENGTH: 107

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 347

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10          15

Lys Ile Thr Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr Ala Trp
                20          25          30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
                35          40          45

Asp Asp Glu Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
50          55          60

Ser Gly Ser Ala Ala Thr Leu Ile Ile Thr Gly Val Arg Val Glu Asp
65          70          75          80

Glu Ala Val Tyr Phe Cys Gly Gly Tyr Asp Gly Arg Thr Tyr Ile Asn
                85          90          95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100          105

```

```

<210> SEQ ID NO 348
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 348

Asp Val Gln Leu Val Glu Ser Gly Gly Val Val Arg Pro Gly Glu
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
20          25          30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
35          40          45

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Pro Ala Ser Val
50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Pro Lys Asn Thr Leu Tyr
65          70          75          80

Leu Gln Met Asn Thr Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85          90          95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
100          105          110

Leu Val Thr Val Ser Ser
115

```

```

<210> SEQ ID NO 349
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 349

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10          15

Lys Ile Ala Cys Ser Gly Gly Ser Tyr Tyr Ser Tyr Tyr Tyr Gly Trp
20          25          30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Leu Val Thr Val Ile Tyr Ser

```

-continued

35	40	45
Asp Asp Lys Arg Pro Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser Ala		
50	55	60
Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Ala Glu Asp		
65	70	75
Glu Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Tyr Ser Ser Tyr Thr Asn		
85	90	95
Asp Phe Gly Ala Gly Thr Thr Leu Thr Val Leu		
100	105	

<210> SEQ ID NO 350
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 350

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu		
1	5	10
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr		
20	25	30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		
35	40	45
Ser Gly Ile Ser Ala Ser Gly Ser Gly Thr Tyr Tyr Gly Asp Ser Val		
50	55	60
Lys Gly Arg Phe Thr Met Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr		
65	70	75
Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys		
85	90	95
Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr		
100	105	110
Leu Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 351
 <211> LENGTH: 105
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 351

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Gly Thr Val		
1	5	10
Glu Ile Thr Cys Ser Gly Gly Ser Ser Ser Tyr Tyr Gly Trp Tyr Gln		
20	25	30
Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Ser Asp Asn		
35	40	45
Gln Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala Ser Asp		
50	55	60
Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Gln Val Glu Asp Glu Ala		
65	70	75
Ile Tyr Tyr Cys Gly Gly Tyr Asp Ser Ser Thr Tyr Ala Asn Thr Phe		
85	90	95
Gly Ala Gly Thr Thr Leu Thr Val Leu		
100	105	

-continued

<210> SEQ ID NO 352
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 352

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Asn
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ala Gly Ile Ser Ala Ser Gly Asp Thr Tyr Tyr Ser Gly Ser Met Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80
 Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95
 Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
 100 105 110
 Val Thr Val Ser Ser
 115

<210> SEQ ID NO 353
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 353

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15
 Lys Ile Thr Cys Ser Gly Gly Ser Asp Ser Tyr Tyr Tyr Gly Trp His
 20 25 30
 Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Ser Asp
 35 40 45
 Asp Gln Arg Pro Pro Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala Ser
 50 55 60
 Gly Ser Thr Thr Thr Leu Thr Ile Thr Gly Val Arg Ala Glu Asp Glu
 65 70 75 80
 Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Tyr Ser Thr Tyr Thr Asn Thr
 85 90 95
 Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 354
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 354

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Arg Phe Ser Ser Tyr

-continued

20	25	30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		
35	40	45
Ser Gly Ile Ser Ser Asp Ser Asp Ala Tyr Tyr Val Asp Ser Val Lys		
50	55	60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu		
65	70	80
Gln Val Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
85	90	95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Met		
100	105	110
Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 355
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 355

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Glu Thr Val		
1	5	10
Lys Ile Thr Cys Ser Gly Gly Asp Ser Ser Ser His Tyr Tyr Gly Trp		
20	25	30
Phe Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Ser		
35	40	45
Asp Asp Glu Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala		
50	55	60
Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp		
65	70	80
Glu Ala Ile Tyr Phe Cys Gly Ala Tyr Asp Gly Ser Thr Tyr Thr Asn		
85	90	95
Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu		
100	105	

<210> SEQ ID NO 356
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 356

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu		
1	5	10
Ser Leu Arg Leu Ser Cys Glu Ala Ser Gly Phe Thr Phe Ser Ser Asp		
20	25	30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		
35	40	45
Ser Gly Ile Ser Ser Gly Ser Ser Thr Tyr Tyr Gly Gly Ser Val Lys		
50	55	60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu		
65	70	80
Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
85	90	95

-continued

Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
 100 105 110

Val Thr Val Ser Ser
 115

<210> SEQ ID NO 357
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 357

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Ser Pro Gly Glu Thr Val
 1 5 10 15

Glu Ile Thr Cys Ser Gly Gly Ser Asp Ser Ser Tyr Tyr Tyr Gly Trp
 20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Ser
 35 40 45

Asp Asn Lys Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60

Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
 65 70 75 80

Glu Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Tyr Ser Thr Tyr Thr Asn
 85 90 95

Pro Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 358
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 358

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Arg Phe Ser Ser Tyr
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Val Asn Ser Leu Thr Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
 85 90 95

Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
 100 105 110

Val Thr Val Ser Ser
 115

<210> SEQ ID NO 359
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 359

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Glu Thr Val
 1 5 10 15
 Glu Ile Thr Cys Ser Gly Gly Asp Ser Ser Ser His Tyr Tyr Gly Trp
 20 25 30
 Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Ser
 35 40 45
 Asp Asp Glu Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60
 Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
 65 70 75 80
 Glu Ala Ile Tyr Tyr Cys Gly Ala Tyr Asp Gly Ser Thr Tyr Ala Asn
 85 90 95
 Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 360

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 360

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45
 Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
 65 70 75 80
 Leu Gln Val Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 361

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 361

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15
 Lys Ile Thr Cys Ser Gly Gly Gly Tyr Tyr Ser Thr Tyr Tyr Gly Trp
 20 25 30
 Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
 35 40 45
 Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60

-continued

Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
65 70 75 80

Glu Ala Val Tyr Tyr Cys Ala Gly Tyr Asp Gly Arg Thr Tyr Ile Asn
85 90 95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 362
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 362

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Arg Phe Ser Ser His
20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Gly Ile Ser Ser Gly Gly Asp Thr Tyr Tyr Val Asp Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Val Asn Ser Leu Thr Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
85 90 95

Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
100 105 110

Val Thr Val Ser Ser
115

<210> SEQ ID NO 363
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 363

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Glu Thr Val
1 5 10 15

Lys Ile Thr Cys Ser Gly Gly Asp Ser Ser Ser His Tyr Tyr Gly Trp
20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile Tyr Ser
35 40 45

Asp Asp Glu Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
50 55 60

Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
65 70 75 80

Glu Ala Ile Tyr Tyr Cys Gly Ala Tyr Asp Gly Ser Thr Tyr Ala Asn
85 90 95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 364
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

-continued

```

<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 364

Asp Val Gln Leu Val Asp Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5              10              15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25              30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
35          40              45
Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
50          55              60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
65          70              75              80
Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90              95
Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
100         105         110
Leu Val Thr Val Ser Ser
115

```

```

<210> SEQ ID NO 365
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 365

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1          5              10              15
Lys Ile Thr Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr Gly Trp
20          25              30
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
35          40              45
Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
50          55              60
Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
65          70              75              80
Glu Ala Val Tyr Phe Cys Gly Ala Tyr Asp Gly Arg Thr Tyr Ile Asn
85          90              95
Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100         105

```

```

<210> SEQ ID NO 366
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 366

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5              10              15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25              30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
35          40              45

```

-continued

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 367
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 367

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15

Lys Ile Thr Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr Gly Trp
 20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
 35 40 45

Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60

Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
 65 70 75 80

Glu Ala Val Tyr Phe Cys Gly Ala Tyr Asp Gly Arg Thr Tyr Ile Asn
 85 90 95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 368
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 368

Asp Leu Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

-continued

<210> SEQ ID NO 369
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 369

```

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15
Lys Ile Ile Cys Ser Gly Gly Asp Tyr Tyr Ser Thr Tyr Tyr Gly Trp
20          25          30
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
35          40          45
Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
50          55          60
Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
65          70          75          80
Glu Ala Val Tyr Phe Cys Gly Ala Tyr Asp Gly Arg Thr Tyr Ile Asn
85          90          95
Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100         105

```

<210> SEQ ID NO 370
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 370

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
35          40          45
Ala Gly Ile Ser Ala Gly Gly Ser Asp Ala Tyr Tyr Ile Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
100         105         110
Leu Val Thr Val Ser Ser
115

```

<210> SEQ ID NO 371
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 371

```

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1           5           10           15

```


-continued

Lys Ile Thr Cys Ser Gly Gly Asp Trp Tyr Ser Thr Tyr Tyr Gly Trp
 20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
 35 40 45

Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60

Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
 65 70 75 80

Glu Ala Val Tyr Tyr Cys Ala Gly Tyr Asp Gly Arg Thr Tyr Ile Asn
 85 90 95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 372
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 372

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 373
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 373

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15

Lys Ile Thr Cys Ser Gly Gly Gly Tyr Tyr Ser Thr Tyr Tyr Gly Trp
 20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
 35 40 45

Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60

Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
 65 70 75 80

Glu Ala Val Tyr Tyr Cys Ala Gly Tyr Asp Ala Arg Thr Tyr Ile Asn
 85 90 95

-continued

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 374
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 374

Asp Val Gln Leu Val Glu Ser Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15
Ser Leu Arg Leu Ser Cys Glu Ala Ser Gly Phe Thr Phe Ser Ser Asn
20 25 30
Ala Met Ser Trp Ala Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ala Gly Ile Ser Ser Gly Ser Asp Thr Tyr Tyr Gly Asp Ser Val Lys
50 55 60
Gly Arg Leu Thr Ile Ser Arg Asp Asn Ser Lys Asn Ile Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr Leu
100 105 110
Val Thr Val Ser Ser
115

<210> SEQ ID NO 375
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 375

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1 5 10 15
Lys Ile Thr Cys Ser Gly Gly Ile Tyr Ser Ser Tyr Tyr Tyr Ala Trp
20 25 30
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Ser
35 40 45
Asp Asp Lys Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Ala
50 55 60
Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Ala Glu Asp
65 70 75 80
Glu Ala Val Tyr Phe Cys Gly Gly Tyr Asp Gln Ser Ser Tyr Thr Asn
85 90 95
Pro Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 376
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 376

-continued

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Asn
          20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
          35          40          45
Ala Gly Ile Ser Ala Gly Asp Ser Asp Thr Tyr Tyr Pro Ala Ser Val
          50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Pro Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Thr Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95
Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
          100          105          110
Leu Val Thr Val Ser Ser
          115

```

```

<210> SEQ ID NO 377
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 377

```

```

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1          5          10          15
Lys Ile Ala Cys Ser Gly Gly Ser Tyr Tyr Ser Tyr Tyr Tyr Gly Trp
          20          25          30
Tyr Gln Gln Lys Ser Pro Gly Ser Ala Leu Val Thr Val Ile Tyr Ser
          35          40          45
Asp Asp Lys Arg Pro Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser Ala
          50          55          60
Ser Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Arg Ala Glu Asp
65          70          75          80
Glu Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Tyr Ser Ser Tyr Thr Asn
          85          90          95
Asp Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
          100          105

```

```

<210> SEQ ID NO 378
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 378

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
          20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
          35          40          45
Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
          50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
65          70          75          80

```

-continued

Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 379

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 379

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1 5 10 15

Lys Ile Thr Cys Ser Gly Gly Asp Trp Tyr Ser Thr Tyr Tyr Gly Trp
20 25 30

Tyr Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
35 40 45

Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
50 55 60

Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
65 70 75 80

Glu Ala Val Tyr Tyr Cys Ala Gly Tyr Asp Gly Arg Thr Tyr Ile Asn
85 90 95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100 105

<210> SEQ ID NO 380

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 380

Asp Val Gln Leu Val Glu Ser Gly Gly Ala Val Val Arg Pro Gly Glu
1 5 10 15

Thr Leu Arg Leu Ser Cys Thr Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Gly Ile Ser Ala Ser Gly Ser Asp Thr Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Ser Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Arg Met Ser Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 381

<211> LENGTH: 106

<212> TYPE: PRT

-continued

```

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 381

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1          5          10         15

Lys Ile Thr Cys Ser Gly Gly Gly Thr Ser Ser Tyr Tyr Gly Trp Tyr
20         25         30

Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile His Ser Asp
35         40         45

Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala Ser
50         55         60

Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Gln Val Glu Asp Glu
65         70         75         80

Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Tyr Thr Thr Tyr Val Asn Thr
85         90         95

Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
100        105

```

```

<210> SEQ ID NO 382
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 382

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10         15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20         25         30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
35         40         45

Ala Gly Ile Ser Ala Gly Gly Ser Asp Thr Tyr Tyr Ile Asp Ser Val
50         55         60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
65         70         75         80

Leu Gln Met Asn Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85         90         95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
100        105        110

Leu Val Thr Val Ser Ser
115

```

```

<210> SEQ ID NO 383
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 383

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
1          5          10         15

Lys Ile Thr Cys Ser Gly Gly Gly Tyr Tyr Ser Thr Tyr Tyr Gly Trp
20         25         30

Tyr Gln Gln Arg Ser Pro Gly Ser Ala Pro Val Thr Val Ile His Ser
35         40         45

```

-continued

Asp Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala
 50 55 60

Ser Gly Ser Ala Ala Thr Leu Thr Ile Thr Gly Val Arg Val Glu Asp
 65 70 75 80

Glu Ala Val Tyr Tyr Cys Ala Gly Tyr Asp Gly Arg Thr Tyr Leu Asn
 85 90 95

Thr Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 384
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 384

Asp Val Gln Leu Val Glu Ser Gly Gly Ala Val Val Arg Pro Gly Glu
 1 5 10 15

Thr Leu Arg Leu Ser Cys Thr Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Gly Ile Ser Ala Ser Gly Ser Asp Thr Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Ser Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Arg Met Ser Ser Leu Thr Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Thr Trp Asn His Leu Phe Asp Tyr Trp Gly Leu Gly Thr
 100 105 110

Leu Val Thr Leu Ser Ser
 115

<210> SEQ ID NO 385
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 385

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Pro Gly Glu Thr Val
 1 5 10 15

Lys Ile Thr Cys Ser Gly Gly Gly Thr Ser Ser Tyr Tyr Gly Trp Tyr
 20 25 30

Gln Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile His Ser Asp
 35 40 45

Asp Lys Arg Pro Ser Asp Ile Pro Ser Arg Phe Ser Gly Ser Ala Ser
 50 55 60

Gly Ser Thr Ala Thr Leu Thr Ile Thr Gly Val Gln Ala Glu Asp Glu
 65 70 75 80

Ala Val Tyr Tyr Cys Gly Gly Tyr Asp Tyr Thr Thr Tyr Val Asn Thr
 85 90 95

Phe Gly Ala Gly Thr Thr Leu Thr Val Leu
 100 105

-continued

```

<210> SEQ ID NO 386
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 386

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Ser
          20          25          30

Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
          35          40          45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Thr Tyr Tyr Ala Asp Ser Val
          50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95

Ala Lys Glu Asn Asp Arg Tyr Arg Phe Phe Asp Tyr Trp Gly Leu Gly
          100          105          110

Thr Leu Val Thr Val Ser Ser
          115

```

```

<210> SEQ ID NO 387
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 387

```

```

Glu Thr Val Leu Thr Gln Ser Pro Gly Ile Leu Ser Leu Ser Pro Gly
1          5          10          15

Glu Arg Ala Thr Leu Thr Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
          20          25          30

Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
          35          40          45

Tyr Gly Ala Ser Ser Arg Ala Ala Gly Ile Pro Asp Arg Phe Ser Gly
          50          55          60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65          70          75          80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
          85          90          95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
          100          105

```

```

<210> SEQ ID NO 388
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 388

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
          20          25          30

```

-continued

Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45
 Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110
 Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 389
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 389

Lys Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Thr Cys Arg Ala Asp Gln Ser Val Tyr Thr Tyr
 20 25 30
 Leu Ala Trp Tyr Gln Glu Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Asp Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
 85 90 95
 Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 390
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 390

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45
 Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Tyr Trp Gly Leu Gly

-continued

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Thr Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
 20 25 30

Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
 85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 394
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 394

Asp Val Gln Leu Val Glu Ser Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 395
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 395

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Asn Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
 20 25 30

Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Glu Ser

-continued

65	70	75	80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro			
	85	90	95
Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys			
	100	105	

<210> SEQ ID NO 396
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 396

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu			
1	5	10	15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr			
	20	25	30
Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val			
	35	40	45
Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val			
	50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr			
65	70	75	80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys			
	85	90	95
Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Leu Gly			
	100	105	110
Thr Leu Val Thr Val Ser Ser			
	115		

<210> SEQ ID NO 397
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 397

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly			
1	5	10	15
Glu Arg Ala Thr Leu Asn Cys Arg Ala Ser Gln Ser Val Tyr Ser Tyr			
	20	25	30
Leu Ala Trp Tyr Gln Glu Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile			
	35	40	45
Tyr Gly Ala Ser Thr Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly			
	50	55	60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Glu Ser			
65	70	75	80
Asp Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro			
	85	90	95
Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys			
	100	105	

<210> SEQ ID NO 398
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 398

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Ser
 20 25 30

Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Thr Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95

Ala Lys Glu Asn Asn Met Tyr Arg Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 399

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 399

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Arg Leu Leu Ile
 35 40 45

His Ala Ala Arg Asn Arg Ala Ala Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
 85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 400

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 400

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val

-continued

50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr		
65	70	75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys		
	85	90 95
Ala Lys Glu Asp Asn Arg Tyr Arg Phe Phe Asp Tyr Trp Gly Leu Gly		
	100	105 110
Ala Leu Val Thr Val Ser Ser		
	115	

<210> SEQ ID NO 401
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 401

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly
1 5 10 15
Glu Arg Thr Thr Leu Thr Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
20 25 30
Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45
Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
85 90 95
Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 402
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 402

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30
Asp Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35 40 45
Ser Leu Ile Ser Gly Ser Gly Glu Ile Thr Tyr Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Tyr Trp Gly Leu Gly
100 105 110
Thr Leu Val Thr Val Ser Ser
115

-continued

<210> SEQ ID NO 403
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 403

Glu Thr Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1 5 10 15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
20 25 30
Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45
Tyr Ala Ala Ser Thr Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
85 90 95
Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 404
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 404

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr
20 25 30
Ala Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35 40 45
Ser Arg Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Lys Gln Tyr Asp Trp Asn Gly Phe Phe Asp Tyr Trp Gly Leu Gly
100 105 110
Ala Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 405
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 405

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1 5 10 15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys

-continued

20	25	30
Leu Ala Trp His Gln Gln Lys	Pro Gly Gln Ala Pro Arg Leu Leu Ile	
35	40	45
Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly		
50	55	60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Gly Leu Gln Thr		
65	70	75
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro		
85	90	95
Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys		
100	105	

<210> SEQ ID NO 406
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 406

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu	
1	15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser His	
20	30
Ala Met Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val	
35	45
Ser Arg Ile Asn Ser Gly Gly Gly Gly Thr Asp Tyr Ala Glu Ser Val	
50	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Glu Asn Thr Leu Tyr	
65	80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys	
85	95
Ala Lys Gln Tyr Asp Trp Asn Gly Phe Phe Asp Tyr Trp Gly Leu Gly	
100	110
Ala Leu Val Thr Val Ser Ser	
115	

<210> SEQ ID NO 407
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 407

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly	
1	15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys	
20	30
Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile	
35	45
Tyr Asp Ala Thr Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly	
50	60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Thr	
65	80
Glu Asp Ser Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Tyr Trp Pro Pro	
85	95

-continued

Tyr Arg Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 408
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 408

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Arg Pro Gly Glu
1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Thr Asn Tyr
20 25 30
 Ala Val Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35 40 45
 Ser Leu Ile Ser Val Thr Gly Thr Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Ser Thr Leu Tyr Leu
65 70 75 80
 Gln Met Asn Gly Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
 Lys Val Thr Trp Lys Asn Val Phe Asp Tyr Trp Gly Leu Gly Thr Leu
100 105 110
 Val Thr Val Ser Ser
115

<210> SEQ ID NO 409
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 409

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Val Ser Pro Gly
1 5 10 15
 Glu Arg Ala Thr Phe Ser Cys Arg Ala Ser Gln Thr Val Gly Ser Lys
20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45
 Tyr Asp Ala Asn Thr Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln His Thr Asp Trp Pro Pro
85 90 95
 Tyr Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 410
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 410

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Arg Pro Gly Glu

-continued

1	5	10	15
Ser Leu Arg	Leu Ser Cys Ala Ala	Ser Gly Phe Thr	Phe Thr Asn Tyr
	20	25	30
Ala Val Thr	Trp Val Arg Gln Ala Pro	Gly Glu Gly	Leu Glu Trp Val
	35	40	45
Ser Leu Ile	Ser Val Thr Gly Thr Thr	Tyr Tyr Ala Asp	Ser Val Lys
	50	55	60
Gly Arg Phe	Thr Ile Ser Arg Asp Asn Ser	Lys Ser Thr Leu	Tyr Leu
	65	70	75
Gln Met Asn	Gly Leu Arg Ala Glu Asp	Thr Ala Val Tyr	Tyr Cys Ala
	85	90	95
Lys Val Thr	Trp Lys Asn Val Phe Asp	Tyr Trp Gly	Leu Gly Thr Leu
	100	105	110
Val Thr Val	Ser Ser		
	115		

<210> SEQ ID NO 411
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 411

Glu Ile Val	Leu Thr Gln Ser Pro	Gly Thr Leu Ser	Val Ser Pro Gly
1	5	10	15
Glu Arg Ala	Thr Phe Ser Cys Arg Ala	Ser Gln Thr Val	Gly Ser Lys
	20	25	30
Leu Ala Trp	Tyr Gln Gln Lys Pro	Gly Gln Ala Pro	Arg Leu Leu Ile
	35	40	45
Tyr Asp Ala	Asn Thr Arg Ala Thr	Gly Ile Pro Ala	Arg Phe Ser Gly
	50	55	60
Ser Arg Ser	Gly Thr Glu Phe Thr	Leu Thr Ile Ser	Ser Leu Gln Ser
	65	70	75
Glu Asp Phe	Ala Val Tyr Tyr Cys	Gln Gln His Thr	Asp Trp Pro Pro
	85	90	95
Tyr Thr Phe	Gly Gly Gly Thr Lys	Val Glu Ile Lys	
	100	105	

<210> SEQ ID NO 412
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 412

Asp Val Gln	Leu Val Glu Ser Gly	Gly Gly Val Val	Arg Pro Gly Glu
1	5	10	15
Ser Leu Arg	Leu Ser Cys Ala Ala	Ser Gly Phe Thr	Phe Ser Asp His
	20	25	30
Asp Met Ser	Trp Val Arg Gln Gly	Pro Gly Glu Gly	Leu Glu Trp Val
	35	40	45
Ala Gly Ile	Ser Leu Arg Gly Gly	Val Thr Trp Tyr	Ala Asp Ser Val
	50	55	60
Lys Gly Arg	Phe Thr Ile Ser Arg	Asp Asn Ser Lys	Asn Thr Leu Tyr
	65	70	75

-continued

Leu Arg Leu Phe Ser Leu Arg Thr Glu Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95

Ala Arg Glu Ser Trp Asn Thr Phe Phe Asp Tyr Trp Gly Leu Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 413
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 413

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Thr Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Val Arg Ser Asn
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Asn Gly Pro Pro
 85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 414
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 414

Asp Val Gln Leu Val Glu Ser Glu Gly Gly Val Val Arg Pro Gly Glu
 1 5 10 15

Ser Leu Arg Leu Ser Cys Glu Ala Ser Gly Phe Thr Phe Arg Asn Ser
 20 25 30

Tyr Met Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
 35 40 45

Ser Thr Ile Ser Asp Thr Gly Asp Thr Ala Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Ser Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95

Ala Lys Ser Trp Ile Trp Trp Thr Phe Phe Asp Tyr Trp Gly Leu Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 415
 <211> LENGTH: 103
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 415

Ala Leu Thr Gln Pro Ala Ser Val Ser Ala Asn Leu Gly Gly Thr Val
 1 5 10 15
 Glu Ile Thr Cys Ser Gly Gly Asn Ser Asn His Tyr Gly Trp Tyr Gln
 20 25 30
 Gln Lys Ser Pro Gly Ser Ala Pro Val Thr Leu Ile Tyr Ala Asp Thr
 35 40 45
 Asn Arg Pro Ser Asn Ile Pro Ser Arg Phe Ser Gly Ser Thr Ser Gly
 50 55 60
 Ser Thr Ser Thr Leu Thr Ile Thr Gly Val Gln Ala Glu Asp Glu Ala
 65 70 75 80
 Val Tyr Tyr Cys Gly Gly Ser Ser Thr Gly Asp Gly Ile Phe Gly Ala
 85 90 95
 Gly Thr Thr Leu Thr Val Leu
 100

<210> SEQ ID NO 416

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 416

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
 1 5 10 15
 Thr Ala Arg Ile Thr Cys Ser Gly Gly Ala Tyr Ser Ser Tyr Tyr Tyr
 20 25 30
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
 35 40 45
 Ser Asp Ser Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
 50 55 60
 Ser Ser Gly Thr Thr Val Thr Leu Thr Ile Ser Gly Val Gln Ala Glu
 65 70 75 80
 Asp Glu Ala Asp Tyr Tyr Cys Gly Gly Tyr Asp Gln Ser Ser Tyr Thr
 85 90 95
 Asn Pro Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 417

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 417

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Thr Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
 50 55 60

-continued

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 418
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 418

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Thr Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 419
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 419

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Tyr Thr Tyr
20 25 30

Leu Ala Trp Tyr Gln Glu Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Asp Arg Pro Pro
85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 420
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 420

```

Asp Val Gln Leu Leu Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe
20          25          30
Ala Leu Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35          40          45
Ser Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
50          55          60
Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
65          70          75          80
Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
85          90          95
Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu
100         105         110
Gly Thr Leu Val Thr Val Ser Ser
115         120

```

<210> SEQ ID NO 421

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 421

```

Asp Val Gln Leu Leu Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Asn Phe
20          25          30
Ala Val Thr Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35          40          45
Ser Thr Ile Gly Ser Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
50          55          60
Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
65          70          75          80
Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
85          90          95
Asp Ser Thr Val Ser Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu
100         105         110
Gly Thr Leu Val Thr Val Ser Ser
115         120

```

<210> SEQ ID NO 422

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 422

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr
20          25          30
Ala Leu Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35          40          45

```

-continued

```

Ser Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 50          55          60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
65          70          75          80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
          85          90          95

Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu
100        105        110

Gly Thr Leu Val Thr Val Ser Ser
115        120

```

```

<210> SEQ ID NO 423
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 423

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ile Tyr
20        25        30

Ala Val Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35        40        45

Ser Thr Ile Gly Ala Asp Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
50          55          60

Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln
65          70          75          80

Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys
85          90          95

Asp Ser Thr Val Gly Trp Ser Gly Asp Phe Phe Asp Tyr Trp Gly Leu
100        105        110

Gly Thr Leu Val Thr Val Ser Ser
115        120

```

```

<210> SEQ ID NO 424
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 424

```

```

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20        25        30

Asp Leu Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35        40        45

Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95

Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Leu Gly
100        105        110

```

-continued

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 425
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 425

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Glu
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30
Asp Val Asn Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35 40 45
Ser Leu Ile Ser Gly Ser Gly Glu Ile Ile Tyr Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Lys Glu Asn Asn Arg Tyr Arg Phe Phe Asp Asp Trp Gly Leu Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 426
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 426

Gly Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser
1 5 10 15
Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp
20 25 30
Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro
35 40 45
Val Lys Ala Gly Val Glu Thr Thr Pro Ser Lys Gln Ser Asn Asn
50 55 60
Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys
65 70 75 80
Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val
85 90 95
Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
100 105

547

What is claimed is:

1. An isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide, wherein the antibody comprises:

- (a) a heavy chain variable (VH) domain comprising:
 - (i) an HVR-H1 sequence comprising the amino acid sequence of SYDMN (SEQ ID NO: 270), SYDVN (SEQ ID NO:221), or SYDLN (SEQ ID NO:313);
 - (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIIYYADSVKG (SEQ ID NO: 156); and
 - (iii) an HVR-H3 sequence comprising the amino acid sequence of ENNRYRFFDD (SEQ ID NO: 157); and
- (b) a light chain variable (VL) domain comprising:
 - (i) an HVR-L1 sequence comprising the amino acid sequence of RASQSVYTYLA (SEQ ID NO:158);
 - (ii) an HVR-L2 sequence comprising the amino acid sequence of GASSRAT (SEQ ID NO:159); and
 - (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160).

2. The antibody of claim 1, wherein the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO: 249, 133, 260, or 329; and the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO: 134, 250, or 251.

3. The antibody of claim 1, wherein:

- (a) the VH domain comprises the amino acid sequence of SEQ ID NO: 133, and the VL domain comprises the amino acid sequence of SEQ ID NO:134;
- (b) the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:134;
- (c) the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:134;
- (d) the VH domain comprises the amino acid sequence of SEQ ID NO: 133, and the VL domain comprises the amino acid sequence of SEQ ID NO:250;
- (e) the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:250;
- (f) the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:250;
- (g) the VH domain comprises the amino acid sequence of SEQ ID NO: 133, and the VL domain comprises the amino acid sequence of SEQ ID NO:251;
- (h) the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:251;
- (i) the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:251;
- (j) the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:134;
- (k) the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; or
- (l) the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:251.

4. The antibody of claim 1, wherein the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide.

548

5. The antibody of claim 1, wherein the antibody enhances activation of a dendritic cell expressing a human SIRP- α polypeptide.

6. The antibody of claim 1, wherein the antibody inhibits in vivo growth of a tumor that expresses CD47.

7. The antibody of claim 1, wherein the antibody does not prevent interactions between a CD47-expressing cell and a T cell.

8. The antibody of claim 1, wherein the antibody is a monoclonal antibody.

9. The antibody of claim 1, wherein the antibody is a scFv-Fc antibody.

10. The antibody of claim 1, wherein the antibody comprises a light chain constant region comprising the amino acid sequence of SEQ ID NO:325, 326, or 426.

11. The antibody of claim 1, wherein the antibody comprises an Fc region.

12. The antibody of claim 11, wherein the antibody comprises an antibody heavy chain polypeptide that comprises the VH domain and a constant region comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 320-324.

13. The antibody of claim 11, wherein the Fc region is a human Fc region selected from the group consisting of an IgG1 Fc region, an IgG2 Fc region, and an IgG4 Fc region.

14. The antibody of claim 11, wherein the Fc region comprises a human IgG1 Fc region comprising one or more mutations selected from the group consisting of L234A, L235A, G237A, and N297A, according to EU numbering.

15. The antibody of claim 11, wherein the Fc region comprises a human IgG2 Fc region comprising one or more mutations selected from the group consisting of A330S, P331S and N297A, according to EU numbering.

16. The antibody of claim 11, wherein the Fc region comprises a human IgG4 Fc region comprising one or more mutations selected from the group consisting of S228P, E233P, F234V, L235A, L235E, delG236, and N297A, according to EU numbering.

17. The antibody of claim 1, wherein the antibody is an antibody fragment selected from the group consisting of a Fab, F(ab')₂, Fab'-SH, Fv, and scFv fragment.

18. The antibody of claim 1, wherein the antibody is conjugated to a cytotoxic agent or label.

19. The antibody of claim 1, wherein the antibody is a bispecific antibody.

20. The antibody of claim 19, wherein the antibody comprises a first antigen binding domain comprising the VH and VL domain that bind an extracellular domain of a human SIRP- α polypeptide and a second antigen binding domain comprising a second VH domain and second VL domain that bind an antigen expressed by a cancer cell.

21. The antibody of claim 20, wherein the antigen expressed by the cancer cell is selected from the group consisting of CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PD-L1, PTK7, STEAP1, Endothelin B Receptor, mesothelin, EGFRVIII, ENPP3, SLC44A4, GNCMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le^x, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-

549

4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7.

22. The antibody of claim 1, wherein the antibody is a human antibody.

23. A polynucleotide encoding the antibody of claim 1.

24. A vector comprising the polynucleotide of claim 23.

25. A host cell comprising the polynucleotide of claim 23.

26. A method of producing an antibody, the method comprising culturing the host cell of claim 25 such that the antibody that binds the extracellular domain of a human SIRP- α polypeptide is produced.

27. The method of claim 26, further comprising recovering the antibody from the host cell.

28. The antibody of claim 1, wherein the antibody binds an extracellular domain of a human SIRP- α v1 polypeptide comprising the amino acid sequence of

(SEQ ID NO: 5)
EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIY
NQKEGHFPRVTTVSDLTNRNNMDFSIRIGNITPADAGTYTCVKFRKGSPD
DVEFKSGAGTELSVRAKPS.

29. The antibody of claim 1, wherein the antibody binds an extracellular domain of a human SIRP- α v2 polypeptide comprising the amino acid sequence of

(SEQ ID NO: 6)
EEELQVIQPDKSVSAAGESAILHCTVTSLIPVGPIQWFRGAGPARELI
YNQKEGHFPRVTTVSESTKRENMDFSISISNITPADAGTYTCVKFRKGS
PDTEFKSGAGTELSVRAKPS.

30. The antibody of claim 1, wherein the antibody is capable of binding an extracellular domain of a human

550

SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPRV

TTVSDLTNRNNMDFSIRIGNIT-

5 PADAGTYTCVKFRKGSPDDVEFKSGAG-

TELSVRAKPS (SEQ ID NO:5) and is capable of binding an extracellular domain of a human SIRP- α v2 polypeptide comprising the amino acid sequence of

(SEQ ID NO: 6)
EEELQVIQPDKSVSAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIY
NQKEGHFPRVTTVSESTKRENMDFSISISNITPADAGTYTCVKFRKGSPD
TEFKSGAGTELSVRAKPS.

31. The antibody of claim 1, wherein the antibody binds an extracellular domain of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:11.

32. The antibody of claim 1, wherein the antibody binds an extracellular domain of a murine SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:7.

33. The antibody of claim 1, wherein the antibody binds the extracellular domain of a human SIRP- β polypeptide comprising the amino acid sequence of SEQ ID NO:13.

34. The antibody of claim 1, wherein the antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

35. A host cell comprising the vector of claim 24.

36. A method of producing an antibody, the method comprising culturing the host cell of claim 35 such that the antibody that binds the extracellular domain of a human SIRP- α polypeptide is produced.

37. The method of claim 36, further comprising recovering the antibody from the host cell.

* * * * *